Package ‘openfactsheet’

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Author	Andrés A. Bagnasco, OpenFactsheet contributors
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Built with	Typst 0.15.0, from the source code of this version; do not edit by hand

Every metric, function and decision in this manual has a stable link target, e.g. `openfactsheet-manual.pdf#nameddest=metric-sharpe_ratio`. OpenFactsheet explains measurements; it does not give investment advice.

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Overview and usage

Introduction

OpenFactsheet answers one question: *what does this investment's performance actually look like under one transparent, standardized methodology?*

Start here

- Getting started — install, analyze, read a report.
- Methodology — the pipeline, the policy layer and the anti-cherry-picking rules.
- Assumptions — every canonical convention of methodology 0.1.0, with links to the reasoning.

Reference

Exact, auditable specification of everything OpenFactsheet computes and decides:

- Exact formulas — the precise mathematical definition of every metric, as implemented (generated from the registry).
- Notation — symbols, wealth/drawdown and quantile definitions used by all formulas.
- Decision criteria — every rule, in evaluation order: input parsing, validation, missing data, frequency inference, percent/decimal detection, data-quality findings, applicability, confidence and status, benchmark alignment, comparison, interpretation, report structure, determinism.
- Policies, thresholds and constants — every numeric threshold and canonical constant, read directly from the code (generated).
- JSON report schema — every field of the report.
- Metric catalog — all canonical metrics (generated from the registry). Each metric links to a page in `interpretation/` answering the twelve documentation questions: what it measures, formula, inputs, direction, plain and professional interpretation, assumptions, when it misleads, minimum data, reference implementations, sources and the OpenFactsheet convention — followed by its calculation criteria: when a value is produced, exactly when it is undefined, the confidence rules that apply to it, and the assumptions recorded on every result.
- Methodology decisions — one note per convention where definitions vary.
- Reference implementations & validation — how results are cross-checked against PerformanceAnalytics, empyrical and QuantStats, and every known convention difference.
- Reference manual (PDF) — all of the above as one CRAN-style PDF, generated from the code, with a link target for every metric, function and decision (how it is built). Every result in every report links to the topic that defines it (decision 0035).

Principles in one paragraph

Compute broadly and never hide an unfavorable metric; never tune periods, thresholds or confidence levels per fund; make every assumption deterministic, versioned and visible; never return an unexplained NaN; separate calculation from applicability, confidence and presentation; and distinguish what was observed from what is estimated or model-dependent. OpenFactsheet explains measurements — it does not give investment advice.

Important information

For information and education only — not investment advice. Investing involves risk; past performance does not predict future results. Verify results and consult a qualified financial adviser. See the full disclaimer, the metric attribution and the third-party notices. OpenFactsheet is a public alpha: interfaces, metrics and documentation may evolve; validate results before investment decisions.

Getting started

Install

```
git clone https://github.com/bagnascotrader/openfactsheet.git
cd openfactsheet
pip install -e ".[excel]"
```

Core dependencies: NumPy, pandas, SciPy, statsmodels. Excel input needs the `excel` extra (openpyxl).

Analyze a return history

```
import pandas as pd
import openfactsheet as ofs

returns = pd.read_csv("examples/monthly_table.csv")  # wide Year x Jan..Dec table
report = ofs.analyze(returns, name="Example fund")
print(report.summary())
```

Accepted inputs:

Input	Notes
`list`, `tuple`, `numpy.ndarray`	Undated: frequency is unknown, so nothing is annualized unless you declare `frequency="monthly"` (recorded as a declaration).
`pandas.Series`	A `DatetimeIndex` or `PeriodIndex` lets OpenFactsheet infer frequency from calendar spacing.
`pandas.DataFrame`	One return column ⟶ `analyze`; several columns ⟶ `compare`. A date column is detected by name or content.
CSV / Excel path	Same layouts as DataFrames, including wide monthly tables.
Pasted text	A multi-line string, e.g. a monthly table copied from a PDF.
Screenshot or scanned PDF	`ofs.analyze("table.png")` with `pip install "openfactsheet[ocr]"`: recognized locally (no network) and accepted only when every year reconciles with its printed YTD; otherwise refused with the table as read. `ofs.load(path, ocr_engine="tesseract")` uses Tesseract instead.
PDF factsheet	`ofs.analyze("factsheet.pdf")` with `pip install "openfactsheet[pdf]"`: the monthly-return table is read from the embedded text, checked against its YTD column, and refused (with the reconstructed table) if anything is inconsistent. Pages without embedded text are recognized by OCR when the `ocr` extra is installed.

Percent vs decimal scale is detected deterministically. If the values are genuinely ambiguous, OpenFactsheet raises `AmbiguousReturnScaleError` instead of guessing; declare `return_scale="percent"` or `"decimal"`.

Read a report

```
report.input_summary  # what OpenFactsheet believes it received
report.returns_frame()  # the exact decimal returns analyzed
report.diagnostics  # data-quality findings, smoothing risk
report.metrics  # every canonical metric
report.assumptions  # the methodology (version + conventions)
report.interpretations["sharpe_ratio"].plain
```

Each metric is a `MetricResult`:

```
m = report.metrics["sharpe_ratio"]
m.value  # float, or None when not computable
m.status  # applicable | low_confidence | supplementary | not_applicable | insufficient_data | error
m.confidence  # high | moderate | low | not_assessable (estimation reliability, not quality)
m.reason  # why there is no value
m.warnings  # data-specific caveats
m.assumptions  # e.g. risk-free rate 0%, annualization factor
```

Values are raw `float64`; rounding happens only in rendered reports.

Benchmarks

```
report = ofs.analyze(fund, benchmark=benchmark)
report.benchmark_summary  # aligned observations, dropped periods, overlap
```

Fund and benchmark are aligned on their common periods (monthly data by calendar month, so month-start and month-end dates match). Missing benchmark returns are never filled. Benchmark-relative metrics report their own observation count. Without a benchmark they are `not_applicable` with a reason — OpenFactsheet never assumes a default benchmark.

Compare investments

```
cmp = ofs.compare({"Fund A": fund_a, "Fund B": fund_b})
cmp.table()  # metric x investment values
cmp.track_records()  # history lengths, frequency, smoothing risk
cmp.table(common_period=True)  # same metrics on the common period, if histories differ
```

Comparisons are never ranked.

Factor models

Supply factor returns explicitly (OpenFactsheet never downloads them): a dated DataFrame, CSV/Excel file or table text with one column per factor. A column named `RF` is treated as the risk-free rate and subtracted from fund returns.

```
factors = pd.read_csv("my_factors.csv", index_col="date", parse_dates=True)
# columns: Mkt-RF, SMB, HML, RF  (percent or decimal: one scale is detected for all)
report = ofs.analyze(fund_returns, factors=factors)
report.factor_analysis.fit.loadings  # every loading with Newey-West SE, t, p, 95% CI
report.metrics["factor_alpha"]  # annualized alpha; details hold the interval
```

Only the months where the fund and every factor have a value are used (never filled). Standard errors are Newey-West with a fixed lag rule (decision 0028).

Export

```
report.to_json("factsheet.json")  # deterministic, NaN-free, methodology embedded
report.to_markdown("factsheet.md")
report.to_html("factsheet.html")  # self-contained, with the canonical charts
report.charts()  # {"wealth": "<svg ...>", "underwater": ..., ...}
```

The HTML report contains eight canonical charts (growth of 1, calendar-year returns, monthly heatmap, rolling one-year volatility, underwater chart, return distribution, rolling one-year return and, with a benchmark, growth over the aligned overlap), each with a caption and a data table. Every chart spans the full history and follows the fixed presentation rules of decision 0026; a chart that cannot be drawn for the data is replaced by the reason.

Explore the methodology

```
ofs.methodology().describe()  # canonical conventions
ofs.metric_catalog()  # DataFrame of all metrics and their definitions
```

Follow a number to its definition

Every result links to the page of the reference manual that defines it — formula, inputs, assumptions, when it misleads, and the papers and reference implementations it comes from (decision 0035):

```
report.metrics["sharpe_ratio"].documentation
# '.../openfactsheet-manual.pdf#nameddest=metric-sharpe_ratio'

report.documentation.manual  # the manual of the version that produced the report
report.documentation.catalog  # the metric catalog at that version
```

The Markdown and HTML reports link every metric name, the JSON carries the link on every metric and a `documentation` block, and the terminal report gives the address of the manual. The links are pinned to the version that computed the numbers, so an archived report always points at the conventions it actually used, not at whatever the documentation says later.

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The short notice does not replace context-specific warnings: every report also carries its data-quality findings, confidence labels, assumptions and the reasons a metric is unavailable.

Related documents

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Functions and classes

The public interface of `import openfactsheet as ofs`. Each topic lists the usage, arguments, value and examples, taken from the docstrings of the code.

analyze

Description

Produce the canonical OpenFactsheet report for one return history.

Usage

```
ofs.analyze(returns, benchmark=None, *, factors=None, name=None, return_scale=None, frequency=None)
```

Arguments

returns	A list/tuple/array of returns, a pandas Series (ideally with a DatetimeIndex), a one-column DataFrame, a CSV/Excel path, pasted table text, or a `ReturnSeries` from `load()`. `Any`
benchmark	Optional benchmark returns in any of the same forms. Benchmark-relative metrics are computed only when a benchmark is supplied. `Any`
factors	Optional factor returns, one column per factor (a DataFrame, a CSV/Excel path or table text, e.g. Fama-French factors). A column named `RF` is the risk-free rate and is subtracted from fund returns. Factor-model metrics are computed only when factors are supplied; OpenFactsheet never downloads factor data. `Any`
name	Display name for the report. `str | None`
return_scale	Declare `"decimal"` or `"percent"` only when automatic detection reports ambiguity. Applies to fund, benchmark and factors. `DeclaredScale | None`
frequency	Declare the observation frequency of *undated* values. It is checked against dates when dates exist. Applies to fund, benchmark and factors. `str | ObservationFrequency | None`

Value

A `FactsheetReport` containing every canonical metric with its status, confidence, warnings and assumptions.

Examples

```
>>> report = ofs.analyze(returns)
>>> print(report.summary())
>>> report.metrics["sharpe_ratio"].value
>>> report = ofs.analyze(returns, benchmark=benchmark_returns)
>>> report.to_html("factsheet.html")
```

compare

Description

Analyze several return histories under one identical methodology.

Usage

```
ofs.compare(funds, benchmark=None, *, factors=None, return_scale=None, frequency=None)
```

Arguments

funds	A mapping of name to returns, a DataFrame with one column per investment, a sequence of return series, or a `ReturnTable`. `Mapping[str, Any] | pd.DataFrame | Sequence[Any] | ReturnTable`
benchmark	Optional benchmark returns, used for every investment. `Any`
factors	Optional factor returns, used for every investment. `Any`
return_scale	Declare `"decimal"` or `"percent"` only when automatic detection reports ambiguity. `DeclaredScale | None`
frequency	Declare the observation frequency of *undated* values. `str | ObservationFrequency | None`

Value

A `Comparison` holding the full report of every investment.

Details

Metrics are shown side by side; nothing is ranked. When the histories cover different periods, an additional common-period analysis (the intersection of all histories) is provided so that track-record differences are visible. The same benchmark and factor returns, if supplied, are used for every investment.

Examples

```
>>> comparison = ofs.compare({"Fund A": fund_a, "Fund B": fund_b, "ETF": etf})
>>> print(comparison.summary())
>>> comparison.table(common_period=True)
```

`load`

Description

Parse and normalize input without computing metrics.

Usage

```
ofs.load(source, *, name=None, return_scale=None, frequency=None, ocr_engine=None)
```

Arguments

`source`	Returns in any form `analyze()` accepts, including CSV, Excel, pasted monthly tables, PDF factsheets and screenshots. `Any`
`name`	Display name for a single series. `str | None`
`return_scale`	Declare `"decimal"` or `"percent"` only when automatic detection reports ambiguity. `DeclaredScale | None`
`frequency`	Declare the observation frequency of *undated* values. `str | ObservationFrequency | None`
`ocr_engine`	`"rapidocr"` (default) or `"tesseract"` for images and scanned PDF pages. It never changes an accepted value, because recognized tables must reconcile with their printed totals. `str | None`

Value

A `ReturnSeries` for a single series, or a `ReturnTable` when the source contains several return columns (for example a fund and the benchmark printed in a factsheet).

Details

Inspect `series.input_diagnostics` to see exactly what OpenFactsheet believes it received before analyzing it.

Examples

```
>>> series = ofs.load("monthly_returns.csv")
>>> print(series.input_diagnostics.to_text())
```

`methodology`

Description

The canonical, versioned methodology used by every report.

Usage

```
ofs.methodology()
```

Value

The methodology object; `.version` is its version and `.to_json()` the complete set of canonical conventions embedded in every report.

Examples

```
>>> ofs.methodology().get("tail.var_confidence")
0.95
```

`metric_catalog`

Description

Every canonical metric with its definition, conventions and data requirements.

Usage

```
ofs.metric_catalog()
```

Value

A DataFrame indexed by metric id, one row per canonical metric.

Examples

```
>>> catalog = ofs.metric_catalog()
>>> catalog.loc["sharpe_ratio", "formula"]
```

`Comparison`

Description

Several investments analyzed under one identical methodology, returned by `compare()`.

Details

Nothing is ranked or scored. Each investment keeps its full canonical report.

Fields

`reports`	The full-history `FactsheetReport` of each investment, by name.
`common_period_reports`	Reports restricted to the period all histories share, or `None` when the histories cover the same period or share fewer than two.
`common_period`	First and last date of the common period, or `None`.
`notes`	Explanations of track-record and alignment differences.

Usage

```
x.status_table(*, common_period=False)
x.summary()
x.table(*, common_period=False)
x.to_dict()
x.to_json(indent=2)
x.track_records()
```

Methods and properties

`.status_table(*, common_period=False)` method

Confidence / status labels by investment.

`common_period`	Use the common-period reports instead of the full histories. `bool`

`.summary()` method

Track records and every metric side by side, as plain text.

`.table(*, common_period=False)` method

Raw metric values (rows) by investment (columns).

`common_period`	Use the common-period reports instead of the full histories. `bool`

Value: A DataFrame indexed by metric id with a `name` column and one column per investment; unavailable values are missing (see `status_table`).

`.to_dict()` method

Every report (and common-period report) as JSON-compatible Python objects.

`.to_json(indent=2)` method

The comparison as deterministic JSON.

`indent`	Indentation of the JSON text. `int`

`.track_records()` method

Start, end, length, frequency, smoothing risk and benchmark overlap by investment.

`FactsheetReport`

Description

The canonical OpenFactsheet report, returned by `analyze()`.

Details

Every canonical metric is present, each with a status, confidence, warnings and the assumptions it used. The report is a plain data object; calculations know nothing about rendering, and every renderer (text, Markdown, HTML, JSON) shows the same content. The report and every metric in it link to the documentation of their own version (decision 0035).

Fields

`name`	Display name of the analyzed series.
`returns`	The `ReturnSeries` exactly as analyzed.
`input_summary`	What OpenFactsheet believes it received: detected frequency, return scale, observations, date range, parsing warnings.
`diagnostics`	Data-quality findings computed before any metric (missing periods, stale returns, smoothing risk, ...).
`metrics`	Every canonical metric as a `MetricResult`, indexable by metric id (`report.metrics["sharpe_ratio"]`).
`assumptions`	The versioned methodology (all canonical conventions).
`interpretations`	Plain-language and professional explanations by metric id.
`benchmark_summary`	Alignment of fund and benchmark (overlap, method), or `None` without a benchmark.
`benchmark_input_summary`	Input diagnostics of the benchmark, or `None`.
`provenance`	Package and methodology versions, input hashes, transformations and, for documents, how the table was extracted.
`drawdown_episodes`	Peak-to-recovery drawdown episodes of the wealth path.
`benchmark_returns`	The benchmark series as analyzed, or `None`.
`factor_analysis`	Factor-model regressions when factors were supplied.
`document`	What a source document (PDF, image) says about the fund, or `None`.
`documentation`	Links to the documentation of the version that produced the report: the PDF reference manual, the metric catalog and a topic per metric.

Usage

```
x.charts()
x.documentation
x.methodology_version
x.returns_frame()
x.summary()
x.to_dict()
x.to_html(path=None)
x.to_json(path=None, *, indent=2)
x.to_markdown(path=None)
```

Methods and properties

`.charts()` method

Canonical charts as self-contained SVG strings, keyed by chart id. Charts that cannot be drawn for these data (e.g. a monthly heatmap for weekly returns) are omitted here; `to_html` states why.

`.documentation` property

Links to the documentation of the version that produced this report. `report.documentation.manual` is the PDF reference manual and `report.documentation.metric("sharpe_ratio")` the page defining one metric, which is also on every result as `documentation`.

`.methodology_version` property

Version of the methodology the report was computed with.

`.returns_frame()` method

Exactly the returns OpenFactsheet analyzed (decimal, after normalization).

`.summary()` method

The full canonical report as plain text for a terminal.

`.to_dict()` method

The report as JSON-compatible Python objects (see the JSON report schema).

`.to_html(path=None)` method

The full canonical report as one self-contained HTML page with charts.

`path`	Also write the HTML to this file (UTF-8) when given. `str | Path | None`

Value: The HTML text.

`.to_json(path=None, *, indent=2)` method

The report as deterministic JSON.

`path`	Also write the JSON to this file (UTF-8) when given. `str | Path | None`
`indent`	Indentation of the JSON text. `int`

Value: The JSON text.

`.to_markdown(path=None)` method

The full canonical report as Markdown.

`path`	Also write the Markdown to this file (UTF-8) when given. `str | Path | None`

Value: The Markdown text.

`MetricResult`

Description

Structured outcome of one metric (CLAUDE.md section 10).

Details

`value` is `None` whenever the metric could not be validly computed, and `reason` then says why. A non-finite value can never be stored. Every result links to the page of the reference manual that defines it (decision 0035).

Fields

`id`	Stable metric identifier (e.g. `sharpe_ratio`).
`name`	Display name.
`category`	Report section the metric belongs to.
`value`	Unrounded value (`float64`), or `None` with a `reason`.
`unit`	How to read the value (`decimal_return`, `ratio`, `fraction`, …).
`status`	`applicable`, `low_confidence`, `supplementary`, `not_applicable`, `insufficient_data` or `error`.
`direction`	`higher_is_better`, `lower_is_better`, `context_dependent` or `neutral`.
`confidence`	Reliability of the estimate (`high`, `moderate`, `low`, `not_assessable`), never a judgement of the investment.
`nature`	`observed`, `derived`, `estimated` or `model_dependent`.
`observations`	Number of observations the value is based on (aligned observations for benchmark and factor metrics).
`formula_id`	Versioned formula identifier; it changes when the definition does.
`methodology_version`	Methodology the value was computed under.
`assumptions`	Canonical assumptions used, in words.
`warnings`	Data-quality and confidence warnings.
`reason`	Why there is no value, or why confidence was reduced.
`details`	Supporting quantities (intervals, dates, episode data, …).
`reference_values`	Values from reference implementations, when recorded.
`documentation`	URL of the reference-manual topic defining this metric, for the version that produced the value.

Usage

```
x.documentation
x.has_value
x.to_dict()
```

Methods and properties

`.documentation` property

URL of the manual topic defining this metric: formula, assumptions, references.

`.has_value` property

Whether a value was computed (`value is not None`).

`.to_dict()` method

The result as JSON-compatible Python objects.

`ObservationFrequency`

Description

Observation frequency inferred from calendar spacing (or declared by the caller).

Values

`DAILY`	`daily`
`WEEKLY`	`weekly`
`MONTHLY`	`monthly`
`QUARTERLY`	`quarterly`
`YEARLY`	`yearly`
`IRREGULAR`	`irregular`
`UNKNOWN`	`unknown`

Usage

```
x.is_regular
```

Methods and properties

`.is_regular` property

Whether the frequency has a canonical annualization factor.

`ReturnSeries`

Description

A normalized, immutable return history.

Fields

`name`	Display name of the series.
`values`	Simple decimal returns, `float64`, read-only, no NaN.
`dates`	Period-end timestamps aligned with `values` (`None` if undated).
`frequency`	Observation frequency.
`periods_per_year`	Canonical annualization factor, `None` when annualization is not defensible (irregular/unknown frequency).
`frequency_source`	`"inferred"` from dates, `"declared"` by the caller, or `"none"`.
`source_type`	Where the data came from (`"list"`, `"pandas.Series"`, `"csv"`, ...).
`return_type`	Always `"simple"` in methodology 0.1.
`return_scale`	How percent/decimal scale was established.
`metadata`	Free-form provenance metadata.
`input_diagnostics`	What OpenFactsheet believes it received.

Usage

```
x.n
x.to_pandas()
x.years
```

Methods and properties

`.n` property

Number of observations.

`.to_pandas()` method

The returns OpenFactsheet actually analyzed, as a pandas Series.

`.years` property

Length of history in years, when a periods-per-year factor is defensible.

`ReturnTable`

Description

Several named return series (e.g. columns of a DataFrame, used by `compare`).

Fields

`series`	The `ReturnSeries` by name, in source order.

Exceptions

Description

Errors raised by OpenFactsheet. Input corruption fails early; a metric that cannot be computed never raises but is reported with a status and a reason.

Classes

`AmbiguousReturnScaleError` subclass of `InputError`

Returns could plausibly be either decimals or percentages.

`BenchmarkAlignmentError` subclass of `OpenFactsheetError`

Fund and benchmark returns cannot be aligned without inventing data.

`FactorAlignmentError` subclass of `OpenFactsheetError`

Fund and factor returns cannot be aligned without inventing data.

`FrequencyInferenceError` subclass of `InputError`

Observation frequency could not be established or was declared inconsistently.

`InputError` subclass of `OpenFactsheetError`

The supplied data cannot be interpreted as a return history.

`MetricCalculationError` subclass of `OpenFactsheetError`

Unexpected failure inside a metric calculation (captured as status `error`).

`OCRExtractionError` subclass of `InputError`

Image/PDF extraction failed or produced inadequate confidence (roadmap phase 6).

`OpenFactsheetError` subclass of `Exception`

Base class for all OpenFactsheet errors.

Constants

Description

Version identifiers embedded in every report.

Values

`METHODOLOGY_VERSION`	`0.1.0`
`__version__`	`0.1.0`

Metrics

Every canonical metric of methodology 0.1.0, grouped by report section. Every report computes all of them; one that cannot be computed is reported with a status and a reason, never omitted. For each metric: what it measures, the exact formula, how to read the value, the assumptions, when it misleads, the minimum data, the exact calculation and confidence criteria, reference implementations and sources. Symbols are defined in Notation thresholds in Policies, thresholds and constants.

Return summary

Cumulative return

`cumulative_return` · Return summary · observed · unit `decimal_return` · formula `cumulative_return@v1`

Description

Total compounded return over the full supplied history.

Formula

$$\mathrm{CR} = W_n - 1 = \prod_{t=1}^{n}(1 + r_t) - 1$$

Plain text: `prod_{t=1..n}(1 + r_t) - 1`. Symbols: see Notation.

Usage

```
report.metrics["cumulative_return"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`observed_point`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. How much 1 unit invested at the start would have gained or lost by the end, with all returns reinvested.

Professional. Path-independent summary of realized compounded performance over the full track record; not comparable across different history lengths.

Assumptions

- Returns are reinvested each period.
- Returns are net of whatever fees the supplied series reflects.

When it can mislead

- Histories of different lengths are compared.
- The track record starts at a favorable date (OpenFactsheet always uses the full supplied history, but the supplier chose the inception date).

Minimum data

Any non-empty history. (policy `observed_point`: Observed fact about the supplied history; any non-empty history suffices.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.

2. At least 1 observations (mathematical minimum of policy `observed_point`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence is always **high** (an observed fact about the supplied data).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `cumulative_return@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric compounding of simple returns; no annualization.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Return.cumulative(geometric=TRUE)`	yes	
empyrical-reloaded 0.5.12	`cum_returns_final`	yes	
QuantStats 0.0.81	`stats.comp`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Annualized return (CAGR)

`cagr` · Return summary · derived · unit `decimal_return` · formula `cagr@v1`

Description

Constant yearly growth rate that would produce the observed cumulative return.

Formula

$$\mathrm{CAGR} = \begin{cases} W_n^{P/n} - 1 & W_n > 0 \\ -1 & W_n \le 0 \end{cases}$$

Plain text: `(prod(1 + r_t))^(P / n) - 1, where P = periods per year and n = observations`. Symbols: see Notation.

Usage

```
report.metrics["cagr"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`location`
Canonical parameters	annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The average yearly growth rate, taking compounding and losses into account.

Professional. Geometric mean return annualized with the canonical periods-per-year factor; the standard GIPS-style annualized return. Always below the annualized arithmetic mean when returns vary.

Assumptions

- Observation frequency is correctly established.

When it can mislead

- The history is shorter than one year: annualizing extrapolates a partial year (GIPS prohibits presenting annualized returns for periods under one year).
- The history contains missing periods.

Minimum data

One observation mathematically; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
4. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `cagr@v1`; the version increments if the definition changes.

OpenFactsheet convention

Observation count (not calendar span) with the canonical periods-per-year factor (decision 0004).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Return.annualized(geometric=TRUE)`	yes	
empyrical-reloaded 0.5.12	`annual_return`	yes	
QuantStats 0.0.81	`stats.cagr(periods=12)`	yes	Agrees when periods=12 is passed for monthly data; the default (periods=252) treats every observation as a trading day.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]
- CFA Institute (2020). Global Investment Performance Standards (GIPS) for Firms, provisions on annualized returns. https://www.gipsstandards.org/ [bibliography]

See also

Decision 0004: Annual return: geometric CAGR by observation count; Decision criteria

---

Mean return per period

`arithmetic_mean` · Return summary · estimated · unit `decimal_return` · formula `arithmetic_mean@v1`

Description

Average single-period return.

Formula

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t$$

Plain text: `(1/n) * sum(r_t)`. Symbols: see Notation.

Usage

```
report.metrics["arithmetic_mean"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`location`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The typical return in one period (e.g. one month), ignoring compounding.

Professional. Sample mean; the numerator of Sharpe-type ratios. An estimate of the expected return with standard error roughly sigma / sqrt(n).

Assumptions

- Returns are identically distributed over the history.

When it can mislead

- Returns are volatile: the arithmetic mean overstates compounded growth.
- Few observations are available: sampling error dominates.

Minimum data

One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `arithmetic_mean@v1`; the version increments if the definition changes.

OpenFactsheet convention

Plain sample mean of per-period returns.

Reference implementations

Library	Function	Agrees	Note
NumPy	`mean`	yes	
PerformanceAnalytics 2.0.4 (R)	`mean.arithmetic`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Geometric mean return per period

`geometric_mean_return` · Return summary · derived · unit `decimal_return` · formula `geometric_mean_return@v1`

Description

Constant per-period growth rate that reproduces the cumulative return.

Formula

$$\bar{r}_g = \begin{cases} W_n^{1/n} - 1 & W_n > 0 \\ -1 & W_n \le 0 \end{cases}$$

Plain text: `(prod(1 + r_t))^(1/n) - 1`. Symbols: see Notation.

Usage

```
report.metrics["geometric_mean_return"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`location`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value
Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation
Plain language. The average compounded return per period (e.g. per month).

Professional. Per-period geometric mean; CAGR is its annualized form. Lies below the arithmetic mean by roughly half the variance (volatility drag). Needs no frequency.

Assumptions
- Returns are reinvested each period.

When it can mislead
- Histories of different lengths or frequencies are compared.

Minimum data
One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria
When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `geometric_mean_return@v1`; the version increments if the definition changes.

OpenFactsheet convention
Geometric compounding of simple returns; -100% after a total loss.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`mean.geometric`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

Annualized mean return (arithmetic)
`annualized_arithmetic_mean` · Return summary · estimated · unit `decimal_return` · formula `annualized_arithmetic_mean@v1`

Description

Average per-period return scaled to a year without compounding.

Formula

$$\bar{r}_{\text{ann}} = P\,\bar{r}$$

Plain text: `P * mean(r_t)`. Symbols: see Notation.

Usage

```
report.metrics["annualized_arithmetic_mean"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`location`
Canonical parameters	annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The average return per year if yearly results were simply added up; usually higher than the compounded annual return.

Professional. Linear annualization of the sample mean; the return component of the canonical annualized Sharpe ratio. Differs from CAGR by roughly half the variance.

Assumptions

- Linear time scaling of the mean.

When it can mislead

- Used as a proxy for realized compounded growth.

Minimum data

One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `annualized_arithmetic_mean@v1`; the version increments if the definition changes.

OpenFactsheet convention

mean x periods per year (decision 0004).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Return.annualized(geometric=FALSE)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0004: Annual return: geometric CAGR by observation count; Decision criteria

---

Median return per period

`median_return` · Return summary · estimated · unit `decimal_return` · formula `median_return@v1`

Description

Middle value of the per-period returns.

Formula

$$\operatorname{med}(r) = \begin{cases} r_{((n+1)/2)} & n \text{ odd} \\ \frac{1}{2}\left(r_{(n/2)} + r_{(n/2+1)}\right) & n \text{ even} \end{cases}$$

Plain text: `median(r_t)`. Symbols: see Notation.

Usage

```
report.metrics["median_return"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`location`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. Half of the periods did better than this and half did worse.

Professional. Robust location estimate, insensitive to a few extreme periods. A median well above the mean indicates occasional large losses (negative skew).

Assumptions

None beyond the general inputs.

When it can mislead

- Many periods have identical or zero returns.

Minimum data

One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `median_return@v1`; the version increments if the definition changes.

OpenFactsheet convention

Sample median (average of the two middle values when n is even).

Reference implementations

Library	Function	Agrees	Note
NumPy	`median`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Best period

`best_period` · Return summary · observed · unit `decimal_return` · formula `best_period@v1`

Description

Highest single-period return in the history.

Formula

$$\max_{1 \le t \le n} r_t = r_{(n)}$$

Plain text: `max(r_t)`. Symbols: see Notation.

Usage

```
report.metrics["best_period"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`observed_point`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value
Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation
Plain language. The best single period (e.g. month) the investment had.

Professional. Observed upper extreme; context for the dispersion and tail statistics.

Assumptions
None beyond the general inputs.

When it can mislead
- Histories of different lengths are compared.

Minimum data
Any non-empty history. (policy `observed_point`: Observed fact about the supplied history; any non-empty history suffices.)

Calculation criteria
When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `observed_point`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence is always **high** (an observed fact about the supplied data).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `best_period@v1`; the version increments if the definition changes.

OpenFactsheet convention
Per observation period of the supplied data (not calendar aggregated).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.best`	yes	
NumPy	`max`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision criteria

Worst period
`worst_period` · Return summary · observed · unit `decimal_return` · formula `worst_period@v1`

Description
Lowest single-period return in the history.

Formula

$$\min_{1 \le t \le n} r_t = r_{(1)}$$

Plain text: `min(r_t)`. Symbols: see Notation.

Usage

```
report.metrics["worst_period"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`observed_point`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The worst single period (e.g. month) the investment had.

Professional. Observed lower extreme; a model-free loss statistic that is always shown.

Assumptions

None beyond the general inputs.

When it can mislead

- Histories of different lengths are compared; longer histories tend to contain worse periods.

Minimum data

Any non-empty history. (policy `observed_point`: Observed fact about the supplied history; any non-empty history suffices.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `observed_point`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence is always **high** (an observed fact about the supplied data).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).

- Formula id `worst_period@v1`; the version increments if the definition changes.

OpenFactsheet convention
Per observation period; signed (negative = loss).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.worst`	yes	
NumPy	`min`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision criteria

Gain-to-pain ratio
`gain_to_pain_ratio` · Return summary · derived · unit `ratio` · formula `gain_to_pain_ratio@v1`

Description
Sum of all returns relative to the sum of losing-period returns.

Formula

$$\mathrm{GPR} = \frac{\sum_{t=1}^{n} r_t}{\left|\sum_{t:\, r_t<0} r_t\right|}$$

Plain text: `sum(r_t) / |sum(r_t for r_t < 0)|`. Symbols: see Notation.

Usage
```
report.metrics["gain_to_pain_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	—		

Inputs
Periodic simple returns.

Value
Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation
Plain language. How much net return was earned for each unit of losses suffered along the way.

Professional. Schwager's gain-to-pain ratio on per-period returns; a simple, distribution-free complement to volatility-based ratios.

Assumptions
- Per-period returns are summed (not compounded).

When it can mislead
- Very few losing periods exist: the denominator is tiny and the ratio unstable.
- Compared across data frequencies (monthly vs daily values differ).

Minimum data
At least 2 observations and at least one losing period; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria
When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:

- No period has a negative return (zero denominator).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `gain_to_pain_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Computed on the supplied observation frequency; undefined (not applicable) when no period is negative.

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.gain_to_pain_ratio`	yes	Agrees on the monthly test fixtures.

References

- Schwager, J. D. (2012). Market Sense and Nonsense. Wiley (Gain-to-Pain ratio). [bibliography]

See also

Decision criteria

Payoff ratio (average win / average loss)

`payoff_ratio` · Return summary · derived · unit `ratio` · formula `payoff_ratio@v1`

Description

Size of the average winning period relative to the average losing period.

Formula

$$\text{Payoff} = \frac{\frac{1}{|G|}\sum_{t\in G} r_t}{\left|\frac{1}{|L|}\sum_{t\in L} r_t\right|}, \quad G = \{t : r_t > 0\},\ L = \{t : r_t < 0\}$$

Plain text: `mean(r_t | r_t > 0) / |mean(r_t | r_t < 0)|`. Symbols: see Notation.

Usage

```
report.metrics["payoff_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How big the typical gain is compared with the typical loss. Read it together with the share of positive periods: a low payoff with a high hit rate can still be profitable, and vice versa.

Professional. Win/loss (payoff) ratio on the supplied frequency; zero-return periods are excluded from both averages. The expectation per period equals hit rate x average win - loss rate x average loss.

Assumptions

- Zero returns count as neither wins nor losses.

When it can mislead

- Very few winning or losing periods.
- Returns are smoothed (many small gains, rare large losses).

Minimum data

At least 2 observations with one gain and one loss; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No positive period or no negative period exists.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `payoff_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic averages of strictly positive and strictly negative periods.

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.payoff_ratio`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0017: Drawdown-based ratios: Burke, Sterling, Pain; payoff ratio; Decision criteria

Profit factor (gross gains / gross losses)

`profit_factor` · Return summary · derived · unit `ratio` · formula `profit_factor@v1`

Description

Sum of all positive period returns relative to the absolute sum of all negative period returns.

Formula

$$\mathrm{PF} = \frac{\sum_{t \in G} r_t}{\left|\sum_{t \in L} r_t\right|} = 1 + \mathrm{GPR}, \quad G = \{t : r_t > 0\},\ L = \{t : r_t < 0\}$$

Plain text: `sum(r_t | r_t > 0) / |sum(r_t | r_t < 0)|`. Symbols: see Notation.

Usage

```
report.metrics["profit_factor"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How many units of gains the history produced for every unit of losses; above 1 means total gains exceeded total losses.

Professional. Gross gains over gross losses on per-period returns (a trading-system statistic). It equals one plus the gain-to-pain ratio, so it adds no information beyond it; reported because it is widely quoted. Not compounded and frequency dependent.

Assumptions

- Per-period returns are summed (not compounded).

When it can mislead

- Very few losing periods exist: the denominator is tiny and the ratio unstable.
- Compared across data frequencies (monthly vs daily values differ).

Minimum data

At least 2 observations and at least one losing period; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No negative period exists (the denominator is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).

- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `profit_factor@v1`; the version increments if the definition changes.

OpenFactsheet convention

Computed on the supplied observation frequency; zero returns count in neither sum (decision 0029).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.profit_factor`	yes	Agrees on the test fixtures.

References

- Pardo, R. (2008). The Evaluation and Optimization of Trading Strategies (2nd ed.). Wiley (profit factor: gross profit / gross loss). [bibliography]
- Schwager, J. D. (2012). Market Sense and Nonsense. Wiley (Gain-to-Pain ratio). [bibliography]

See also

Decision 0029: Serial correlation and annualization: variance ratio, Lo-adjusted Sharpe, profit factor; Decision criteria

Risk summary

Volatility per period

`volatility` · Risk summary · estimated · unit `decimal_return` · formula `volatility@v1`

Description

Dispersion of single-period returns around their mean.

Formula

$$s = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

Plain text: `sqrt( sum((r_t - mean)^2) / (n - 1) )`. Symbols: see Notation.

Usage

```
report.metrics["volatility"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	sample std (n-1)		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. How much returns typically swing from one period to the next; higher means a bumpier ride.
Professional. Sample standard deviation of per-period returns; symmetric, treats upside and downside deviations alike.

Assumptions

- Returns are identically distributed over the history.

When it can mislead

- Returns are smoothed or stale (illiquid assets, appraisal pricing): measured volatility is understated.
- Returns are skewed or fat-tailed: volatility does not describe tail losses.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Sample standard deviation with the n - 1 denominator.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `volatility@v1`; the version increments if the definition changes.

OpenFactsheet convention

Sample standard deviation (n - 1), decision 0003.

Reference implementations

Library	Function	Agrees	Note
NumPy	`std(ddof=1)`	yes	
PerformanceAnalytics 2.0.4 (R)	`StdDev`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0003: Standard deviation: sample (n – 1) estimator; Decision criteria

---

Annualized volatility

`annualized_volatility` · Risk summary · estimated · unit `decimal_return` · formula `annualized_volatility@v1`

Description

Per-period volatility scaled to one year.

Formula

$$\sigma_{\text{ann}} = s\sqrt{P}$$

Plain text: `std(r_t, ddof=1) * sqrt(P)`. Symbols: see Notation.

Usage

```
report.metrics["annualized_volatility"]
```

At a glance

Direction	Lower is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	sample std (n-1), annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. The typical size of yearly swings implied by the period-to-period variability.

Professional. Square-root-of-time scaling assumes serially uncorrelated returns; positive autocorrelation makes it understate true annual volatility (Lo, 2002).

Assumptions

- Returns are serially uncorrelated (square-root-of-time rule).

When it can mislead

- Returns are smoothed or stale (illiquid assets, appraisal pricing): measured volatility is understated.
- Returns are serially correlated or volatility changes over time.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Sample standard deviation with the n - 1 denominator.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.

- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `annualized_volatility@v1`; the version increments if the definition changes.

OpenFactsheet convention

Sample standard deviation x sqrt(periods per year).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`StdDev.annualized`	yes	
empyrical-reloaded 0.5.12	`annual_volatility`	yes	
QuantStats 0.0.81	`stats.volatility`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36-52. [bibliography]

See also

Decision 0003: Standard deviation: sample (n − 1) estimator; Decision criteria

---

Mean absolute deviation

`mean_absolute_deviation` · Risk summary · estimated · unit `decimal_return` · formula `mean_absolute_deviation@v1`

Description

Average absolute distance of single-period returns from their mean.

Formula

$$\mathrm{MAD} = \frac{1}{n}\sum_{t=1}^{n}|r_t - \bar{r}|$$

Plain text: `(1/n) * sum |r_t - mean|`. Symbols: see Notation.

Usage

```
report.metrics["mean_absolute_deviation"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. The typical size of a period's deviation from the average, without squaring (so single extreme periods weigh less than in volatility).

Professional. Mean absolute deviation around the mean (the L1 dispersion used in Konno-Yamazaki portfolio models); for normal returns MAD = sqrt(2/pi) sigma. Per period.

Assumptions

- Returns are identically distributed over the history.

When it can mislead

- Returns are smoothed or stale.
- Compared across data frequencies (per-period quantity).

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `mean_absolute_deviation@v1`; the version increments if the definition changes.

OpenFactsheet convention

Deviation from the sample mean, denominator n, per period.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`MeanAbsoluteDeviation`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

---

Interquartile range

`interquartile_range` · Risk summary · estimated · unit `decimal_return` · formula `interquartile_range@v1`

Description

Spread of the middle 50% of single-period returns.

Formula

$$\mathrm{IQR} = Q_{0.75}(r) - Q_{0.25}(r)$$

Plain text: `Q_0.75(r) - Q_0.25(r), linear interpolation`. Symbols: see Notation.

Usage

```
report.metrics["interquartile_range"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`

Canonical parameters	—

Inputs
Periodic simple returns.

Value
Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation
Plain language. How far apart typical good and typical bad periods are, ignoring the extremes.

Professional. Robust dispersion measure, insensitive to the tails (for normal returns IQR = 1.349 sigma). Contrast with volatility to see how much the tails drive risk. Per period.

Assumptions
- Linear-interpolated empirical quantiles (Hyndman-Fan type 7).

When it can mislead
- It deliberately ignores tail risk; never use it alone.
- Compared across data frequencies (per-period quantity).

Minimum data
At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria
When a value is produced (checked in this order)
1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status
- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result
- None beyond simple decimal returns as supplied.

Output
- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `interquartile_range@v1`; the version increments if the definition changes.

OpenFactsheet convention
Linear quantiles, per period (decision 0025).

Reference implementations
Library	Function	Agrees	Note
NumPy	`quantile(method='linear')`	yes	
PerformanceAnalytics 2.0.4 (R)	`R IQR()`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

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Downside deviation (annualized)

`downside_deviation` · Risk summary · estimated · unit `decimal_return` · formula `downside_deviation@v1`

Description

Variability of returns below the target (0% per period), scaled to a year.

Formula

$$\mathrm{DD}_{\mathrm{ann}} = \sqrt{P}\,\mathrm{DD}, \quad \mathrm{DD} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} \min(r_t - \tau, 0)^2}, \quad \tau = 0$$

Plain text: `sqrt( (1/n) * sum(min(r_t - MAR, 0)^2) ) * sqrt(P), MAR = 0`. Symbols: see Notation.

Usage

```
report.metrics["downside_deviation"]
```

At a glance

Direction	Lower is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	MAR = 0, annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. How large the losing periods typically are; gains do not count as risk here.

Professional. Lower partial moment of order 2 (root), full-sample denominator as in Sortino & Price (1994); the Sortino ratio denominator.

Assumptions

- Target (MAR) of 0% per period.
- Square-root-of-time scaling.

When it can mislead

- Very few losing periods exist: the estimate rests on a handful of observations.
- Returns are smoothed, which dampens measured losses.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).

- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Minimum acceptable return (target) tau = 0% per period; the downside deviation divides by all n observations.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `downside_deviation@v1`; the version increments if the definition changes.

OpenFactsheet convention

MAR = 0; denominator n (all observations), not the number of losing periods (decision 0006).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`DownsideDeviation(MAR=0, method='full') * sqrt(P)`	yes	
empyrical-reloaded 0.5.12	`downside_risk(required_return=0)`	yes	
QuantStats 0.0.81	`stats.sortino (denominator)`	yes	

References

- Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. Journal of Investing, 3(3), 59-64. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0006: Sortino ratio and downside deviation: MAR 0, full-sample denominator; Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

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Semi-deviation (annualized)

`semi_deviation` · Risk summary · estimated · unit `decimal_return` · formula `semi_deviation@v1`

Description

Variability of returns below their own mean, scaled to a year.

Formula

$$\mathrm{SemiDev}_{\mathrm{ann}} = \sqrt{P}\sqrt{\frac{1}{n}\sum_{t=1}^{n}\min(r_t - \bar{r}, 0)^2}$$

Plain text: `sqrt( (1/n) * sum over r_t < mean of (r_t - mean)^2 ) * sqrt(P)`. Symbols: see Notation.

Usage

```
report.metrics["semi_deviation"]
```

At a glance

Direction	Lower is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. How large the below-average periods typically are.

Professional. Downside deviation with the sample mean as target (Markowitz semi-variance, square root), full-sample denominator. Unlike downside deviation it measures dispersion below the mean, not losses.

Assumptions

- Target equals the sample mean.
- Square-root-of-time scaling.

When it can mislead

- Returns are smoothed, which compresses deviations.
- Compared with semi-deviations that divide by the number of below-mean periods.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.

- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `semi_deviation@v1`; the version increments if the definition changes.

OpenFactsheet convention

Full-sample denominator n, target = mean, annualized by sqrt(P) (decision 0016).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`SemiDeviation x sqrt(P)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

---

Downside potential (LPM 1)

`downside_potential` · Risk summary · estimated · unit `decimal_return` · formula `downside_potential@v1`

Description

Average shortfall below the 0% target per period (first lower partial moment).

Formula

$$\mathrm{LPM}_1 = \frac{1}{n}\sum_{t=1}^{n}\max(\tau - r_t, 0), \quad \tau = 0$$

Plain text: `(1/n) * sum max(tau - r_t, 0), tau = 0`. Symbols: see Notation.

Usage

```
report.metrics["downside_potential"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	MAR = 0		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. The average loss per period, counting gains as zero.

Professional. First lower partial moment at target 0; the denominator of the Omega ratio (Omega = 1 + mean / LPM1 at threshold 0). Per period, not annualized.

Assumptions

- Target (MAR) of 0% per period.

When it can mislead

- Compared across data frequencies (per-period quantity).

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Minimum acceptable return (target) tau = 0% per period; the downside deviation divides by all n observations.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `downside_potential@v1`; the version increments if the definition changes.

OpenFactsheet convention

Target 0, full-sample denominator n, per period (decision 0016).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`DownsideDeviation(MAR=0, potential=TRUE)`	yes	

References

- Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. Journal of Investing, 3(3), 59-64. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

Drawdowns

Maximum drawdown

`max_drawdown` · Drawdowns · observed · unit `decimal_return` · formula `max_drawdown@v1`

Description

Largest peak-to-trough decline of compounded wealth.

Formula

$$\mathrm{MDD} = \min_{0 \le t \le n} D_t, \quad D_t = \frac{W_t}{M_t} - 1, \quad M_t = \max_{0 \le u \le t} W_u, \quad W_0 = 1$$

Plain text: `min_t( W_t / max_{s<=t} W_s - 1 ), W_0 = 1, W_t = prod_{i<=t}(1 + r_i)`. Symbols: see Notation.

Usage

```
report.metrics["max_drawdown"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The worst loss an investor would have suffered buying at a high point and selling at the following low point.

Professional. Path-dependent realized loss; always shown. Expected maximum drawdown grows with history length and volatility (Magdon-Ismail & Atiya, 2004), so compare only across similar horizons.

Assumptions

- Wealth index starts at 1 before the first return.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Returns are smoothed, which reduces measured drawdowns.
- Data frequency is low: intra-period declines are invisible.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `max_drawdown@v1`; the version increments if the definition changes.

OpenFactsheet convention

Signed (negative = loss); the inception value counts as a peak so a first-period loss is a drawdown (decision 0007).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`maxDrawdown`	yes	PerformanceAnalytics reports a positive magnitude.
empyrical-reloaded 0.5.12	`max_drawdown`	yes	
QuantStats 0.0.81	`stats.max_drawdown`	yes	

References

- Magdon-Ismail, M., & Atiya, A. (2004). Maximum Drawdown. Risk, 17(10), 99-102. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0007: Drawdown conventions; Decision criteria

Average drawdown (per episode)

`average_drawdown` · Drawdowns · observed · unit `decimal_return` · formula `average_drawdown@v1`

Description

Mean depth of all distinct drawdown episodes.

Formula

$$\overline{\mathrm{DD}} = \begin{cases} \frac{1}{K}\sum_{k=1}^{K} \delta_k & K > 0 \\ 0 & K = 0 \end{cases}, \quad \delta_k = \min_{t \in E_k} D_t$$

Plain text: `(1/K) * sum_k depth_k over K peak-to-recovery episodes`. Symbols: see Notation.

Usage

```
report.metrics["average_drawdown"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. How deep the typical decline from a high point has been.

Professional. Episode-based average (PerformanceAnalytics AverageDrawdown convention); less dominated by a single event than maximum drawdown. Compare with the Pain Index, which averages over time instead of over episodes.

Assumptions

- Every decline below a prior peak, however small, is an episode.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Many tiny episodes dilute the average.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `average_drawdown@v1`; the version increments if the definition changes.

OpenFactsheet convention

Mean of episode depths, signed (negative = loss); 0 when no drawdown occurred.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`AverageDrawdown`	yes	PerformanceAnalytics reports a positive magnitude.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0007: Drawdown conventions; Decision criteria

Current drawdown

`current_drawdown` · Drawdowns · observed · unit `decimal_return` · formula `current_drawdown@v1`

Description

Decline of wealth at the end of the history relative to its highest point.

Formula

$$D_n = \frac{W_n}{M_n} - 1$$

Plain text: `W_n / max_{s<=n} W_s - 1`. Symbols: see Notation.

Usage

```
report.metrics["current_drawdown"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better	Annualized	no

	than −25% (the higher, less negative value is better)		
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`observed_point`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. How far below its previous high the investment stands at the end of the data.

Professional. State variable at the reporting date; zero when the series ends at a high-water mark.

Assumptions

- Wealth index starts at 1 before the first return.

When it can mislead

- The history ends at an arbitrary date.

Minimum data

Any non-empty history. (policy `observed_point`: Observed fact about the supplied history; any non-empty history suffices.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `observed_point`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence is always **high** (an observed fact about the supplied data).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `current_drawdown@v1`; the version increments if the definition changes.

OpenFactsheet convention

Signed (negative = loss).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.to_drawdown_series (last value)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Longest drawdown duration

`longest_drawdown_duration` · Drawdowns · observed · unit `periods` · formula `longest_drawdown_duration@v1`

Description

Longest time from a peak until wealth regained that peak (or until the end of the history, if not yet recovered).

Formula

$$L_{\max} = \max_{1 \le k \le K} \ell_k, \quad \ell_k = \begin{cases} \rho_k - \pi_k & \text{episode } k \text{ recovered} \\ n - \pi_k & \text{episode } k \text{ ongoing} \end{cases}, \quad L_{\max} = 0 \text{ if } K = 0$$

Plain text: `max_k (recovery_k - peak_k) in periods; ongoing episodes use n - peak_k`. Symbols: see Notation.

Usage

```
report.metrics["longest_drawdown_duration"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Lower is better.

Interpretation

Plain language. The longest stretch an investor would have waited to get back to a previous high.

Professional. Peak-to-recovery duration in observation periods; an ongoing episode is a lower bound and is flagged as such.

Assumptions

- Recovery means regaining the prior peak of compounded wealth.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- The longest episode is still ongoing (a lower bound).

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `longest_drawdown_duration@v1`; the version increments if the definition changes.

OpenFactsheet convention

Measured in observation periods from the peak to the recovery period inclusive; details report the equivalent in years when the frequency is regular.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns (length)`	yes	Equal for recovered episodes; PerformanceAnalytics counts an episode still open at the end of the data one period longer.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Number of drawdown episodes

`drawdown_count` · Drawdowns · observed · unit `count` · formula `drawdown_count@v1`

Description

How many distinct peak-to-recovery declines occurred.

Formula

$$K = \#\{\text{maximal runs } t_1..t_2 \subseteq 1..n \text{ with } D_t < 0 \ \forall t \in t_1..t_2\}$$

Plain text: `K = number of maximal runs with W_t below its running peak`. Symbols: see Notation.

Usage

```
report.metrics["drawdown_count"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs
Periodic simple returns.

Value
Count; displayed as an integer. Neutral (descriptive).

Interpretation
Plain language. How often the investment fell below a previous high.

Professional. Context for average drawdown and durations; depends strongly on data frequency and history length.

Assumptions
- Every decline below a prior peak counts, however small.

When it can mislead
- Compared across frequencies or history lengths.

Minimum data
One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria
When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Count; displayed as an integer. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `drawdown_count@v1`; the version increments if the definition changes.

OpenFactsheet convention
An ongoing episode at the end of the data is counted.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns (negative episodes)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision criteria

Time under water

`time_underwater_pct` · Drawdowns · observed · unit `fraction` · formula `time_underwater_pct@v1`

Description

Share of periods in which wealth was below its previous peak.

Formula

$$\mathrm{TUW} = \frac{1}{n}\#\{t \in 1..n : D_t < 0\}$$

Plain text: `(1/n) * #{t : D_t < 0}`. Symbols: see Notation.

Usage

```
report.metrics["time_underwater_pct"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Lower is better.

Interpretation

Plain language. The share of the time the investment was below a previous high.

Professional. Complements depth measures with persistence; high values with shallow drawdowns indicate grinding rather than crash risk.

Assumptions

- Wealth index starts at 1 before the first return.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `time_underwater_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention

Counted over observations 1..n (the inception point is excluded).

Reference implementations

No established library implementation was identified.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Ulcer Index

`ulcer_index` · Drawdowns · observed · unit `decimal_return` · formula `ulcer_index@v1`

Description

Root-mean-square drawdown: combines depth and duration of declines.

Formula

$$\mathrm{UI} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} D_t^2}$$

Plain text: `sqrt( (1/n) * sum_t D_t^2 )`. Symbols: see Notation.

Usage

```
report.metrics["ulcer_index"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. A single number for how deep and how long-lasting the declines were; 0 means the investment never fell below a previous high.

Professional. Martin & McCann's Ulcer Index on the canonical drawdown path; the denominator of the Martin ratio (Ulcer Performance Index).

Assumptions

- Drawdowns measured from running peaks of compounded wealth.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Compared across data frequencies.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `ulcer_index@v1`; the version increments if the definition changes.

OpenFactsheet convention

Non-negative magnitude in decimal units, computed from the canonical drawdown path over observations 1..n.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`UlcerIndex`	no (documented difference)	PerformanceAnalytics 2.0.4 builds drawdowns with DrawdownPeak, which divides returns by 100 (assumes percent inputs); agreement is verified against sqrt(mean(Drawdowns(R)^2)) instead.

References

- Martin, P. G., & McCann, B. B. (1989). The Investor's Guide to Fidelity Funds. Wiley (Ulcer Index and Ulcer Performance Index). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0007: Drawdown conventions; Decision criteria

Pain Index

`pain_index` · Drawdowns · observed · unit `decimal_return` · formula `pain_index@v1`

Description

Average drawdown over time (mean absolute drawdown across all periods).

Formula

$$\mathrm{PI} = \frac{1}{n}\sum_{t=1}^{n}|D_t|$$

Plain text: `(1/n) * sum_t |D_t|`. Symbols: see Notation.

Usage

```
report.metrics["pain_index"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Lower is better.

Interpretation

Plain language. How far below its previous high the investment was, on average, over the whole history.

Professional. Zephyr Pain Index (see Bacon); time-weighted, unlike the episode-based average drawdown.

Assumptions

- Drawdowns measured from running peaks of compounded wealth.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).

- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `pain_index@v1`; the version increments if the definition changes.

OpenFactsheet convention
Non-negative magnitude in decimal units over observations 1..n.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`PainIndex`	no (documented difference)	Same DrawdownPeak percent-input issue as UlcerIndex; verified against mean(|Drawdowns(R)|).

References
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision 0007: Drawdown conventions; Decision criteria

Median drawdown (per episode)
`median_drawdown` · Drawdowns · observed · unit `decimal_return` · formula `median_drawdown@v1`

Description
Median depth of the distinct drawdown episodes.

Formula
$$\widetilde{\mathrm{DD}} = \mathrm{med}(\delta_1, \ldots, \delta_K), \quad 0 \text{ if } K = 0$$

Plain text: `median_k depth_k over K episodes (0 if K = 0)`. Symbols: see Notation.

Usage
```
report.metrics["median_drawdown"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs
Periodic simple returns.

Value
Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation
Plain language. The depth of a typical decline from a previous high; unlike the average it is not pulled down by one extreme episode.

Professional. Robust counterpart of the episode-based average drawdown; many shallow episodes make it close to zero.

Assumptions
- Every decline below a prior peak, however small, is an episode.

When it can mislead
- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Few episodes (the median of 1-3 values is fragile).

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `median_drawdown@v1`; the version increments if the definition changes.

OpenFactsheet convention

Median of episode depths, signed (decision 0025).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns / table.Drawdowns`	yes	Agrees on the episode table; PerformanceAnalytics counts an episode still open at the end of the data one period longer.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

Average drawdown duration

`average_drawdown_duration` · Drawdowns · observed · unit `periods` · formula `average_drawdown_duration@v1`

Description

Mean time from a peak to recovery (or to the end of the data for an open episode), across all drawdown episodes.

Formula

$$\bar{\ell} = \frac{1}{K}\sum_{k=1}^{K} \ell_k, \quad \ell_k = \begin{cases} \rho_k - \pi_k & \text{recovered} \\ n - \pi_k & \text{ongoing} \end{cases}, \quad 0 \text{ if } K = 0$$

Plain text: `(1/K) * sum_k duration_k, duration as in longest_drawdown_duration (0 if K = 0)`. Symbols: see Notation.

Usage

```
report.metrics["average_drawdown_duration"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Lower is better.

Interpretation

Plain language. How long, on average, the investment took to get back to a previous high.

Professional. Episode-weighted average duration in observation periods; an open episode contributes a lower bound (flagged).

Assumptions

- Recovery means regaining the prior peak of compounded wealth.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- The longest episodes are still open (lower bound).

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).

- Formula id `average_drawdown_duration@v1`; the version increments if the definition changes.

OpenFactsheet convention
Peak to recovery inclusive; open episodes measured to the last observation (decision 0025).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns / table.Drawdowns`	yes	Agrees on the episode table; PerformanceAnalytics counts an episode still open at the end of the data one period longer.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

---

Longest decline (peak to trough)
`max_peak_to_trough_duration` · Drawdowns · observed · unit `periods` · formula `max_peak_to_trough_duration@v1`

Description
Longest time from a peak to the lowest point of the same drawdown episode.

Formula
$$\max_{1 \le k \le K} (\tau_k - \pi_k), \quad \tau_k = \arg\min_{t \in E_k} D_t \text{ (first occurrence)}, \quad 0 \text{ if } K = 0$$

Plain text: `max_k (trough_k - peak_k) in periods (0 if K = 0)`. Symbols: see Notation.

Usage
```
report.metrics["max_peak_to_trough_duration"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs
Periodic simple returns.

Value
Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Lower is better.

Interpretation
Plain language. The longest stretch during which losses kept deepening before the turnaround.
Professional. Maximum peak-to-trough length in observation periods; complements the total duration, which also includes the recovery.

Assumptions
- The trough is the first period with the episode's minimum drawdown.

When it can mislead
- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- The deepest point of an open episode may still lie ahead.

Minimum data
One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `max_peak_to_trough_duration@v1`; the version increments if the definition changes.

OpenFactsheet convention

Periods from the peak to the trough (decision 0025).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns (peaktotrough)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

---

Average recovery time (trough to recovery)

`average_recovery_duration` · Drawdowns · observed · unit `periods` · formula `average_recovery_duration@v1`

Description

Mean time from the trough of a drawdown back to the previous peak, over recovered episodes.

Formula

$$\bar{R} = \frac{1}{|\,K_r|} \sum_{k \in K_r} (\rho_k - \tau_k), \quad K_r = \{k : \text{episode } k \text{ recovered}\}$$

Plain text: `mean_k (recovery_k - trough_k) over recovered episodes`. Symbols: see Notation.

Usage

```
report.metrics["average_recovery_duration"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no

Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Lower is better.

Interpretation

Plain language. Once the bottom was reached, how long it typically took to climb back to the previous high.
Professional. Average recovery length of completed episodes only; open episodes are excluded because their recovery time is unknown (disclosed in details).

Assumptions

- Recovery means regaining the prior peak of compounded wealth.

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- The deepest episode is still open and therefore excluded.

Minimum data

One recovered episode; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown episode has recovered (none occurred, or all are open).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `average_recovery_duration@v1`; the version increments if the definition changes.

OpenFactsheet convention

Recovered episodes only (decision 0025).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`findDrawdowns (recovery)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

---

Drawdown-at-Risk (95%)

`drawdown_at_risk_95` · Drawdowns · estimated · unit `decimal_return` · formula `drawdown_at_risk_95@v1`

Description

Drawdown level that was exceeded in the worst 5% of periods.

Formula

$$\mathrm{DaR}_{95} = Q_{0.05}(D_1, \ldots, D_n)$$

Plain text: `Q_0.05(D_1..D_n), linear interpolation (signed)`. Symbols: see Notation.

Usage

```
report.metrics["drawdown_at_risk_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	wealth starts at 1, 95% confidence		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. In about 1 period out of 20 the investment was at least this far below its previous high.

Professional. Drawdown-at-Risk (Chekhlov, Uryasev & Zabarankin 2005) on the compounded drawdown path, with the same linear quantile as historical VaR.

Assumptions

- Drawdowns of all periods 1..n form the sample (inception excluded).

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Drawdown observations are strongly serially dependent: the 5% tail of the drawdown path often comes from a single episode.

Minimum data

At least 2 observations; low confidence below n = 100. (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.

3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- 95% confidence; historical (empirical) quantile with linear interpolation (Hyndman-Fan type 7); no distributional assumption; signed return (negative = loss).

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `drawdown_at_risk_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Compounded drawdowns, linear quantile, signed (decision 0021).

Reference implementations

Library	Function	Agrees	Note
skfolio 1.0.6	`measures.drawdown_at_risk (compounded drawdowns)`	no (documented difference)	skfolio uses the lower order statistic (inverted-CDF quantile) and, by default, uncompounded drawdowns.

References

- Chekhlov, A., Uryasev, S., & Zabarankin, M. (2005). Drawdown Measure in Portfolio Optimization. International Journal of Theoretical and Applied Finance, 8(1), 13-58 (Drawdown-at-Risk and Conditional Drawdown-at-Risk). [bibliography]

See also

Decision 0021: Drawdown tail (DaR, CDaR, EDaR) and entropic VaR; Decision criteria

Conditional Drawdown-at-Risk (95%)

`cdar_95` · Drawdowns · estimated · unit `decimal_return` · formula `cdar_95@v1`

Description

Average drawdown in the worst 5% of periods.

Formula

$$\mathrm{CDaR}_{95} = \frac{1}{\mid T \mid} \sum_{t \in T} D_t, \quad T = \{t \in 1..n : D_t \leq Q_{0.05}(D)\}$$

Plain text: `mean(D_t | D_t <= Q_0.05(D))`. Symbols: see Notation.

Usage

```
report.metrics["cdar_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	wealth starts at 1, 95% confidence		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. How deep the drawdowns were, on average, during the worst 5% of periods.

Professional. CDaR (Chekhlov, Uryasev & Zabarankin 2005): Expected Shortfall of the drawdown path, a coherent path-dependent risk measure between DaR and the maximum drawdown. Estimated exactly like historical ES (decision 0010).

Assumptions

- Drawdowns of all periods 1..n form the sample (inception excluded).

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Drawdown observations are strongly serially dependent: the 5% tail of the drawdown path often comes from a single episode.

Minimum data

At least 2 observations; low confidence below n = 100. (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- 95% confidence; historical (empirical) quantile with linear interpolation (Hyndman-Fan type 7); no distributional assumption; signed return (negative = loss).

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `cdar_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Compounded drawdowns; mean of observations at or below the linear 5% quantile; signed (decision 0021).

Reference implementations

Library	Function	Agrees	Note
skfolio 1.0.6	`measures.cdar (compounded drawdowns)`	yes	Agrees when (1 - 0.95) x n is an integer; otherwise skfolio's fractional Rockafellar-Uryasev weighting differs slightly.
PerformanceAnalytics 2.0.4 (R)	`CDD`	no (documented difference)	PerformanceAnalytics' 'Conditional Drawdown' is the 5% quantile over all findDrawdowns runs (episodes and zero runs), not a conditional expectation of the drawdown path.

References

- Chekhlov, A., Uryasev, S., & Zabarankin, M. (2005). Drawdown Measure in Portfolio Optimization. International Journal of Theoretical and Applied Finance, 8(1), 13-58 (Drawdown-at-Risk and Conditional Drawdown-at-Risk). [bibliography]

See also

Decision 0010: Historical VaR, Expected Shortfall and tail ratio; Decision 0021: Drawdown tail (DaR, CDaR, EDaR) and entropic VaR; Decision criteria

---

Entropic Drawdown-at-Risk (95%)

`edar_95` · Drawdowns · estimated · unit `decimal_return` · formula `edar_95@v1`

Description

Entropic (exponential-moment) upper bound on the drawdown tail at 95%.

Formula

$$\mathrm{EDaR}_{95} = -\mathrm{EVaR}_{0.05}(-D), \quad \mathrm{EVaR}_{a}(X) = \inf_{z>0} z \ln\left(\frac{1}{a\,n}\sum_{t=1}^{n} e^{X_t/z}\right)$$

Plain text: `-EVaR_0.05(-D), EVaR_a(X) = inf_{z>0} z * ln( mean(exp(X/z)) / a )`. Symbols: see Notation.

Usage

```
report.metrics["edar_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`

Canonical parameters	wealth starts at 1, 95% entropic

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. A conservative measure of how deep drawdowns get, sitting between the Conditional Drawdown-at-Risk and the maximum drawdown.

Professional. EDaR (Cajas 2021): Entropic VaR (Ahmadi-Javid 2012) of the drawdown magnitudes; the tightest Chernoff-type bound on DaR, coherent and more tail-sensitive than CDaR. Computed by one-dimensional convex minimization.

Assumptions

- Drawdowns of all periods 1..n form the sample (inception excluded).

When it can mislead

- Short histories: drawdown statistics grow with the length of the track record and cannot reveal risks that have not yet materialized.
- Drawdown observations are strongly serially dependent: the 5% tail of the drawdown path often comes from a single episode.
- With 20 or fewer observations (n <= 1/0.05) the empirical EDaR equals the maximum drawdown exactly.

Minimum data

At least 2 observations; low confidence below n = 100. (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- 95% confidence; entropic VaR of the empirical distribution (exponential-moment bound), one-dimensional convex minimization; signed return.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.

- Formula id `edar_95@v1`; the version increments if the definition changes.

OpenFactsheet convention
Compounded drawdowns, 95%, signed (decision 0021).

Reference implementations

Library	Function	Agrees	Note
skfolio 1.0.6	`measures.edar (compounded drawdowns)`	yes	Agrees to solver tolerance (about 1e-8).

References

- Cajas, D. (2021). Entropic Portfolio Optimization: A Disciplined Convex Programming Framework. SSRN working paper 3792520 (Entropic Drawdown-at-Risk). [bibliography]
- Ahmadi-Javid, A. (2012). Entropic Value-at-Risk: A New Coherent Risk Measure. Journal of Optimization Theory and Applications, 155(3), 1105-1123. [bibliography]

See also
Decision 0021: Drawdown tail (DaR, CDaR, EDaR) and entropic VaR; Decision criteria

Risk-adjusted performance

Sharpe ratio
`sharpe_ratio` · Risk-adjusted performance · estimated · unit `ratio` · formula `sharpe_ratio@v1`

Description
Average return per unit of total volatility, annualized.

Formula
$$\mathrm{SR} = \frac{\bar{r} - r_f}{s}\sqrt{P}, \quad r_f = 0$$

Plain text: `(mean(r_t) - rf) / std(r_t, ddof=1) * sqrt(P), rf = 0`. Symbols: see Notation.

Usage
```
report.metrics["sharpe_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`ratio_standard`
Canonical parameters	risk-free = 0, sample std (n-1), annualization factor		

Inputs
Periodic simple returns with a regular frequency.

Value
Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation
Plain language. The Sharpe ratio compares excess return with total return variability. Higher values indicate more return per unit of measured volatility, but the statistic can be misleading when returns are smoothed, strongly skewed, serially correlated, or have significant tail risk.

Professional. Ex-post Sharpe (1994) with arithmetic mean and square-root-of-time annualization. Serial dependence biases the annualized value (Lo, 2002); smoothed returns inflate it (Getmansky, Lo & Makarov, 2004). Standard error is roughly sqrt((1 + SR^2/2) / years) under i.i.d. normality.

Assumptions
- Risk-free rate 0% (methodology 0.1.0).
- Volatility is an adequate risk measure (approximately symmetric returns).
- Serially uncorrelated returns for annualization.

When it can mislead
- Returns are smoothed or stale (hedge funds, private assets): volatility is understated and the ratio overstated.

- Returns are negatively skewed or fat-tailed (e.g. option-selling strategies).
- The history is short: sampling error is large.
- Cash yields are materially positive (rf = 0 overstates the excess return).

Minimum data

At least 2 observations; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Sample standard deviation is exactly zero (constant returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.
- Sample standard deviation with the n - 1 denominator.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `sharpe_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic mean / sample std x sqrt(P); rf = 0 (decisions 0005, 0008).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`SharpeRatio.annualized(geometric=FALSE)`	yes	The PerformanceAnalytics default (geometric=TRUE) uses the annualized geometric return in the numerator.
empyrical-reloaded 0.5.12	`sharpe_ratio`	yes	
QuantStats 0.0.81	`stats.sharpe`	yes	

References

- Sharpe, W. F. (1966). Mutual Fund Performance. Journal of Business, 39(1), 119-138. [bibliography]

- Sharpe, W. F. (1994). The Sharpe Ratio. Journal of Portfolio Management, 21(1), 49-58. [bibliography]
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36-52. [bibliography]
- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also

Decision 0005: Risk-free rate: zero in methodology 0.1.0, always disclosed; Decision 0008: Sharpe ratio: arithmetic mean over sample volatility, square-root-of-time; Decision criteria

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Sortino ratio

`sortino_ratio` · Risk-adjusted performance · estimated · unit `ratio` · formula `sortino_ratio@v1`

Description

Average return in excess of the target per unit of downside deviation, annualized.

Formula

$$\text{Sortino} = \frac{P\,(\bar{r} - \tau)}{\text{DD}\,\sqrt{P}} = \frac{\bar{r} - \tau}{\text{DD}}\sqrt{P}, \quad \tau = 0$$

Plain text: `P * mean(r_t - MAR) / ( sqrt((1/n) sum min(r_t - MAR, 0)^2) * sqrt(P) ), MAR = 0`. Symbols: see Notation.

Usage

```
report.metrics["sortino_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`ratio_standard`
Canonical parameters	risk-free = 0, MAR = 0, annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Like the Sharpe ratio, but only losing periods count as risk.

Professional. Sortino & Price (1994) with MAR = 0 and full-sample downside deviation. Downside deviation from few losses is noisy; the ratio is undefined without losses.

Assumptions

- Target (MAR) of 0% per period.
- Square-root-of-time scaling.

When it can mislead

- Very few losing periods: the denominator is unstable.
- Returns are smoothed or stale.

Minimum data

At least 2 observations and at least one period below the target; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No return is below the 0% target (downside deviation is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.
- Minimum acceptable return (target) tau = 0% per period; the downside deviation divides by all n observations.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `sortino_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

MAR = 0 per period; full-sample denominator; annualized by sqrt(P) (decision 0006).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`SortinoRatio(MAR=0) * sqrt(P)`	yes	PerformanceAnalytics reports the per-period ratio.
empyrical-reloaded 0.5.12	`sortino_ratio`	yes	
QuantStats 0.0.81	`stats.sortino`	yes	

References

- Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. Journal of Investing, 3(3), 59-64. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0006: Sortino ratio and downside deviation: MAR 0, full-sample denominator; Decision criteria

Calmar ratio

`calmar_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `calmar_ratio@v1`

Description

Annualized return per unit of maximum drawdown.

Formula

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{MDD}|}$$

Plain text: `CAGR / |max drawdown|`. Symbols: see Notation.

Usage

```
report.metrics["calmar_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	annualization factor, wealth starts at 1, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How much yearly return was earned relative to the worst peak-to-trough loss.

Professional. Young (1991) defined Calmar on a trailing 36-month window; OpenFactsheet canonically uses the full history so that no window is chosen. Maximum drawdown grows with history length, so values fall as track records lengthen.

Assumptions

- Full-history window.
- Risk-free rate 0%.

When it can mislead

- Histories of different lengths are compared.
- The maximum drawdown is a single unrepresentative event.

Minimum data

At least 2 observations and a non-zero drawdown; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown occurred (maximum drawdown is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `calmar_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Full history, geometric CAGR, signed max drawdown in absolute value (decision 0009).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`CalmarRatio`	yes	
empyrical-reloaded 0.5.12	`calmar_ratio`	yes	
QuantStats 0.0.81	`stats.calmar`	no (documented difference)	QuantStats' default CAGR assumes 252 periods per year whatever the data frequency, so monthly results differ unless periods are adjusted.

References

- Young, T. W. (1991). Calmar Ratio: A Smoother Tool. Futures, 20(1), 40. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0009: Calmar ratio over the full history; Decision criteria

---

Martin ratio (Ulcer Performance Index)

`martin_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `martin_ratio@v1`

Description

Annualized return per unit of Ulcer Index.

Formula

$$\text{Martin} = \frac{\text{CAGR} - r_f}{\text{UI}}, \quad r_f = 0$$

Plain text: `(CAGR - rf) / UlcerIndex, rf = 0`. Symbols: see Notation.

Usage

```
report.metrics["martin_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	annualization factor, wealth starts at 1, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Return relative to how deep and prolonged the declines were.

Professional. Ulcer Performance Index (Martin & McCann, 1989); penalizes both depth and duration of drawdowns rather than volatility.

Assumptions

- Risk-free rate 0%.

When it can mislead

- Short histories with few drawdowns.

Minimum data

At least 2 observations and at least one drawdown; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown occurred (Ulcer Index is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.
- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `martin_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric CAGR over the canonical Ulcer Index.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`MartinRatio`	no (documented difference)	Inherits the UlcerIndex percent-input issue in PerformanceAnalytics 2.0.4; agreement verified with the drawdown-path Ulcer Index.

References

- Martin, P. G., & McCann, B. B. (1989). The Investor's Guide to Fidelity Funds. Wiley (Ulcer Index and Ulcer Performance Index). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0007: Drawdown conventions; Decision 0009: Calmar ratio over the full history; Decision criteria

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Omega ratio

`omega_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `omega_ratio@v1`

Description

Probability-weighted gains above the threshold relative to probability-weighted losses below it.

Formula

$$\Omega(L) = \frac{\sum_{t=1}^{n} \max(r_t - L, 0)}{\sum_{t=1}^{n} \max(L - r_t, 0)}, \quad L = 0$$

Plain text: `sum(max(r_t - L, 0)) / sum(max(L - r_t, 0)), L = 0`. Symbols: see Notation.

Usage

```
report.metrics["omega_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	threshold = 0		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Total gains divided by total losses; above 1 means gains outweighed losses.

Professional. Keating & Shadwick (2002) Omega at threshold 0, using the full empirical distribution (all moments). Equals 1 + mean / lower partial moment of order 1.

Assumptions

- Threshold of 0% per period.

When it can mislead

- Few losing periods: the denominator is tiny.
- Compared across data frequencies.

Minimum data

At least 2 observations and at least one loss; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No return is below the 0% threshold (zero denominator).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - ▸ None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Threshold L = 0% per period.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `omega_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Threshold 0 per period; simple empirical estimator; not annualized.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Omega(L=0, method='simple')`	yes	
empyrical-reloaded 0.5.12	`omega_ratio(required_return=0)`	yes	
QuantStats 0.0.81	`stats.omega`	yes	

References

- Keating, C., & Shadwick, W. F. (2002). A Universal Performance Measure. Journal of Performance Measurement, 6(3), 59-84. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Kappa 3 ratio

`kappa_3_ratio` · Risk-adjusted performance · estimated · unit `ratio` · formula `kappa_3_ratio@v1`

Description

Mean excess return over the target per unit of third-order lower partial moment, penalizing large losses more than the Sortino ratio does.

Formula

$$K_3 = \frac{\bar{r} - \tau}{\left(\frac{1}{n}\sum_{t=1}^{n} \max(\tau - r_t, 0)^3\right)^{1/3}}, \quad \tau = 0$$

Plain text: `(mean(r_t) - tau) / ( (1/n) sum max(tau - r_t, 0)^3 )^(1/3), tau = 0`. Symbols: see Notation.

Usage

```
report.metrics["kappa_3_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`ratio_standard`
Canonical parameters	MAR = 0		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Return per unit of downside risk, where large losses count much more than small ones.

Professional. Kaplan & Knowles (2004) Kappa of order 3 (Sortino is Kappa 2; Omega - 1 is Kappa 1). The cubic lower partial moment emphasizes tail losses. Per-period, not annualized.

Assumptions

- Target (MAR) of 0% per period.

When it can mislead

- Very few losing periods: the lower partial moment rests on a handful of values.
- Compared across data frequencies (per-period ratio).

Minimum data

At least 2 observations and one loss; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No return is below the 0% target (zero lower partial moment).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Minimum acceptable return (target) tau = 0% per period; the downside deviation divides by all n observations.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `kappa_3_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Target 0, full-sample denominator n, per period (decision 0016).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Kappa(MAR=0, l=3)`	yes	

References

- Kaplan, P. D., & Knowles, J. A. (2004). Kappa: A Generalized Downside Risk-Adjusted Performance Measure. Journal of Performance Measurement, 8(3), 42-54. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

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Upside potential ratio

`upside_potential_ratio` · Risk-adjusted performance · estimated · unit `ratio` · formula `upside_potential_ratio@v1`

Description

Average gain above the target per unit of downside deviation.

Formula

$$\mathrm{UPR} = \frac{\frac{1}{n}\sum_{t=1}^{n} \max(r_t - \tau, 0)}{\mathrm{DD}}, \quad \tau = 0$$

Plain text: `( (1/n) sum max(r_t - tau, 0) ) / sqrt( (1/n) sum min(r_t - tau, 0)^2 ), tau = 0`. Symbols: see Notation.

Usage

```
report.metrics["upside_potential_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`ratio_standard`
Canonical parameters	MAR = 0		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How much upside the investment delivered for each unit of downside it exposed investors to. *Professional.* Sortino, van der Meer & Plantinga (1999): first upper partial moment over the square root of the second lower partial moment, both at target 0 with full-sample denominators. Per-period, not annualized.

Assumptions

- Target (MAR) of 0% per period.

When it can mislead

- Very few losing periods: the denominator is unstable.
- Compared across data frequencies (per-period ratio).

Minimum data

At least 2 observations and one loss; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No return is below the 0% target (downside deviation is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Minimum acceptable return (target) tau = 0% per period; the downside deviation divides by all n observations.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `upside_potential_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Target 0, full-sample denominators (decision 0016).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`UpsidePotentialRatio(MAR=0, method='full')`	yes	The PerformanceAnalytics default method='subset' divides by the number of gains / losses instead of n.

References

- Sortino, F. A., van der Meer, R., & Plantinga, A. (1999). The Dutch Triangle. Journal of Portfolio Management, 26(1), 50-57 (upside potential ratio). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators; Decision criteria

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Burke ratio

`burke_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `burke_ratio@v1`

Description

Annualized return per unit of the root sum of squared drawdown depths.

Formula

$$\text{Burke} = \frac{\text{CAGR} - r_f}{\sqrt{\sum_{k=1}^{K} \delta_k^2}}, \quad r_f = 0$$

Plain text: `(CAGR - rf) / sqrt( sum_k depth_k^2 ), rf = 0`. Symbols: see Notation.

Usage

```
report.metrics["burke_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	annualization factor, wealth starts at 1, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Return relative to the drawdowns suffered, where deep drawdowns weigh more than shallow ones.

Professional. Burke (1994): penalizes the square of every drawdown episode rather than only the worst one (Calmar) or their time-average (Martin, Pain).

Assumptions

- Risk-free rate 0%.
- Every peak-to-recovery episode counts once.

When it can mislead

- Short histories with few drawdown episodes.
- Compared across frequencies or history lengths (the sum grows with both).

Minimum data

At least 2 observations and one drawdown; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown occurred (K = 0).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.
- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `burke_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Episode depths from the canonical drawdown engine; geometric CAGR (decision 0017).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`BurkeRatio`	no (documented difference)	PerformanceAnalytics 2.0.4 treats runs of negative returns as drawdowns and multiplies returns by 0.01 (percent-input assumption); agreement is verified against CAGR / sqrt(sum of findDrawdowns depths squared).

References

- Burke, G. (1994). A Sharper Sharpe Ratio. Futures, 23(3), 56. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0017: Drawdown-based ratios: Burke, Sterling, Pain; payoff ratio; Decision criteria

Sterling ratio

`sterling_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `sterling_ratio@v1`

Description

Annualized return per unit of maximum drawdown plus a fixed 10% adjustment.

Formula

$$\text{Sterling} = \frac{\text{CAGR}}{|\text{MDD}| + e}, \quad e = 0.10$$

Plain text: `CAGR / ( |max drawdown| + 0.10 )`. Symbols: see Notation.

Usage

```
report.metrics["sterling_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	annualization factor, wealth starts at 1, +10% adjustment, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Like the Calmar ratio, but the fixed 10% added to the worst loss keeps the ratio from exploding for investments with tiny drawdowns.

Professional. Sterling-type ratio in the form popularized by Kestner (1996) and Bacon, using the full-history maximum drawdown. The 10% adjustment is a historical convention, not an estimate; definitions in the literature vary (e.g. averages of annual maximum drawdowns).

Assumptions

- Full-history maximum drawdown.
- Fixed 10% adjustment.

When it can mislead

- Compared with Sterling ratios computed from averaged annual drawdowns.
- Histories of different lengths are compared.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be `>= -100%` (a valid wealth path); otherwise `not_applicable`.
3. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
4. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.

- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.
- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- Sterling adjustment of 10% added to the absolute maximum drawdown (a historical convention, not an estimate).
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `sterling_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

CAGR / (|MDD| + 10%), full history (decision 0017).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`SterlingRatio(excess=0.1)`	yes	

References

- Kestner, L. N. (1996). Getting a Handle on True Performance. Futures, 25(1), 44-46 (Sterling ratio). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0017: Drawdown-based ratios: Burke, Sterling, Pain; payoff ratio; Decision criteria

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Pain ratio

`pain_ratio` · Risk-adjusted performance · derived · unit `ratio` · formula `pain_ratio@v1`

Description

Annualized return per unit of Pain Index (average drawdown over time).

Formula

$$\mathrm{PR} = \frac{\mathrm{CAGR} - r_f}{\mathrm{PI}}, \quad r_f = 0$$

Plain text: `(CAGR - rf) / PainIndex, rf = 0`. Symbols: see Notation.

Usage

```
report.metrics["pain_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	annualization factor, wealth starts at 1, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Return relative to how far below its previous highs the investment typically was.

Professional. Zephyr Pain ratio (see Bacon): like the Martin ratio but with the mean rather than the root-mean-square drawdown, so deep drawdowns weigh less.

Assumptions

- Risk-free rate 0%.

When it can mislead

- Short histories with few drawdowns.

Minimum data

At least 2 observations and one drawdown; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown occurred (Pain Index is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.
- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `pain_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric CAGR over the canonical Pain Index (decision 0017).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`PainRatio`	no (documented difference)	Inherits the PainIndex percent-input issue in PerformanceAnalytics 2.0.4; agreement is verified against CAGR / mean(|Drawdowns(R)|).

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0017: Drawdown-based ratios: Burke, Sterling, Pain; payoff ratio; Decision criteria

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Recovery factor

`recovery_factor` · Risk-adjusted performance · derived · unit `ratio` · formula `recovery_factor@v1`

Description

Total compounded return relative to the maximum drawdown.

Formula

$$\mathrm{RF} = \frac{W_n - 1}{|\mathrm{MDD}|}$$

Plain text: `cumulative return / |max drawdown|`. Symbols: see Notation.

Usage

```
report.metrics["recovery_factor"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	derived	Minimum-data policy	`ratio_standard`
Canonical parameters	wealth starts at 1		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How many times over the investment earned back its worst decline.

Professional. Net compounded profit over the maximum drawdown (a practitioner measure used in trading-system evaluation). Not annualized, so it grows with the length of the track record; compare only across equal histories.

Assumptions

- Full history.

When it can mislead

- Histories of different lengths are compared.
- Compared with implementations that use the arithmetic sum of returns.

Minimum data

At least 2 observations and one drawdown; low confidence below 3 years. (policy `ratio_standard`: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 2 observations (mathematical minimum of policy `ratio_standard`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No drawdown occurred (maximum drawdown is zero).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).

- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Wealth index W_0 = 1 before the first return counts as the first peak; drawdowns are measured from running peaks of compounded wealth.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `recovery_factor@v1`; the version increments if the definition changes.

OpenFactsheet convention

Compounded cumulative return over |MDD| (decision 0025).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.recovery_factor`	no (documented difference)	QuantStats uses the arithmetic sum of returns in the numerator.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0025: Episode durations, recovery factor and path summaries; Decision criteria

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Sharpe ratio (serial-correlation adjusted, Lo 2002)

`sharpe_ratio_autocorrelation_adjusted` · Risk-adjusted performance · estimated · unit `ratio` · formula `sharpe_ratio_autocorrelation_adjusted@v1`

Description

The annualized Sharpe ratio with the square-root-of-time factor replaced by Lo's (2002) factor, which accounts for serial correlation up to one year.

Formula

$$\mathrm{SR}_{\mathrm{Lo}} = \frac{\bar{r} - r_f}{s}\,\eta(q), \quad \eta(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\hat{\rho}_k}} = \sqrt{\frac{q}{\mathrm{VR}(q)}}, \quad q = \mathrm{round}(P), \quad r_f = 0$$

Plain text: `(mean(r_t) / std(r_t, ddof=1)) * q / sqrt(q + 2 sum_{k=1}^{q-1} (q - k) rho_k), q = round(P)`. Symbols: see Notation.

Usage

```
report.metrics["sharpe_ratio_autocorrelation_adjusted"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`long_horizon_dependence`
Canonical parameters	risk-free = 0, sample std (n-1), annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. The Sharpe ratio after allowing for gains or losses that persist from one period to the next. When returns are smoothed (common for hedge funds and private assets) this is usually lower than the conventional Sharpe ratio.

Professional. Lo (2002) annualization of the per-period Sharpe ratio: eta(q) replaces sqrt(q); with no autocorrelation the two coincide. Positive autocorrelation (for example from smoothing) lowers it. The adjustment rests on q - 1 estimated autocorrelations and is itself noisy for short histories; details report eta, sqrt(q) and the variance ratio.

Assumptions

- Risk-free rate 0% (methodology 0.1.0).
- Stationary returns; sample autocorrelations up to lag q - 1.

When it can mislead

- The history is short relative to one year (the autocorrelations are noisy).
- Serial correlation changes over time.
- Returns are skewed or fat-tailed (as for the conventional Sharpe ratio).

Minimum data

At least two years of observations; low confidence below 5 years, moderate below 10 years. (policy `long_horizon_dependence`: One-year-horizon serial-correlation statistics need at least two years of data (undefined below); their standard error is about sqrt(4 / (3 y)) for y years, so confidence is low below 5 years and moderate below 10 years.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 4 observations (mathematical minimum of policy `long_horizon_dependence`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The frequency is not regular or there is one observation per year (q = 1).
   - Fewer than two years of observations (n < 2q).
   - Sample standard deviation is exactly zero (constant returns).
   - q + 2 sum (q - k) rho_k <= 0 (estimated autocorrelations imply a non-positive one-year variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 10`; **moderate** if `5 <= y < 10`; **low** if `y < 5` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.
- Sample standard deviation with the n - 1 denominator.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.

- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `sharpe_ratio_autocorrelation_adjusted@v1`; the version increments if the definition changes.

OpenFactsheet convention

Horizon one year (q = round(P)); per-period Sharpe ratio as in decision 0008; autocorrelations of the dependence metrics (decision 0029).

Reference implementations

No established library implementation was identified.

References

- Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36-52. [bibliography]
- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]
- Sharpe, W. F. (1994). The Sharpe Ratio. Journal of Portfolio Management, 21(1), 49-58. [bibliography]

See also

Decision 0008: Sharpe ratio: arithmetic mean over sample volatility, square-root-of-time; Decision 0029: Serial correlation and annualization: variance ratio, Lo-adjusted Sharpe, profit factor; Decision criteria

Tail behavior

Value at Risk (historical, 95%)

`var_historical_95` · Tail behavior · estimated · unit `decimal_return` · formula `var_historical_95@v1`

Description

Return threshold that was breached in the worst 5% of periods.

Formula

$$\mathrm{VaR}_{95} = Q_{0.05}(r), \quad Q_p(r) = r_{(j)} + (h-j)\big(r_{(j+1)} - r_{(j)}\big),\ h = (n-1)p + 1,\ j = \lfloor h \rfloor$$

Plain text: `Q_0.05(r), empirical quantile with linear interpolation (Hyndman-Fan type 7)`. Symbols: see Notation.

Usage

```
report.metrics["var_historical_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	95% confidence		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. In about 1 period out of 20, the return was this bad or worse.

Professional. Historical-simulation VaR at 95% for one observation period; says nothing about losses beyond the threshold (see Expected Shortfall).

Assumptions

- The supplied history is representative of the tail.

When it can mislead

- Few observations fall in the 5% tail: with monthly data a 95% estimate rests on roughly n/20 points.

- The history does not contain a stress period.
- Returns are smoothed, which compresses the tail.

Minimum data

At least 2 observations; low confidence when fewer than 5 observations are expected in the tail (n < 100). (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- 95% confidence; historical (empirical) quantile with linear interpolation (Hyndman-Fan type 7); no distributional assumption; signed return (negative = loss).

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `var_historical_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

95% confidence, per observation period, signed return (negative = loss), linear interpolation (decision 0010).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`VaR(p=0.95, method='historical')`	yes	Reports the negated quantile (loss as a positive number in 2.0.4 internals, negative in the public wrapper).
NumPy	`quantile(method='linear')`	yes	
QuantStats 0.0.81	`stats.value_at_risk`	no (documented difference)	QuantStats defaults to a parametric (normal) VaR.

References

- Jorion, P. (2006). Value at Risk (3rd ed.). McGraw-Hill. [bibliography]
- Hyndman, R. J., & Fan, Y. (1996). Sample Quantiles in Statistical Packages. The American Statistician, 50(4), 361-365 (definition 7 = linear interpolation). [bibliography]

See also

Decision 0010: Historical VaR, Expected Shortfall and tail ratio; Decision criteria

Expected Shortfall / CVaR (historical, 95%)

`expected_shortfall_historical_95` · Tail behavior · estimated · unit `decimal_return` · formula `expected_shortfall_historical_95@v1`

Description

Average return in the periods at or below the 95% VaR threshold.

Formula

$$\mathrm{ES}_{95} = \frac{1}{\mid T \mid} \sum_{t \in T} r_t, \quad T = \{t : r_t \le Q_{0.05}(r)\}$$

Plain text: `mean(r_t | r_t <= Q_0.05(r))`. Symbols: see Notation.

Usage

```
report.metrics["expected_shortfall_historical_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	95% confidence		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. When things went badly (the worst 5% of periods), this was the average return.

Professional. Historical Expected Shortfall (CVaR); a coherent risk measure (Acerbi & Tasche, 2002) that accounts for the severity of tail losses.

Assumptions

- The supplied history is representative of the tail.

When it can mislead

- Few observations fall in the 5% tail: with monthly data a 95% estimate rests on roughly n/20 points.
- The history does not contain a stress period.
- Returns are smoothed, which compresses the tail.

Minimum data

At least 2 observations; low confidence when fewer than 5 observations are expected in the tail (n < 100). (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- 95% confidence; historical (empirical) quantile with linear interpolation (Hyndman-Fan type 7); no distributional assumption; signed return (negative = loss).

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `expected_shortfall_historical_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Mean of returns at or below the linear-interpolated 5% quantile; signed.

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`ES(p=0.95, method='historical')`	yes	Uses returns strictly below the quantile; identical unless an observation equals the quantile exactly.
QuantStats 0.0.81	`stats.cvar`	no (documented difference)	QuantStats conditions on a parametric VaR threshold.
empyrical-reloaded 0.5.12	`conditional_value_at_risk`	yes	Uses returns at or below the 5th percentile.

References

- Acerbi, C., & Tasche, D. (2002). On the Coherence of Expected Shortfall. Journal of Banking & Finance, 26(7), 1487-1503. [bibliography]
- Jorion, P. (2006). Value at Risk (3rd ed.). McGraw-Hill. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0010: Historical VaR, Expected Shortfall and tail ratio; Decision criteria

Tail ratio (95th / 5th percentile)

`tail_ratio` · Tail behavior · estimated · unit `ratio` · formula `tail_ratio@v1`

Description

Size of the right tail relative to the left tail.

Formula

$$\mathrm{TR} = \frac{|Q_{0.95}(r)|}{|Q_{0.05}(r)|}$$

Plain text: `|Q_0.95(r)| / |Q_0.05(r)|`. Symbols: see Notation.

Usage

```
report.metrics["tail_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Above 1: the best periods were larger than the worst periods were bad.

Professional. Quantile-based asymmetry measure, robust to single extremes but noisy in short samples.

Assumptions

- Linear-interpolated empirical quantiles.

When it can mislead

- Few observations fall in the 5% tail: with monthly data a 95% estimate rests on roughly n/20 points.
- The history does not contain a stress period.

Minimum data

At least 2 observations; low confidence below n = 100. (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The 5th percentile return is exactly zero.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `tail_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Absolute values of the 95th and 5th percentiles (decision 0010).

Reference implementations

Library	Function	Agrees	Note
empyrical-reloaded 0.5.12	`tail_ratio`	yes	
QuantStats 0.0.81	`stats.tail_ratio`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0010: Historical VaR, Expected Shortfall and tail ratio; Decision criteria

Value at Risk (Gaussian, 95%)

`gaussian_var_95` · Tail behavior · model-dependent · unit `decimal_return` · formula `gaussian_var_95@v1`

Description

The 5% quantile of a normal distribution with the series' mean and variance.

Formula

$$\mathrm{VaR}_{95}^{N} = \bar{r} + z_{0.05}\sqrt{m_2}, \quad z_{0.05} = \Phi^{-1}(0.05) \approx -1.645$$

Plain text: `mean + z_0.05 * sqrt(m2), m2 = (1/n) sum (r_t - mean)^2, z_0.05 = -1.645`. Symbols: see Notation.

Usage

```
report.metrics["gaussian_var_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	yes
Nature	model-dependent	Minimum-data policy	`dispersion`
Canonical parameters	parametric (population moments)		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The 1-in-20 loss threshold if returns followed a bell curve. Compare it with the historical VaR: a large gap means the bell curve does not fit.

Professional. Parametric (delta-normal) VaR with population moments. Supplementary and model-dependent: it ignores skewness and fat tails, which the historical and Cornish-Fisher VaR capture.

Assumptions

- Normally distributed, i.i.d. returns.

When it can mislead

- Returns are skewed or fat-tailed (typical for hedge funds and options strategies).
- Returns are smoothed, which understates the variance.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- Model-dependent: return distribution approximated from population moments (normal, or Cornish-Fisher expansion with skewness and excess kurtosis); not a historical estimate.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Model-dependent: valid only under the stated model.
- Formula id `gaussian_var_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Population standard deviation sqrt(m2), signed return, per period (decision 0018).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`VaR(p=0.95, method='gaussian')`	yes	
QuantStats 0.0.81	`stats.value_at_risk`	no (documented difference)	QuantStats uses the sample (n - 1) standard deviation.

References

- Jorion, P. (2006). Value at Risk (3rd ed.). McGraw-Hill. [bibliography]
- Boudt, K., Peterson, B. G., & Croux, C. (2008). Estimation and Decomposition of Downside Risk for Portfolios with Non-Normal Returns. Journal of Risk, 11(2), 79-103. [bibliography]

See also

Decision 0018: Parametric VaR: Gaussian and Cornish-Fisher with population moments; Decision criteria

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Modified VaR (Cornish-Fisher, 95%)

`modified_var_95` · Tail behavior · model-dependent · unit `decimal_return` · formula `modified_var_95@v1`

Description

Parametric 95% VaR adjusted for the skewness and excess kurtosis of returns.

Formula

$$\mathrm{mVaR}_{95} = \bar{r} + h\sqrt{m_2}, \quad h = z + \frac{(z^2-1)S}{6} + \frac{(z^3-3z)K}{24} - \frac{(2z^3-5z)S^2}{36}, \quad z = \Phi^{-1}(0.05),\ S = \frac{m_3}{m_2^{3/2}},\ K = \frac{m_4}{m_2^2} - 3$$

Plain text: `mean + h * sqrt(m2), h = z + (z^2-1) S/6 + (z^3-3z) K/24 - (2z^3-5z) S^2/36`. Symbols: see Notation.

Usage

```
report.metrics["modified_var_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	yes
Nature	model-dependent	Minimum-data policy	`kurtosis`
Canonical parameters	parametric (population moments)		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. A 1-in-20 loss threshold that accounts for lopsided and fat-tailed returns.

Professional. Cornish-Fisher modified VaR (Zangari 1996; Favre & Galeano 2002) with population moments as in Boudt, Peterson & Croux (2008). Reduces to the Gaussian VaR when S = K = 0. Supplementary and model-dependent.

Assumptions

- The Cornish-Fisher expansion adequately approximates the quantile.

When it can mislead

- Skewness or kurtosis are extreme: the expansion can become non-monotonic and unreliable.
- Samples are short: higher moments are estimated imprecisely.

Minimum data

At least 4 observations; low confidence below 120 observations (kurtosis is unstable). (policy `kurtosis`: Requires at least 4 observations; kurtosis estimates are highly unstable, so low confidence below 120 and moderate below 240 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `kurtosis`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Return variance is zero or underflows in floating point (e.g. identical returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 240`; **moderate** if `120 <= n < 240`; **low** if `n < 120`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- Model-dependent: return distribution approximated from population moments (normal, or Cornish-Fisher expansion with skewness and excess kurtosis); not a historical estimate.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Model-dependent: valid only under the stated model.
- Formula id `modified_var_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Population moments, signed return, per period (decision 0018).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`VaR(p=0.95, method='modified')`	yes	

References

- Zangari, P. (1996). A VaR Methodology for Portfolios That Include Options. RiskMetrics Monitor, First Quarter, 4-12 (Cornish-Fisher VaR). [bibliography]
- Favre, L., & Galeano, J.-A. (2002). Mean-Modified Value-at-Risk Optimization with Hedge Funds. Journal of Alternative Investments, 5(2), 21-25. [bibliography]
- Boudt, K., Peterson, B. G., & Croux, C. (2008). Estimation and Decomposition of Downside Risk for Portfolios with Non-Normal Returns. Journal of Risk, 11(2), 79-103. [bibliography]

See also

Decision 0018: Parametric VaR: Gaussian and Cornish-Fisher with population moments; Decision 0022: Modified (Cornish-Fisher) Expected Shortfall; Decision criteria

Modified Expected Shortfall (Cornish-Fisher, 95%)

`modified_es_95` · Tail behavior · model-dependent · unit `decimal_return` · formula `modified_es_95@v1`

Description

Parametric 95% Expected Shortfall adjusted for skewness and excess kurtosis.

Formula

$$\mathrm{mES}_{95} = \bar{r} - \frac{\sqrt{m_2}}{a}\,\varphi(h)\left[1 + \frac{h^3 S}{6} + \frac{(h^6 - 9h^4 + 9h^2 + 3)S^2}{72} + \frac{(h^4 - 2h^2 - 1)K}{24}\right], \quad a = 0.05,\ h = \text{CF quantile of modified_ var_95}$$

Plain text: `mean - sqrt(m2)/a * phi(h) * [1 + h^3 S/6 + (h^6 - 9h^4 + 9h^2 + 3) S^2/72 + (h^4 - 2h^2 - 1) K/24], h = Cornish-Fisher quantile, a = 0.05`. Symbols: see Notation.

Usage

```
report.metrics["modified_es_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	yes
Nature	model-dependent	Minimum-data policy	`kurtosis`
Canonical parameters	parametric (population moments)		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The average loss in the worst 5% of periods implied by a skew- and fat-tail-adjusted distribution. Compare with the historical Expected Shortfall.

Professional. Modified ES of Boudt, Peterson & Croux (2008), the Cornish-Fisher analogue of the historical ES, with population moments and the full second-order expansion. Supplementary and model-dependent; for extreme skewness or kurtosis the expansion can make it less severe than the modified VaR, which is flagged.

Assumptions

- The Cornish-Fisher expansion adequately approximates the tail.

When it can mislead

- Skewness or kurtosis are extreme (the expansion breaks down).
- Samples are short: higher moments are estimated imprecisely.

Minimum data

At least 4 observations; low confidence below 120 observations. (policy `kurtosis`: Requires at least 4 observations; kurtosis estimates are highly unstable, so low confidence below 120 and moderate below 240 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `kurtosis`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Return variance is zero or underflows in floating point (e.g. identical returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 240`; **moderate** if `120 <= n < 240`; **low** if `n < 120`.

- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- Model-dependent: return distribution approximated from population moments (normal, or Cornish-Fisher expansion with skewness and excess kurtosis); not a historical estimate.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Model-dependent: valid only under the stated model.
- Formula id `modified_es_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Population moments, full expansion, signed return, per period (decision 0022).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`ES(p=0.95, method='modified')`	yes	

References

- Boudt, K., Peterson, B. G., & Croux, C. (2008). Estimation and Decomposition of Downside Risk for Portfolios with Non-Normal Returns. Journal of Risk, 11(2), 79-103. [bibliography]

See also

Decision 0022: Modified (Cornish-Fisher) Expected Shortfall; Decision criteria

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Entropic Value at Risk (95%)

`evar_95` · Tail behavior · estimated · unit `decimal_return` · formula `evar_95@v1`

Description

Entropic (exponential-moment) bound on the 5% tail of single-period returns.

Formula

$$\mathrm{EVaR}_{95} = -\mathrm{EVaR}_{0.05}(-r), \quad \mathrm{EVaR}_{a}(X) = \inf_{z>0} z \ln\left(\frac{1}{a\,n}\sum_{t=1}^{n} e^{X_t/z}\right)$$

Plain text: `-EVaR_0.05(-r), EVaR_a(X) = inf_{z>0} z * ln( mean(exp(X/z)) / a )`. Symbols: see Notation.

Usage

```
report.metrics["evar_95"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`tail`
Canonical parameters	95% entropic		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. A conservative loss threshold for bad periods, at least as severe as the Expected Shortfall and never worse than the worst period.

Professional. Entropic VaR (Ahmadi-Javid 2012) of the empirical loss distribution: the tightest Chernoff bound on VaR, coherent, and more sensitive to extreme losses than ES. Computed by one-dimensional convex minimization; signed return.

Assumptions

- The supplied history is representative of the tail.

When it can mislead

- Few observations fall in the 5% tail: with monthly data a 95% estimate rests on roughly n/20 points.
- The history does not contain a stress period.
- Returns are smoothed, which compresses the tail.
- With 20 or fewer observations (n <= 1/0.05) the empirical EVaR equals the worst period exactly.

Minimum data

At least 2 observations; low confidence below n = 100. (policy `tail`: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `tail`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the expected number of tail observations `n x 0.05`: **high** if `>= 10`; **moderate** if `5 <= . < 10`; **low** if `< 5`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- 95% confidence; entropic VaR of the empirical distribution (exponential-moment bound), one-dimensional convex minimization; signed return.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `evar_95@v1`; the version increments if the definition changes.

OpenFactsheet convention

Empirical distribution of per-period returns, 95%, signed (decision 0021).

Reference implementations

Library	Function	Agrees	Note
skfolio 1.0.6	`measures.evar`	yes	Agrees to solver tolerance (about 1e-8).

References

- Ahmadi-Javid, A. (2012). Entropic Value-at-Risk: A New Coherent Risk Measure. Journal of Optimization Theory and Applications, 155(3), 1105-1123. [bibliography]

See also

Decision 0021: Drawdown tail (DaR, CDaR, EDaR) and entropic VaR; Decision criteria

Distribution

Skewness

`skewness` · Distribution · estimated · unit `ratio` · formula `skewness@v1`

Description

Asymmetry of the return distribution.

Formula

$$G_1 = \frac{\sqrt{n(n-1)}}{n-2}\,\frac{m_3}{m_2^{3/2}}, \quad m_k = \frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^k$$

Plain text: `G1 = sqrt(n(n-1))/(n-2) * m3 / m2^(3/2), m_k = (1/n) sum (r_t - mean)^k`. Symbols: see Notation.

Usage

```
report.metrics["skewness"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`skewness`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Dimensionless ratio; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. Negative values mean occasional large losses and many small gains; positive values mean occasional large gains.

Professional. Bias-corrected sample skewness (Joanes & Gill type G1). Negative skew combined with a high Sharpe ratio is a classic signature of hidden tail risk.

Assumptions

- Returns are identically distributed over the history.

When it can mislead

- Samples are short: standard error is roughly sqrt(6/n) (about 0.3 at n = 60).
- A single extreme observation dominates the estimate.

Minimum data

At least 3 observations; low confidence below 60, moderate below 120. (policy `skewness`: Requires at least 3 observations; standard error is roughly sqrt(6/n), so low confidence below 60 and moderate below 120 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 3 observations (mathematical minimum of policy `skewness`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Return variance is zero or underflows in floating point (e.g. identical returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 120`; **moderate** if `60 <= n < 120`; **low** if `n < 60`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `skewness@v1`; the version increments if the definition changes.

OpenFactsheet convention

Adjusted Fisher-Pearson G1, as in pandas and Excel SKEW (decision 0011).

Reference implementations

Library	Function	Agrees	Note
SciPy	`stats.skew(bias=False)`	yes	
pandas	`Series.skew`	yes	
QuantStats 0.0.81	`stats.skew`	yes	
PerformanceAnalytics 2.0.4 (R)	`skewness(method='sample')`	no (documented difference)	PerformanceAnalytics 'sample' uses the population standard deviation, giving G1 * (n/(n-1))^1.5; its 'moment' method equals the biased g1.

References

- Joanes, D. N., & Gill, C. A. (1998). Comparing Measures of Sample Skewness and Kurtosis. Journal of the Royal Statistical Society D, 47(1), 183-189. [bibliography]

See also

Decision 0011: Skewness and kurtosis estimators; Decision 0020: Mean-return inference and the Jarque-Bera normality test; Decision criteria

---

Excess kurtosis

`excess_kurtosis` · Distribution · estimated · unit `ratio` · formula `excess_kurtosis@v1`

Description

Tail heaviness relative to a normal distribution (normal = 0).

Formula

$$G_2 = \frac{n-1}{(n-2)(n-3)}\left[(n+1)\frac{m_4}{m_2^2} - 3(n-1)\right]$$

Plain text: `G2 = (n-1)/((n-2)(n-3)) * ((n+1) * m4/m2^2 - 3(n-1))`. Symbols: see Notation.

Usage

```
report.metrics["excess_kurtosis"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`kurtosis`
Canonical parameters	—		

Inputs
Periodic simple returns.

Value
Dimensionless ratio; displayed with 2 decimals. Context dependent.

Interpretation
Plain language. Positive values mean extreme periods (good or bad) happen more often than a bell curve would suggest.

Professional. Bias-corrected sample excess kurtosis (G2). Fat tails make volatility-based risk measures understate tail risk.

Assumptions
- Returns are identically distributed over the history.

When it can mislead
- Samples are short: kurtosis estimates are extremely unstable.
- A single extreme observation dominates the estimate.

Minimum data
At least 4 observations; low confidence below 120, moderate below 240. (policy `kurtosis`: Requires at least 4 observations; kurtosis estimates are highly unstable, so low confidence below 120 and moderate below 240 observations.)

Calculation criteria
When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `kurtosis`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Return variance is zero or underflows in floating point (e.g. identical returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 240`; **moderate** if `120 <= n < 240`; **low** if `n < 120`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `excess_kurtosis@v1`; the version increments if the definition changes.

OpenFactsheet convention
Excess (normal = 0) G2, as in pandas and Excel KURT (decision 0011).

Reference implementations

Library	Function	Agrees	Note
SciPy	`stats.kurtosis(fisher=True, bias=False)`	yes	
pandas	`Series.kurt`	yes	
QuantStats 0.0.81	`stats.kurtosis`	yes	
PerformanceAnalytics 2.0.4 (R)	`kurtosis(method='sample_excess')`	yes	

References

- Joanes, D. N., & Gill, C. A. (1998). Comparing Measures of Sample Skewness and Kurtosis. Journal of the Royal Statistical Society D, 47(1), 183-189. [bibliography]

See also

Decision 0011: Skewness and kurtosis estimators; Decision criteria

---

Jarque-Bera normality p-value

`jarque_bera_pvalue` · Distribution · model-dependent · unit `probability` · formula `jarque_bera_pvalue@v1`

Description

Evidence against normally distributed returns from skewness and kurtosis.

Formula

$$JB = \frac{n}{6}\left(g_1^2 + \frac{g_2^2}{4}\right), \quad g_1 = \frac{m_3}{m_2^{3/2}},\ g_2 = \frac{m_4}{m_2^2} - 3, \quad p = 1 - F_{\chi^2_2}(JB)$$

Plain text: `JB = n/6 * (g1^2 + g2^2 / 4), g1 = m3/m2^1.5, g2 = m4/m2^2 - 3; p = 1 - F_chi2(2)`. Symbols: see Notation.

Usage

```
report.metrics["jarque_bera_pvalue"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	yes
Nature	model-dependent	Minimum-data policy	`skewness`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Probability between 0 and 1; displayed with 3 decimals. Context dependent.

Interpretation

Plain language. A small value (below 0.05) indicates the returns are not bell-shaped, so volatility-based statistics tell only part of the story.

Professional. Jarque & Bera (1987) test with population moments and its asymptotic chi-squared(2) p-value. A diagnostic, not a performance measure: in small samples the test has low power and poor size; in very large samples trivial departures are significant.

Assumptions

- i.i.d. returns; asymptotic chi-squared distribution of the statistic.

When it can mislead

- Samples are short (the asymptotic p-value is unreliable).
- Returns are serially correlated or heteroskedastic.

Minimum data

At least 3 observations; low confidence below 60, moderate below 120. (policy `skewness`: Requires at least 3 observations; standard error is roughly sqrt(6/n), so low confidence below 60 and moderate below 120 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 3 observations (mathematical minimum of policy `skewness`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Return variance is zero or underflows in floating point (e.g. identical returns).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 120`; **moderate** if `60 <= n < 120`; **low** if `n < 60`.

- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Probability between 0 and 1; displayed with 3 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `jarque_bera_pvalue@v1`; the version increments if the definition changes.

OpenFactsheet convention

Population (biased) skewness and excess kurtosis, as in the original test (decision 0020).

Reference implementations

Library	Function	Agrees	Note
SciPy	`stats.jarque_bera`	yes	

References

- Jarque, C. M., & Bera, A. K. (1987). A Test for Normality of Observations and Regression Residuals. International Statistical Review, 55(2), 163-172. [bibliography]

See also

Decision 0020: Mean-return inference and the Jarque-Bera normality test; Decision criteria

Consistency

Positive periods

`positive_period_pct` · Consistency · observed · unit `fraction` · formula `positive_period_pct@v1`

Description

Share of periods with a strictly positive return (hit rate).

Formula

$$\text{Pos} = \frac{1}{n}\#\{t : r_t > 0\}$$

Plain text: `#{t : r_t > 0} / n`. Symbols: see Notation.

Usage

```
report.metrics["positive_period_pct"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`location`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Higher is better.

Interpretation

Plain language. How often the investment made money in a period.

Professional. Win rate on the supplied frequency; says nothing about the size of wins and losses (see gain-to-pain, Omega). A very high hit rate with small gains can mask rare large losses.

Assumptions
- Zero returns count as neither positive nor negative.

When it can mislead
- Returns are smoothed or stale, which inflates the share of small positive periods.

Minimum data
One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria
When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status
- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result
- None beyond simple decimal returns as supplied.

Output
- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `positive_period_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention
Denominator includes all observations (zeros count as not positive).

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.win_rate`	no (documented difference)	QuantStats excludes zero-return periods from the denominator.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision criteria

Negative periods
`negative_period_pct` · Consistency · observed · unit `fraction` · formula `negative_period_pct@v1`

Description
Share of periods with a strictly negative return.

Formula
$$\text{Neg} = \frac{1}{n}\#\{t : r_t < 0\}$$

Plain text: `#{t : r_t < 0} / n`. Symbols: see Notation.

Usage

```
report.metrics["negative_period_pct"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`location`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Lower is better.

Interpretation

Plain language. How often the investment lost money in a period.

Professional. Loss frequency on the supplied frequency; together with positive periods it also reveals the share of exactly-zero periods.

Assumptions

- Zero returns count as neither positive nor negative.

When it can mislead

- Returns are smoothed or stale.

Minimum data

One observation; low confidence below 3 years. (policy `location`: Mean-type estimates: low confidence below 3 years, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `location`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `negative_period_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention

Denominator includes all observations.

Reference implementations

No established library implementation was identified.

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Longest run of positive periods

`longest_winning_streak` · Consistency · observed · unit `periods` · formula `longest_winning_streak@v1`

Description

Maximum number of consecutive periods with positive returns.

Formula

$$\max\{j - i + 1 : 1 \le i \le j \le n,\ r_u > 0\ \forall u \in i..j\}\ (0 \text{ if no } r_t > 0)$$

Plain text: `max length of consecutive runs with r_t > 0`. Symbols: see Notation.

Usage

```
report.metrics["longest_winning_streak"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Neutral (descriptive).

Interpretation

Plain language. The longest stretch of back-to-back gaining periods.

Professional. Descriptive path statistic; unusually long streaks relative to the hit rate can indicate smoothing.

Assumptions

None beyond the general inputs.

When it can mislead

- Compared across frequencies or history lengths.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `longest_winning_streak@v1`; the version increments if the definition changes.

OpenFactsheet convention

Strictly positive returns only.

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.consecutive_wins`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Longest run of negative periods

`longest_losing_streak` · Consistency · observed · unit `periods` · formula `longest_losing_streak@v1`

Description

Maximum number of consecutive periods with negative returns.

Formula

$$\max\{j - i + 1 : 1 \le i \le j \le n,\ r_u < 0\ \forall u \in i..j\}\ (0 \text{ if no } r_t < 0)$$

Plain text: `max length of consecutive runs with r_t < 0`. Symbols: see Notation.

Usage

```
report.metrics["longest_losing_streak"]
```

At a glance

Direction	Lower is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Lower is better.

Interpretation

Plain language. The longest stretch of back-to-back losing periods.

Professional. Descriptive path statistic complementing drawdown duration.

Assumptions

None beyond the general inputs.

When it can mislead

- Compared across frequencies or history lengths.

Minimum data

One observation; low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `longest_losing_streak@v1`; the version increments if the definition changes.

OpenFactsheet convention

Strictly negative returns only.

Reference implementations

Library	Function	Agrees	Note
QuantStats 0.0.81	`stats.consecutive_losses`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision criteria

Best calendar year

`best_calendar_year` · Consistency · observed · unit `decimal_return` · formula `best_calendar_year@v1`

Description

Highest compounded return of any complete calendar year.

Formula

$$\max_{y \in Y} R_y, \quad R_y = \prod_{t \in y}(1 + r_t) - 1 \text{ for each complete calendar year } y \in Y$$

Plain text: `max over complete years y of prod_{t in y} (1 + r_t) - 1`. Symbols: see Notation.

Usage

```
report.metrics["best_calendar_year"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	—		

Inputs

Dated periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The best full January-to-December year.

Professional. Maximum complete-calendar-year return; partial first and last years are excluded and listed in the details, together with every complete year's return.

Assumptions

- Only complete calendar years are used.

When it can mislead

- Few complete calendar years: one or two years say little about consistency.
- Calendar boundaries are arbitrary: a loss straddling two years is split between them.

Minimum data

One complete calendar year; low confidence below 3 years of history. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No complete calendar year is covered by the history.
   - The returns are undated or the frequency is irregular.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `best_calendar_year@v1`; the version increments if the definition changes.

OpenFactsheet convention

Complete years only; compounded monthly/quarterly/daily returns (decision 0023).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`apply.yearly(R, Return.cumulative) on complete years`	yes	Agrees on complete years; OpenFactsheet excludes partial first and last years.

References

- CFA Institute (2020). Global Investment Performance Standards (GIPS) for Firms, provisions on annualized returns. https://www.gipsstandards.org/ [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0023: Calendar-year statistics: complete years only; Decision criteria

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Worst calendar year

`worst_calendar_year` · Consistency · observed · unit `decimal_return` · formula `worst_calendar_year@v1`

Description

Lowest compounded return of any complete calendar year.

Formula

$$\min_{y \in Y} R_y, \quad R_y = \prod_{t \in y}(1 + r_t) - 1 \text{ for each complete calendar year } y \in Y$$

Plain text: `min over complete years y of prod_{t in y} (1 + r_t) - 1`. Symbols: see Notation.

Usage

```
report.metrics["worst_calendar_year"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	—		

Inputs

Dated periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The worst full January-to-December year.

Professional. Minimum complete-calendar-year return; always shown, including when it is unfavorable.

Assumptions

- Only complete calendar years are used.

When it can mislead

- Few complete calendar years: one or two years say little about consistency.
- Calendar boundaries are arbitrary: a loss straddling two years is split between them.

Minimum data

One complete calendar year; low confidence below 3 years of history. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No complete calendar year is covered by the history.
   - The returns are undated or the frequency is irregular.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `worst_calendar_year@v1`; the version increments if the definition changes.

OpenFactsheet convention

Complete years only (decision 0023).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`apply.yearly(R, Return.cumulative) on complete years`	yes	Agrees on complete years; OpenFactsheet excludes partial first and last years.

References

- CFA Institute (2020). Global Investment Performance Standards (GIPS) for Firms, provisions on annualized returns. https://www.gipsstandards.org/ [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0023: Calendar-year statistics: complete years only; Decision criteria

---

Positive calendar years

`positive_calendar_years_pct` · Consistency · observed · unit `fraction` · formula `positive_calendar_years_pct@v1`

Description

Share of complete calendar years with a positive compounded return.

Formula

$$\frac{\#\{y \in Y : R_y > 0\}}{\mid Y \mid}, \quad R_y = \prod_{t \in y}(1 + r_t) - 1 \text{ for each complete calendar year } y \in Y$$

Plain text: `#{y : R_y > 0} / #Y over complete calendar years`. Symbols: see Notation.

Usage

```
report.metrics["positive_calendar_years_pct"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`

Canonical parameters	—

Inputs
Dated periodic simple returns with a regular frequency.

Value
Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Higher is better.

Interpretation
Plain language. How many full years ended with a gain.

Professional. Calendar-year hit rate; the details list every complete year's return and every excluded partial year.

Assumptions
- Only complete calendar years are used.

When it can mislead
- Few complete calendar years: one or two years say little about consistency.
- Calendar boundaries are arbitrary: a loss straddling two years is split between them.

Minimum data
One complete calendar year; low confidence below 3 years of history. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria
When a value is produced (checked in this order)
1. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
2. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - No complete calendar year is covered by the history.
   - The returns are undated or the frequency is irregular.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status
- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result
- None beyond simple decimal returns as supplied.

Output
- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `positive_calendar_years_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention
Complete years only; a zero year counts as not positive (decision 0023).

Reference implementations
Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`apply.yearly(R, Return.cumulative) on complete years`	yes	Agrees on complete years; OpenFactsheet excludes partial first and last years.

References

- CFA Institute (2020). Global Investment Performance Standards (GIPS) for Firms, provisions on annualized returns. https://www.gipsstandards.org/ [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0023: Calendar-year statistics: complete years only; Decision criteria

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Best rolling one-year return

`best_rolling_year` · Consistency · observed · unit `decimal_return` · formula `best_rolling_year@v1`

Description

Highest compounded return over any window of one year of consecutive observations.

Formula

$$\max_{w \le t \le n} R_t^{(w)}, \quad R_t^{(w)} = \prod_{i=t-w+1}^{t} (1 + r_i) - 1, \quad t = w, \ldots, n, \quad w = \text{round}(P)$$

Plain text: `max_t prod_{i=t-w+1..t} (1 + r_i) - 1, w = round(P)`. Symbols: see Notation.

Usage

```
report.metrics["best_rolling_year"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The best twelve months (or one year of periods) an investor could have experienced, whatever the start date.

Professional. Maximum of all overlapping one-year compounded returns; every start date is considered, so no favorable start date can be chosen.

Assumptions

- Window = one year of observations, step = one observation.

When it can mislead

- Overlapping windows share observations, so rolling statistics are strongly autocorrelated and overstate the amount of independent evidence.
- Gaps in the history make a window span more than one calendar year.

Minimum data

One full window (n >= P); low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fewer observations than one year (n < w).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `best_rolling_year@v1`; the version increments if the definition changes.

OpenFactsheet convention

Window round(P) observations, step 1 (decision 0024).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`rollapply(R, P, Return.cumulative)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0024: Rolling one-year windows; Decision 0026: Canonical charts and the drawdown-episode table; Decision criteria

Worst rolling one-year return

`worst_rolling_year` · Consistency · observed · unit `decimal_return` · formula `worst_rolling_year@v1`

Description

Lowest compounded return over any window of one year of consecutive observations.

Formula

$$\min_{w \le t \le n} R_t^{(w)}, \quad R_t^{(w)} = \prod_{i=t-w+1}^{t} (1 + r_i) - 1, \quad t = w, \ldots, n, \quad w = \text{round}(P)$$

Plain text: `min_t prod_{i=t-w+1..t} (1 + r_i) - 1, w = round(P)`. Symbols: see Notation.

Usage

```
report.metrics["worst_rolling_year"]
```

At a glance

Direction	Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better)	Annualized	yes

Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	annualization factor		

Inputs

Periodic simple returns with a regular frequency.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).

Interpretation

Plain language. The worst twelve months (or one year of periods) an investor could have experienced, whatever the start date.

Professional. Minimum of all overlapping one-year compounded returns; a start-date-free worst case that is often much worse than the worst calendar year.

Assumptions

- Window = one year of observations, step = one observation.

When it can mislead

- Overlapping windows share observations, so rolling statistics are strongly autocorrelated and overstate the amount of independent evidence.
- Gaps in the history make a window span more than one calendar year.

Minimum data

One full window (n >= P); low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fewer observations than one year (n < w).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).

- Direction: Smaller loss is better: losses are negative numbers, so −10% is better than −25% (the higher, less negative value is better).
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `worst_rolling_year@v1`; the version increments if the definition changes.

OpenFactsheet convention
Window round(P) observations, step 1 (decision 0024).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`rollapply(R, P, Return.cumulative)`	yes	

References
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also
Decision 0024: Rolling one-year windows; Decision 0026: Canonical charts and the drawdown-episode table; Decision criteria

---

Positive rolling one-year windows
`positive_rolling_year_pct` · Consistency · observed · unit `fraction` · formula `positive_rolling_year_pct@v1`

Description
Share of all one-year windows (every start date) with a positive compounded return.

Formula
$$\frac{\#\{t : R_t^{(w)} > 0\}}{n - w + 1}, \quad R_t^{(w)} = \prod_{i=t-w+1}^{t} (1 + r_i) - 1, \quad t = w, \ldots, n, \quad w = \mathrm{round}(P)$$

Plain text: `#{t : R_t^(w) > 0} / (n - w + 1), w = round(P)`. Symbols: see Notation.

Usage
```
report.metrics["positive_rolling_year_pct"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`path_risk`
Canonical parameters	annualization factor		

Inputs
Periodic simple returns with a regular frequency.

Value
Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Higher is better.

Interpretation
Plain language. If you had invested at any point and held for one year, how often you would have ended with a gain.

Professional. Rolling one-year hit rate over all start dates. Windows overlap, so it is a descriptive statistic, not a probability estimate.

Assumptions
- Window = one year of observations, step = one observation.

When it can mislead
- Overlapping windows share observations, so rolling statistics are strongly autocorrelated and overstate the amount of independent evidence.
- Gaps in the history make a window span more than one calendar year.

Minimum data

One full window (n >= P); low confidence below 3 years. (policy `path_risk`: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. Every return must be >= -100% (a valid wealth path); otherwise `not_applicable`.
3. At least 1 observations (mathematical minimum of policy `path_risk`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fewer observations than one year (n < w).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - A -100% return (total loss) adds a warning.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `positive_rolling_year_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention

Window round(P) observations, step 1; zero counts as not positive (decision 0024).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`rollapply(R, P, Return.cumulative)`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0024: Rolling one-year windows; Decision 0026: Canonical charts and the drawdown-episode table; Decision criteria

Serial dependence & smoothing

Autocorrelation (lag 1)

`autocorrelation_lag1` · Serial dependence & smoothing · estimated · unit `correlation` · formula `autocorrelation_lag1@v1`

Description

Correlation between each return and the return 1 period(s) earlier.

Formula

$$\hat{\rho}_1 = \frac{\sum_{t=2}^{n}(r_t - \bar{r})(r_{t-1} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

Plain text: `rho_1 = sum_(t>1) (r_t - m)(r_(t-1) - m) / sum_t (r_t - m)^2`. Symbols: see Notation.

Usage

```
report.metrics["autocorrelation_lag1"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`autocorrelation`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Correlation coefficient between -1 and 1; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. Whether good or bad periods tend to be followed by similar periods. Values near zero are typical for liquid, market-priced investments.

Professional. Sample autocorrelation (full-sample variance denominator). Significant positive low-lag autocorrelation is a hallmark of return smoothing or illiquidity (Getmansky, Lo & Makarov, 2004) and biases square-root-of-time annualization. Approximate 95% significance bound: +/- 1.96 / sqrt(n).

Assumptions

- Stationary returns.

When it can mislead

- Samples are short: standard error is roughly 1/sqrt(n).
- Volatility regimes change over the history.

Minimum data

At least 5 observations; low confidence below 36, moderate below 60. (policy `autocorrelation`: Requires at least 5 observations; standard error is roughly 1/sqrt(n), so low confidence below 36 and moderate below 60 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 5 observations (mathematical minimum of policy `autocorrelation`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - n <= 2 observations.
   - All returns are identical (zero variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Correlation coefficient between -1 and 1; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `autocorrelation_lag1@v1`; the version increments if the definition changes.

OpenFactsheet convention

Standard biased-denominator estimator; not bias-corrected.

Reference implementations

Library	Function	Agrees	Note
statsmodels	`tsa.stattools.acf(adjusted=False)`	yes	

References

- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also

Decision 0014: Return-smoothing risk: diagnose and warn, never unsmooth; Decision criteria

---

Autocorrelation (lag 2)

`autocorrelation_lag2` · Serial dependence & smoothing · estimated · unit `correlation` · formula `autocorrelation_lag2@v1`

Description

Correlation between each return and the return 2 period(s) earlier.

Formula

$$\hat{\rho}_2 = \frac{\sum_{t=3}^{n}(r_t - \bar{r})(r_{t-2} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

Plain text: `rho_2 = sum_(t>2) (r_t - m)(r_(t-2) - m) / sum_t (r_t - m)^2`. Symbols: see Notation.

Usage

```
report.metrics["autocorrelation_lag2"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`autocorrelation`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Correlation coefficient between -1 and 1; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. Whether good or bad periods tend to be followed by similar periods. Values near zero are typical for liquid, market-priced investments.

Professional. Sample autocorrelation (full-sample variance denominator). Significant positive low-lag autocorrelation is a hallmark of return smoothing or illiquidity (Getmansky, Lo & Makarov, 2004) and biases square-root-of-time annualization. Approximate 95% significance bound: +/- 1.96 / sqrt(n).

Assumptions

- Stationary returns.

When it can mislead

- Samples are short: standard error is roughly 1/sqrt(n).
- Volatility regimes change over the history.

Minimum data

At least 5 observations; low confidence below 36, moderate below 60. (policy `autocorrelation`: Requires at least 5 observations; standard error is roughly 1/sqrt(n), so low confidence below 36 and moderate below 60 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 5 observations (mathematical minimum of policy `autocorrelation`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - n <= 3 observations.
   - All returns are identical (zero variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Correlation coefficient between -1 and 1; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `autocorrelation_lag2@v1`; the version increments if the definition changes.

OpenFactsheet convention

Standard biased-denominator estimator; not bias-corrected.

Reference implementations

Library	Function	Agrees	Note
statsmodels	`tsa.stattools.acf(adjusted=False)`	yes	

References

- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also

Decision criteria

Autocorrelation (lag 3)

`autocorrelation_lag3` · Serial dependence & smoothing · estimated · unit `correlation` · formula `autocorrelation_lag3@v1`

Description

Correlation between each return and the return 3 period(s) earlier.

Formula

$$\hat{\rho}_3 = \frac{\sum_{t=4}^{n}(r_t - \bar{r})(r_{t-3} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

Plain text: `rho_3 = sum_(t>3) (r_t - m)(r_(t-3) - m) / sum_t (r_t - m)^2`. Symbols: see Notation.

Usage

```
report.metrics["autocorrelation_lag3"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`autocorrelation`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Correlation coefficient between -1 and 1; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. Whether good or bad periods tend to be followed by similar periods. Values near zero are typical for liquid, market-priced investments.

Professional. Sample autocorrelation (full-sample variance denominator). Significant positive low-lag autocorrelation is a hallmark of return smoothing or illiquidity (Getmansky, Lo & Makarov, 2004) and biases square-root-of-time annualization. Approximate 95% significance bound: +/- 1.96 / sqrt(n).

Assumptions

- Stationary returns.

When it can mislead

- Samples are short: standard error is roughly 1/sqrt(n).
- Volatility regimes change over the history.

Minimum data

At least 5 observations; low confidence below 36, moderate below 60. (policy `autocorrelation`: Requires at least 5 observations; standard error is roughly 1/sqrt(n), so low confidence below 36 and moderate below 60 observations.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 5 observations (mathematical minimum of policy `autocorrelation`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - n <= 4 observations.
   - All returns are identical (zero variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Correlation coefficient between -1 and 1; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `autocorrelation_lag3@v1`; the version increments if the definition changes.

OpenFactsheet convention

Standard biased-denominator estimator; not bias-corrected.

Reference implementations

Library	Function	Agrees	Note
statsmodels	`tsa.stattools.acf(adjusted=False)`	yes	

References

- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also

Decision criteria

Ljung-Box p-value (3 lags)

`ljung_box_pvalue` · Serial dependence & smoothing · model-dependent · unit `probability` · formula `ljung_box_pvalue@v1`

Description

Joint test that the first three autocorrelations are all zero.

Formula

$$Q = n(n+2)\sum_{k=1}^{3}\frac{\hat{\rho}_k^2}{n-k}, \quad p = 1 - F_{\chi^2_3}(Q)$$

Plain text: `Q = n(n+2) sum_{k=1..3} rho_k^2 / (n - k); p-value from chi-squared(3)`. Symbols: see Notation.

Usage

```
report.metrics["ljung_box_pvalue"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	yes
Nature	model-dependent	Minimum-data policy	`ljung_box`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Probability between 0 and 1; displayed with 3 decimals. Context dependent.

Interpretation

Plain language. A small value (below 0.05) suggests returns are not independent from one period to the next.

Professional. Ljung-Box portmanteau test with 3 lags (matching the reported autocorrelations). A diagnostic, not a performance measure; with very long samples trivial dependence becomes 'significant'.

Assumptions

- Asymptotic chi-squared distribution of Q.

When it can mislead

- Very short samples (low power) or very long samples (tiny effects significant).
- Heteroskedastic returns (the test assumes constant variance).

Minimum data

At least 10 observations; low confidence below 36, moderate below 60. (policy `ljung_box`: Requires at least 10 observations for a 3-lag test; low below 36, moderate below 60.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 10 observations (mathematical minimum of policy `ljung_box`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - All returns are identical (zero variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Probability between 0 and 1; displayed with 3 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `ljung_box_pvalue@v1`; the version increments if the definition changes.

OpenFactsheet convention

3 lags, no degrees-of-freedom adjustment; supplementary status.

Reference implementations

Library	Function	Agrees	Note
statsmodels	`stats.diagnostic.acorr_ljungbox`	yes	

References

- Ljung, G. M., & Box, G. E. P. (1978). On a Measure of Lack of Fit in Time Series Models. Biometrika, 65(2), 297-303. [bibliography]

See also

Decision 0014: Return-smoothing risk: diagnose and warn, never unsmooth; Decision criteria

---

Zero-return periods

`zero_return_pct` · Serial dependence & smoothing · observed · unit `fraction` · formula `zero_return_pct@v1`

Description

Share of periods with a return of exactly zero.

Formula

$$Z = \frac{1}{n}\#\{t : r_t = 0\}$$

Plain text: `#{t : r_t = 0} / n`. Symbols: see Notation.

Usage

```
report.metrics["zero_return_pct"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	no
Nature	observed	Minimum-data policy	`observed_point`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Context dependent.

Interpretation

Plain language. Many exactly-zero periods can mean prices were not updated (stale pricing).

Professional. Stale-price indicator; exact zeros are rare for marked-to-market portfolios and suppress measured volatility.

Assumptions
None beyond the general inputs.

When it can mislead
- Cash-like or closed strategies legitimately report zero returns.
- Returns were rounded to few decimals.

Minimum data
Any non-empty history. (policy `observed_point`: Observed fact about the supplied history; any non-empty history suffices.)

Calculation criteria
When a value is produced (checked in this order)

1. At least 1 observations (mathematical minimum of policy `observed_point`); otherwise `insufficient_data`.
2. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status
- Base confidence is always **high** (an observed fact about the supplied data).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result
- None beyond simple decimal returns as supplied.

Output
- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `zero_return_pct@v1`; the version increments if the definition changes.

OpenFactsheet convention
Exact equality with 0.0 after normalization.

Reference implementations
No established library implementation was identified.

References

- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also
Decision criteria

Variance ratio (one-year horizon)
`variance_ratio_1y` · Serial dependence & smoothing · estimated · unit `ratio` · formula `variance_ratio_1y@v1`

Description
Variance of one-year sums of returns relative to what independent periods would imply; 1 when returns are serially uncorrelated.

Formula
$$\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\hat{\rho}_k, \quad q = \mathrm{round}(P), \quad z = \frac{\mathrm{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qn)}}$$

Plain text: `VR(q) = 1 + 2 sum_{k=1}^{q-1} (1 - k/q) rho_k, q = round(P)`. Symbols: see Notation.

Usage

```
report.metrics["variance_ratio_1y"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	returns only	Supplementary	yes
Nature	estimated	Minimum-data policy	`long_horizon_dependence`
Canonical parameters	—		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. Whether risk over a whole year is larger or smaller than monthly (or daily) volatility suggests. Above 1: gains and losses tend to persist, so yearly swings are bigger than the annualized volatility shows; below 1: they tend to reverse.

Professional. Lo-MacKinlay variance ratio at the annualization horizon, written in terms of the sample autocorrelations, so it equals (q + 2 sum (q - k) rho_k) / q, the quantity that corrects square-root-of-time annualization (Lo 2002). Details report the horizon, the asymptotic i.i.d. standard error, z-statistic and p-value. A diagnostic of the annualization assumption, not a performance measure.

Assumptions

- Stationary returns; sample autocorrelations with the full-sample denominator.
- Homoskedastic i.i.d. null for the standard error.

When it can mislead

- The history is short relative to one year: the standard error is about sqrt(4 / (3 years)).
- Volatility regimes change (heteroskedasticity inflates apparent significance).

Minimum data

At least two years of observations; low confidence below 5 years, moderate below 10 years. (policy `long_horizon_dependence`: One-year-horizon serial-correlation statistics need at least two years of data (undefined below); their standard error is about sqrt(4 / (3 y)) for y years, so confidence is low below 5 years and moderate below 10 years.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `long_horizon_dependence`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The frequency is not regular (P undefined) or there is one observation per year (q = 1).
   - Fewer than two years of observations (n < 2q).
   - All returns are identical (zero variance).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 10`; **moderate** if `5 <= y < 10`; **low** if `y < 5` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - None apply to this metric.
- Status: always `supplementary` when a value exists (diagnostic statistic).

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `variance_ratio_1y@v1`; the version increments if the definition changes.

OpenFactsheet convention

Autocorrelation form of the overlapping variance ratio on simple returns, horizon one year, i.i.d. standard error; supplementary (decision 0029).

Reference implementations

Library	Function	Agrees	Note
statsmodels	`tsa.stattools.acf(adjusted=False)`	yes	Autocorrelations.

References

- Lo, A. W., & MacKinlay, A. C. (1988). Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test. Review of Financial Studies, 1(1), 41-66. [bibliography]
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36-52. [bibliography]
- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. Journal of Financial Economics, 74(3), 529-609. [bibliography]

See also

Decision 0029: Serial correlation and annualization: variance ratio, Lo-adjusted Sharpe, profit factor; Decision criteria

Benchmark-relative analysis

Beta

`beta` · Benchmark-relative analysis · estimated · unit `beta` · formula `beta@v1`

Description

Sensitivity of fund returns to benchmark returns.

Formula

$$\hat{\beta} = \frac{\sum_{t=1}^{m}(r_t - \bar{r})(b_t - \bar{b})}{\sum_{t=1}^{m}(b_t - \bar{b})^2} \quad (\text{OLS slope of } r_t = \alpha + \beta b_t + \varepsilon_t)$$

Plain text: `slope b in OLS: r_fund,t = a + b * r_bench,t + e_t`. Symbols: see Notation.

Usage

```
report.metrics["beta"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Regression coefficient; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. A beta of 1 means the fund tended to move one-for-one with the benchmark; 0.5 means about half as much.

Professional. OLS beta against the supplied benchmark with rf = 0; details include the standard error and t-statistic (conventional i.i.d. errors).

Assumptions

- Linear, stable relationship over the overlap.
- Risk-free rate 0%.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Returns are smoothed: lagged benchmark exposure is missed and beta understated.
- The relationship is nonlinear (e.g. option-like strategies).

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant over the aligned periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.

Output

- Unit: Regression coefficient; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `beta@v1`; the version increments if the definition changes.

OpenFactsheet convention

Total returns (rf = 0), OLS with intercept (decision 0012).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`CAPM.beta`	yes	
empyrical-reloaded 0.5.12	`beta`	yes	
statsmodels	`OLS`	yes	

References

- Jensen, M. C. (1968). The Performance of Mutual Funds in the Period 1945-1964. Journal of Finance, 23(2), 389-416. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

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Alpha (annualized, CAPM regression)

`alpha` · Benchmark-relative analysis · model-dependent · unit `decimal_return` · formula `alpha@v1`

Description

Average return not explained by benchmark exposure, annualized.

Formula

$$\alpha_{\text{ann}} = P\,\hat{\alpha}, \quad \hat{\alpha} = \bar{r} - \hat{\beta}\,\bar{b}$$

Plain text: `P * a, with a the intercept of OLS: r_fund,t = a + b * r_bench,t + e_t`. Symbols: see Notation.

Usage

```
report.metrics["alpha"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	model-dependent	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The part of the fund's yearly return that the benchmark does not explain; it depends heavily on which benchmark is used.

Professional. Jensen-style single-index alpha against the supplied benchmark with rf = 0, annualized arithmetically. Details report the per-period intercept, its standard error, t-statistic and p-value. Statistical significance, not the point estimate, indicates whether alpha is distinguishable from zero.

Assumptions

- Single-index (CAPM-style) model with the supplied benchmark.
- Risk-free rate 0%.
- Arithmetic annualization of the intercept.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Omitted risk factors (size, value, momentum, credit, illiquidity) masquerade as alpha.
- Returns are smoothed or the history is short.

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant over the aligned periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `alpha@v1`; the version increments if the definition changes.

OpenFactsheet convention

Per-period OLS intercept x periods per year (decision 0012).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`CAPM.alpha * P`	yes	PerformanceAnalytics reports the per-period intercept.
empyrical-reloaded 0.5.12	`alpha`	no (documented difference)	empyrical annualizes geometrically: (1 + a)^P - 1.

References

- Jensen, M. C. (1968). The Performance of Mutual Funds in the Period 1945-1964. Journal of Finance, 23(2), 389-416. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

R-squared

`r_squared` · Benchmark-relative analysis · estimated · unit `fraction` · formula `r_squared@v1`

Description

Share of fund return variance explained by the benchmark.

Formula

$$R^2 = 1 - \frac{\sum_{t=1}^{m} \hat{\varepsilon}_t^2}{\sum_{t=1}^{m} (r_t - \bar{r})^2}, \quad \hat{\varepsilon}_t = r_t - \hat{\alpha} - \hat{\beta} b_t$$

Plain text: `1 - SSR/SST of the OLS regression (equals correlation^2 for one regressor)`. Symbols: see Notation.

Usage

```
report.metrics["r_squared"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Neutral (descriptive).

Interpretation

Plain language. How closely the fund's ups and downs followed the benchmark (1 = perfectly).

Professional. Low R-squared means beta and alpha say little about the fund; the benchmark may be inappropriate.

Assumptions

- Linear relationship.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant over the aligned periods.
   - Fund returns are constant over the aligned periods (no variance to explain).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `r_squared@v1`; the version increments if the definition changes.

OpenFactsheet convention

Regression with intercept.

Reference implementations

Library	Function	Agrees	Note
statsmodels	`OLS.rsquared`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

---

Correlation with benchmark

`correlation` · Benchmark-relative analysis · estimated · unit `correlation` · formula `correlation@v1`

Description

Linear co-movement of fund and benchmark returns.

Formula

$$\rho_{rb} = \frac{\sum_{t=1}^{m}(r_t - \bar{r})(b_t - \bar{b})}{\sqrt{\sum_{t=1}^{m}(r_t - \bar{r})^2 \sum_{t=1}^{m}(b_t - \bar{b})^2}}$$

Plain text: `Pearson correlation of aligned returns`. Symbols: see Notation.

Usage

```
report.metrics["correlation"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Correlation coefficient between -1 and 1; displayed with 2 decimals. Neutral (descriptive).

Interpretation

Plain language. +1 means the fund moved exactly with the benchmark, 0 no linear relation, -1 exactly opposite.

Professional. Pearson correlation; sensitive to outliers and to regime changes.

Assumptions

- Linear dependence.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Dependence is nonlinear or tail-specific.

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fund or benchmark returns are constant over the aligned periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.

Output

- Unit: Correlation coefficient between -1 and 1; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `correlation@v1`; the version increments if the definition changes.

OpenFactsheet convention

Pearson product-moment correlation.

Reference implementations

Library	Function	Agrees	Note
NumPy	`corrcoef`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

---

Tracking error (annualized)

`tracking_error` · Benchmark-relative analysis · estimated · unit `decimal_return` · formula `tracking_error@v1`

Description

Volatility of the difference between fund and benchmark returns, annualized.

Formula

$$\mathrm{TE} = s_a\sqrt{P}, \quad s_a = \sqrt{\frac{1}{m-1}\sum_{t=1}^{m}(a_t - \bar{a})^2}, \quad a_t = r_t - b_t$$

Plain text: `std(r_fund,t - r_bench,t, ddof=1) * sqrt(P)`. Symbols: see Notation.

Usage

```
report.metrics["tracking_error"]
```

At a glance

Direction	Context dependent	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, sample std (n-1), annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Context dependent.

Interpretation

Plain language. How much the fund's results differed from the benchmark from period to period.

Professional. Ex-post tracking error (active risk); the Information Ratio denominator.

Assumptions

- Square-root-of-time scaling.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Returns are smoothed.

Minimum data

At least 2 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Sample standard deviation with the n - 1 denominator.

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `tracking_error@v1`; the version increments if the definition changes.

OpenFactsheet convention

Sample standard deviation of arithmetic active returns x sqrt(P).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`TrackingError`	yes	
empyrical-reloaded 0.5.12	`tracking_error`	yes	

References

- Grinold, R. C., & Kahn, R. N. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

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Active return (annualized, arithmetic)

`active_return` · Benchmark-relative analysis · estimated · unit `decimal_return` · formula `active_return@v1`

Description

Average fund-minus-benchmark return, annualized.

Formula

$$\mathrm{AR} = P\,\bar{a} = \frac{P}{m}\sum_{t=1}^{m}(r_t - b_t)$$

Plain text: `P * mean(r_fund,t - r_bench,t)`. Symbols: see Notation.

Usage

```
report.metrics["active_return"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. How much the fund returned above (or below) the benchmark per year, on average.

Professional. Arithmetic active return, consistent with the Information Ratio numerator; differs from the difference of geometric annualized returns (the PerformanceAnalytics ActivePremium).

Assumptions

- Linear annualization of the mean.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.

- The overlap with the benchmark is short.

Minimum data

At least 2 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `active_return@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic mean of active returns x periods per year (decision 0012).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`ActivePremium`	no (documented difference)	Difference of geometric annualized returns.

References

- Grinold, R. C., & Kahn, R. N. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

Information ratio

`information_ratio` · Benchmark-relative analysis · estimated · unit `ratio` · formula `information_ratio@v1`

Description

Active return per unit of tracking error, annualized.

Formula

$$\mathrm{IR} = \frac{P\,\bar{a}}{s_a\sqrt{P}} = \frac{\bar{a}}{s_a}\sqrt{P}$$

Plain text: `P * mean(a_t) / (std(a_t, ddof=1) * sqrt(P)), a_t = r_fund,t - r_bench,t`. Symbols: see Notation.

Usage

```
report.metrics["information_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, sample std (n-1), annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How consistently the fund beat (or lagged) its benchmark relative to how much it deviated from it.
Professional. Arithmetic ex-post Information Ratio (Grinold & Kahn), equal to the Sharpe ratio of active returns. PerformanceAnalytics divides the difference of geometric annualized returns instead.

Assumptions

- Square-root-of-time scaling.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Returns are smoothed.
- Tracking error is tiny.

Minimum data

At least 2 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Tracking error is exactly zero (fund equals benchmark plus a constant).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.

  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Sample standard deviation with the n - 1 denominator.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `information_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic active mean over tracking error (decision 0012).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`InformationRatio`	no (documented difference)	Uses ActivePremium (geometric) / TrackingError.
QuantStats 0.0.81	`stats.information_ratio`	no (documented difference)	QuantStats reports the non-annualized per-period ratio.

References

- Grinold, R. C., & Kahn, R. N. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill. [bibliography]
- Goodwin, T. H. (1998). The Information Ratio. Financial Analysts Journal, 54(4), 34-43. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

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Treynor ratio

`treynor_ratio` · Benchmark-relative analysis · model-dependent · unit `ratio` · formula `treynor_ratio@v1`

Description

Annualized return per unit of benchmark beta.

Formula

$$\mathrm{TR} = \frac{P\,(\bar{r}^{(m)} - r_f)}{\hat{\beta}}, \quad r_f = 0, \quad \bar{r}^{(m)} = \frac{1}{m}\sum_{t=1}^{m} r_t (\text{aligned periods})$$

Plain text: `P * mean(r_fund,t - rf) / beta, rf = 0`. Symbols: see Notation.

Usage

```
report.metrics["treynor_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	model-dependent	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS, risk-free = 0, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Return earned per unit of market (benchmark) risk taken.

Professional. Treynor (1965) with arithmetic annualized mean on the aligned sample; only meaningful for well-diversified portfolios with a stable, clearly positive beta.

Assumptions

- Risk-free rate 0%.
- Beta is the relevant risk measure.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Beta is close to zero or negative: the ratio explodes or changes sign.
- The portfolio is not diversified (idiosyncratic risk is ignored).

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant (beta undefined).
   - Estimated beta is exactly zero.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `treynor_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic annualized mean / OLS beta on aligned periods (decision 0012).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`TreynorRatio`	no (documented difference)	Uses the geometric annualized return in the numerator.

References

- Treynor, J. L. (1965). How to Rate Management of Investment Funds. Harvard Business Review, 43(1), 63-75. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

---

Upside capture ratio

`upside_capture` · Benchmark-relative analysis · derived · unit `ratio` · formula `upside_capture@v1`

Description

Fund return in benchmark up periods relative to the benchmark's, compounded.

Formula

$$\mathrm{UC} = \frac{G(U, r)}{G(U, b)}, \quad U = \{t : b_t > 0\}, \quad G(S, x) = \begin{cases} \left(\prod_{t \in S}(1 + x_t)\right)^{P/|\,S\,|} - 1 & \prod_{t \in S}(1 + x_t) > 0 \\ -1 & \text{otherwise} \end{cases}$$

Plain text: `annualized geometric fund return over {t : r_bench,t > 0} / same for benchmark`. Symbols: see Notation.

Usage

```
report.metrics["upside_capture"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	derived	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Above 1 (100%): the fund gained more than the benchmark when the benchmark rose.

Professional. Morningstar-style geometric upside capture; read together with downside capture.

Assumptions

- Up periods defined by strictly positive benchmark returns.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.

- The overlap with the benchmark is short.
- Few up periods in the overlap.

Minimum data

At least 2 aligned observations and one up period; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The benchmark has no up periods in the overlap.
   - The benchmark's compounded up-period return is exactly zero.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `upside_capture@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric, annualized over the up periods (decision 0012).

Reference implementations

Library	Function	Agrees	Note
empyrical-reloaded 0.5.12	`up_capture / down_capture`	yes	
PerformanceAnalytics 2.0.4 (R)	`UpDownRatios(method='Capture')`	no (documented difference)	PerformanceAnalytics divides sums of returns and counts zero-benchmark periods as down periods.

References

- Morningstar, Inc. Upside/Downside Capture Ratio methodology (geometric compounding of returns in up/down benchmark periods). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

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Downside capture ratio

`downside_capture` · Benchmark-relative analysis · derived · unit `ratio` · formula `downside_capture@v1`

Description

Fund return in benchmark down periods relative to the benchmark's, compounded.

Formula

$$\mathrm{DC} = \frac{G(D, r)}{G(D, b)}, \quad D = \{t : b_t < 0\} \quad (G \text{ as for upside capture})$$

Plain text: `annualized geometric fund return over {t : r_bench,t < 0} / same for benchmark`. Symbols: see Notation.

Usage

```
report.metrics["downside_capture"]
```

At a glance

Direction	Lower is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	derived	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Lower is better.

Interpretation

Plain language. Below 1 (100%): the fund lost less than the benchmark when the benchmark fell. Negative: the fund gained on average when the benchmark fell.

Professional. Morningstar-style geometric downside capture; read together with upside capture.

Assumptions

- Down periods defined by strictly negative benchmark returns.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Few down periods in the overlap.

Minimum data

At least 2 aligned observations and one down period; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The benchmark has no down periods in the overlap.

- The benchmark's compounded down-period return is exactly zero.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Lower is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `downside_capture@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric, annualized over the down periods (decision 0012).

Reference implementations

Library	Function	Agrees	Note
empyrical-reloaded 0.5.12	`up_capture / down_capture`	yes	
PerformanceAnalytics 2.0.4 (R)	`UpDownRatios(method='Capture')`	no (documented difference)	PerformanceAnalytics divides sums of returns and counts zero-benchmark periods as down periods.

References

- Morningstar, Inc. Upside/Downside Capture Ratio methodology (geometric compounding of returns in up/down benchmark periods). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0012: Benchmark alignment and benchmark-relative metrics; Decision criteria

Capture ratio (upside / downside capture)

`capture_ratio` · Benchmark-relative analysis · derived · unit `ratio` · formula `capture_ratio@v1`

Description

Upside capture divided by downside capture: how much of the benchmark's gains the fund kept per unit of the benchmark's losses it took.

Formula

$$\mathrm{CR} = \frac{\mathrm{UC}}{\mathrm{DC}}, \quad \mathrm{DC} > 0$$

Plain text: `upside_capture / downside_capture`. Symbols: see Notation.

Usage

```
report.metrics["capture_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	derived	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Above 1: the fund participated more in rising markets than in falling ones. A fund with 80% upside and 60% downside capture has a capture ratio of about 1.33.

Professional. Ratio of the Morningstar-style geometric capture ratios of this report. It hides the level of participation (a fund with 10% / 5% capture has the same ratio as one with 120% / 60%); always read it with both components.

Assumptions

- Up / down periods defined by the sign of the benchmark return.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Few up or down periods in the overlap.
- Participation levels differ greatly: the ratio discards them.

Minimum data

At least 2 aligned observations with at least one up and one down period; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Upside or downside capture is undefined (no up or no down benchmark periods, or a zero compounded benchmark return on one side).
   - Downside capture is zero or negative (the fund gained on average when the benchmark fell); the ratio's sign would then be meaningless. Upside and downside capture are still reported.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.

  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `capture_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Geometric annualized capture ratios; downside capture must be positive (decision 0027).

Reference implementations

Library	Function	Agrees	Note
empyrical-reloaded 0.5.12	`up_down_capture`	yes	
PerformanceAnalytics 2.0.4 (R)	`UpDownRatios(method='Capture')`	no (documented difference)	Ratio of PerformanceAnalytics' arithmetic (summed) capture ratios.

References

- Morningstar, Inc. Upside/Downside Capture Ratio methodology (geometric compounding of returns in up/down benchmark periods). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

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Batting average (share of periods beating the benchmark)

`batting_average` · Benchmark-relative analysis · observed · unit `fraction` · formula `batting_average@v1`

Description

Share of aligned periods in which the fund's return exceeded the benchmark's.

Formula

$$\mathrm{BA} = \frac{1}{m}\sum_{t=1}^{m} \mathbf{1}[\,r_t > b_t\,]; \quad \mathrm{BA}^{\uparrow} = \frac{\#\{t : r_t > b_t,\, b_t > 0\}}{\#\{t : b_t > 0\}}, \mathrm{BA}^{\downarrow} = \frac{\#\{t : r_t > b_t,\, b_t < 0\}}{\#\{t : b_t < 0\}}$$

Plain text: `(1/m) * #{t : r_fund,t > r_bench,t}`. Symbols: see Notation.

Usage

```
report.metrics["batting_average"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	benchmark	Supplementary	no
Nature	observed	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Higher is better.

Interpretation

Plain language. How often the fund beat its benchmark: 0.6 means in 6 periods out of 10. It says nothing about by how much it won or lost.

Professional. Frequency of positive active returns (ties count as not beating). Details give the up- and down-market batting averages (PerformanceAnalytics UpDownRatios 'Percent'). A high batting average with a negative active return indicates small frequent wins and large occasional losses.

Assumptions

- A period counts only if the fund's return strictly exceeds the benchmark's.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- The size of wins and losses differs greatly (see active return).

Minimum data

At least 2 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
3. Once the conditions above hold, the value is always defined.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Observed: a fact about the supplied path (no estimation).
- Formula id `batting_average@v1`; the version increments if the definition changes.

OpenFactsheet convention

Strict outperformance; all aligned periods in the denominator (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`UpDownRatios(method='Percent', side='Up' / 'Down')`	yes	Agrees for the up- and down-market components in the details.

References

- Morningstar, Inc. Batting average: share of periods in which a fund outperformed its benchmark (Morningstar Direct data definitions). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

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Residual (specific) volatility (annualized)

`residual_volatility` · Benchmark-relative analysis · model-dependent · unit `decimal_return` · formula `residual_volatility@v1`

Description

Volatility of the part of fund returns not explained by the benchmark (regression residuals), annualized.

Formula

$$\hat{\sigma}_\varepsilon\sqrt{P}, \quad \hat{\sigma}_\varepsilon = \sqrt{\frac{1}{m-2}\sum_{t=1}^{m}\hat{\varepsilon}_t^2}, \quad \hat{\varepsilon}_t = r_t - \hat{\alpha} - \hat{\beta}b_t$$

Plain text: `sqrt( sum(e_t^2) / (m - 2) ) * sqrt(P), e_t the OLS residuals`. Symbols: see Notation.

Usage

```
report.metrics["residual_volatility"]
```

At a glance

Direction	Context dependent	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	model-dependent	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Context dependent.

Interpretation

Plain language. How much the fund moved for reasons unrelated to its benchmark. Near zero means the fund behaves like the benchmark (scaled by beta).

Professional. Specific (idiosyncratic, non-systematic) risk: the regression standard error of the single-index model, annualized by sqrt(P). Details give the systematic risk |beta| x benchmark volatility. Unlike tracking error it is measured after scaling by beta.

Assumptions

- Linear single-index model with the supplied benchmark.
- Square-root-of-time scaling.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Returns are smoothed: lagged benchmark exposure appears as residual risk.

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.

3. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant over the aligned periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `residual_volatility@v1`; the version increments if the definition changes.

OpenFactsheet convention

Regression standard error (m - 2 degrees of freedom) x sqrt(P) (decision 0027).

Reference implementations

Library	Function	Agrees	Note
statsmodels	`OLS.mse_resid`	yes	
PerformanceAnalytics 2.0.4 (R)	`SpecificRisk`	no (documented difference)	PerformanceAnalytics divides the residual sum of squares by m instead of m - 2.

References

- Treynor, J. L., & Black, F. (1973). How to Use Security Analysis to Improve Portfolio Selection. Journal of Business, 46(1), 66-86 (appraisal ratio). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

Appraisal ratio

`appraisal_ratio` · Benchmark-relative analysis · model-dependent · unit `ratio` · formula `appraisal_ratio@v1`

Description

Alpha per unit of residual (specific) risk, annualized.

Formula

$$\mathrm{AppR} = \frac{P\,\hat{\alpha}}{\hat{\sigma}_\varepsilon\sqrt{P}} = \frac{\hat{\alpha}}{\hat{\sigma}_\varepsilon}\sqrt{P}$$

Plain text: `(P * a) / (sigma_e * sqrt(P)) = a / sigma_e * sqrt(P)`. Symbols: see Notation.

Usage

```
report.metrics["appraisal_ratio"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	model-dependent	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS, risk-free = 0, annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. How much benchmark-independent return the fund earned for each unit of benchmark-independent risk it took.

Professional. Treynor & Black (1973) appraisal ratio: annualized OLS alpha over annualized residual volatility (the Information ratio of the beta-adjusted active return). It is approximately the alpha t-statistic divided by the square root of the overlap in years (exactly so when the benchmark's mean return is zero); details give that t-statistic and its p-value.

Assumptions

- Single-index (CAPM-style) model with the supplied benchmark.
- Risk-free rate 0%.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Omitted risk factors masquerade as alpha.
- Returns are smoothed or the history is short.

Minimum data

At least 3 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Benchmark returns are constant over the aligned periods.
   - Residual volatility is zero to floating-point precision (below sqrt(machine epsilon) x fund volatility, i.e. 1 - R-squared below machine epsilon): the benchmark explains the fund exactly.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `appraisal_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic annualized OLS alpha / residual volatility, rf = 0 (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`AppraisalRatio(method='appraisal')`	no (documented difference)	PerformanceAnalytics uses Jensen's alpha from geometric annualized returns and the m-denominator SpecificRisk.

References

- Treynor, J. L., & Black, F. (1973). How to Use Security Analysis to Improve Portfolio Selection. Journal of Business, 46(1), 66-86 (appraisal ratio). [bibliography]
- Grinold, R. C., & Kahn, R. N. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

M-squared (Modigliani risk-adjusted return)

`m_squared` · Benchmark-relative analysis · derived · unit `decimal_return` · formula `m_squared@v1`

Description

The fund's annualized return had it been leveraged or de-leveraged to the benchmark's volatility.

Formula

$$M^2 = P\,\frac{\bar{r}^{(m)} - r_f}{s_r^{(m)}}\,s_b + P\,r_f, \quad r_f = 0; \quad M^2 - P\bar{b}\ \text{(details)}$$

Plain text: `P * (mean(r_fund,t) / std(r_fund,t)) * std(r_bench,t), aligned periods, rf = 0`. Symbols: see Notation.

Usage

```
report.metrics["m_squared"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	benchmark	Supplementary	no
Nature	derived	Minimum-data policy	`benchmark_simple`
Canonical parameters	supplied benchmark, risk-free = 0, sample std (n-1), annualization factor		

Inputs

Fund and benchmark returns on aligned periods with a regular frequency. A benchmark is required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. Puts the fund and the benchmark on the same risk footing: the return the fund would have delivered at the benchmark's volatility. Compare it with the benchmark's own return (details).

Professional. Modigliani & Modigliani (1997) M-squared with rf = 0 on the aligned overlap: the aligned-sample Sharpe ratio times the benchmark's annualized volatility. It ranks funds exactly as the Sharpe ratio does, but in return units. Details give the benchmark's annualized mean and the difference (M-squared excess).

Assumptions

- Risk-free rate 0%: leverage is assumed costless.
- Volatility is the relevant risk measure.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Returns are smoothed or strongly non-normal (inherits the Sharpe ratio's caveats).
- Borrowing costs are material (rf = 0 understates the cost of leverage).

Minimum data

At least 2 aligned observations; low confidence below 3 years of overlap. (policy `benchmark_simple`: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least 2 aligned observations (mathematical minimum of policy `benchmark_simple`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fund returns are constant over the aligned periods (zero volatility).
   - Benchmark returns are constant over the aligned periods (zero volatility).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.
- Sample standard deviation with the n - 1 denominator.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Derived: a deterministic combination of observed quantities.
- Formula id `m_squared@v1`; the version increments if the definition changes.

OpenFactsheet convention

Arithmetic annualization, sample standard deviations on aligned periods, rf = 0 (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`Modigliani * P`	yes	PerformanceAnalytics reports the per-period value.

References

- Modigliani, F., & Modigliani, L. (1997). Risk-Adjusted Performance. Journal of Portfolio Management, 23(2), 45-54 (M-squared). [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

Bull-market beta

`bull_beta` · Benchmark-relative analysis · estimated · unit `beta` · formula `bull_beta@v1`

Description

Sensitivity of fund returns to benchmark returns in periods when the benchmark rose.

Formula

$$\hat{\beta}^{+} = \text{OLS slope of } r_t = \alpha^{+} + \beta^{+} b_t + \varepsilon_t \text{ over } \{t : b_t > 0\}$$

Plain text: `slope of OLS r_fund,t = a + b * r_bench,t + e_t over {t : r_bench,t > 0}`. Symbols: see Notation.

Usage

```
report.metrics["bull_beta"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Regression coefficient; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. How strongly the fund followed the benchmark when the benchmark went up.

Professional. Single-index beta estimated on the benchmark-up subsample (Fabozzi & Francis 1977), rf = 0; compare with bear-market beta for asymmetric participation. Details give the standard error and number of periods.

Assumptions

- Linear relationship within each market regime.
- Risk-free rate 0%.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Few up or down periods: the subset regression is very noisy.
- Conditioning on the sign of the benchmark truncates the regressor, which biases subset betas towards zero when the true relationship is noisy.

Minimum data

At least 3 benchmark up periods; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fewer than 3 benchmark up periods in the overlap.
   - Benchmark returns are constant over the up periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.

Output

- Unit: Regression coefficient; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `bull_beta@v1`; the version increments if the definition changes.

OpenFactsheet convention

OLS with intercept on periods with benchmark return > 0 (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`CAPM.beta.bull`	yes	

References

- Fabozzi, F. J., & Francis, J. C. (1977). Stability Tests for Alphas and Betas over Bull and Bear Market Conditions. Journal of Finance, 32(4), 1093-1099. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

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Bear-market beta

`bear_beta` · Benchmark-relative analysis · estimated · unit `beta` · formula `bear_beta@v1`

Description

Sensitivity of fund returns to benchmark returns in periods when the benchmark fell.

Formula

$$\hat{\beta}^{-} = \text{OLS slope of } r_t = \alpha^{-} + \beta^{-} b_t + \varepsilon_t \text{ over } \{t : b_t < 0\}$$

Plain text: `slope of OLS r_fund,t = a + b * r_bench,t + e_t over {t : r_bench,t < 0}`. Symbols: see Notation.

Usage

```
report.metrics["bear_beta"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Regression coefficient; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. How strongly the fund followed the benchmark when the benchmark went down; lower means better protection in falling markets.

Professional. Single-index beta estimated on the benchmark-down subsample (Fabozzi & Francis 1977), rf = 0. Details give the standard error and number of periods.

Assumptions

- Linear relationship within each market regime.

- Risk-free rate 0%.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Few up or down periods: the subset regression is very noisy.
- Conditioning on the sign of the benchmark truncates the regressor, which biases subset betas towards zero when the true relationship is noisy.

Minimum data

At least 3 benchmark down periods; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Fewer than 3 benchmark down periods in the overlap.
   - Benchmark returns are constant over the down periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.

Output

- Unit: Regression coefficient; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `bear_beta@v1`; the version increments if the definition changes.

OpenFactsheet convention

OLS with intercept on periods with benchmark return < 0 (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`CAPM.beta.bear`	yes	

References

- Fabozzi, F. J., & Francis, J. C. (1977). Stability Tests for Alphas and Betas over Bull and Bear Market Conditions. Journal of Finance, 32(4), 1093-1099. [bibliography]
- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

---

Timing ratio (bull beta / bear beta)

`timing_ratio` · Benchmark-relative analysis · estimated · unit `ratio` · formula `timing_ratio@v1`

Description

Bull-market beta divided by bear-market beta: asymmetry of market participation.

Formula

$$\mathrm{TR}_{\text{timing}} = \frac{\hat{\beta}^{+}}{\hat{\beta}^{-}}, \quad \hat{\beta}^{-} > 0$$

Plain text: `bull_beta / bear_beta, bear_beta > 0`. Symbols: see Notation.

Usage

```
report.metrics["timing_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	benchmark	Supplementary	no
Nature	estimated	Minimum-data policy	`benchmark_regression`
Canonical parameters	supplied benchmark, OLS		

Inputs

Fund and benchmark returns on aligned periods. A benchmark is required.

Value

Dimensionless ratio; displayed with 2 decimals. Higher is better.

Interpretation

Plain language. Above 1: the fund was more sensitive to rising than to falling markets.

Professional. Ratio of subsample betas (PerformanceAnalytics TimingRatio). Both betas are noisy, so the ratio is very noisy; it is descriptive, not evidence of timing skill. Formal timing tests (Treynor-Mazuy, Henriksson-Merton) are not part of methodology 0.1.0.

Assumptions

- Linear relationship within each market regime.
- Risk-free rate 0%.

When it can mislead

- The benchmark is not representative of the strategy's opportunity set.
- The overlap with the benchmark is short.
- Few up or down periods: the subset regression is very noisy.
- Conditioning on the sign of the benchmark truncates the regressor, which biases subset betas towards zero when the true relationship is noisy.
- Option-like or dynamically hedged strategies: asymmetry need not reflect skill.

Minimum data

At least 3 benchmark up and 3 down periods; low confidence below 3 years of overlap. (policy `benchmark_regression`: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. A benchmark must be supplied explicitly; otherwise status `not_applicable` (no default benchmark is ever assumed).
2. At least 3 aligned observations (mathematical minimum of policy `benchmark_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:

- Bull-market or bear-market beta is undefined (fewer than 3 up or down periods, or a constant benchmark within a regime).
- Bear-market beta is zero or negative; the ratio's sign would be meaningless. Both betas are still reported.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Benchmark overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and benchmark paired by position (undated) adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Only the m aligned periods (intersection of fund and benchmark) are used; missing returns are never filled; the report states m and the alignment method.
- Ordinary least squares with an intercept (statsmodels), risk-free rate 0%, conventional i.i.d. standard errors with m - 2 degrees of freedom.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `timing_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Ratio of OLS subsample betas; bear beta must be positive (decision 0027).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`TimingRatio`	yes	

References

- Bacon, C. R. (2021). Practical Risk-Adjusted Performance Measurement (2nd ed.). Wiley. [bibliography]
- Fabozzi, F. J., & Francis, J. C. (1977). Stability Tests for Alphas and Betas over Bull and Bear Market Conditions. Journal of Finance, 32(4), 1093-1099. [bibliography]

See also

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas; Decision criteria

Factor analysis

Factor-model alpha (annualized)

`factor_alpha` · Factor analysis · model-dependent · unit `decimal_return` · formula `factor_alpha@v1`

Description

Average return not explained by the supplied factors, annualized.

Formula

$$\alpha_{\text{ann}} = P\,\hat{\alpha}, \quad y_t = \alpha + \sum_{j=1}^{k} \beta_j f_{j,t} + \varepsilon_t, \quad y_t = r_t - r_t^{\text{RF}}, \quad t = 1..m$$

Plain text: `P * a, with a the intercept of OLS: y_t = a + sum_j b_j f_j,t + e_t`. Symbols: see Notation.

Usage

```
report.metrics["factor_alpha"]
```

At a glance

Direction	Higher is better	Annualized	yes
Needs	factor returns	Supplementary	no
Nature	model-dependent	Minimum-data policy	`factor_regression`
Canonical parameters	supplied factors, OLS + Newey-West, annualization factor		

Inputs

Fund returns and caller-supplied factor returns (optionally with an RF column) on aligned periods with a regular frequency. Factor returns are required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Higher is better.

Interpretation

Plain language. The part of the fund's yearly return that the supplied factors (for example the market, size and value) do not explain. It depends entirely on which factors are used.

Professional. Multi-factor (Jensen-style) alpha, annualized arithmetically. Details report the per-period intercept with its Newey-West standard error, t-statistic, p-value and 95% confidence interval (annualized). Judge alpha by its uncertainty, not its point estimate; with smoothed returns it is biased upward.

Assumptions

- Linear factor model with constant loadings over the aligned periods.
- Factor returns supplied by the caller, aligned by period, never filled.
- Dependent variable: fund return minus the supplied RF column, or the fund return when no RF column is supplied (risk-free rate 0%).
- Newey-West (HAC) covariance with the canonical lag rule; Student t inference.
- Arithmetic annualization of the intercept.

When it can mislead

- The factors do not span the fund's true risk exposures (omitted factors appear as alpha).
- Exposures change over time; one full-history regression averages them.
- Returns are smoothed or stale, which biases loadings toward zero and alpha upward.
- The factor returns are not investable at zero cost (alpha is measured against paper portfolios before trading costs).
- The history is short.

Minimum data

At least k + 2 aligned observations for k factors; low confidence below 3 years of overlap, moderate below 5 years. (policy `factor_regression`: Requires at least 3 aligned observations and at least k + 2 for k factors (k + 1 coefficients and one residual degree of freedom); low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. Factor returns must be supplied explicitly; otherwise status `not_applicable` (factor data are never downloaded).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least max(3, k + 2) aligned observations for k factors (mathematical minimum of policy `factor_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - A factor is constant over the aligned periods.
   - The factor returns are perfectly collinear over the aligned periods.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).

- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Factor overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and factors paired by position (undated) adds a warning.
  - Smoothing risk **high** sets confidence to **low**; **moderate** caps at **moderate** (smoothed returns bias loadings toward zero and alpha upward).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- OLS with an intercept of y_t (fund return minus a supplied RF column, or the fund return when none is supplied) on the m aligned factor returns; Newey-West (HAC) covariance with lag floor(4 (m/100)^(2/9)), Bartlett kernel, small-sample factor m / (m - k - 1) and Student t with m - k - 1 degrees of freedom; periods missing for the fund or any factor are dropped, never filled.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `factor_alpha@v1`; the version increments if the definition changes.

OpenFactsheet convention

Per-period OLS intercept x periods per year (decision 0028).

Reference implementations

Library	Function	Agrees	Note
R lm + sandwich::NeweyWest + lmtest::coeftest	`coef(lm)[1] * P`	yes	Validated on the factor fixture.
statsmodels	`OLS(...).fit(cov_type='HAC')`	yes	

References

- Jensen, M. C. (1968). The Performance of Mutual Funds in the Period 1945-1964. Journal of Finance, 23(2), 389-416. [bibliography]
- Fama, E. F., & French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. Journal of Financial Economics, 33(1), 3-56. [bibliography]
- Carhart, M. M. (1997). On Persistence in Mutual Fund Performance. Journal of Finance, 52(1), 57-82. [bibliography]
- Fama, E. F., & French, K. R. (2015). A Five-Factor Asset Pricing Model. Journal of Financial Economics, 116(1), 1-22. [bibliography]
- Newey, W. K., & West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. Econometrica, 55(3), 703-708. [bibliography]
- Newey, W. K., & West, K. D. (1994). Automatic Lag Selection in Covariance Matrix Estimation. Review of Economic Studies, 61(4), 631-653 (lag rule floor(4 (T/100)^(2/9))). [bibliography]
- Asness, C., Krail, R., & Liew, J. (2001). Do Hedge Funds Hedge? Journal of Portfolio Management, 28(1), 6-19 (stale prices bias betas down and alphas up). [bibliography]

See also

Decision 0028: Factor models with caller-supplied factor returns; Decision criteria

Factor-model alpha t-statistic (Newey-West)

`factor_alpha_t_statistic` · Factor analysis · model-dependent · unit `ratio` · formula `factor_alpha_t_statistic@v1`

Description

How many (Newey-West) standard errors the factor-model alpha lies from zero.

Formula

$$t_\alpha = \frac{\hat{\alpha}}{\widehat{\mathrm{SE}}_{\mathrm{HAC}}(\hat{\alpha})}, \quad \hat{V}_{\mathrm{HAC}} = \frac{m}{m-k-1}(X^\top X)^{-1}\hat{S}(X^\top X)^{-1}, \quad \hat{S} = \sum_{\ell=-q}^{q}\Big(1-\frac{|\,\ell\,|}{q+1}\Big)\hat{\Gamma}_\ell, \quad q = \lfloor 4(m/100)^{2/9} \rfloor$$

Plain text: `a / SE_HAC(a)`. Symbols: see Notation.

Usage

```
report.metrics["factor_alpha_t_statistic"]
```

At a glance

Direction	Context dependent	Annualized	no
Needs	factor returns	Supplementary	no
Nature	model-dependent	Minimum-data policy	`factor_regression`
Canonical parameters	supplied factors, OLS + Newey-West		

Inputs

Fund returns and caller-supplied factor returns on aligned periods. Factor returns are required.

Value

Dimensionless ratio; displayed with 2 decimals. Context dependent.

Interpretation

Plain language. A rough gauge of whether the unexplained return could be luck: values beyond about +2 or -2 are unlikely to arise by chance alone if the model is right.

Professional. Student-t statistic of the intercept with m - k - 1 degrees of freedom and heteroskedasticity- and autocorrelation-consistent standard errors; the p-value and interval are in the alpha details. Multiple testing across funds or models inflates apparent significance.

Assumptions

- Linear factor model with constant loadings over the aligned periods.
- Factor returns supplied by the caller, aligned by period, never filled.
- Dependent variable: fund return minus the supplied RF column, or the fund return when no RF column is supplied (risk-free rate 0%).
- Newey-West (HAC) covariance with the canonical lag rule; Student t inference.

When it can mislead

- The factors do not span the fund's true risk exposures (omitted factors appear as alpha).
- Exposures change over time; one full-history regression averages them.
- Returns are smoothed or stale, which biases loadings toward zero and alpha upward.
- The factor returns are not investable at zero cost (alpha is measured against paper portfolios before trading costs).
- Many funds or factor sets were tried and the best t-statistic is reported.

Minimum data

At least k + 2 aligned observations for k factors; low confidence below 3 years of overlap, moderate below 5 years. (policy `factor_regression`: Requires at least 3 aligned observations and at least k + 2 for k factors (k + 1 coefficients and one residual degree of freedom); low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. Factor returns must be supplied explicitly; otherwise status `not_applicable` (factor data are never downloaded).
2. At least max(3, k + 2) aligned observations for k factors (mathematical minimum of policy `factor_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - A factor is constant or the factors are perfectly collinear.
   - The factors reproduce the fund exactly (residuals zero or below floating-point resolution), so the standard error is zero.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Factor overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and factors paired by position (undated) adds a warning.
  - Smoothing risk **high** sets confidence to **low**; **moderate** caps at **moderate** (smoothed returns bias loadings toward zero and alpha upward).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- OLS with an intercept of y_t (fund return minus a supplied RF column, or the fund return when none is supplied) on the m aligned factor returns; Newey-West (HAC) covariance with lag floor(4 (m/100)^(2/9)), Bartlett kernel, small-sample factor m / (m - k - 1) and Student t with m - k - 1 degrees of freedom; periods missing for the fund or any factor are dropped, never filled.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Context dependent.
- Nature: Model-dependent: valid only under the stated model.
- Formula id `factor_alpha_t_statistic@v1`; the version increments if the definition changes.

OpenFactsheet convention

Newey-West lag floor(4 (m/100)^(2/9)), Bartlett kernel, small-sample factor m / (m - k - 1), no prewhitening (decision 0028).

Reference implementations

Library	Function	Agrees	Note
R lm + sandwich::NeweyWest + lmtest::coeftest	`coeftest(lm, vcov = NeweyWest(lm, lag = L, prewhite = FALSE))`	yes	
statsmodels	`OLS(...).fit(cov_type='HAC', use_t=True).tvalues[0]`	yes	

References

- Newey, W. K., & West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. Econometrica, 55(3), 703-708. [bibliography]
- Newey, W. K., & West, K. D. (1994). Automatic Lag Selection in Covariance Matrix Estimation. Review of Economic Studies, 61(4), 631-653 (lag rule floor(4 (T/100)^(2/9))). [bibliography]
- Fama, E. F., & French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. Journal of Financial Economics, 33(1), 3-56. [bibliography]
- Carhart, M. M. (1997). On Persistence in Mutual Fund Performance. Journal of Finance, 52(1), 57-82. [bibliography]
- Fama, E. F., & French, K. R. (2015). A Five-Factor Asset Pricing Model. Journal of Financial Economics, 116(1), 1-22. [bibliography]

See also

Decision 0028: Factor models with caller-supplied factor returns; Decision 0034: The reference manual; Decision criteria

Factor-model R-squared

`factor_r_squared` · Factor analysis · estimated · unit `fraction` · formula `factor_r_squared@v1`

Description

Share of the variance of fund (excess) returns explained by the factors.

Formula

$$R^2 = 1 - \frac{\sum_{t=1}^{m} \hat{\varepsilon}_t^2}{\sum_{t=1}^{m} (y_t - \bar{y})^2}, \quad \bar{R}^2 = 1 - (1 - R^2)\frac{m-1}{m-k-1}$$

Plain text: `1 - SSR / SST of the factor regression`. Symbols: see Notation.

Usage

```
report.metrics["factor_r_squared"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	factor returns	Supplementary	no
Nature	estimated	Minimum-data policy	`factor_regression`
Canonical parameters	supplied factors, OLS + Newey-West		

Inputs

Fund returns and caller-supplied factor returns on aligned periods. Factor returns are required.

Value

Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Neutral (descriptive).

Interpretation

Plain language. How much of the fund's ups and downs the factors account for (1 = all of it).

Professional. A low R-squared means the factor loadings and alpha describe little of the fund's risk; the factor set may be inappropriate for the strategy. Adjusted R-squared is in the details.

Assumptions

- Linear factor model with constant loadings over the aligned periods.
- Factor returns supplied by the caller, aligned by period, never filled.
- Dependent variable: fund return minus the supplied RF column, or the fund return when no RF column is supplied (risk-free rate 0%).

When it can mislead

- Adding factors always raises R-squared; compare adjusted R-squared across models.
- Returns are smoothed (explained variance is understated).

Minimum data

At least k + 2 aligned observations for k factors; low confidence below 3 years of overlap, moderate below 5 years. (policy `factor_regression`: Requires at least 3 aligned observations and at least k + 2 for k factors (k + 1 coefficients and one residual degree of freedom); low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. Factor returns must be supplied explicitly; otherwise status `not_applicable` (factor data are never downloaded).
2. At least max(3, k + 2) aligned observations for k factors (mathematical minimum of policy `factor_regression`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - A factor is constant or the factors are perfectly collinear.
   - The dependent variable is constant (no variance to explain).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Factor overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and factors paired by position (undated) adds a warning.

    - Smoothing risk **high** sets confidence to **low**; **moderate** caps at **moderate** (smoothed returns bias loadings toward zero and alpha upward).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- OLS with an intercept of y_t (fund return minus a supplied RF column, or the fund return when none is supplied) on the m aligned factor returns; Newey-West (HAC) covariance with lag floor(4 (m/100)^(2/9)), Bartlett kernel, small-sample factor m / (m - k - 1) and Student t with m - k - 1 degrees of freedom; periods missing for the fund or any factor are dropped, never filled.

Output

- Unit: Share of observations between 0 and 1; displayed as a percentage with 1 decimal. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `factor_r_squared@v1`; the version increments if the definition changes.

OpenFactsheet convention

Centred R-squared of the OLS fit with an intercept (decision 0028).

Reference implementations

Library	Function	Agrees	Note
R lm + sandwich::NeweyWest + lmtest::coeftest	`summary(lm)$r.squared`	yes	
statsmodels	`OLS(...).fit().rsquared`	yes	

References

- Fama, E. F., & French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. Journal of Financial Economics, 33(1), 3-56. [bibliography]
- Carhart, M. M. (1997). On Persistence in Mutual Fund Performance. Journal of Finance, 52(1), 57-82. [bibliography]
- Fama, E. F., & French, K. R. (2015). A Five-Factor Asset Pricing Model. Journal of Financial Economics, 116(1), 1-22. [bibliography]
- Fung, W., & Hsieh, D. A. (2004). Hedge Fund Benchmarks: A Risk-Based Approach. Financial Analysts Journal, 60(5), 65-80. [bibliography]

See also

Decision 0028: Factor models with caller-supplied factor returns; Decision criteria

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Factor-model residual volatility (annualized)

`factor_residual_volatility` · Factor analysis · estimated · unit `decimal_return` · formula `factor_residual_volatility@v1`

Description

Volatility of the return not explained by the factors (specific risk).

Formula

$$\sigma_\varepsilon = \sqrt{\frac{\sum_{t=1}^{m} \hat{\varepsilon}_t^2}{m-k-1}}\sqrt{P}$$

Plain text: `sqrt(SSR / (m - k - 1)) * sqrt(P)`. Symbols: see Notation.

Usage

```
report.metrics["factor_residual_volatility"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	yes
Needs	factor returns	Supplementary	no
Nature	estimated	Minimum-data policy	`factor_regression`
Canonical parameters	supplied factors, OLS + Newey-West, annualization factor		

Inputs

Fund returns and caller-supplied factor returns on aligned periods with a regular frequency. Factor returns are required.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Neutral (descriptive).

Interpretation

Plain language. How much the fund moves for reasons the factors do not capture; the size of the manager's independent bets, whether they pay off or not.

Professional. Regression standard error annualized by the square root of time; the denominator of a factor-model appraisal ratio. Understated when returns are smoothed.

Assumptions

- Linear factor model with constant loadings over the aligned periods.
- Factor returns supplied by the caller, aligned by period, never filled.
- Dependent variable: fund return minus the supplied RF column, or the fund return when no RF column is supplied (risk-free rate 0%).
- Square-root-of-time annualization.

When it can mislead

- Returns are smoothed or stale (volatility suppressed).
- Exposures change over time (time-varying betas appear as residual risk).

Minimum data

At least k + 2 aligned observations for k factors; low confidence below 3 years of overlap, moderate below 5 years. (policy `factor_regression`: Requires at least 3 aligned observations and at least k + 2 for k factors (k + 1 coefficients and one residual degree of freedom); low confidence below 3 years of overlap, moderate below 5 years.)

Calculation criteria

When a value is produced (checked in this order)

1. Factor returns must be supplied explicitly; otherwise status `not_applicable` (factor data are never downloaded).
2. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
3. At least max(3, k + 2) aligned observations for k factors (mathematical minimum of policy `factor_regression`); otherwise `insufficient_data`.
4. The value is undefined, and reported as `not_applicable` with the reason, when:
   - A factor is constant or the factors are perfectly collinear.

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
  - Factor overlap below 90% of the fund's observations caps at **moderate**.
  - Fund and factors paired by position (undated) adds a warning.
  - Smoothing risk **high** sets confidence to **low**; **moderate** caps at **moderate** (smoothed returns bias loadings toward zero and alpha upward).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- OLS with an intercept of y_t (fund return minus a supplied RF column, or the fund return when none is supplied) on the m aligned factor returns; Newey-West (HAC) covariance with lag floor(4 (m/100)^(2/9)),

Bartlett kernel, small-sample factor m / (m - k - 1) and Student t with m - k - 1 degrees of freedom; periods missing for the fund or any factor are dropped, never filled.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.

Output
- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `factor_residual_volatility@v1`; the version increments if the definition changes.

OpenFactsheet convention
Residual degrees of freedom m - k - 1, square-root-of-time annualization (decision 0028).

Reference implementations

Library	Function	Agrees	Note
R lm + sandwich::NeweyWest + lmtest::coeftest	`summary(lm)$sigma * sqrt(P)`	yes	

References

- Fama, E. F., & French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. Journal of Financial Economics, 33(1), 3-56. [bibliography]
- Carhart, M. M. (1997). On Persistence in Mutual Fund Performance. Journal of Finance, 52(1), 57-82. [bibliography]
- Fama, E. F., & French, K. R. (2015). A Five-Factor Asset Pricing Model. Journal of Financial Economics, 116(1), 1-22. [bibliography]

See also
Decision 0028: Factor models with caller-supplied factor returns; Decision criteria

Statistical confidence

Probabilistic Sharpe Ratio (vs 0)
`probabilistic_sharpe_ratio` · Statistical confidence · estimated · unit `probability` · formula `probabilistic_sharpe_ratio@v1`

Description
Estimated probability that the true Sharpe ratio is above zero, given the track-record length, skewness and kurtosis.

Formula

$$\mathrm{PSR} = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - g_3\widehat{SR} + \frac{g_4-1}{4}\widehat{SR}^2}}\right), \quad \widehat{SR} = \frac{\bar{r}}{s},\ g_3 = \frac{m_3}{m_2^{3/2}},\ g_4 = \frac{m_4}{m_2^2},\ SR^* = 0$$

Plain text: `Phi( (SR - SR*) * sqrt(n - 1) / sqrt(1 - g3*SR + (g4 - 1)/4 * SR^2) ), SR* = 0`. Symbols: see Notation.

Usage
```
report.metrics["probabilistic_sharpe_ratio"]
```

At a glance

Direction	Higher is better	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`asymptotic_inference`
Canonical parameters	i.i.d. asymptotics, risk-free = 0		

Inputs
Periodic simple returns.

Value

Probability between 0 and 1; displayed with 3 decimals. Higher is better.

Interpretation

Plain language. How confident one can be, statistically, that the investment's risk-adjusted return is genuinely positive rather than luck. Values near 1 mean strong evidence; 0.5 means no evidence either way.

Professional. Bailey & Lopez de Prado (2012) PSR with benchmark SR* = 0 on the per-period (non-annualized) Sharpe ratio, using moment skewness and non-excess kurtosis. Unlike PerformanceAnalytics' default, kurtosis is not ignored: fat tails reduce confidence.

Assumptions

- Independent, identically distributed returns.
- Risk-free rate 0%.

When it can mislead

- Returns are serially correlated or smoothed: the i.i.d. variance understates the true uncertainty (Lo, 2002).
- Samples are short: the asymptotic normal approximation is poor.
- Many strategies were tried before this one was shown (selection bias; see the Deflated Sharpe Ratio, which needs the number of trials and is not computable from one series).

Minimum data

At least 4 observations; low confidence below 36, moderate below 60. (policy `asymptotic_inference`: Asymptotic Sharpe-ratio inference needs at least 4 observations for the moments it uses; the normal approximation is poor below 36 observations (low) and 60 (moderate).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `asymptotic_inference`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Sample standard deviation is exactly zero.
   - Return variance underflows in floating point (moments not computable).
   - The variance factor is not positive (degenerate distribution).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Asymptotic normal approximation for independent, identically distributed returns, allowing for skewness and kurtosis (Mertens 2002; Bailey & Lopez de Prado 2012); serial correlation is not modelled.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Probability between 0 and 1; displayed with 3 decimals. Values are stored unrounded (float64).
- Direction: Higher is better.
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `probabilistic_sharpe_ratio@v1`; the version increments if the definition changes.

OpenFactsheet convention

Reference Sharpe ratio 0, per-period Sharpe, population moments (decision 0019).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`ProbSharpeRatio(refSR=0, ignore_kurtosis=FALSE)`	yes	Agrees with ignore_kurtosis=FALSE; the PerformanceAnalytics default sets kurtosis to 3.

References

- Bailey, D. H., & Lopez de Prado, M. (2012). The Sharpe Ratio Efficient Frontier. Journal of Risk, 15(2), 3-44 (Probabilistic Sharpe Ratio, Minimum Track Record Length). [bibliography]
- Mertens, E. (2002). Comments on Variance of the IID Estimator in Lo (2002). Working paper, University of Basel. [bibliography]

See also

Decision 0019: Sharpe-ratio inference: PSR, standard error, minimum track record length; Decision criteria

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Sharpe ratio standard error (annualized)

`sharpe_ratio_standard_error` · Statistical confidence · estimated · unit `ratio` · formula `sharpe_ratio_standard_error@v1`

Description

Sampling uncertainty of the annualized Sharpe ratio; details give the 95% confidence interval.

Formula

$$\mathrm{SE}(\widehat{SR}_{\mathrm{ann}}) = \sqrt{P}\sqrt{\frac{1 - g_3\widehat{SR} + \frac{g_4 - 1}{4}\widehat{SR}^2}{n-1}}, \quad \mathrm{CI}_{95} = \widehat{SR}\sqrt{P} \pm z_{0.975}\,\mathrm{SE}$$

Plain text: `sqrt(P) * sqrt( (1 - g3*SR + (g4 - 1)/4 * SR^2) / (n - 1) ); CI = SR_ann +/- z_0.975 * SE`. Symbols: see Notation.

Usage

```
report.metrics["sharpe_ratio_standard_error"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	yes
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`asymptotic_inference`
Canonical parameters	i.i.d. asymptotics, annualization factor, risk-free = 0		

Inputs

Periodic simple returns with a regular frequency.

Value

Dimensionless ratio; displayed with 2 decimals. Neutral (descriptive).

Interpretation

Plain language. How precisely the Sharpe ratio is known. A Sharpe ratio of 1.0 with a standard error of 0.5 could plausibly be anywhere from about 0 to 2.

Professional. Non-normal i.i.d. asymptotic standard error (Mertens 2002; Opdyke 2007), scaled by sqrt(P) like the annualized Sharpe ratio. The details report the two-sided 95% interval (z = 1.96); PerformanceAnalytics' interval uses the one-sided quantile and is therefore a 90% interval.

Assumptions

- Independent, identically distributed returns.
- Square-root-of-time scaling of the per-period standard error.

When it can mislead

- Returns are serially correlated or smoothed: the i.i.d. variance understates the true uncertainty (Lo, 2002).
- Samples are short: the asymptotic normal approximation is poor.
- Many strategies were tried before this one was shown (selection bias; see the Deflated Sharpe Ratio, which needs the number of trials and is not computable from one series).

Minimum data

At least 4 observations; low confidence below 36, moderate below 60. (policy `asymptotic_inference`: Asymptotic Sharpe-ratio inference needs at least 4 observations for the moments it uses; the normal approximation is poor below 36 observations (low) and 60 (moderate).)

Calculation criteria

When a value is produced (checked in this order)

1. The observation frequency must be regular so that P is defined; otherwise `not_applicable`.
2. At least 4 observations (mathematical minimum of policy `asymptotic_inference`); otherwise `insufficient_data`.
3. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Sample standard deviation is exactly zero.
   - Return variance underflows in floating point (moments not computable).
   - The variance factor is not positive (degenerate distribution).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
  - Missing values or missing expected periods in the history cap at **moderate**.
  - Fewer than 1 year(s) of data sets confidence to **low** (annualizing extrapolates a partial year).
  - A caller-declared (not date-inferred) frequency adds a warning.
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Asymptotic normal approximation for independent, identically distributed returns, allowing for skewness and kurtosis (Mertens 2002; Bailey & Lopez de Prado 2012); serial correlation is not modelled.
- Annualized with the canonical periods-per-year factor P of the detected frequency (daily business 252, daily calendar 365, weekly 52, monthly 12, quarterly 4, yearly 1); irregular or unknown frequency is never annualized.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Dimensionless ratio; displayed with 2 decimals. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `sharpe_ratio_standard_error@v1`; the version increments if the definition changes.

OpenFactsheet convention

Two-sided 95% interval, annualized by sqrt(P), population moments (decision 0019).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`ProbSharpeRatio(...)$sr_confidence_interval`	no (documented difference)	Same variance formula; PerformanceAnalytics uses qnorm(0.95) (a 90% two-sided interval), a per-period Sharpe ratio and

Library	Function	Agrees	Note
			ignores kurtosis by default.

References

- Mertens, E. (2002). Comments on Variance of the IID Estimator in Lo (2002). Working paper, University of Basel. [bibliography]
- Opdyke, J. D. (2007). Comparing Sharpe Ratios: So Where Are the p-Values? Journal of Asset Management, 8(5), 308-336. [bibliography]
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36-52. [bibliography]

See also

Decision 0019: Sharpe-ratio inference: PSR, standard error, minimum track record length; Decision criteria

Minimum track record length

`minimum_track_record_length` · Statistical confidence · estimated · unit `periods` · formula `minimum_track_record_length@v1`

Description

Number of observations needed for the observed Sharpe ratio to be statistically above zero at 95% confidence.

Formula

$$\mathrm{MinTRL} = 1 + \left(1 - g_3\widehat{SR} + \frac{g_4 - 1}{4}\widehat{SR}^2\right)\left(\frac{z_{0.95}}{\widehat{SR} - SR^*}\right)^2, \quad SR^* = 0$$

Plain text: `1 + (1 - g3*SR + (g4 - 1)/4 * SR^2) * ( z_0.95 / (SR - SR*) )^2, SR* = 0`. Symbols: see Notation.

Usage

```
report.metrics["minimum_track_record_length"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`asymptotic_inference`
Canonical parameters	i.i.d. asymptotics, risk-free = 0		

Inputs

Periodic simple returns.

Value

Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Neutral (descriptive).

Interpretation

Plain language. How long a track record this investment would need before its risk-adjusted return could be told apart from luck. If it exceeds the actual history, the record is too short to be conclusive.

Professional. Bailey & Lopez de Prado (2012) MinTRL at 95% one-sided confidence with SR* = 0, in observation periods (details give years and whether the actual history suffices). Depends on the observed Sharpe ratio, skewness and kurtosis.

Assumptions

- Independent, identically distributed returns.

When it can mislead

- Returns are serially correlated or smoothed: the i.i.d. variance understates the true uncertainty (Lo, 2002).
- Samples are short: the asymptotic normal approximation is poor.
- Many strategies were tried before this one was shown (selection bias; see the Deflated Sharpe Ratio, which needs the number of trials and is not computable from one series).

Minimum data

At least 4 observations; low confidence below 36, moderate below 60. (policy `asymptotic_inference`: Asymptotic Sharpe-ratio inference needs at least 4 observations for the moments it uses; the normal approximation is poor below 36 observations (low) and 60 (moderate).)

Calculation criteria

When a value is produced (checked in this order)

1. At least 4 observations (mathematical minimum of policy `asymptotic_inference`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - The observed Sharpe ratio is not above the reference 0 (no track-record length can make it significant).
   - Sample standard deviation is exactly zero.
   - Return variance underflows in floating point (moments not computable).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from the number of observations `n`: **high** if `n >= 60`; **moderate** if `36 <= n < 60`; **low** if `n < 36`.
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- Asymptotic normal approximation for independent, identically distributed returns, allowing for skewness and kurtosis (Mertens 2002; Bailey & Lopez de Prado 2012); serial correlation is not modelled.
- Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive.

Output

- Unit: Number of observation periods; displayed as an integer with the equivalent in years when the frequency is regular. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).
- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `minimum_track_record_length@v1`; the version increments if the definition changes.

OpenFactsheet convention

95% one-sided, SR* = 0, per-period Sharpe, population moments (decision 0019).

Reference implementations

Library	Function	Agrees	Note
PerformanceAnalytics 2.0.4 (R)	`MinTrackRecord(refSR=0, p=0.95, ignore_kurtosis=FALSE)`	yes	Agrees with ignore_kurtosis=FALSE.

References

- Bailey, D. H., & Lopez de Prado, M. (2012). The Sharpe Ratio Efficient Frontier. Journal of Risk, 15(2), 3-44 (Probabilistic Sharpe Ratio, Minimum Track Record Length). [bibliography]

See also

Decision 0019: Sharpe-ratio inference: PSR, standard error, minimum track record length; Decision criteria

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Mean return standard error (per period)

`mean_return_standard_error` · Statistical confidence · estimated · unit `decimal_return` · formula `mean_return_standard_error@v1`

Description

Sampling uncertainty of the average per-period return; details give the 95% confidence interval and a test of the mean against zero.

Formula

$$\mathrm{SE}(\bar{r}) = \frac{s}{\sqrt{n}}, \quad \mathrm{CI}_{95} = \bar{r} \pm t_{0.975,\,n-1}\,\frac{s}{\sqrt{n}}, \quad t = \frac{\bar{r}}{\mathrm{SE}(\bar{r})}$$

Plain text: `s / sqrt(n); CI = mean +/- t_(0.975, n-1) * s / sqrt(n)`. Symbols: see Notation.

Usage

```
report.metrics["mean_return_standard_error"]
```

At a glance

Direction	Neutral (descriptive)	Annualized	no
Needs	returns only	Supplementary	no
Nature	estimated	Minimum-data policy	`dispersion`
Canonical parameters	—		

Inputs

Periodic simple returns.

Value

Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Neutral (descriptive).

Interpretation

Plain language. How precisely the average return is known. If the confidence interval includes zero, the history cannot distinguish the average return from zero.

Professional. Student-t interval and two-sided t-test for the mean with n - 1 degrees of freedom (exact under normal i.i.d. returns, asymptotically valid otherwise). Serial correlation widens the true interval.

Assumptions

- Independent, identically distributed returns.

When it can mislead

- Returns are serially correlated or smoothed (the interval is too narrow).
- Returns are strongly skewed and the sample is short.

Minimum data

At least 2 observations; low confidence below 3 years. (policy `dispersion`: Requires at least 2 observations; low confidence below 3 years, moderate below 5.)

Calculation criteria

When a value is produced (checked in this order)

1. At least 2 observations (mathematical minimum of policy `dispersion`); otherwise `insufficient_data`.
2. The value is undefined, and reported as `not_applicable` with the reason, when:
   - Sample standard deviation is exactly zero (no sampling variability).

An unexpected calculation failure or a non-finite result is reported as `error` with the value withheld; it never appears as NaN or infinity.

Confidence and status

- Base confidence from years of history `y = n / P` (aligned observations for benchmark metrics): **high** if `y >= 5`; **moderate** if `3 <= y < 5`; **low** if `y < 3` or if `y` cannot be established (no regular frequency).
- Then, in this order, data-quality modifiers (a cap lowers confidence to at most the stated level; warnings are attached to the result):
  - Smoothing risk **high** (lag-1 autocorrelation >= 0.3 and z = rho_1 sqrt(n) >= 1.96) sets confidence to **low**.
  - Smoothing risk **moderate** (rho_1 >= 0.15 and z >= 1.645) caps at **moderate**.
  - At least 10% exactly-zero returns caps at **moderate**.
  - With n >= 24 and skewness <= -1 or excess kurtosis >= 3: warning only (no confidence change).
- Status: `low_confidence` if the final confidence is **low**, otherwise `applicable`. Low confidence never hides the value.

Assumptions recorded on every result

- None beyond simple decimal returns as supplied.

Output

- Unit: Decimal return (0.01 = 1%); displayed as a percentage with 2 decimals. Values are stored unrounded (float64).
- Direction: Neutral (descriptive).

- Nature: Estimated: a sample statistic subject to sampling error.
- Formula id `mean_return_standard_error@v1`; the version increments if the definition changes.

OpenFactsheet convention

Student-t, two-sided 95%, per period (decision 0020).

Reference implementations

Library	Function	Agrees	Note
SciPy	`stats.t / stats.ttest_1samp`	yes	

References

- Student [Gosset, W. S.] (1908). The Probable Error of a Mean. Biometrika, 6(1), 1-25. [bibliography]

See also

Decision 0020: Mean-return inference and the Jarque-Bera normality test; Decision criteria

Methodology

Canonical methodology

OpenFactsheet produces one standardized report for any return history. The report is determined by a versioned methodology, not by the person supplying the data. This page describes how that methodology is organized. The specific conventions are listed in assumptions.md and justified in the decision notes.

Current version: **OpenFactsheet Methodology 0.1.0** (`ofs.METHODOLOGY_VERSION`).

Pipeline

```
INPUT → NORMALIZATION → DATA QUALITY & FREQUENCY → METRIC CALCULATION
      → APPLICABILITY / CONFIDENCE POLICY → INTERPRETATION → REPORTING
```

Stage	Module	What it does
Input	`openfactsheet.input`	Parses lists, arrays, pandas objects, CSV, Excel, pasted text and wide monthly tables into raw values and dates. Never mutates caller data.
Normalization	`openfactsheet.normalization`	Rejects infinite values and duplicate timestamps, sorts dates, trims and drops (never fills) missing values, infers frequency, detects percent vs decimal scale. Produces a `ReturnSeries` and `InputDiagnostics`.
Data quality	`openfactsheet.diagnostics`	Missing periods, stale and repeated returns, impossible and extreme returns, short history, serial correlation, smoothing risk. Produces a `DataQualityReport` before any metric is computed.
Calculation	`openfactsheet.metrics`	One registry of metric functions. Functions only calculate; they raise `UndefinedMetric` when a value is mathematically undefined. Shared quantities (wealth index, drawdown path, episodes, moments, aligned benchmark) are cached once per report.
Policy	`openfactsheet.policy`	Decides applicability, status and confidence, and attaches assumptions and warnings.
Interpretation	`openfactsheet.interpretation`	Plain-language and professional explanations plus context notes (for example, a smoothing caveat on the Sharpe ratio). Never grades or advises.
Reporting	`openfactsheet.report`	Builds the `FactsheetReport` and renders it as JSON, terminal text, Markdown or HTML. Calculations know nothing about rendering.

Compute everything

The metric engine evaluates every registered metric for every report. There is no public way to select a subset. Each metric produces a structured `MetricResult`; none is omitted because its value is unattractive.

Policy layer: applicability, confidence, status

Three separate questions are answered for each metric:

1. **Applicability** (before calculation, `policy/applicability.py`): can the metric be computed at all?
   - benchmark required but none supplied ⟶ `not_applicable`;
   - annualized metric without a regular frequency ⟶ `not_applicable`;
   - wealth-path metric with returns below -100% ⟶ `not_applicable`;
   - fewer observations than the mathematical minimum ⟶ `insufficient_data`.
2. **Calculation outcome**: a value that is mathematically undefined (for example zero volatility in a ratio denominator) ⟶ `not_applicable` with the reason; an unexpected failure or non-finite result ⟶ `error` with the reason (the value is withheld).
3. **Confidence** (after calculation, `policy/confidence.py`): how reliable is the estimate? Labels are `high`, `moderate`, `low` and `not_assessable` (no value). They come from the metric's minimum-data policy (observations, years, or expected tail observations) and are then capped by data quality: smoothing risk, stale returns, missing periods, less than one year of data for an annualized metric, and limited benchmark overlap (0015, 0014).

The resulting **status** is one of:

Status	Value shown	Meaning
`applicable`	yes	Computed; confidence `high` or `moderate`.
`low_confidence`	yes	Computed; the estimate is unreliable for the stated reasons.

Status	Value shown	Meaning
`supplementary`	yes	Diagnostic or model-dependent statistic shown for context.
`not_applicable`	no	Cannot be meaningfully computed for this input; reason given.
`insufficient_data`	no	Below the mathematical minimum; reason given.
`error`	no	Unexpected failure; reason given.

Low confidence never hides a value. Confidence describes the reliability of an estimate, never the quality of the investment.

Observed, derived, estimated, model-dependent

Every metric definition carries a `nature` tag, shown in the output:

Nature	Meaning	Examples
observed	A fact about the supplied path.	cumulative return, best/worst period, maximum drawdown, time under water, streaks
derived	A deterministic transformation of observed quantities.	CAGR, Calmar, Martin, Omega, gain-to-pain, capture ratios
estimated	A sample estimate of a population quantity, subject to sampling error.	mean, volatility, Sharpe, Sortino, VaR, ES, skewness, autocorrelation, beta
model_dependent	Depends on an explicit statistical or economic model.	CAPM alpha, Treynor ratio, Ljung-Box p-value

Assumptions are explicit, not absent

Parameter-less does not mean assumption-less. Every assumption that affects a metric is recorded on that metric's `assumptions` field (annualization factor and its source, 0% risk-free rate, 0% target return, 95% confidence level, sample standard deviation, drawdown starting point, benchmark alignment, regression model). The full methodology object is embedded in every report and returned by `ofs.methodology()`.

Versioning

- The methodology version is independent of the package version.
- Any change to a canonical value, status, confidence rule, threshold or convention increments `METHODOLOGY_VERSION`, adds a CHANGELOG entry and a decision note, and requires conscious regeneration of the golden reports.
- The meaning of a metric is never changed silently: its `formula_id` (for example `sharpe_ratio@v1`) changes when its formula changes.
- Future research APIs may allow overrides; results produced with overrides must be labeled non-canonical (`provenance.canonical = false`).

Anti-cherry-picking rules

Implemented from CLAUDE.md section 43:

- Negative periods, maximum drawdown, and downside, path and tail statistics are always part of the report when the data allow them.
- The full supplied history is used; no start date, window, target, confidence level, benchmark or risk-free rate is chosen per fund.
- Track-record length, frequency and annualization are always disclosed.
- Benchmark overlap (aligned observations and dropped periods) is always disclosed; benchmarks are never inferred.
- Material data-quality findings appear in the diagnostics section and on the affected metrics.
- Renderers have no flags to remove canonical sections; unavailable sections are shown with the reason (for example "no benchmark supplied").
- `ofs.compare()` applies the identical methodology to every investment, adds a common-period analysis when histories differ, and does not rank.
- There is no composite score.

Provenance

Each report records the package version, methodology version, input type, a SHA-256 fingerprint of the analyzed returns (and benchmark), the date range, frequency, annualization statement, number of observations and every

transformation applied (sorting, trimming, percent conversion, table reshaping). The JSON output contains no timestamps, so identical inputs produce identical JSON.

Related documents

- Assumptions for methodology 0.1.0
- Methodology decision notes
- Reference implementations and validation
- Full specification: exact formulas, notation, decision criteria, policies and thresholds, JSON report schema

---

Canonical assumptions — methodology 0.1.0

This page answers each methodological question listed in CLAUDE.md section 41 for methodology **0.1.0**. The machine-readable source of truth is `src/openfactsheet/policy/defaults.py` (`ofs.methodology()` returns it, and every report embeds it). Rationale and library comparisons are in the decision notes.

Items marked **Deferred** have no canonical answer yet; the report's behavior in those cases is described so that nothing is hidden.

Question	Methodology 0.1.0 answer	Decision
Arithmetic vs geometric annual return	Headline annualized return is geometric (CAGR by observation count). The arithmetic `P x mean` is reported separately as `annualized_arithmetic_mean`.	0004
Sample vs population standard deviation	Sample (n − 1) everywhere.	0003
Annualization under serial correlation	The canonical annualized ratios keep square-root-of-time scaling (never silently corrected). Beside them, the one-year variance ratio and Lo's (2002) serial-correlation-adjusted Sharpe ratio (horizon q = round(P), sample autocorrelations, at least two years of data) show the size of the bias.	0029
Risk-free treatment	0% per period for every excess-return metric, disclosed on each metric. Currency is never inferred from the locale.	0005
Sortino MAR / target	0% per period.	0006
Omega threshold	0% per period, simple empirical estimator `sum(max(r, 0)) / sum(max(-r, 0))`, not annualized.	`policy/defaults.py` (`omega`)
VaR confidence	95%, historical, per observation period.	0010
ES confidence	95%, historical: mean of returns at or below the 5% quantile.	0010
Exact quantile interpolation	Linear interpolation (Hyndman-Fan type 7; R and NumPy default).	0010
Downside deviation denominator	All n observations (full sample).	0006
Minimum track record	`minimum_track_record_length` (Bailey & López de Prado 2012): observations needed for the observed Sharpe ratio to exceed 0 at 95% one-sided confidence, keeping skewness and kurtosis; `not_applicable` when the Sharpe ratio is not positive. Independently, confidence labels treat most estimates as `low` below 3 years and `moderate` below 5.	0019, 0015
Sharpe-ratio uncertainty	Non-normal i.i.d. asymptotics on the per-period Sharpe ratio with population skewness and non-excess kurtosis (kurtosis is never ignored). Probabilistic	0019

Question	Methodology 0.1.0 answer	Decision
	Sharpe Ratio against a reference of 0; standard error of the annualized Sharpe ratio with a two-sided 95% interval. Serial correlation is not modelled (smoothing lowers confidence). The Deflated Sharpe Ratio is **not computable** from a single series because it needs the number of strategies tried.	
Mean-return uncertainty	Standard error `s / sqrt(n)` with a two-sided 95% Student-t interval (n − 1 degrees of freedom) and a t-test against zero.	0020
Normality testing	Jarque-Bera with population moments and the asymptotic chi-squared(2) p-value; supplementary diagnostic only.	0020
Lower-partial-moment targets and denominators	Target 0% per period and full-sample denominator n for Kappa 3, upside potential ratio and downside potential (all per period); semi-deviation uses the sample mean as target, denominator n, annualized by sqrt(P); mean absolute deviation is per period.	0016
Drawdown-ratio conventions	Burke uses the depths of peak-to-recovery episodes; Sterling = CAGR / (| max drawdown| + 10%), the 10% being a disclosed historical convention; Pain ratio = CAGR / Pain Index; full history, rf = 0. Payoff ratio excludes zero-return periods.	0017
Parametric VaR	Gaussian and Cornish-Fisher VaR at 95% with population moments (Boudt, Peterson & Croux 2008); supplementary and model-dependent; the historical VaR remains the headline tail estimate.	0018
Benchmark alignment	Intersection of calendar periods (monthly/quarterly/yearly) or exact dates (daily/weekly); never filled; overlap disclosed; different frequencies raise an error.	0012
Factor models	Only with factor returns supplied by the caller (never downloaded). A column named `RF` is subtracted from fund returns; otherwise the risk-free rate is 0%. One return scale for all factor columns. OLS with Newey-West standard errors (lag floor(4 (m/100)^(2/9)), Bartlett kernel, no prewhitening, small-sample factor, Student t); 95% intervals; periods missing for any series are dropped, never filled.	0028
Charts	Eight fixed charts over the full history: linear scales, zero baselines for bars, symmetric heatmap scale, partial years labelled, every return in the histogram, benchmark only on the aligned overlap; the 10 deepest drawdown episodes tabulated.	0026

Question	Methodology 0.1.0 answer	Decision
Missing values	Never replaced by zero or forward-filled. Leading and trailing missing values are trimmed; interior missing values are removed, counted and disclosed, and appear as missing expected periods. Wealth-path and annualized metrics are capped at `moderate` confidence when gaps exist.	0002, 0015
PDF factsheets	Embedded text only; values assigned to month columns by horizontal position; every year with a YTD / total cell must reconcile (compounded, within worst-case rounding) or the table is refused; blank months inside the history are missing, never zero; scanned PDFs refused.	0030
OCR (screenshots, scanned PDFs)	Local engine only (RapidOCR with bundled models, or Tesseract); cells below 0.90 confidence refuse the table; every year needs a reconciling YTD / total cell; documented character repairs and up to three inferred month headers are recorded and lower confidence.	0031
Fund profile of a document	Fixed fields with evidence; identifiers validated by check digit and never corrected, label-only codes only after their label; the profile never changes a metric and a named benchmark is never used as benchmark returns.	0032
Benchmark returns printed in a document	Returned by `ofs.load()` as a separate series, never used automatically; `analyze()` uses the fund series and says so. Relative rows are ignored. The returns table is attributed to a share class only by printed evidence; otherwise ambiguous.	0033
Monthly table partial years	Empty cells are missing, never zero. Empty cells after the last reported month (future months) are trimmed; empty cells inside the history are interior missing values. Annual total columns (YTD, Year, Total) are ignored for calculation and the omission is recorded as a transformation. Calendar-year statistics use **complete years only** (all 12 months; all 4 quarters; daily/weekly within 7 days of both year ends); partial years are listed in details and warnings and never used.	`input/monthly_table.py`, 0023
Drawdown tail and entropic measures	DaR and CDaR at 95% on the compounded drawdown path D_1..D_n with the historical VaR / ES estimators (linear quantile; mean at or below it); EDaR and EVaR at 95% by one-dimensional convex minimization. With 20 or fewer observations EVaR equals the worst period and EDaR the maximum drawdown.	0021
Modified Expected Shortfall	Cornish-Fisher ES (Boudt, Peterson & Croux 2008) at 95% with population	0022

Question	Methodology 0.1.0 answer	Decision
	moments and the full expansion; supplementary and model-dependent; a note is attached when it is less severe than the modified VaR.	
Rolling-window convention	One-year windows of round(P) consecutive observations, advanced one observation at a time; every start date is used; never user-selected.	0024
Drawdown durations and recovery	Durations in observation periods from peak to recovery; open episodes count to the last observation in the average duration (lower bound) and are excluded from the average recovery time. Recovery factor = compounded cumulative return / |max drawdown|.	0025
Return scale detection	Deterministic classifier: declared scale, `%` labels, then median absolute value against frequency-dependent bounds; ambiguous cases raise `AmbiguousReturnScaleError` instead of guessing.	0013
-100% return behavior	Valid (total loss). Wealth becomes zero, cumulative return and CAGR are -100%, maximum drawdown is -100%, and a critical finding plus metric warnings are shown.	0001, 0007
Returns below -100%	Impossible for ordinary simple returns: kept (never clipped), flagged as critical, and every wealth-path metric is `not_applicable`. Moment-based metrics are still computed.	0001
Non-finite return behavior	Infinite values raise `InputError` before any calculation. NaN means a missing observation (see above). No metric result can hold NaN or infinity: undefined values are `not_applicable` with a reason, unexpected failures are `error` with a reason. A result that overflows to plus or minus infinity (a denominator non-zero but below floating-point resolution, e.g. a single subnormal loss) is reported as `not_applicable` with an overflow explanation; a NaN result remains `error`.	0001, `metrics/engine.py`
Numerically constant returns	Shape statistics built on moments (skewness, kurtosis, Jarque-Bera, Cornish-Fisher VaR / ES, Sharpe-ratio inference) are `not_applicable` when the returns are identical or their variance is at or below floating-point resolution (`ptp(r) == 0` or `m2 <= 16 (eps · max|r|)^2`), instead of reporting rounding noise.	`metrics/context.py` (`_standardized_moment`)
Compounding after total loss	Wealth stays at zero; later returns cannot rebuild it and the drawdown never recovers.	0007
Irregular time series	Never annualized: annualized metrics are `not_applicable` with a reason. Non-annualized metrics are computed.	0002

Question	Methodology 0.1.0 answer	Decision
Undated values	Frequency is unknown and nothing is annualized unless the caller declares the frequency; a declaration is recorded and warned.	0002
Crypto / non-business-day data	Daily series with weekend observations use 365 periods per year; business-day series use 252. **Deferred:** mixed calendars (e.g. a 365-day fund against a 252-day benchmark, which currently aligns only on common dates), intraday data, and holiday calendars by exchange.	0002
Frequency changes within a series	Detected as irregular when fewer than half of the spacings equal one period; no resampling. **Deferred:** splitting a history into regular segments.	0002
Drawdown starting point	Initial wealth 1 counts as the first peak, so a first-period loss is a drawdown.	0007
Skewness / kurtosis estimators	Bias-corrected G1 and excess G2.	0011
Smoothed returns	Diagnosed (lag-1 autocorrelation with significance), warned and downgraded; never unsmoothed.	0014

What the caller may and may not set

The canonical API has no parameters for windows, thresholds, confidence levels, target returns, risk-free rates or metric subsets. The only optional arguments describe the input when the data cannot establish it:

- `return_scale="decimal" | "percent"` — only needed when detection reports ambiguity;
- `frequency=` — the observation frequency of undated values (checked against dates when dates exist);
- `benchmark=` — an explicitly supplied benchmark; none is ever inferred;
- `name=` — a display name.

Declarations are recorded in the input diagnostics and warned on the affected metrics.

Decision criteria reference

Every rule OpenFactsheet applies between receiving data and printing a report, stated exactly: thresholds, comparison operators and order of evaluation, with a pointer to the implementing code. All rules are part of methodology 0.1.0; changing any of them changes results and requires a methodology version bump (see methodology decisions).

Related references: exact metric formulas are in formulas.md, minimum-data policies in policies.md, notation in notation.md, and the JSON output in report_schema.md.

Notation used below: `r` is the vector of analyzed simple decimal returns, `n` its length, `P` the periods-per-year factor, `years = n / P`.

---

1. Pipeline and order of operations

Step	What happens	Code
1. Input	Any supported source is parsed into one or more `RawReturns` (values may contain NaN and may be in percent).	input/dispatcher.py `to_raw`
2. Normalization	Validation, sorting, missing-data handling, frequency inference, scale detection ⟶ `ReturnSeries` + `InputDiagnostics`.	normalization/returns.py `normalize_returns`
3. Diagnostics	Data-quality checks ⟶ `DataQualityReport` (runs before any metric).	diagnostics/data_quality.py `run_diagnostics`
4. Benchmark alignment	Only if a benchmark is supplied.	compare/alignment.py `align_benchmark`

Step	What happens	Code
5. Metrics	Every registered metric, in canonical order: applicability pre-check → calculation → outcome handling.	metrics/engine.py `compute_all`, `evaluate`
6. Policy	Status, confidence, warnings and assumptions for each computed value.	policy/confidence.py `assess`
7. Interpretation	Plain and professional text plus context notes.	interpretation/engine.py
8. Report	Assembly and rendering (terminal, Markdown, HTML, JSON).	report/builder.py, report/

`ofs.analyze(returns, benchmark=None, *, name=None, return_scale=None, frequency=None)` (api.py) runs steps 1–8. `return_scale` and `frequency` are *input declarations*, applied to both fund and benchmark, and are recorded in the report; they are not tuning parameters.

---

2. Input parsing criteria

2.1 Source dispatch — input/dispatcher.py

Checked in this order:

Source	Treatment
`RawReturns`	Used as is.
`pandas.Series`	`from_series`.
`pandas.DataFrame`	`from_frame` (may yield several series).
`pathlib.Path`	By file suffix (below).
`str` containing a newline after stripping	Pasted table text.
other `str`	Treated as a file path.
`numpy.ndarray`, `list`, `tuple` (or other sequence)	`from_sequence` (undated).
anything else	`InputError`.

File suffixes (case-insensitive): `.csv`, `.txt`, `.tsv` → CSV reader; `.xlsx`, `.xlsm`, `.xls` → Excel reader; `.png .jpg .jpeg .gif .bmp .tif .tiff .webp` → OCR reader (2.9); `.pdf` → embedded-text PDF reader (2.8), with OCR for pages that have no text (2.9); any other suffix → `InputError`; a non-existent path → `InputError("File not found")`. When a file yields exactly one series whose name is the default `"Returns"` and no `name` was passed, the series is named after the file stem. The file name is stored in metadata.

`analyze()` requires exactly one series after parsing; more than one raises `InputError` suggesting `ofs.compare()`.

A `ReturnSeries` passed to `analyze()` that carries its `input_diagnostics` (e.g. from `ofs.load`) is reused unchanged when no declaration is passed (only renamed if `name` differs). Otherwise it is re-normalized with `return_scale="decimal"` unless another scale is declared.

2.2 Cell parsing — input/values.py `parse_cell`

Cell	Result
`None`	missing (NaN)
`bool`	`InputError` (a boolean is not a return)
`int`, `float`, NumPy number	its float value; carries no percent evidence
string, after stripping, whose lower-case form is one of `""`, `-`, `--`, –, —, `na`, `n/a`, `nan`, `none`, `null`, `#n/a`	missing (NaN)
other string	parsed as below

String parsing, in order:

1. `%` anywhere in the string sets the percent flag for that cell; all `%` characters are removed.
2. The unicode minus − becomes `-`; spaces and non-breaking spaces are removed.
3. Parentheses around the whole value, `(1.2)`, mean a negative number.
4. If the string contains `,` and no `.`, the comma is a decimal separator (`1,5` → `1.5`). Thousands separators are not supported because they are not plausible in return data.

5. The remainder must match `^[+-]?(\d+(\.\d*)?|\.\d+)([eE][+-]?\d+)?$`, otherwise `InputError`.

Percent hint for a whole column (`parse_cells`): `True` if any cell contained `%`; `False` if cells were strings but none contained `%`; `None` if no cell was a string (numeric input carries no label evidence).

2.3 Sequences — input/arrays.py

A 2-D array with one dimension equal to 1 is flattened; any other non-1-D shape raises `InputError`. Sequences are undated. Default name `"Returns"`.

2.4 pandas Series — input/pandas_input.py and normalization/dates.py

Index handling (`coerce_dates`):

Index	Result
`PeriodIndex`	Converted to period-end dates, normalized to midnight (recorded as a transformation).
`DatetimeIndex`	Used as dates.
`RangeIndex` or any numeric dtype (integer or float)	Undated: numeric labels are positions, never dates (converting them would produce 1970-epoch timestamps).
other (e.g. strings)	`pd.to_datetime`; failure ⟶ `InputError`.
timezone-aware	Timezone removed, wall-clock dates kept (recorded).
any `NaT`	`InputError`.

The series name becomes the report name unless `name` is given.

2.5 pandas DataFrame layouts — `from_frame`

Evaluated in this order:

1. Empty frame ⟶ `InputError`.
2. **Wide monthly table** if the column labels contain at least 12 distinct month names (`is_monthly_table`). Month labels are matched after removing non-letters and lower-casing: `jan`/`january`, `feb`/`february`, …, `sep`/`sept`/`september`, …, `dec`/`december`. See 2.6.
3. If the index is not a `DatetimeIndex`/`PeriodIndex`, a **date column** is searched for: first, a column whose stripped name matches `^(date|dates|period|month|time|timestamp|as of|asof|day)$` (case-insensitive) *and* looks like dates; otherwise the first column if it looks like dates. "Looks like dates" (`_looks_like_dates`): datetime or period dtype ⟶ yes; numeric dtype ⟶ no; otherwise every non-null value must parse with `pd.to_datetime(format="mixed")`. The date column becomes the index.
4. Remaining columns are return series. Duplicate column names ⟶ `InputError`; no columns ⟶ `InputError`.

Series naming: with one column, the `name` argument if given; otherwise `"Returns"` if the header is generic (after removing `% ( ) [ ]`, lower-cased: `return`, `returns`, `ret`, `rets`, `r`, `value`, `values`, `performance`, `monthly return(s)`, `daily return(s)`, `total return`); otherwise the column header. With several columns, each series is named by its header.

A `%` in a column header sets the percent hint to `True` for that column.

2.6 Wide monthly tables — input/monthly_table.py

- Exactly one column per month is required (a repeated month ⟶ `InputError`; missing months ⟶ `InputError`).
- Year labels: a column named `year`, `yr`, `years` or `fy` (case-insensitive), otherwise the index. Labels must convert to integers in `[1800, 2200]` and be unique; otherwise `InputError`.
- Every cell becomes the return for the month-end date of its (year, month). Empty cells stay missing, so trailing future months are trimmed later and are never treated as zero.
- Other columns (e.g. `YTD`, `Total`) are ignored for calculation and listed in the transformations.
- Percent hint: `True` if any column header contains `%`, else the cell-based hint.

2.7 CSV, Excel and pasted text

- **CSV** (input/csv.py): read as text with delimiter sniffing (`sep=None`, Python engine, `skipinitialspace`) so that `%` labels and accounting negatives survive; all-empty columns dropped; then `from_frame`.
- **Excel** (input/excel.py): requires `openpyxl` (else `InputError` naming the `excel` extra); **only the first sheet** is read; all-empty rows and columns dropped; then `from_frame`.
- **Pasted text** (input/clipboard.py): lines are stripped and blank lines dropped. Delimiter: tab if every line has a tab; else `;` if every line has one; else `,` if every line has a comma *and* commas do not look like decimal separators

(they do when the data rows contain no . and do contain whitespace); else runs of whitespace. If the header has one field fewer than the data rows (blank top-left cell), the first column is restored and named `Year`.

2.8 PDF factsheets — input/pdf.py (decision 0030)

Requires the `pdf` extra (pypdf); otherwise `OCRExtractionError` naming it.

1. Each page's embedded text is extracted **with layout** (character positions kept). No page has any text → `OCRExtractionError` (scanned PDF; OCR is not part of the core library).
2. **Header line**: its tokens include exactly one month name for each month, January to December, left to right (month names as in 2.5). A token to the right of December whose letters are `ytd`, `year`, `total`, `annual`, `yr`, `fy`, `cum` or `cumulative` is the total column. A detached `%` token joins the number before it.
3. **Year rows**: the following lines whose first token is a year `^(18|19|20|21)\d{2}$`; up to two blank lines are skipped; any other line ends the table. No header followed by a year row anywhere → `OCRExtractionError`.
4. **Column assignment**: each value goes to the column (12 months + total) whose header centre is nearest to the value's centre, if within half the median month spacing. Problems (any one refuses the table): a value under no column; two values under one column; a token that is not a return (2.2); a year in two rows; a table with no monthly values.
5. **Reconciliation** of each year with a total cell: the months compound (as percentages; also as decimals unless the table is labelled in percent) to the total within `0.5·10^−d_total + 0.5·10^−d_cells × months` (worst-case rounding) → check passed; else if their arithmetic sum matches → warning (the total is a sum); else a **problem** (refused).
6. **Warnings** (confidence `moderate`, table still used): no total column (no cross-check possible); months missing inside the history (they stay missing, never zero); totals that are sums. No problems and no warnings → confidence `high`.
7. A refused table raises `OCRExtractionError` listing every problem and the reconstructed table. Accepted tables become a wide monthly table (2.6); the percent hint is `True` if a cell or one of the three lines above the header contains `%`. Several tables → several series (named `<name> (table k, page p)`).
8. Recorded: a transformation naming the page, engine version, value count and confidence; the warnings in `InputDiagnostics.warnings`; and `provenance.extraction` (engine, source SHA-256, page, confidence, checks, warnings, reconstructed table).

2.9 OCR of images and scanned pages — input/image.py, documents/ocr (decision 0031)

Requires the `ocr` extra (RapidOCR, default) or `tesseract` extra; otherwise `OCRExtractionError` naming it. Engines use only local models (RapidOCR's bundled files, passed as explicit paths) and never access the network.

1. Images are converted to RGB; scanned PDF pages (no text layer) are rendered with pypdfium2 at `ocr.dpi` (300).
2. Words (pixel boxes, confidence 0–1) are grouped into lines: a word joins a line when it overlaps the line's vertical band by at least half the smaller height.
3. Table reconstruction as in 2.8, with these differences:
   - **Header**: at least `ocr.header_min_months` (9) month names read in order and evenly spaced (steps within 15% of the median); missing months are placed by that spacing (warning).
   - **Repairs**: a numeric token or year label that does not parse is retried with `ocr.numeric_repairs` (O/o→0, l/I/| →1) and dash variants → `-`; a repair is kept only if the result parses (warning, recorded per cell).
   - **Problems** (refused): any cell with confidence `< ocr.min_cell_confidence` (0.90); no total column; any year with values but no total cell; any failed reconciliation.
4. Recorded: `provenance.extraction` gains `ocr` (engine name, version, model files and SHA-256), `image_sha256`, `repairs`, `min_cell_confidence` and, for PDFs, `renderer` and `dpi`.

2.10 Fund profile of a document — documents/profile.py (decision 0032)

For PDF and image inputs, the page layouts are also searched for:

Field	Rule	Confidence
Identifiers	ISIN anywhere if its check digit passes; CUSIP / SEDOL / LEI / FIGI only on a line with their label; a labelled code failing its check digit is kept as `valid: false` with a note, never corrected; WKN / Valor / Bloomberg ticker / RIC only immediately after their label; a code equal to the national code embedded in an ISIN gets `linked_isin`	—
Fund name	text after a name label (`fund name`, `Fondsname`, `nom du fonds`, …)	high
Fund name (fallback)	a page-1 line (first 15) with a fund word, at most 12 words, no colon, not a table row; tallest wins	moderate

Field	Rule	Confidence
Objective	text after an objective / policy heading up to a blank line, another heading or a table, at most 8 lines	high
Benchmark name(s)	text after a benchmark label at the start of a line (duplicates removed)	high
Currency	first ISO 4217 code after a currency label	high
Launch date	ISO as printed; dotted day-first; slash dates converted only when day and month cannot be swapped, else kept as printed	high / low

Labels are matched at the start of a line, case- and accent-insensitively, longest first (vocabulary in documents/labels.py). The profile never changes a metric; a named benchmark adds a note to the benchmark section when no benchmark returns were supplied.

2.11 Table roles and share classes — documents/tables.py, documents/share_classes.py (decision 0033)

- **Row labels**: words left of the January column (centre minus half the column spacing) that are not numbers; a labelled line without a year continues the previous year; each label forms its own series (validated separately).
- **Roles** (first match): row label with relative words (`relative`, `difference`, `excess`, `active`, …) ⟶ `relative`; with benchmark or index words (`benchmark`, `index`, `MSCI`, `S&P`, …) and no fund word ⟶ `benchmark`; with fund words (`fund`, `class`, `portfolio`, …) ⟶ `fund`; other label ⟶ `unknown`; unlabelled series of a labelled table ⟶ `fund`; single series ⟶ `benchmark` only if the caption (nearest three non-blank lines within six above the header) says benchmark without a fund word, else `fund`.
- **Use**: `relative` series are ignored (note); an unreliable `benchmark` series is left out (note); any other unreliable series refuses the document. `analyze(document)` uses the single non-benchmark series and notes that benchmark returns were not used; no non-benchmark series ⟶ `InputError`; several ⟶ `InputError` (choose with `ofs.load`).
- **Share classes**: a class-table header (class word and `ISIN`, no ISIN on the line) opens rows until one without a valid ISIN; a line matching `class <label>` with a valid ISIN on it or on the next line also defines a class.
- **Attribution** (all evaluated): caption + footer (three lines after the table) mention exactly one class ⟶ rule 1; performance heading (within ten lines above) mentions exactly one ⟶ rule 2; exactly one class ⟶ rule 3; exactly one ISO currency in the caption matching exactly one class ⟶ rule 4. One distinct answer ⟶ `attributed` (first rule); several ⟶ `ambiguous`; none with several classes ⟶ `ambiguous` with all classes; no classes ⟶ `no_share_classes`.

---

3. Validation criteria (input corruption fails early)

Applied by `normalize_returns` in this order (normalization/validation.py):

Condition	Outcome
No observations	`InputError`
Any `+inf` or `-inf`	`InputError` listing up to 5 positions
All values missing	`InputError`
Duplicate timestamps (checked on all supplied dates, including rows whose value is missing)	`InputError` listing up to 5 dates
Timestamps not in increasing order	Stable sort; `was_sorted = false`; transformation and warning recorded

Returns below −100% are **not** rejected and not clipped: they are counted (`impossible_return_count`) and make wealth-path metrics not applicable (see 7).

---

4. Missing-data criteria

normalization/returns.py:

1. Leading and trailing missing values are trimmed and counted (`leading_missing_trimmed`, `trailing_missing_trimmed`).
2. Interior missing values are **dropped, never filled** with zero or a previous value; their count is `missing_values`, and an input warning is recorded.
3. Frequency is inferred from the **grid dates**: the dates after trimming but *before* dropping interior missing rows, so a missing value is treated as a missing observation rather than as irregular spacing.

4. `missing_expected_periods` = calendar gaps in the grid dates (from frequency inference) + the number of interior missing values. It is `null` when the frequency does not support counting (business-daily, irregular, unknown, undated).

Metrics are computed on the remaining observations; compounding and annualization skip the missing periods, and affected metrics are capped at moderate confidence (section 9).

---

5. Frequency inference

normalization/frequency.py `infer_frequency`. Only calendar spacing is evidence; the number of observations never is.

Let `g` be the day gaps between consecutive grid dates and `m = median(g)`.

Condition	Outcome
fewer than 2 dates	`unknown` (no annualization)
`25 ≤ m ≤ 35`	monthly candidate; steps = differences of `12·year + month`
`80 ≤ m ≤ 100`	quarterly candidate; steps = differences of `4·year + quarter`
`300 ≤ m ≤ 400`	yearly candidate; steps = differences of `year`
`5 ≤ m ≤ 9`	weekly candidate (below)
`m ≤ 4`	daily candidate (below)
any other `m`	`irregular`

Bands are inclusive and tested in the order monthly, quarterly, yearly, weekly, daily. Comparing calendar periods (not days) makes month-start, month-end and business-month-end dates equivalent.

Period candidates (monthly, quarterly, yearly). With `steps` as above:

- any step equal to 0 (two observations in the same calendar period) ⟶ `irregular`, and the count is reported as `duplicate_periods`;
- else if the share of steps equal to exactly 1 is `< 0.5` ⟶ `irregular`;
- else **regular**; `missing_expected_periods = Σ(steps − 1)`.

Weekly candidate. `weeks = round(g / 7)`. The series is weekly only if every gap satisfies `|g − 7·weeks| ≤ 1` (one-day holiday shifts allowed), every `weeks ≥ 1`, and the share of `weeks == 1` is `≥ 0.5`; missing periods = `Σ(weeks − 1)`. Otherwise `irregular`.

Daily candidate. If the share of gaps `≤ 4` days is `< 0.8` ⟶ `irregular`. Otherwise:

- if any date falls on a Saturday or Sunday ⟶ **calendar-day** daily, `P = 365`, `missing_expected_periods = Σ(g − 1)`;
- else ⟶ **business-day** daily, `P = 252`, missing periods not counted (exchange holidays are legitimate gaps); the evidence notes the number of gaps longer than 7 days.

Annualization factors (policy/annualization.py): business daily 252, calendar daily 365, weekly 52, monthly 12, quarterly 4, yearly 1; `irregular` and `unknown` ⟶ `P = null` (never annualized).

Declared frequency (`frequency=` argument):

Situation	Outcome
Declared value not one of daily, weekly, monthly, quarterly, yearly	`FrequencyInferenceError`
Dated input, inferred regular frequency ≠ declared	`FrequencyInferenceError`
Dated input, inferred `irregular`	`FrequencyInferenceError`
Dated input with a single timestamp (inferred `unknown`)	Declared frequency used (`frequency_source = "declared"`); one timestamp cannot contradict it
Dated input, inferred = declared	Inferred frequency used (`frequency_source = "inferred"`)
Undated input	Declared frequency used (`frequency_source = "declared"`); declared daily assumes business days (`P = 252`) with a warning
Undated input, nothing declared	`unknown`, `frequency_source = "none"`, warning that nothing is annualized

For dated input without a declaration, `frequency_source` is `"inferred"` when the inferred frequency is regular and `"none"` otherwise.

---

6. Return-scale classifier (percent vs decimal)

normalization/scaling.py `detect_return_scale`, thresholds in policy/defaults.py `return_scale`. Runs after missing values are removed and after frequency is established.

Upper bound `U` for the median absolute return of a plausible *decimal* series:

Frequency	`U`	Confident decimal bound `U / 4`
daily	0.05	0.0125
weekly	0.10	0.025
monthly	0.20	0.05
quarterly	0.30	0.075
yearly	0.60	0.15
irregular	0.30	0.075
unknown	0.30	0.075

Let `M` = median of `|r|` over the **non-zero** raw values. Rules, first match wins:

1. `return_scale` declared ⟶ `declared_decimal` (×1) or `declared_percent` (×0.01). Any other declared value ⟶ `InputError`.
2. Percent hint is `True` (`%` labels in the source) ⟶ `detected_percent`.
3. All values are exactly zero ⟶ `detected_decimal` (scale immaterial).
4. `M > U`:
   - if `M / 100 > U` ⟶ `InputError` (implausible in both scales; when every value is positive the message suggests the data are prices/NAVs that must be converted explicitly);
   - else ⟶ `detected_percent`.
5. `M ≤ U / 4` ⟶ `detected_decimal`. Values below −1 stay decimal and are flagged as impossible, not rescaled.
6. Ambiguous zone `U/4 < M ≤ U`:
   - if any value `< −1` (impossible as a decimal return) ⟶ `detected_percent`;
   - else ⟶ `AmbiguousReturnScaleError` asking the caller to declare the scale.

Percent values are multiplied by 0.01 and the conversion is recorded as a transformation.

---

7. Data-quality diagnostics

diagnostics/data_quality.py `run_diagnostics` and diagnostics/missingness.py. Diagnostics never remove or alter data.

7.1 Findings

id	Severity	Exact trigger	Effect on metrics
`missing_values`	warning	interior missing values `> 0`	wealth-path and annualized metrics capped at moderate, with a warning
`missing_periods`	warning	`missing_expected_periods > 0`	same as above
`duplicate_periods`	critical	two or more observations in the same calendar period (frequency becomes irregular)	annualized metrics not applicable (no `P`)
`unsorted_dates`	info	input was not chronologically sorted	none (data sorted)
`frequency_declared`	info	`frequency_source == "declared"`	warning on annualized metrics
`frequency_not_established`	warning	not declared and frequency is `irregular` or `unknown`	annualized metrics not applicable
`percent_scale_detected`	info	`return_scale == "detected_percent"`	none
`very_short_history`	warning	`years` known and `years < 1`	none directly (years-based policies already give low confidence)
`short_history`	warning	`years` known and `1 ≤ years < 3`	none directly
`very_few_observations`	warning	`years` unknown and `n < 12`	none directly

id	Severity	Exact trigger	Effect on metrics
`impossible_returns`	critical	any `r_t < -1`	every wealth-path metric not applicable; drawdown episodes omitted from the report
`total_loss`	critical	any `r_t = -1` exactly	warning on wealth-path metrics (wealth is 0 afterwards)
`suspicious_large_returns`	warning	robust outlier count `> 0` (7.2)	none (values are kept)
`excessive_zero_returns`	warning	share of exact zeros `≥ 0.10`	volatility-based metrics capped at moderate, with a warning
`repeated_returns`	warning	longest run of identical consecutive returns `≥ 3`	none
`smoothing_risk`	warning	smoothing risk is `moderate` or `high` (7.4)	volatility-based metrics: high ⟶ low confidence, moderate ⟶ capped at moderate
`serial_correlation`	info	smoothing risk is neither moderate nor high, Ljung-Box computed and `p < 0.05`	none

History findings are mutually exclusive and evaluated in the order shown.

7.2 Outliers — diagnostics/outliers.py

$$z_t = \frac{|\, r_t - \text{median}(r)|}{1.4826 \cdot \text{MAD}}, \qquad \text{MAD} = \text{median}(|\, r - \text{median}(r)|)$$

An observation is flagged when `z_t > 5`. No observation is flagged when `n < 5` or `MAD = 0`. Flagged values are kept in every calculation.

7.3 Stale returns — diagnostics/stale_returns.py

- zero-return share = `#{r_t = 0} / n` (exact equality after normalization);
- longest identical run = longest run of consecutive, exactly equal returns (1 when nothing repeats; 0 for an empty series).

7.4 Serial dependence and smoothing

- **Autocorrelation** (diagnostics/autocorrelation.py), lags 1–3, statsmodels `acf(adjusted=False)`:

  $$\hat{\rho}_k = \frac{\sum_{t=k+1}^{n}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

  undefined (`null`) when `n ≤ k + 1` or all returns are identical.
- **Significance bound** reported as `±1.96 / √n` when `n > 3`.
- **Ljung-Box** with 3 lags (statsmodels `acorr_ljungbox`), computed only when `n ≥ 10` and returns are not constant.
- **Smoothing risk** (diagnostics/smoothing.py), with `z = ρ̂ · √n`:

Condition (first match)	Risk
`n < 12`	`not_assessable`
`ρ̂` undefined	`not_assessable`
`ρ̂ ≥ 0.30` and `z ≥ 1.96`	`high`
`ρ̂ ≥ 0.15` and `z ≥ 1.645`	`moderate`
otherwise	`none`

  The thresholds are a documented heuristic (decision 0014). Returns are never unsmoothed.

---

8. Applicability criteria

policy/applicability.py `precheck`, evaluated before calculation, first match wins:

#	Condition	Status	Reason (abridged)
1	metric requires a benchmark and none was supplied	`not_applicable`	no benchmark supplied; no default benchmark is ever assumed
2	metric requires factor returns and none were supplied	`not_applicable`	no factor returns supplied; factor data are never downloaded

#	Condition	Status	Reason (abridged)
3	metric is annualized and `P` is `null`	`not_applicable`	annualization requires a regular frequency
4	metric uses the wealth path and any `r_t < −1`	`not_applicable`	compounded wealth path not meaningful
5	observations < the policy's mathematical minimum (for factor metrics `max(3, k + 2)` with *k* factors)	`insufficient_data`	requires at least *N* observations

The observation count is `n` for single-series metrics and the number of **aligned** observations for benchmark and factor metrics. For benchmark and factor metrics, check 3 uses the fund's `P`. Mathematical minimums per policy are listed in policies.md.

A metric is never withheld because its value is unfavorable or because its confidence is low.

---

9. Metric-function outcomes

metrics/engine.py `evaluate`:

Outcome of the metric function	Result
Raises `UndefinedMetric(reason)` (e.g. zero volatility, no losing periods)	`not_applicable`, `value = null`, `reason` = the metric's reason, `confidence = not_assessable`
Raises any other exception	`error`, value withheld, reason names the exception and asks for a bug report
Returns `+inf` or `-inf`	`not_applicable`, value withheld, reason: the result overflows because a denominator is effectively zero (below floating-point resolution)
Returns `NaN`	`error`, value withheld (treated as a defect)
Returns a finite number	passed to the confidence engine (10)

Calculations run with NumPy floating-point warnings suppressed; the finiteness check above is the guard. Moment-based shape statistics (skewness, kurtosis and everything built on them: Cornish-Fisher VaR/ES, Jarque-Bera, Sharpe inference) are undefined when returns are numerically constant: identical values, or a population variance `m2 <= 16 (eps · max|r|)^2` (below floating-point resolution relative to the size of the returns; metrics/context.py `_standardized_moment`). Results without a value still carry their assumptions. Each metric's undefined conditions are listed on its page in `interpretation/`.

---

10. Confidence and status algorithm

policy/confidence.py `assess` with policy/thresholds.py. Confidence describes the reliability of an estimate, not the quality of the investment. Ranks: `high > moderate > low`. "Cap at X" means `confidence = min(confidence, X)`.

Step 1 — base confidence from the metric's minimum-data policy. Inputs: `n` (aligned `n` for benchmark metrics), `years = n / P` (`null` if `P` is `null`), tail probability `p = 1 − 0.95 = 0.05`.

Policy basis	Rule
`none`	`high`
`years`	`years` null ⟶ `low`; `years < low` ⟶ `low`; `years < moderate` ⟶ `moderate`; else `high`
`observations`	`n < low` ⟶ `low`; `n < moderate` ⟶ `moderate`; else `high`
`tail_observations`	`e = n · p`; `e < low` ⟶ `low`; `e < moderate` ⟶ `moderate`; else `high`

Every reduction adds a warning stating the reason. Thresholds per policy are in policies.md.

Step 2 — modifiers, in code order.

#	Applies to	Trigger	Effect
a	volatility-based metrics	smoothing risk `high`	set to `low`, warning
b	volatility-based metrics	smoothing risk `moderate`	cap at `moderate`, warning
c	volatility-based metrics	zero-return share `≥ 0.10`	cap at `moderate`, warning

#	Applies to	Trigger	Effect
d	volatility-based metrics	`n ≥ 24`, skewness and excess kurtosis both defined, and (`skewness ≤ −1` or `excess kurtosis ≥ 3`)	warning only, no confidence change
e	wealth-path or annualized metrics	`missing_expected_periods > 0` or `missing_values > 0`	cap at `moderate`, warning
f	wealth-path metrics	a −100% return occurs	warning only
g	annualized metrics	`years` known and `years < 1` (`confidence.annualization_min_years`)	set to `low`, warning (annualizing a partial year)
h	annualized metrics	`frequency_source == "declared"`	warning only
i	benchmark metrics	benchmark covers `< 90%` of fund observations (`confidence.benchmark_min_coverage`)	cap at `moderate`, warning
j	benchmark metrics	alignment method starts with "by position"	warning only
k	factor metrics	factors cover `< 90%` of fund observations (`confidence.factor_min_coverage`)	cap at `moderate`, warning
l	factor metrics	alignment method starts with "by position"	warning only
m	factor metrics	smoothing risk `high` / `moderate`	set to `low` / cap at `moderate`, warning (smoothed returns bias loadings toward zero and alpha upward)

Thresholds in modifiers a–k are read from the canonical methodology (policy/defaults.py: `stale_returns`, `distribution_warning`, `confidence`), so they are embedded in every report.

Which metrics are volatility-based, wealth-path, annualized, benchmark-relative or supplementary is listed in the code-generated metric flags table (and on each metric page), so the membership can never drift from the registry.

Notes produced by the metric function itself (e.g. "the longest drawdown has not recovered") come first in the warning list. Warnings are de-duplicated, preserving first occurrence.

Step 3 — status.

Condition (first match)	Status
metric is flagged supplementary (metric flags)	`supplementary`
confidence is `low`	`low_confidence`
otherwise	`applicable`

A value is always shown when it exists.

Assumptions attached to each result come from the metric's declared assumption keys: annualization statement, risk-free rate 0%, MAR 0%, Omega threshold 0%, 95% historical tail convention, drawdown convention, sample standard deviation, the benchmark name and alignment, the OLS convention, i.i.d. asymptotics (statistical-confidence metrics), the parametric-moment convention (Gaussian / Cornish-Fisher VaR), the Sterling 10% adjustment and the factor model (model, factors, aligned periods, dependent variable, Newey-West lag). The keys per metric are listed in the metric flags table.

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11. Benchmark alignment criteria

compare/alignment.py `align_benchmark`. Benchmark returns are normalized with the same rules as the fund. Missing benchmark returns are never filled.

Case	Rule
Both dated, both regular, different frequencies	`BenchmarkAlignmentError` (no resampling in 0.1.0)
Both dated otherwise	Key type chosen once for both series: the calendar period (`M`, `Q`, `Y`) of the fund's frequency if it is monthly/quarterly/yearly, else of the benchmark's; if neither has one, exact normalized dates. Duplicate keys in either series →

Case	Rule
	`BenchmarkAlignmentError`. The intersection of keys is used, in fund order; dates are the fund's. Method: `calendar <frequency> period` or `exact date`.
Neither dated	Equal lengths ⟶ paired by position with a warning; different lengths ⟶ `BenchmarkAlignmentError`.
Exactly one dated	Equal lengths ⟶ the undated series is assumed to share the other's dates (warning); different lengths ⟶ `BenchmarkAlignmentError`.

Reported: fund, benchmark and aligned observation counts, `fund_only_periods = n_fund − n_aligned`, `benchmark_only_periods = n_bench − n_aligned`, `fund_coverage = n_aligned / n_fund`, overlap start/end. Warnings: fund observations without a benchmark return (when `n_aligned < n_fund`), and "no periods in common" when `n_aligned = 0`. Benchmark metrics use the fund's `P`.

Regression (`AlignedBenchmark.ols`): statsmodels OLS of aligned fund returns on a constant and aligned benchmark returns (risk-free 0%), conventional i.i.d. standard errors, residual degrees of freedom `n − 2`. Undefined (metric not applicable) when `n < 3` or the benchmark returns are constant. Standard errors, t-statistics and p-values are `null` when non-finite.

Factor returns (decision 0028)
factors/data.py `load_factor_set` and factors/regressions.py:

Step	Rule
Parsing	Same parsers as fund returns; one column per factor; duplicate names ⟶ `InputError`. A column named `RF` (case-insensitive, spaces ignored) is the risk-free rate, not a factor; only an RF column ⟶ `InputError`.
Frequency	Every column normalized separately (missing values dropped, never filled); different frequencies across columns ⟶ `InputError`.
Scale	The classifier of section 6 runs once on the pooled non-RF values (`%` labels in any column count as evidence); the result, or a declared `return_scale`, is applied to every column including RF.
Alignment	Both dated: key type as for benchmarks (calendar period of the fund's frequency if monthly/quarterly/yearly, else of the factors', else exact date); different regular frequencies ⟶ `FactorAlignmentError`; duplicate keys ⟶ `FactorAlignmentError`; the intersection of the fund's keys with the keys present in **every** column is used, in fund order. Otherwise equal lengths are paired by position (warning), else `FactorAlignmentError`.
Dependent variable	`y_t = r_t − RF_t` when an RF column exists, else `y_t = r_t`.
Estimation	statsmodels OLS with intercept; HAC covariance, Bartlett kernel, lag `floor(4 (m/100)^(2/9))`, small-sample factor `m/(m−k−1)`; Student t with `m − k − 1` df; 95% intervals.
Not identified (`not_applicable`)	a factor is constant over the aligned periods; the design matrix has rank `< k + 1` (collinear factors).
Exact fit	constant `y`, or residual sum of squares `≤ m (16 ε max|y|)²`: loadings as solved, standard errors 0, t / p `null`; R^2 undefined (constant `y`) or 1.
Fit warnings	maximum VIF `≥ 10` (`factor_model.vif_warning`); residual Ljung-Box p `< 0.05` with `m ≥ 10`.

Reported (`FactorAnalysis`): model label, factor names, dependent variable, factor return scale and evidence, alignment counts and overlap, and the fit (or the reason it could not be estimated).

---

12. Comparison criteria

compare/comparison.py:

- Accepted inputs: mapping name ⟶ returns, `DataFrame` (one series per column), `ReturnTable`, or a sequence (named by each item's `.name` or `Series i`).
- At least two investments and unique names, else `InputError`.
- Every investment is analyzed by the same `analyze()` call with the same optional benchmark and declarations: the methodology is identical.
- Nothing is ranked.
- Note emitted when observation frequencies differ.

- **Common-period analysis** is produced when all investments are dated, share one regular frequency and their (start, end) ranges differ: the intersection of their calendar-period (or exact-date) keys is taken; if it has at least 2 periods, each investment is re-analyzed on it (recorded as the transformation "restricted to the common period of all compared histories"; the fund subset keeps its original scale evidence and the benchmark is normalized once, as for the full-history reports), otherwise a note says the histories have fewer than 2 periods in common.

---

13. Interpretation rules

interpretation/templates.py `RULES`. Each metric gets its plain-language and professional text from the registry. When the metric has no value, its reason is the first context note. Then the rules below are applied in order. No rule grades an investment ("good", "excellent") or gives advice.

Rule	Applies to	Exact trigger	Note (summary)
`_smoothing`	`sharpe_ratio`, `sortino_ratio`, `information_ratio`, `annualized_volatility`, `volatility` with a value	smoothing risk `moderate` or `high`	reports lag-1 autocorrelation; treat the figure as an optimistic bound, not evidence of skill
`_hidden_tail`	`sharpe_ratio` with a value	skewness `≤ −1` or excess kurtosis `≥ 3` (each only if computed)	volatility-based ratios do not reflect such tail risk; compare with maximum drawdown and Expected Shortfall
`_alpha_significance`	`alpha` with a value and both p-value and t-statistic available	`p ≥ 0.05` / `p < 0.05`	not distinguishable from zero at 5% / significant under the single-index model, omitted factors may still explain it
`_short_path`	every drawdown-category metric with a value	`years` known and `years < 3`	drawdowns observed so far are likely milder than a longer history would reveal
`_hit_rate_smoothing`	`positive_period_pct` with a value	smoothing risk `high` and value `≥ 0.75`	very high hit rate with strong serial correlation is a pattern associated with smoothed returns
`_annualized_short`	`cagr`, `annualized_arithmetic_mean` with a value	`years` known and `years < 1`	annualizing extrapolates a partial year; GIPS prohibits annualized returns for periods under one year
`_sharpe_significance`	`probabilistic_sharpe_ratio` with a value	value `≥ 0.95` / value `< 0.95`	the Sharpe ratio is statistically above zero at 95% under i.i.d. assumptions (no guarantee of future performance) / not distinguishable from zero, the track record cannot separate skill from luck
`_track_record_sufficiency`	`minimum_track_record_length` with a value	`details.sufficient` true / false	the history is at least as long as the minimum track record length / states how many periods are missing (`details.additional_periods_needed`)
`_serial_correlation_adjustment`	`sharpe_ratio_autocorrelation_adjusted` with a value, when `sharpe_ratio` has a value	always; wording by `details.variance_ratio > 1` / `< 1` / `= 1`	states both annualized Sharpe ratios and the variance ratio: persistence (VR > 1) means square-root-of-time annualization overstates the conventional ratio's magnitude, reversals (VR < 1) that it understates it; the

Rule	Applies to	Exact trigger	Note (summary)
			adjustment rests on estimated autocorrelations

---

14. Report structure and display rules

report/tables.py and the renderers (terminal, markdown, html).

The canonical report has exactly 14 sections; none can be removed by a flag:

1. Input & methodology
2. Data-quality diagnostics — including the serial dependence & smoothing metrics (`autocorrelation_lag1–3`, `ljung_box_pvalue`, `zero_return_pct`)
3. Return summary
4. Risk summary
5. Drawdowns
6. Risk-adjusted performance
7. Tail behavior
8. Distribution
9. Consistency
10. Benchmark-relative analysis (without a benchmark: the reason and the list of metrics that are not applicable)
11. Factor analysis — without factor returns: the reason and the list of factor metrics that are not applicable; with them: the model summary, alignment, a coefficient table (alpha per period and every loading with Newey-West standard error, t, p and 95% interval), fit statistics and the `factors` category metrics
12. Statistical confidence — counts of confidence labels and statuses, then the `statistical` category metrics (`probabilistic_sharpe_ratio`, `sharpe_ratio_standard_error`, `minimum_track_record_length`, `mean_return_standard_error`); the Deflated Sharpe Ratio is stated as not computable (it needs the number of strategies tried)
13. Metric catalog (pointer)
14. Assumptions & caveats (methodology description and all warning/critical findings)

Charts and episode table (decision 0026, report/charts.py): the HTML report draws eight fixed charts in their sections (growth of 1, calendar-year returns and the monthly heatmap in 3; rolling one-year volatility in 4; the underwater chart in 5; the return distribution in 8; the rolling one-year return in 9; growth over the benchmark overlap in 10). A chart that cannot be drawn is replaced by its reason: returns below −100% (wealth-path charts), no regular frequency or fewer than `window + 1` observations (rolling charts), undated or non-monthly data (heatmap), fewer than 5 or identical returns (histogram), no benchmark or fewer than 2 aligned periods (benchmark chart). Section 5 of every renderer lists the `charts.episode_table_rows` (10) deepest drawdown episodes, deepest first, ties by earlier peak.

Display rounding (presentation only; stored values are raw `float64`):

Unit	Display
`decimal_return`	percent, 2 decimals (`-27.32%`)
`fraction`	percent, 1 decimal (`54.2%`)
`probability`	3 decimals
`periods`	integer, plus years in parentheses when `details.years` is available (`54 (4.5y)`)
`count`	integer
`ratio`, `beta`, `correlation`	2 decimals
no value	–

Status column: the confidence label when a value exists (`supplementary (<confidence>)` for supplementary metrics); otherwise `not applicable`, `insufficient data` or `error`.

Notes: each result's reason (when it has no value) and warnings are shown as numbered footnotes; identical texts share one number across the whole report.

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15. Determinism and provenance

report/json.py, report/builder.py:

- JSON is written with sorted keys and `allow_nan=False`; NumPy types are converted to plain Python and any non-finite float in nested data becomes `null`. No timestamps are embedded, so identical inputs give identical JSON.
- **Input fingerprint** `input_sha256`: SHA-256 of the compact JSON `{"dates": [...], "values": [...]}` (separators `,` and `:`), where dates are `YYYY-MM-DD` strings (or `null` if undated) and values are Python `repr` of each analyzed float (shortest round-trip form). `benchmark_sha256` uses the same construction for the benchmark.
- **Versions**: `package_version` is the software release; `methodology_version` identifies the conventions that produced the numbers. A package release that changes no result keeps the methodology version.
- `canonical` is `true` for every report produced by the public API.

Notation

Symbols used in the exact formulas and on every metric page. All quantities are computed in `float64` without intermediate rounding.

Returns and time

Symbol	Meaning
r_t	Simple decimal return of period t after normalization (0.01 = +1%). Percent inputs have been divided by 100; missing values have been removed, never filled.
$t = 1, \ldots, n$	Observation index in chronological order.
n	Number of analyzed observations (after trimming and removing missing values).
$r_{(1)} \le \ldots \le r_{(n)}$	Order statistics (returns sorted ascending).
P	Canonical periods per year for the detected frequency: 252 (business-day daily), 365 (calendar-day daily), 52, 12, 4, 1. Undefined for irregular or unknown frequency, in which case no annualized metric is computed. See policies.
$y = n/P$	Years of history used by the confidence policy.
$\#\{\cdot\}$	Number of elements of a set.

Moments

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad s = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2}, \qquad m_k = \frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^k$$

s is the sample standard deviation (denominator $n - 1$); m_k are central sample moments (denominator n) used by skewness and kurtosis.

Wealth and drawdowns

$$W_0 = 1, \qquad W_t = \prod_{i=1}^{t}(1 + r_i), \qquad M_t = \max_{0 \le u \le t} W_u, \qquad D_t = \frac{W_t}{M_t} - 1 \le 0$$

- $W_0 = 1$ is the wealth before the first return and counts as the first peak, so a loss in the first period is a drawdown.
- A **drawdown episode** k is a maximal run of consecutive periods $E_k \subseteq \{1..n\}$ with $D_t < 0$. Its **peak index** π_k is the period just before the run (possibly 0), its **trough** is the $\arg\min_{t \in E_k} D_t$ (first occurrence), its **depth** is $\delta_k = \min_{t \in E_k} D_t$, and its **recovery index** ρ_k is the first period after the run with $D_t = 0$, if any. K is the number of episodes.
- Episodes that have not recovered by period n are **ongoing**.
- If some $r_t < -1$, the wealth path is invalid and every wealth-path metric is `not_applicable`; if some $r_t = -1$, wealth is 0 from then on (total loss).

Quantiles

$Q_p(r)$ is the empirical quantile with linear interpolation (Hyndman & Fan definition 7, the default of NumPy and R):

$$h = (n-1)p + 1, \quad j = \lfloor h \rfloor, \quad Q_p(r) = r_{(j)} + (h - j)\left(r_{(j+1)} - r_{(j)}\right)$$

with $r_{(n+1)} := r_{(n)}$ when $j = n$.

Benchmark-relative quantities

Symbol	Meaning
b_t	Benchmark return in the same period as r_t, after alignment.
m	Number of aligned periods (intersection of fund and benchmark; never filled). All benchmark-relative sums run over these m periods, and $\bar{r}$, $\bar{b}$ are means over them.
$a_t = r_t - b_t$	Active (excess-over-benchmark) return.
$\bar{a}$, s_a	Mean and sample standard deviation of a_t.
$\hat{\alpha}, \hat{\beta}$	OLS intercept and slope of $r_t = \alpha + \beta b_t + \varepsilon_t$.
$\hat{\varepsilon}_t$	OLS residuals $r_t - \hat{\alpha} - \hat{\beta} b_t$.

Factor-model quantities

Symbol	Meaning
$f_{j,t}$	Return of supplied factor $j = 1, \ldots, k$ in the same period as r_t, after alignment (never filled).
k	Number of supplied factors (the risk-free column is not a factor).
m	Number of periods where the fund and every factor (and the RF column, if supplied) have a return. Factor-model sums run over these periods.
r_t^{RF}	Value of a supplied factor-file column named `RF` (risk-free rate); taken as 0 when no such column is supplied.
$y_t = r_t - r_t^{\mathrm{RF}}$	Dependent variable of the factor regression.
$\hat{\alpha}, \hat{\beta}_j$	OLS intercept and loadings of $y_t = \alpha + \sum_j \beta_j f_{j,t} + \varepsilon_t$.
X	$m \times (k+1)$ design matrix: a column of ones and the factor returns.
$\hat{\Gamma}_\ell$	$\frac{1}{m} \sum_{t=\ell+1}^{m} \hat{\varepsilon}_t \hat{\varepsilon}_{t-\ell}\, x_t x_{t-\ell}^\top$ (and its transpose for $\ell < 0$), with x_t the t-th row of X.
q	Newey-West lag $\lfloor 4(m/100)^{2/9} \rfloor$.

Canonical constants (methodology 0.1.0)

Symbol	Value	Meaning
r_f	0	Risk-free rate per period.
τ	0	Minimum acceptable return (Sortino target) per period.
L	0	Omega threshold per period.
p	0.05	Tail probability for 95% VaR / ES.

The authoritative values are in policies (generated from the code).

Policies, thresholds and constants

All numeric policy values of methodology 0.1.0, read directly from `src/openfactsheet/policy/defaults.py` and `src/openfactsheet/policy/thresholds.py`. They are conventions, versioned with the methodology and never tuned per fund. The rules that use them are described in criteria.

Minimum-data policies

Below the mathematical minimum a metric is `insufficient_data`; the statistical thresholds set the base confidence (low / moderate / high).

Policy	Mathematical minimum	Basis	Low below	Moderate below	Metrics
`observed_point`	1 obs	none (always high)	—	—	`cumulative_return`, `best_period`, `worst_period`, `current_drawdown`, `zero_return_pct`
`path_risk`	1 obs	years of history	3	5	`max_drawdown`, `average_drawdown`, `longest_drawdown_duration`, `drawdown_count`, `time_underwater_pct`,

Policy	Mathematical minimum	Basis	Low below	Moderate below	Metrics
					ulcer_index, pain_index, median_drawdown, average_drawdown_duration, max_peak_to_trough_duration, average_recovery_duration, longest_winning_streak, longest_losing_streak, best_calendar_year, worst_calendar_year, positive_calendar_years_pct, best_rolling_year, worst_rolling_year, positive_rolling_year_pct
location	1 obs	years of history	3	5	cagr, arithmetic_mean, geometric_mean_return, annualized_arithmetic_mean, median_return, positive_period_pct, negative_period_pct
dispersion	2 obs	years of history	3	5	volatility, annualized_volatility, mean_absolute_deviation, interquartile_range, downside_deviation, semi_deviation, downside_potential, gaussian_var_95, mean_return_standard_error
ratio_standard	2 obs	years of history	3	5	gain_to_pain_ratio, payoff_ratio, profit_factor, sharpe_ratio, sortino_ratio, calmar_ratio, martin_ratio, omega_ratio, kappa_3_ratio, upside_potential_ratio, burke_ratio, sterling_ratio, pain_ratio, recovery_factor
skewness	3 obs	observations	60	120	skewness, jarque_bera_pvalue
kurtosis	4 obs	observations	120	240	modified_var_95, modified_es_95, excess_kurtosis
tail	2 obs	expected tail observations	5	10	drawdown_at_risk_95, cdar_95, edar_95, var_historical_95, expected_shortfall_historical_95, tail_ratio, evar_95

Policy	Mathematical minimum	Basis	Low below	Moderate below	Metrics
autocorrelation	5 obs	observations	36	60	autocorrelation_lag1, autocorrelation_lag2, autocorrelation_lag3
ljung_box	10 obs	observations	36	60	ljung_box_pvalue
asymptotic_inference	4 obs	observations	36	60	probabilistic_sharpe_ratio, sharpe_ratio_standard_error, minimum_track_record_length
benchmark_regression	3 obs	years of history	3	5	beta, alpha, r_squared, correlation, treynor_ratio, residual_volatility, appraisal_ratio, bull_beta, bear_beta, timing_ratio
benchmark_simple	2 obs	years of history	3	5	tracking_error, active_return, information_ratio, upside_capture, downside_capture, capture_ratio, batting_average, m_squared
long_horizon_dependence	4 obs	years of history	5	10	sharpe_ratio_autocorrelation_adjusted, variance_ratio_1y
factor_regression	3 obs	years of history	3	5	factor_alpha, factor_alpha_t_statistic, factor_r_squared, factor_residual_volatility

Policy descriptions:

- observed_point: Observed fact about the supplied history; any non-empty history suffices.
- path_risk: Path statistics (drawdowns, streaks) are computable from one observation but are unrepresentative below 3 years (low) and 5 years (moderate); longer histories tend to reveal deeper drawdowns.
- location: Mean-type estimates: low confidence below 3 years, moderate below 5 years.
- dispersion: Requires at least 2 observations; low confidence below 3 years, moderate below 5.
- ratio_standard: Risk-adjusted ratios: at least 2 observations; low confidence below 3 years, moderate below 5 years.
- skewness: Requires at least 3 observations; standard error is roughly sqrt(6/n), so low confidence below 60 and moderate below 120 observations.
- kurtosis: Requires at least 4 observations; kurtosis estimates are highly unstable, so low confidence below 120 and moderate below 240 observations.
- tail: Requires at least 2 observations; low confidence when fewer than 5 observations are expected in the 5% tail (n < 100), moderate when fewer than 10 (n < 200).
- autocorrelation: Requires at least 5 observations; standard error is roughly 1/sqrt(n), so low confidence below 36 and moderate below 60 observations.
- ljung_box: Requires at least 10 observations for a 3-lag test; low below 36, moderate below 60.
- asymptotic_inference: Asymptotic Sharpe-ratio inference needs at least 4 observations for the moments it uses; the normal approximation is poor below 36 observations (low) and 60 (moderate).
- benchmark_regression: Requires at least 3 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.
- benchmark_simple: Requires at least 2 aligned observations; low confidence below 3 years of overlap, moderate below 5 years.

- `long_horizon_dependence`: One-year-horizon serial-correlation statistics need at least two years of data (undefined below); their standard error is about sqrt(4 / (3 y)) for y years, so confidence is low below 5 years and moderate below 10 years.
- `factor_regression`: Requires at least 3 aligned observations and at least k + 2 for k factors (k + 1 coefficients and one residual degree of freedom); low confidence below 3 years of overlap, moderate below 5 years.

Metric flags

Flags that decide which applicability checks and confidence modifiers apply to each metric (see criteria).

Metric	Category	Volatility-based	Wealth path	Annualized	Benchmark	Factors	Supplementary	Signed loss	Minimum-data policy	Assumption keys
cumulative_return	returns		yes						observed_point	—
cagr	returns		yes	yes					location	annualization
arithmetic_mean	returns								location	—
geometric_mean_return	returns		yes						location	—
annualized_arithmetic_mean	returns			yes					location	annualization
median_return	returns								location	—
best_period	returns								observed_point	—
worst_period	returns							yes	observed_point	—
gain_to_pain_ratio	returns								ratio_standard	—
payoff_ratio	returns								ratio_standard	—
profit_factor	returns								ratio_standard	—
volatility	volatility	yes							dispersion	sample_std
annualized_volatility	volatility	yes		yes					dispersion	sample_std, annualization
mean_absolute_deviation	volatility	yes							dispersion	—
interquartile_range	volatility	yes							dispersion	—
downside_deviation	volatility	yes		yes					dispersion	mar, annualization
semi_deviation	volatility	yes		yes					dispersion	annualization
downside_potential	volatility								dispersion	mar
max_drawdown	drawdown		yes					yes	path_risk	drawdown
average_drawdown	drawdown		yes					yes	path_risk	drawdown
current_drawdown	drawdown		yes					yes	observed_point	drawdown
longest_drawdown_duration	drawdown		yes						path_risk	drawdown
drawdown_count	drawdown		yes						path_risk	drawdown
time_underwater_pct	drawdown		yes						path_risk	drawdown
ulcer_index	drawdown		yes						path_risk	drawdown
pain_index	drawdown		yes						path_risk	drawdown
median_drawdown	drawdown		yes					yes	path_risk	drawdown
average_drawdown_duration	drawdown		yes						path_risk	drawdown
max_peak_to_	drawdown		yes						path_risk	drawdown

Metric	Category	Volatility-based	Wealth path	Annualized	Benchmark	Factors	Supplementary	Signed loss	Minimum-data policy	Assumption keys
trough_duration										
average_recovery_duration	drawdown		yes						path_risk	drawdown
drawdown_at_risk_95	drawdown		yes					yes	tail	drawdown, tail
cdar_95	drawdown		yes					yes	tail	drawdown, tail
edar_95	drawdown		yes					yes	tail	drawdown, entropic
sharpe_ratio	risk_adjusted	yes		yes					ratio_standard	risk_free, sample_std, annualization
sortino_ratio	risk_adjusted	yes		yes					ratio_standard	risk_free, mar, annualization
calmar_ratio	risk_adjusted		yes	yes					ratio_standard	annualization, drawdown, risk_free
martin_ratio	risk_adjusted		yes	yes					ratio_standard	annualization, drawdown, risk_free
omega_ratio	risk_adjusted								ratio_standard	omega_threshold
kappa_3_ratio	risk_adjusted	yes							ratio_standard	mar
upside_potential_ratio	risk_adjusted	yes							ratio_standard	mar
burke_ratio	risk_adjusted		yes	yes					ratio_standard	annualization, drawdown, risk_free
sterling_ratio	risk_adjusted		yes	yes					ratio_standard	annualization, drawdown, sterling_excess, risk_free
pain_ratio	risk_adjusted		yes	yes					ratio_standard	annualization, drawdown, risk_free
recovery_factor	risk_adjusted		yes						ratio_standard	drawdown
sharpe_ratio_autocorrelation_adjusted	risk_adjusted			yes					long_horizon_dependence	risk_free, sample_std, annualization
var_historical_95	tail							yes	tail	tail
expected_shortfall_historical_95	tail							yes	tail	tail
tail_ratio	tail								tail	—
gaussian_var_95	tail	yes					yes	yes	dispersion	parametric
modified_var_95	tail	yes					yes	yes	kurtosis	parametric
modified_es_95	tail	yes					yes	yes	kurtosis	parametric
evar_95	tail							yes	tail	entropic
skewness	distribution								skewness	—
excess_kurtosis	distribution								kurtosis	—

Metric	Category	Volatility-based	Wealth path	Annualized	Benchmark	Factors	Supplementary	Signed loss	Minimum-data policy	Assumption keys
jarque_bera_pvalue	distribution						yes		skewness	—
positive_period_pct	consistency								location	—
negative_period_pct	consistency								location	—
longest_winning_streak	consistency								path_risk	—
longest_losing_streak	consistency								path_risk	—
best_calendar_year	consistency		yes						path_risk	—
worst_calendar_year	consistency		yes					yes	path_risk	—
positive_calendar_years_pct	consistency		yes						path_risk	—
best_rolling_year	consistency		yes	yes					path_risk	annualization
worst_rolling_year	consistency		yes	yes				yes	path_risk	annualization
positive_rolling_year_pct	consistency		yes	yes					path_risk	annualization
autocorrelation_lag1	dependence								autocorrelation	—
autocorrelation_lag2	dependence								autocorrelation	—
autocorrelation_lag3	dependence								autocorrelation	—
ljung_box_pvalue	dependence						yes		ljung_box	—
zero_return_pct	dependence								observed_point	—
variance_ratio_1y	dependence						yes		long_horizon_dependence	—
beta	benchmark				yes				benchmark_regression	benchmark, regression
alpha	benchmark			yes	yes				benchmark_regression	benchmark, regression, annualization
r_squared	benchmark				yes				benchmark_regression	benchmark, regression
correlation	benchmark				yes				benchmark_regression	benchmark
tracking_error	benchmark	yes		yes	yes				benchmark_simple	benchmark, sample_std, annualization
active_return	benchmark			yes	yes				benchmark_simple	benchmark, annualization
information_ratio	benchmark	yes		yes	yes				benchmark_simple	benchmark, sample_std, annualization
treynor_ratio	benchmark			yes	yes				benchmark_regression	benchmark, regression,

Metric	Category	Volatility-based	Wealth path	Annualized	Benchmark	Factors	Supplementary	Signed loss	Minimum-data policy	Assumption keys
										risk_free, annualization
upside_capture	benchmark			yes	yes				benchmark_simple	benchmark, annualization
downside_capture	benchmark			yes	yes				benchmark_simple	benchmark, annualization
capture_ratio	benchmark			yes	yes				benchmark_simple	benchmark, annualization
batting_average	benchmark				yes				benchmark_simple	benchmark
residual_volatility	benchmark	yes		yes	yes				benchmark_regression	benchmark, regression, annualization
appraisal_ratio	benchmark	yes		yes	yes				benchmark_regression	benchmark, regression, risk_free, annualization
m_squared	benchmark	yes		yes	yes				benchmark_simple	benchmark, risk_free, sample_std, annualization
bull_beta	benchmark				yes				benchmark_regression	benchmark, regression
bear_beta	benchmark				yes				benchmark_regression	benchmark, regression
timing_ratio	benchmark				yes				benchmark_regression	benchmark, regression
factor_alpha	factors			yes		yes			factor_regression	factor_model, annualization
factor_alpha_t_statistic	factors					yes			factor_regression	factor_model
factor_r_squared	factors					yes			factor_regression	factor_model
factor_residual_volatility	factors			yes		yes			factor_regression	factor_model, annualization
probabilistic_sharpe_ratio	statistical	yes							asymptotic_inference	iid_asymptotics, risk_free
sharpe_ratio_standard_error	statistical	yes		yes					asymptotic_inference	iid_asymptotics, annualization, risk_free
minimum_track_record_length	statistical	yes							asymptotic_inference	iid_asymptotics, risk_free
mean_return_standard_error	statistical	yes							dispersion	—

Signed loss: the value is a loss stated as a negative return (for example a maximum drawdown of −25%). Its `direction` is `higher_is_better` because −10% is a smaller loss than −25%; the documentation says “smaller loss is better”.

Annualization factors

Frequency	Periods per year P
daily business	252
daily calendar	365
weekly	52

Frequency	Periods per year P
monthly	12
quarterly	4
yearly	1

Irregular: not annualized. Unknown: not annualized unless the caller declares the observation frequency. Scaling: square-root-of-time for volatility; linear for arithmetic means.

Return-scale classifier bounds

A series is confidently decimal when its median absolute non-zero value is at most `upper / divisor`; percent when it exceeds `upper`; the zone in between is ambiguous (percent only if some value is below -1, otherwise the caller must declare the scale).

Confident divisor: 4.

Frequency	Decimal upper bound	Confident-decimal bound
daily	0.05	0.0125
weekly	0.1	0.025
monthly	0.2	0.05
quarterly	0.3	0.075
yearly	0.6	0.15
irregular	0.3	0.075
unknown	0.3	0.075

Diagnostic and confidence thresholds

Rule	Value	Source key
Smoothing risk assessed from	12 observations	`smoothing.min_observations`
High smoothing risk	rho_1 >= 0.3 and z >= 1.96	`smoothing.high`
Moderate smoothing risk	rho_1 >= 0.15 and z >= 1.645	`smoothing.moderate`
Excessive zero returns	share >= 10%	`stale_returns.zero_share_warning`
Repeated returns	run >= 3 identical values	`stale_returns.identical_run_warning`
Suspicious large return	robust z > 5	`outliers.robust_z_threshold`
Distribution-shape warning	n >= 24, skewness <= -1 or excess kurtosis >= 3	`distribution_warning`
Ljung-Box lags	3	`dependence.ljung_box_lags`
Benchmark coverage cap	overlap < 90% of fund observations ⟶ at most moderate	`confidence.benchmark_min_coverage`
Factor coverage cap	overlap < 90% of fund observations ⟶ at most moderate	`confidence.factor_min_coverage`
Short annualization	years < 1 ⟶ low	`confidence.annualization_min_years`

Complete canonical assumptions (machine-readable)

This is the `methodology` object embedded in every JSON report.


```
{
  "conventions": {
    "annual_return": {
      "description": "geometric (CAGR): (prod(1 + r))^(periods_per_year / n) - 1; the arithmetic mean x
periods_per_year is reported separately",
      "method": "geometric"
    },
    "annualization": {
      "irregular": "not annualized",
      "periods_per_year": {
        "daily_business": 252.0,
        "daily_calendar": 365.0,
        "monthly": 12.0,
        "quarterly": 4.0,
        "weekly": 52.0,
        "yearly": 1.0
      },
      "scaling": "square-root-of-time for volatility; linear for arithmetic means",
      "unknown": "not annualized unless the caller declares the observation frequency"
    },
    "benchmark": {
```

```
      "alignment": "intersection of periods; never filled",
      "alpha_annualization": "arithmetic: per-period intercept x periods_per_year",
      "appraisal_ratio": "annualized OLS alpha / annualized residual volatility",
      "batting_average": "share of aligned periods with fund return strictly above benchmark",
      "capture": "geometric, annualized over up (benchmark > 0) or down (benchmark < 0) periods",
      "capture_ratio": "upside capture / downside capture; undefined unless downside capture > 0",
      "description": "periods aligned on their intersection (monthly/quarterly/yearly by calendar period, daily/
weekly by exact date); missing benchmark returns are never filled",
      "information_ratio": "annualized arithmetic mean active return / tracking error",
      "m_squared": "aligned Sharpe ratio x benchmark volatility, arithmetic annualization",
      "regime_betas": "OLS with intercept on benchmark > 0 (bull) or < 0 (bear) periods",
      "regression": "OLS of fund on benchmark returns (risk-free = 0)",
      "residual_volatility": "OLS regression standard error (m - 2 dof) x sqrt(periods_per_year)",
      "timing_ratio": "bull beta / bear beta; undefined unless bear beta > 0"
    },
    "calendar_years": {
      "daily_weekly_edge_tolerance_days": 7,
      "monthly": "all 12 calendar months observed",
      "quarterly": "all 4 calendar quarters observed",
      "yearly": "the year's observation"
    },
    "charts": {
      "bar_baseline": "bars always grow from zero; the value axis always includes zero",
      "benchmark": "compared only over the aligned overlap, both rebased to 1.0 at its start",
      "episode_table": "deepest drawdown episodes first (ties: earliest peak); all in JSON",
      "episode_table_rows": 10,
      "heatmap_scale": "diverging, symmetric around zero at the largest absolute month",
      "histogram_bins": "Freedman-Diaconis width clipped to 10 to 40 bins over the range; edges then aligned at
zero, which can add one bin",
      "histogram_max_bins": 40,
      "histogram_min_bins": 10,
      "history": "every chart spans the full analyzed history; no start date is chosen",
      "partial_years": "shown outlined and labelled partial; never styled as full years",
      "rolling_windows": "the one-year windows of the rolling metrics (rolling.*)",
      "wealth_scale": "linear, starting at 1.0 one period before the first return"
    },
    "confidence": {
      "annualization_min_years": 1.0,
      "benchmark_min_coverage": 0.9,
      "factor_min_coverage": 0.9
    },
    "dependence": {
      "acf_lags": [
        1,
        2,
        3
      ],
      "description": "sample autocorrelation (full-sample variance denominator)",
      "ljung_box_lags": 3
    },
    "dispersion": {
      "ddof": 1,
      "description": "sample standard deviation (n - 1 denominator)"
    },
    "distribution": {
      "description": "bias-corrected sample estimators G1 and G2 (as pandas / Excel)",
      "kurtosis": "bias-corrected sample excess kurtosis G2",
      "skewness": "adjusted Fisher-Pearson G1 (bias-corrected sample skewness)"
    },
    "distribution_warning": {
      "excess_kurtosis_at_or_above": 3.0,
      "min_observations": 24,
      "skewness_at_or_below": -1.0
    },
    "downside": {
      "denominator": "full_sample",
      "description": "target (MAR) = 0% per period; downside deviation = sqrt(sum(min(r - MAR, 0)^2) / n) using
all n observations",
      "mar": 0.0
    },
    "drawdown": {
      "description": "wealth index starts at 1 before the first return, so a first-period loss is a drawdown;
drawdowns reported as signed returns (negative = loss)",
      "initial_wealth_is_peak": true
    },
```

```
    "drawdown_risk": {
      "confidence": 0.95,
      "quantile_method": "linear"
    },
    "entropic": {
      "confidence": 0.95
    },
    "factor_model": {
      "alignment": "periods where the fund and every factor have a return; never filled",
      "alpha_annualization": "arithmetic: per-period intercept x periods_per_year",
      "ci_level": 0.95,
      "covariance": "Newey-West HAC, Bartlett kernel, small-sample factor m / (m - k - 1)",
      "estimator": "OLS with intercept of fund (excess) returns on the supplied factors",
      "hac_lag_rule": "floor(4 * (m / 100) ** (2 / 9))",
      "inference": "Student t with m - k - 1 degrees of freedom",
      "ljung_box_min_observations": 10,
      "risk_free": "a factor column named RF is subtracted from fund returns; without one the dependent variable
is the fund return (risk-free rate 0%)",
      "vif_warning": 10.0
    },
    "lower_partial_moments": {
      "denominator": "full_sample",
      "kappa_order": 3,
      "target": 0.0
    },
    "mean_inference": {
      "ci_level": 0.95
    },
    "missing_data": {
      "description": "missing returns are never replaced by zero or forward-filled; leading and trailing gaps are
trimmed, interior gaps are disclosed and the remaining observations are analyzed"
    },
    "modified_es": {
      "confidence": 0.95,
      "moments": "population"
    },
    "ocr": {
      "default_engine": "rapidocr",
      "dpi": 300,
      "header_inference": "up to three month headers the engine misread are placed by the column spacing of the
others (recorded)",
      "header_min_months": 9,
      "line_grouping": "words overlapping vertically by at least half the smaller height",
      "min_cell_confidence": 0.9,
      "numeric_repairs": {
        "I": "1",
        "O": "0",
        "l": "1",
        "o": "0",
        "|": "1"
      },
      "require_reconciled_totals": true
    },
    "omega": {
      "description": "threshold = 0% per period, simple (empirical) estimator",
      "threshold": 0.0
    },
    "outliers": {
      "description": "robust z-score |r - median| / (1.4826 * MAD) above 5 is flagged (never removed)",
      "robust_z_threshold": 5.0
    },
    "parametric_var": {
      "confidence": 0.95,
      "moments": "population"
    },
    "return_scale": {
      "confident_divisor": 4.0,
      "decimal_median_abs_upper": {
        "daily": 0.05,
        "irregular": 0.3,
        "monthly": 0.2,
        "quarterly": 0.3,
        "unknown": 0.3,
        "weekly": 0.1,
        "yearly": 0.6
```

```
      }
    },
    "return_type": "simple",
    "risk_free": {
      "description": "0% per period in methodology 0.1.0 for all excess-return metrics (Sharpe, Sortino, Calmar,
Martin, alpha, Treynor). Ratios are therefore total-return ratios; this is disclosed on every affected metric.",
      "rate": 0.0
    },
    "rolling": {
      "step_periods": 1,
      "window_years": 1
    },
    "serial_correlation": {
      "horizon_years": 1,
      "min_years": 2,
      "sharpe_adjustment": "Lo (2002): SR_per_period * q / sqrt(q + 2 * sum_{k=1}^{q-1} (q-k) rho_k)",
      "variance_ratio": "1 + 2 * sum_{k=1}^{q-1} (1 - k/q) * rho_k"
    },
    "sharpe_inference": {
      "ci_level": 0.95,
      "min_track_record_confidence": 0.95,
      "moments": "population skewness g1 and non-excess kurtosis m4 / m2^2",
      "reference_sharpe": 0.0
    },
    "smoothing": {
      "high": {
        "lag1_min": 0.3,
        "z_min": 1.96
      },
      "min_observations": 12,
      "moderate": {
        "lag1_min": 0.15,
        "z_min": 1.645
      },
      "reference": "Getmansky, Lo & Makarov (2004), Journal of Financial Economics 74(3)"
    },
    "stale_returns": {
      "identical_run_warning": 3,
      "zero_share_warning": 0.1
    },
    "sterling": {
      "excess": 0.1
    },
    "tail": {
      "description": "historical 95% VaR = 5% empirical quantile (linear interpolation, R type 7); 95% ES = mean
of returns at or below that quantile; both reported as signed returns (negative = loss)",
      "es_confidence": 0.95,
      "quantile_method": "linear",
      "sign": "signed return (negative = loss)",
      "var_confidence": 0.95
    }
  },
  "name": "OpenFactsheet Methodology 0.1.0",
  "version": "0.1.0"
}
```


JSON report schema (schema version 0.1)

`report.to_json()` / `report.to_dict()` produce one JSON object per report (report/json.py `report_to_dict`). The output is deterministic: keys are sorted, no timestamps are embedded, floats are written at full precision, and `NaN`/`Infinity` can never appear (any non-finite float inside nested data is written as `null`).

Complete, always-current examples are the golden reports in `tests/golden/` (e.g. `monthly_fund_10y.json`); the test suite fails if the code and those files disagree.

Conventions used throughout:

- **Returns** are simple decimal returns: `0.01` = +1%.
- **Loss measures are signed**: drawdowns, VaR and Expected Shortfall are negative numbers for losses (`-0.27` = -27%). Ulcer and Pain indices are non-negative magnitudes.
- **Dates** are `YYYY-MM-DD` strings, or `null` when the input is undated.
- **Periods** are observation periods of the supplied frequency (e.g. months).

`ofs.compare(...).to_json()` wraps several reports: `{"reports": {name: report}, "common_period": [start, end] | null, "common_period_reports": {name: report} | null, "notes": [string]}`.

Disclaimer. The JSON output is data for other programs, so it carries no disclaimer field and its schema is unchanged by the disclosure work; a notice is never inserted into the returns or metric values. The same terms apply to it as to every other output: *For information and education only — not investment advice. Investing involves risk; past performance does not predict future results. Verify results and consult a qualified financial adviser.* The full text is DISCLAIMER.md, reachable from the `documentation.repository` link of every report. Applications that display OpenFactsheet results to people should show this notice next to them.

Top level

Key	Type	Meaning
`schema_version`	string	Version of this JSON layout (`"0.1"`). Independent of package and methodology versions.
`canonical`	boolean	`true` for reports produced by the public API with the canonical methodology.
`name`	string	Report / investment name.
`methodology`	object	The full methodology used (methodology).
`input`	object	What OpenFactsheet believes it received (input).
`returns`	array	Exactly the decimal returns analyzed (returns).
`data_quality`	object	Data-quality diagnostics (data_quality).
`metrics`	array	One entry per canonical metric, in canonical order (metrics).
`drawdown_episodes`	array	All drawdown episodes (drawdown_episodes).
`benchmark`	object | null	Benchmark alignment and input summaries; `null` without a benchmark (benchmark).
`document`	object | null	For PDF / image input: the fund profile read from the document (decision 0032): `source`, `name`, `objective`, `benchmarks`, `currency`, `launch_date` (each `{value, confidence, evidence: {page, line, snippet, rule}, note}`), `identifiers` (`kind`, `value`, `valid` — true / false / null without a check digit —, `linked_isin`, `evidence`), `share_classes` (`label`, `isins`, `currency`, `page`, `snippet`), `attribution` (`status`: attributed / ambiguous / no_share_classes, `share_class`, `rule`, `evidence`, `candidates`; decision 0033) and `notes`; `null` otherwise.
`factor_analysis`	object | null	Factor-model regression; `null` without factor returns (factor_analysis).
`interpretations`	object	Map metric id ⟶ interpretation (interpretations).
`provenance`	object	Audit information (provenance).
`documentation`	object	Where the documentation of the version that produced the report lives (documentation); every metric also carries its own topic URL (decision 0035).

---

methodology

models/assumptions.py `Methodology.to_dict`.

Key	Type	Meaning
`version`	string	Methodology version, e.g. `"0.1.0"`.
`name`	string	`"OpenFactsheet Methodology <version>"`.
`conventions`	object	The complete canonical assumptions registry from policy/defaults.py, reproduced in full in policies.

---

input

models/input.py `InputDiagnostics.to_dict`. The benchmark's own input summary has the same shape.

Key	Type	Meaning
`name`	string	Series name.
`source_type`	string	`list`, `tuple`, `numpy.ndarray`, `pandas.Series`, `pandas.DataFrame`, `csv`, `excel`, `pasted text`, with `(monthly table)` appended for reshaped wide tables.

Key	Type	Meaning
`observations`	integer	Number of analyzed returns.
`start`, `end`	string | null	First and last analyzed dates; `null` if undated.
`years`	number | null	`observations / periods_per_year`; `null` when not annualizable.
`frequency`	string	`daily`, `weekly`, `monthly`, `quarterly`, `yearly`, `irregular`, `unknown`.
`frequency_source`	string	`inferred` (from dates), `declared` (by the caller: undated input, or a single timestamp), `none`.
`frequency_evidence`	string	Human-readable reason for the frequency decision.
`periods_per_year`	number | null	Annualization factor P; `null` for irregular/unknown frequency.
`return_scale`	string	`detected_decimal`, `detected_percent`, `declared_decimal`, `declared_percent`.
`return_scale_evidence`	string	Reason for the scale decision.
`missing_values`	integer	Interior missing values dropped (never filled).
`leading_missing_trimmed`, `trailing_missing_trimmed`	integer	Missing values trimmed at the ends.
`missing_expected_periods`	integer | null	Calendar gaps plus interior missing values; `null` when not assessable (business-day daily, irregular, unknown, undated).
`duplicate_periods`	integer	Pairs of observations in the same calendar period (these make the frequency irregular). Identical timestamps are rejected before this point with `InputError`.
`was_sorted`	boolean	`false` if OpenFactsheet had to sort the timestamps.
`transformations`	array of string	Every transformation applied (sorting, trimming, percent→decimal, table reshaping, timezone removal, ...).
`warnings`	array of string	Input-level warnings.

---

returns

Dated input: `[{"date": "YYYY-MM-DD", "value": number}, ...]`. Undated input: `[{"period": 1, "value": number}, ...]` (1-based position). Values are decimal returns after all transformations listed in `input.transformations`.

---

data_quality

models/diagnostics.py `DataQualityReport.to_dict`. Trigger rules for every field are in criteria §7.

Key	Type	Meaning
`observations`	integer	`n`.
`years`	number | null	History length in years.
`frequency`	string	Observation frequency.
`missing_values`	integer	Interior missing values.
`missing_expected_periods`	integer | null	Missing periods (see input).
`duplicate_periods`	integer	Observations sharing a calendar period.
`zero_return_count`	integer	Returns exactly equal to 0.
`zero_return_pct`	number	`zero_return_count / n`, 0–1.
`longest_identical_run`	integer	Longest run of identical consecutive returns.
`impossible_return_count`	integer	Returns below -1 (-100%).
`total_loss`	boolean	Whether a return of exactly -1 occurs.
`suspicious_large_return_count`	integer	Robust outliers (robust z > 5); kept in calculations.
`autocorrelation`	object	`lag1`, `lag2`, `lag3` (number | null) and `approx_95pct_significance_bound` (`1.96/sqrt(n)`, number | null).
`ljung_box_pvalue`	number | null	Ljung-Box p-value with 3 lags; `null` if `n < 10` or returns are constant.
`smoothing_risk`	string	`none`, `moderate`, `high`, `not_assessable`.

Key	Type	Meaning
`smoothing_evidence`	string	Values and rule behind `smoothing_risk`.
`findings`	array	`{"id": string, "severity": "info" | "warning" | "critical", "message": string, "value": number | null}`; every id is listed in criteria §7.1.

---

metrics

models/metric.py `MetricResult.to_dict`. Every canonical metric appears exactly once, in canonical order, whether or not it could be computed.

Key	Type	Meaning
`id`	string	Stable metric identifier (see the metric catalog).
`name`	string	Display name.
`category`	string	`returns`, `volatility`, `drawdown`, `risk_adjusted`, `tail`, `distribution`, `consistency`, `dependence`, `benchmark`, `factors`, `statistical`.
`value`	number | null	Raw `float64` value (never rounded, never non-finite); `null` when not computable.
`unit`	string	`decimal_return`, `fraction` (0–1), `ratio`, `periods`, `count`, `correlation`, `probability`, `beta`.
`status`	string	`applicable`, `low_confidence`, `supplementary`, `not_applicable`, `insufficient_data`, `error`.
`confidence`	string	`high`, `moderate`, `low`; `not_assessable` when `value` is `null`. Describes estimation reliability, not investment quality.
`direction`	string	`higher_is_better`, `lower_is_better`, `context_dependent`, `neutral`. Applies to the signed value. Losses are negative numbers (drawdowns, VaR, expected shortfall, worst period), so for them `higher_is_better` means *smaller loss is better*: −10% beats −25%. These metrics are flagged `signed_loss` in `ofs.metric_catalog()`.
`nature`	string	`observed`, `derived`, `estimated`, `model_dependent`.
`observations`	integer	Observations used: `n`, or the aligned count `m` for benchmark and factor metrics.
`formula_id`	string	`<id>@v<version>`; changes when the formula changes.
`methodology_version`	string	Methodology that produced the value.
`documentation`	string	URL of the reference-manual topic defining this metric (formula, assumptions, interpretation, references), for the version that produced the value. Present even when `value` is `null` (decision 0035).
`assumptions`	array of string	Assumptions behind this value (annualization factor, risk-free rate, targets, benchmark, …). Present even when `value` is `null`.
`warnings`	array of string	Data-specific caveats, de-duplicated.
`reason`	string | null	Why there is no value; always present when `value` is `null`.
`details`	object	Metric-specific supporting values (metric details); `{}` for most metrics.
`reference_values`	object | null	Reserved for reference-implementation values; `null` in methodology 0.1.0.

Status and confidence rules: criteria §8–10 and each metric's page in `interpretation/`.

Metric details

Keys emitted by metric functions in src/openfactsheet/metrics/. Metrics not listed emit `{}`.

Metric	Key	Type	Meaning
`best_period`, `worst_period`	`date`	string | null	Date of the extreme observation (first occurrence).
	`position`	integer	1-based position in the analyzed series.
`max_drawdown`	`peak_date`, `trough_date`	string | null	Peak (`null` if the peak is the inception point or the input is undated) and trough dates of the deepest episode.
	`recovery_date`	string | null	Recovery date; `null` if not recovered (the result then carries an open-loss warning).

Metric	Key	Type	Meaning
	recovered	boolean	Whether wealth regained the peak.
	peak_to_trough_periods	integer	Periods from peak to trough.
	duration_periods	integer	Peak to recovery, or to the end of data if ongoing.
	episodes	integer	Only 0, emitted instead of the keys above when no drawdown occurred.
average_drawdown	episodes	integer	Number of episodes averaged (0 when none).
longest_drawdown_duration	years	number | null	Duration in years (null if not annualizable).
	peak_date, recovery_date	string | null	Episode dates.
	ongoing	boolean	true if the episode has not recovered (the value is a lower bound).
	episodes	integer	Only 0, emitted instead when no drawdown occurred.
drawdown_count	ongoing	integer	Number of episodes still open at the end (0 or 1).
expected_shortfall_historical_95	tail_observations	integer	Returns at or below the 5% quantile that were averaged.
autocorrelation_lag1 … lag3	approx_95pct_bound	number	1.96/sqrt(n).
	significant	boolean	abs(rho) > bound.
ljung_box_pvalue	q_statistic	number	Ljung-Box Q.
	lags	integer	3.
beta	standard_error, t_statistic, p_value	number | null	OLS inference for the slope (conventional i.i.d. errors).
alpha	per_period	number	OLS intercept per period (the value is per_period × P).
	standard_error_per_period, t_statistic, p_value	number | null	OLS inference for the intercept.
treynor_ratio	beta	number	Beta used as the denominator.
upside_capture	up_periods	integer	Aligned periods with benchmark return > 0.
downside_capture	down_periods	integer	Aligned periods with benchmark return < 0.
upside_capture, downside_capture	fund_annualized, benchmark_annualized	number	Annualized geometric returns over those periods (numerator and denominator).
payoff_ratio	average_gain, average_loss	number	Mean of positive and of negative period returns (loss is signed).
pain_ratio	pain_index	number	Pain Index used as the denominator.
modified_var_95	cornish_fisher_quantile	number	Adjusted quantile h (replaces z = -1.645).
	skewness, excess_kurtosis	number	Population moments used by the expansion.
jarque_bera_pvalue	statistic	number	Jarque-Bera statistic (chi-squared with 2 degrees of freedom).
probabilistic_sharpe_ratio	sharpe_per_period	number	Per-period Sharpe ratio used by the test.
	z_statistic	number	Standardized distance of the Sharpe ratio from 0.

Metric	Key	Type	Meaning
	`moment_skewness`, `moment_kurtosis`	number	Population skewness and non-excess kurtosis.
`sharpe_ratio_standard_error`	`sharpe_ratio`	number	Annualized Sharpe ratio (same as the `sharpe_ratio` metric).
	`ci_level`, `ci_lower`, `ci_upper`	number	Two-sided confidence interval of the annualized Sharpe ratio.
	`standard_error_per_period`	number	Standard error of the per-period Sharpe ratio.
`minimum_track_record_length`	`years`	number | null	MinTRL in years (`null` if not annualizable).
	`observed_periods`	integer	Actual number of observations.
	`sufficient`	boolean	Whether the history is at least MinTRL long.
	`additional_periods_needed`	integer	`max(0, ceil(MinTRL - n))`.
`median_drawdown`	`episodes`	integer	Number of episodes.
`average_drawdown_duration`	`years`	number | null	Average duration in years.
	`episodes`, `ongoing`	integer	Episodes averaged; how many are still open (lower bound).
`max_peak_to_trough_duration`	`years`	number | null	Longest decline in years.
	`peak_date`, `trough_date`	string | null	Dates of that decline (or `episodes: 0` when none).
`average_recovery_duration`	`years`	number | null	Average recovery time in years.
	`recovered_episodes`, `excluded_open_episodes`	integer	Episodes used and open episodes excluded.
`cdar_95`	`tail_observations`	integer	Drawdown observations at or below the 5% quantile.
`modified_es_95`	`cornish_fisher_quantile`	number	Cornish-Fisher quantile `h` used by the expansion.
`best_calendar_year`	`year`	integer	Year of the best complete calendar year.
	`calendar_year_returns`	object	Map year (string) ⟶ compounded return of every complete calendar year.
	`excluded_partial_years`	object	Map year (string) ⟶ why it is incomplete.
`worst_calendar_year`	`year`	integer	Year of the worst complete calendar year.
`positive_calendar_years_pct`	`complete_years`, `positive_years`	integer	Counts behind the share.
	`calendar_year_returns`, `excluded_partial_years`	object	As for `best_calendar_year`.
`best_rolling_year`, `worst_rolling_year`	`window_end`	string | null	Last date of the extreme one-year window.
	`windows`, `window_periods`	integer	Number of windows and window length in observations.
`positive_rolling_year_pct`	`windows`, `window_periods`	integer	Number of windows and window length.
`mean_return_standard_error`	`ci_level`, `ci_lower`, `ci_upper`	number	Student-t confidence interval of the per-period mean.
	`t_statistic`, `p_value`	number	Two-sided t-test of the mean against zero.
`factor_alpha`, `factor_alpha_t_statistic`, `factor_r_squared`, `factor_residual_volatility`	`model`, `dependent`	string	Model label and dependent variable of the factor regression.
	`factors`	array of string	Factor names.

Metric	Key	Type	Meaning
	`hac_lags`, `df_resid`	integer	Newey-West lag and residual degrees of freedom.
`factor_alpha`	`per_period`	number	OLS intercept per period (the value is `per_period × P`).
	`standard_error_per_period`, `t_statistic`, `p_value`	number | null	Newey-West inference for the intercept.
	`ci_level`, `ci_lower`, `ci_upper`	number | null	Confidence interval of the **annualized** alpha.
`factor_alpha_t_statistic`	`p_value`	number | null	Two-sided Student-t p-value.
`factor_r_squared`	`adjusted_r_squared`	number | null	Adjusted R-squared.
`factor_residual_volatility`	`per_period`	number	Regression standard error per period.

---

drawdown_episodes

metrics/path.py `DrawdownEpisode.to_dict`. Every peak-to-recovery episode of the canonical wealth path, in chronological order. Empty when no drawdown occurred; also empty when the series contains returns below -100% (the wealth path is then not meaningful — see `data_quality.impossible_return_count`).

Indices refer to the wealth index `W_0 … W_n` (notation), where `W_0 = 1` is the inception point before the first return and `W_k` is wealth after the k-th return.

Key	Type	Meaning
`peak_index`	integer	Index of the peak (0 = inception).
`trough_index`	integer	Index of the lowest wealth in the episode (first occurrence).
`recovery_index`	integer | null	First index where wealth regains the peak; `null` if ongoing.
`depth`	number	Signed drawdown at the trough (≤ 0).
`peak_to_trough_periods`	integer	`trough_index - peak_index`.
`recovery_periods`	integer | null	`recovery_index - trough_index`; `null` if ongoing.
`total_underwater_periods`	integer	Observations with drawdown < 0 in the episode.
`duration_periods`	integer	`recovery_index - peak_index`, or `n - peak_index` if ongoing.
`area`	number	Sum of `abs(D_t)` over the underwater observations (decimal × periods).
`peak_date`, `trough_date`, `recovery_date`	string | null	Dates; `null` for the inception point, undated input or no recovery.
`recovered`	boolean	`recovery_index` is not `null`.

---

benchmark

`null` unless a benchmark was supplied. Otherwise `{"alignment": {...}, "input": {...}}`, where `input` has the shape of input and `alignment` is models/report.py `BenchmarkDiagnostics.to_dict`:

Key	Type	Meaning
`name`	string	Benchmark name.
`fund_observations`	integer	Fund observations before alignment.
`benchmark_observations`	integer	Benchmark observations before alignment.
`aligned_observations`	integer	Periods present in both (`m`, used by all benchmark metrics).
`fund_only_periods`	integer	Fund observations without a benchmark return (excluded, never filled).
`benchmark_only_periods`	integer	Benchmark observations outside the fund history.
`fund_coverage`	number	`aligned_observations / fund_observations`, 0–1.
`overlap_start`, `overlap_end`	string | null	First and last aligned dates.
`alignment_method`	string	`calendar monthly period`, `calendar quarterly period`, `calendar yearly period`, `exact date`, or a `by position (...)` description.
`warnings`	array of string	Alignment warnings.

factor_analysis

`null` unless factor returns were supplied. Otherwise factors/models.py `FactorAnalysis.to_dict` (decision 0028):

Key	Type	Meaning
`model`	string	Descriptive label, e.g. `Fama-French three-factor model` or `Custom factor model (2 factors)`.
`factor_names`	array of string	Factor columns in regression order.
`dependent`	string	Dependent variable: fund return minus RF, or the fund return when no RF column was supplied.
`return_scale`, `return_scale_evidence`	string	Scale applied to every factor column and why.
`alignment`	object	`fund_observations`, `factor_observations` (periods with every factor), `aligned_observations` (`m`), `fund_only_periods`, `fund_coverage`, `overlap_start`, `overlap_end`, `alignment_method`, `warnings`.
`fit`	object | null	The regression, or `null` when it is not identified (see `reason`).
`reason`	string | null	Why the regression could not be estimated.

`fit`: `model`, `factor_names`, `dependent`, `observations` (`m`), `factors` (`k`), `df_resid` (`m − k − 1`), `hac_lags`, `ci_level`, `alpha_per_period` and each entry of `loadings` (`name`, `estimate`, `standard_error`, `t_statistic`, `p_value`, `ci_lower`, `ci_upper`; inference is Newey-West), `r_squared`, `adj_r_squared`, `residual_std_per_period`, `residual_autocorrelation_lag1`, `residual_ljung_box_pvalue`, `max_vif`, `periods_per_year`, `alignment` and `warnings`. Undefined statistics are `null`.

interpretations

Map of metric id → models/report.py `Interpretation.to_dict`:

Key	Type	Meaning
`metric_id`	string	Metric id.
`plain`	string	Plain-language explanation (from the metric registry).
`professional`	string	Professional note (from the metric registry).
`context`	array of string	Data-specific notes: the reason when there is no value, then any rule notes (criteria §13).

provenance

models/report.py `Provenance.to_dict`.

Key	Type	Meaning
`package_version`	string	OpenFactsheet package version.
`methodology_version`	string	Methodology version.
`input_type`	string	Same as `input.source_type`.
`input_sha256`	string	SHA-256 fingerprint of the analyzed dates and values (criteria §15).
`date_range`	[string | null, string | null]	Start and end dates.
`frequency`	string	Observation frequency.
`annualization`	string	Annualization statement (factor and whether it was inferred or declared, or why nothing is annualized).
`observations`	integer	`n`.
`transformations`	array of string	Same as `input.transformations`.
`benchmark_name`	string | null	Benchmark name.
`benchmark_sha256`	string | null	Fingerprint of the analyzed benchmark returns.
`factor_names`	array of string | null	Factor columns supplied (the RF column excluded).
`extraction`	object | null	For document input (PDF, image): `engine` (e.g. `pypdf 6.18.1` or `rapidocr 3.9.2`), `source_sha256` (of the file), `page`, `confidence` (`high` / `moderate`), `monthly_values`, `checks`, `warnings`, `reconstructed_table` (decision 0030). Tables also record `role` (fund / benchmark / unknown), `role_evidence` and `series_label` (decision 0033).

Key	Type	Meaning
		OCR adds `ocr` (`name`, `version`, `models` with `file` and `sha256`), `image_sha256`, `repairs`, `min_cell_confidence` and, for scanned PDFs, `renderer` and `dpi` (decision 0031). `null` otherwise.
`factors_sha256`	string | null	SHA-256 of the compact JSON map column name → fingerprint of that column (same construction as `input_sha256`), RF included.
`canonical`	boolean	Same as the top-level `canonical`.

---

documentation

manual/links.py `DocumentationLinks.to_dict` (decision 0035). The links are pinned to the version that produced the report, never to "latest": a released version points at its own release assets and sources, and an unreleased one at the rolling manual built from `main`.

Key	Type	Meaning
`package_version`	string	The `openfactsheet` version the documentation describes.
`released`	boolean	`true` when the links point at a released version; `false` when they point at the rolling manual built from `main`.
`repository`	string	Repository the documentation is published from.
`manual`	string	URL of the PDF reference manual for this version.
`manual_tag`	string	Release tag whose assets carry that PDF (`manual-latest` before release, otherwise `v<version>`).
`source_ref`	string	Git ref the documentation sources are read at (`main` before release, otherwise `v<version>`).
`docs`	string	URL of the documentation tree at this version.
`catalog`	string	URL of the metric catalog at this version.
`metric_topic`	string	How a metric's link is formed: the manual URL followed by `#nameddest=metric-<id>`.

Example: one metric entry

`ofs.analyze("tests/fixtures/monthly_fund_10y.csv")`, entry for `max_drawdown`:


```
{
  "assumptions": [
    "Wealth index starts at 1 before the first return; drawdowns are measured from running peaks of compounded
wealth."
  ],
  "category": "drawdown",
  "confidence": "high",
  "details": {
    "duration_periods": 54,
    "peak_date": "2020-06-30",
    "peak_to_trough_periods": 24,
    "recovered": false,
    "recovery_date": null,
    "trough_date": "2022-06-30"
  },
  "direction": "higher_is_better",
  "documentation": "https://github.com/bagnascotrader/openfactsheet/releases/download/manual-latest/openfactsheet-
manual.pdf#nameddest=metric-max_drawdown",
  "formula_id": "max_drawdown@v1",
  "id": "max_drawdown",
  "methodology_version": "0.1.0",
  "name": "Maximum drawdown",
  "nature": "observed",
  "observations": 120,
  "reason": null,
  "reference_values": null,
  "status": "applicable",
  "unit": "decimal_return",
  "value": -0.2732106828524845,
  "warnings": [
    "The deepest drawdown has not recovered by the end of the data; the loss is still open."
```

```
  ]
}
```

Methodology decisions

Methodology decisions

Each file records one convention of the canonical OpenFactsheet methodology: the problem, how reference implementations behave, the academic sources, the decision and its consequences. The machine-readable form of every decision is `src/openfactsheet/policy/defaults.py` (conventions) and `src/openfactsheet/policy/thresholds.py` (minimum-data policies).

All decisions below are **Accepted** for methodology **0.1.0**.

#	Decision	Summary
0001	Return type	Simple decimal returns internally; percent converted; prices rejected; values below -100% flagged, never clipped.
0002	Frequency and annualization	Frequency inferred from calendar spacing; 252/365/52/12/4/1; irregular or unknown frequency is never annualized.
0003	Standard deviation	Sample estimator (n − 1) everywhere volatility appears.
0004	Annual return	Geometric CAGR by observation count; arithmetic `P x mean` reported separately; under one year ⟶ low confidence.
0005	Risk-free rate	0% in 0.1.0, disclosed on every affected metric; never inferred from locale.
0006	Sortino / downside deviation	MAR = 0, full-sample denominator, annualized by sqrt(P).
0007	Drawdowns	Initial wealth 1 is a peak; signed depths; peak-to-recovery episodes and durations; one shared engine.
0008	Sharpe ratio	Arithmetic mean / sample std x sqrt(P), rf = 0.
0009	Calmar ratio	Full history (not a trailing 36-month window).
0010	VaR / ES / tail ratio	Historical 95%, linear (type 7) quantile, signed returns, ES over r <= q.
0011	Skewness / kurtosis	Bias-corrected G1 and excess G2 (pandas / Excel).
0012	Benchmark alignment and metrics	Calendar-period intersection, never filled; OLS with rf = 0; arithmetic alpha / IR / Treynor; geometric capture.
0013	Return-scale detection	Deterministic percent/decimal classifier; refuses to guess in the ambiguous zone.
0014	Smoothing risk	Lag-1 autocorrelation + significance thresholds; warn and downgrade, never unsmooth.
0015	Minimum-data policies	Named policies: mathematical minimum vs statistical confidence thresholds.
0016	Lower-partial-moment ratios and dispersion	Kappa 3, upside potential, LPM 1, semi-deviation, MAD: target 0, full-sample denominators; per period except annualized semi-deviation.
0017	Burke, Sterling, Pain, payoff	Canonical episode depths for Burke; Sterling = CAGR / (|MDD| + 10%); Pain ratio = CAGR / Pain Index; payoff excludes zeros.
0018	Parametric VaR	Gaussian and Cornish-Fisher VaR with population moments; supplementary, model-dependent.
0019	Sharpe-ratio inference	PSR vs 0, SE with two-sided 95% CI, MinTRL at 95%; kurtosis kept; i.i.d. asymptotics; no Deflated Sharpe.
0020	Mean inference and normality	Student-t 95% CI and t-test for the mean; Jarque-Bera with population moments (supplementary).
0021	Drawdown tail and entropic measures	DaR / CDaR on the compounded drawdown path with the VaR / ES estimators; EDaR / EVaR by 1-D convex minimization; 95%.
0022	Modified Expected Shortfall	Cornish-Fisher ES (Boudt, Peterson & Croux 2008), population moments; supplementary; flagged when less severe than modified VaR.
0023	Calendar-year statistics	Complete calendar years only (12 months / 4 quarters / 7-day edge tolerance for daily and weekly); partial years listed, never used.
0024	Rolling one-year windows	Window round(P) observations, step 1, every start date; descriptive only.
0025	Episode durations and path summaries	Open episodes in average duration (lower bound) but not in recovery time; recovery factor = compounded return / |MDD|; geometric mean; IQR.

#	Decision	Summary
0026	Canonical charts	Eight fixed charts over the full history, linear scales, zero baselines for bars, symmetric heatmap scale, partial years flagged, benchmark on the aligned overlap only; the 10 deepest drawdown episodes tabulated.
0027	Benchmark-relative additions	Residual volatility (m − 2), appraisal ratio (arithmetic alpha), M^2 (rf = 0), strict batting average, capture and timing ratios need positive denominators; regime betas on benchmark > 0 / < 0; exact fit for a constant fund.
0028	Factor models	Caller-supplied factors only; an `RF` column is subtracted from fund returns; one return scale for all factor columns; OLS with Newey-West errors (lag floor(4 (m/100)^(2/9)), no prewhitening, Student t); never filled; smoothing lowers confidence.
0029	Serial correlation and annualization	One-year horizon q = round(P); variance ratio from the sample autocorrelations (supplementary, with i.i.d. z-test); Lo (2002) adjusted Sharpe = per-period SR × sqrt(q / VR); at least two years; profit factor = gross gains / gross losses.
0030	PDF ingestion	Embedded text only (optional pypdf extra); twelve-month header; values assigned to columns by position; each year reconciled with its YTD; any inconsistency refuses the table and prints the reconstruction; provenance records engine, file hash and checks.
0031	OCR ingestion	Screenshots and scanned pages via a local engine (RapidOCR default with its bundled models, Tesseract optional; no network); same reconstruction as PDF text; any cell below 0.90 confidence, a missing or unreconciled YTD, or an unreadable table refuses it; repairs and inferred headers recorded.
0032	Identifiers and fund profile	Fixed fields with evidence (name, objective, benchmark name, currency, launch date, identifiers); ISIN/CUSIP/SEDOL/LEI/FIGI validated by check digit and never corrected; WKN/Valor/tickers only after their label; national codes linked to ISINs; the profile never changes a metric.
0033	Share classes and table roles	Series-labelled rows become separate series with roles (fund / benchmark / relative / unknown); document benchmark returns are returned by load() but never used; the returns table is attributed to a share class by caption, heading, single-class or currency rules, otherwise reported as ambiguous.
0034	Reference manual	A CRAN-style PDF manual generated from the code on every change: API topics from docstrings, one topic per metric from the registry, every methodology document and decision; offline, reproducible Typst build; a named destination for every topic.
0035	Linking a result to its documentation	Every result carries the URL of the manual topic that defines it; the report carries the manual, catalog and documentation links of its own version; links are pinned to the version that produced the number, never to "latest".

Adding or changing a decision

Follow CLAUDE.md sections 20 and 45 whenever definitions of a metric or convention disagree:

1. Identify the authoritative definition (original paper, recognized textbook or practitioner standard, PerformanceAnalytics documentation, established Python libraries).
2. Compare implementations and record each library's behavior. Mark anything not checked numerically as "reported in library docs, not verified here".
3. Write a note `NNNN-short-slug.md` with the sections: Status, Problem, Known definitions, PerformanceAnalytics behavior, QuantStats behavior, Empyrical behavior, Academic sources, OpenFactsheet decision, Rationale, Consequences.
4. Implement the decision independently from the mathematical definition (never copy GPL code), add edge-case unit tests and a cross-library test that pins agreement or the documented difference.
5. Update `docs/validation/reference_implementations.md` and, if applicable, `docs/assumptions.md`.

Never resolve a disagreement by choosing whichever convention produces the more attractive number.

Versioning rule
Any change that alters a canonical value, status, confidence label, threshold or convention requires:

- incrementing `METHODOLOGY_VERSION` in `src/openfactsheet/policy/methodology_version.py`;
- a CHANGELOG entry;
- a new or superseding decision note (the old note is kept and marked "Superseded by NNNN");
- conscious regeneration and review of the golden reports (`python scripts/update_golden.py`).

Package releases that do not change results keep the same methodology version.

Decision 0001: Return type: simple decimal returns
Status: Accepted — methodology 0.1.0

Problem
Return histories arrive as simple returns, log returns, percentages, decimals, or price/NAV levels. Mixing conventions silently changes every downstream statistic (compounding, drawdowns, moments). One internal representation is required.

Known definitions
- Simple (arithmetic) return: `r_t = P_t / P_(t-1) - 1`. Compounds by multiplication: `W_t = prod(1 + r_i)`.
- Log return: `l_t = ln(P_t / P_(t-1)) = ln(1 + r_t)`. Adds over time but does not aggregate across assets and is not what fund factsheets report.

PerformanceAnalytics behavior
Assumes simple returns by default (`Return.cumulative(geometric = TRUE)` uses `prod(1 + R) - 1`). Some functions accept a `geometric` switch. Values are expected as decimals; `DrawdownPeak` (used by `UlcerIndex`/`PainIndex`) divides returns by 100, i.e. it assumes percent inputs (see 0007).

QuantStats behavior
Works on simple decimal returns in a pandas Series; compounding uses `(1 + r).prod() - 1`.

Empyrical behavior
Simple decimal returns; `cum_returns_final` compounds `prod(1 + r) - 1`.

Academic sources
- Bacon, C. R. (2021). *Practical Risk-Adjusted Performance Measurement* (2nd ed.).
- CFA Institute (2020). GIPS — time-weighted returns are geometrically linked simple returns.

OpenFactsheet decision
- Internally every series is a read-only `float64` array of **simple decimal returns** (`0.01` = +1%). `return_type` is always `"simple"` in 0.1.0.
- Percent inputs are converted by dividing by 100, recorded as a transformation (scale detection: 0013).
- Log returns are not accepted as a separate type in 0.1.0.
- Price/NAV levels are detected as implausible return values and rejected with a message asking for explicit conversion `r_t = P_t / P_(t-1) - 1`.
- Values below -100% are kept (never clipped) and flagged as impossible; wealth-path metrics become `not_applicable`.
- A -100% return (total loss) is valid: wealth is zero from that period and cannot be rebuilt by later returns.

Rationale
Simple returns are what funds report and what investors experience when compounding. One representation removes a whole class of silent mixing errors.

Consequences
- `ReturnSeries.values` is always decimal simple returns; the report exposes them exactly (`report.returns_frame()` and the `returns` block of the JSON).
- A price-to-returns conversion helper with provenance is future work.

Decision 0002: Frequency inference and annualization factors

Status: Accepted — methodology 0.1.0

Problem

Annualized statistics need a periods-per-year factor `P`. Guessing it (for example "many observations, so daily, so 252") can mis-scale volatility by large factors. The factor must come from evidence and be disclosed.

Known definitions

Conventional factors: daily 252 (trading days) or 365 (calendar days), weekly 52, monthly 12, quarterly 4, yearly 1. Volatility scales with `sqrt(P)`, arithmetic means with `P`, geometric returns via `(prod(1 + r))^(P/n) - 1`.

PerformanceAnalytics behavior

`periodicity()` (xts) classifies the index; functions map daily ⟶ 252, weekly ⟶ 52, monthly ⟶ 12, quarterly ⟶ 4, yearly ⟶ 1 (seen in `Return.annualized`, `StdDev.annualized`, `SharpeRatio.annualized`, `CalmarRatio`). There is no 365 option without an explicit `scale`.

QuantStats behavior

Defaults to `periods=252`, overridable by argument; frequency is not inferred for most statistics (reported in library docs, not verified here).

Empyrical behavior

The caller passes `period="daily" | "weekly" | "monthly" | ...` or `annualization=`; the default is daily (252). No inference.

Academic sources

- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *FAJ* 58(4) — square-root-of-time scaling assumes serially uncorrelated returns.

OpenFactsheet decision

Frequency is inferred from **calendar spacing only** (`normalization/frequency.py`):

Median gap between dates	Candidate	Regularity test
25–35 days	monthly	calendar-month steps; >= 50% exactly one month, none zero
80–100 days	quarterly	calendar-quarter steps, same rule
300–400 days	yearly	calendar-year steps, same rule
5–9 days	weekly	every gap a multiple of 7 days (±1), same rule
<= 4 days	daily	>= 80% of gaps <= 4 days
anything else	irregular	—

- Steps larger than one period are gaps: counted as missing expected periods and disclosed. Two observations in the same calendar period ⟶ irregular.
- Inference uses the full date grid, including dates whose value is missing, so a NaN row is a missing observation, not irregular spacing.
- Daily: weekend observations present ⟶ calendar-day series, `P = 365`, missing days counted; otherwise business-day series, `P = 252`, missing days *not* counted (holidays are legitimate), gaps longer than 7 days reported.
- Canonical factors: daily 252/365, weekly 52, monthly 12, quarterly 4, yearly 1 (`policy/defaults.py`).
- **Irregular or unknown frequency is never annualized**: annualized metrics are `not_applicable` with a reason. Undated input has `UNKNOWN` frequency.
- The caller may *declare* the frequency of undated values (`frequency="monthly"`). With dates, a declaration must match the inferred frequency or a `FrequencyInferenceError` is raised. Declared daily without dates assumes 252 and says so. Declarations are recorded and warned on every annualized metric.
- The number of observations is never evidence of frequency.

Rationale

Calendar spacing is objective and reproducible. Refusing to annualize irregular data is "approximately right" rather than precisely wrong.

Consequences

- Every annualized metric carries an assumption string such as "Annualized with 12 periods per year (monthly frequency, inferred from dates)."

- Crypto and other 7-day markets get 365 automatically when weekends appear.
- Resampling between frequencies is not performed in 0.1.0.

Decision 0003: Standard deviation: sample (n – 1) estimator

Status: Accepted — methodology 0.1.0

Problem

Volatility can use the population (`1/n`) or sample (`1/(n-1)`) variance denominator. The difference matters for short histories (about 9% at n = 6) and propagates into Sharpe, tracking error and the Information ratio.

Known definitions

- Population: `sqrt( (1/n) sum (r_t - m)^2 )`.
- Sample (Bessel-corrected): `sqrt( (1/(n-1)) sum (r_t - m)^2 )`.

PerformanceAnalytics behavior

`StdDev` and `StdDev.annualized` use R `sd()`, the sample estimator. `TrackingError` uses `sd.xts`, also sample. Verified: `volatility` and `annualized_volatility` agree to 1e-9 on all fixtures.

QuantStats behavior

Uses pandas `.std()`, sample by default (`ddof=1`). Its Sharpe ratio agrees with OpenFactsheet to 1e-9 on all fixtures, which implies the same estimator.

Empyrical behavior

`annual_volatility` uses `nanstd(ddof=1)`. Verified: agrees to 1e-9.

Academic sources

- Bacon (2021), chapter on standard deviation.
- Sharpe (1994) uses the sample standard deviation of differential returns.

OpenFactsheet decision

All standard deviations (volatility, annualized volatility, Sharpe denominator, tracking error, Information ratio denominator) use `ddof = 1` (`dispersion.ddof = 1` in the methodology). A single observation makes these metrics `insufficient_data`.

Downside deviation is a different quantity (a lower partial moment) with a full-sample `n` denominator; see 0006.

Rationale

The sample estimator is unbiased for the variance, is the default in all reference implementations, and makes results reproducible in Excel (`STDEV.S`).

Consequences

- Zero variance makes volatility 0 and Sharpe `not_applicable` (“Volatility is zero; the Sharpe ratio is undefined.”).
- Metric assumption strings state “Sample standard deviation (n - 1 denominator).”

Decision 0004: Annual return: geometric CAGR by observation count

Status: Accepted — methodology 0.1.0

Problem

“Annualized return” can mean the compound annual growth rate or the arithmetic mean times periods per year. The two differ by roughly half the variance, and CAGR itself can be computed from observation counts or calendar time.

Known definitions

- Geometric (CAGR): `(prod(1 + r_t))^(P / n) - 1`.
- Arithmetic: `P * mean(r_t)`.
- Calendar-time CAGR: `(W_end / W_start)^(365.25 / days) - 1`.

PerformanceAnalytics behavior

`Return.annualized(geometric = TRUE)` computes `prod(1 + R)^(scale / n) - 1`; `geometric = FALSE` computes `mean(R) * scale`. Verified: `cagr` and `annualized_arithmetic_mean` agree to 1e-9 on all fixtures.

QuantStats behavior

`stats.cagr(returns, periods=...)` uses a `periods` argument whose default assumes daily observations. Verified on the monthly fixture: with the default it returns 93.75% (each month treated as a trading day); with `periods=12` it agrees with OpenFactsheet to 1e-9. OpenFactsheet never takes an annualization factor from the caller; it is established from the dates.

Empyrical behavior

`annual_return` computes `ending_value^(1 / (n / P)) - 1`, the same as observation-count CAGR. Verified: agrees to 1e-9.

Academic sources

- Bacon (2021).
- CFA Institute (2020). GIPS: annualized returns are geometric; returns for periods shorter than one year must not be annualized.

OpenFactsheet decision

- `cagr = (prod(1 + r))^(P / n) - 1`, using the canonical `P` from 0002 and the number of observations `n`.
- `annualized_arithmetic_mean = P * mean(r)` is reported separately; it is the return component of the canonical Sharpe ratio (0008).
- If wealth reaches zero or below, CAGR is reported as -100%.
- **Less than one year of data:** annualized metrics are still computed (the value is mathematically valid) but confidence is forced to `low`, with the warning "Annualized from only X years of data; the figure extrapolates a partial year." The interpretation of `cagr` and `annualized_arithmetic_mean` cites the GIPS rule (0015).

Rationale

Observation count ties annualization to the detected frequency and removes ambiguity about whether the first period's start date is known. Reporting both geometric and arithmetic figures prevents choosing the more attractive one.

Consequences

- Differences against QuantStats `cagr` are expected when calendar span and `n / P` differ.
- Missing periods shorten `n` and are disclosed; confidence is capped at `moderate` for annualized and wealth-path metrics when gaps exist.

Decision 0005: Risk-free rate: zero in methodology 0.1.0, always disclosed

Status: Accepted — methodology 0.1.0

Problem

Excess-return metrics (Sharpe, Sortino, Calmar, Martin, alpha, Treynor) need a risk-free rate. The correct rate depends on currency, period and investor, and OpenFactsheet has no reliable, free, deterministic source keyed on metadata it can trust. Choosing a rate per fund would also be a tuning knob.

Known definitions

Sharpe (1994) defines the ratio on differential returns `r_t - rf_t`. Many practitioners use a local T-bill or overnight rate; many libraries default to `rf = 0`.

PerformanceAnalytics behavior

`Rf = 0` default in `SharpeRatio`, `CAPM.alpha`, `CAPM.beta`, `TreynorRatio`, `MartinRatio`. The fixtures were computed with `Rf = 0`.

QuantStats behavior

`rf = 0.0` default in `sharpe`, `sortino` and related functions.

Empyrical behavior

`risk_free = 0` default in `sharpe_ratio`, `alpha_beta`.

Academic sources

- Sharpe, W. F. (1994). The Sharpe Ratio. *JPM* 21(1).

OpenFactsheet decision

- `risk_free.rate = 0.0` per period for every excess-return metric in 0.1.0.

- Every affected metric carries the assumption: "Risk-free rate assumed to be 0% (methodology 0.1.0); the ratio therefore uses total rather than excess returns and is higher than an excess-return version when cash yields are positive."
- Regression-based metrics (beta, alpha, Treynor) regress total returns.
- The investor's currency is never inferred from the computer locale or any other ambient setting.
- There is no per-call override in the canonical API.

Rationale

A disclosed zero is transparent and reproducible, and it matches the default of every reference library, which makes cross-validation direct. A rate chosen per fund could be used to improve presentation.

Consequences

- In periods of positive cash yields, Sharpe-type ratios are overstated relative to excess-return versions; the report says so on each metric.
- Roadmap: an optional, deterministic public risk-free series selected from explicitly supplied currency metadata. Adding it will be a methodology version change, and overridden results will be labeled non-canonical.

Decision 0006: Sortino ratio and downside deviation: MAR 0, full-sample denominator

Status: Accepted — methodology 0.1.0

Problem

Downside deviation has two free choices: the target (minimum acceptable return, MAR) and the denominator (all observations or only those below the target). Both change the Sortino ratio materially, and both could be tuned.

Known definitions

- Full-sample: `DD = sqrt( (1/n) sum min(r_t - MAR, 0)^2 )` (Sortino & Price, 1994).
- Subset: `DD = sqrt( (1/k) sum_{r_t < MAR} (r_t - MAR)^2 )`, k = number of shortfalls.
- Sortino ratio: `(mean(r) - MAR) / DD`, annualized by `sqrt(P)` or reported per period.

PerformanceAnalytics behavior

`DownsideDeviation(R, MAR = 0, method = c("full", "subset"))` defaults to `"full"` (`len = length(R)`). `SortinoRatio(R, MAR = 0)` returns the per-period `mean(R - MAR) / DownsideDeviation(R, MAR)`. Verified: `downside_deviation = DownsideDeviation(full) * sqrt(12)` and `sortino_ratio = SortinoRatio * sqrt(12)`, both to 1e-9.

QuantStats behavior

`stats.sortino` uses `sqrt(sum(r[r < 0]^2) / n)` (full sample) and annualizes by `sqrt(periods)` (reported in library docs, not verified here).

Empyrical behavior

`downside_risk` uses the full-sample denominator and annualizes by `sqrt(P)`; `sortino_ratio` = `mean(r - required) * P / downside_risk`. Verified: both agree to 1e-9 on all fixtures.

Academic sources

- Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. *Journal of Investing* 3(3).
- Bacon (2021).

OpenFactsheet decision

- MAR = 0% per period (`downside.mar = 0.0`).
- Full-sample denominator `n` (`downside.denominator = "full_sample"`).
- `downside_deviation` is reported annualized: `DD * sqrt(P)`.
- `sortino_ratio = mean(r) / DD * sqrt(P)` (equivalently `P * mean / (DD * sqrt(P))`).
- With no return below 0%, downside deviation is 0 and the Sortino ratio is `not_applicable` ("No returns below the 0% target...").

Rationale
The full-sample denominator is the original definition and the default in all three reference libraries. MAR = 0 matches the canonical zero risk-free rate (0005) and avoids a per-fund target.

Consequences
- Series with very few losses give unstable Sortino values; the metric's documentation says so, and it is volatility-based for smoothing warnings.
- A different MAR would be a methodology change, not a parameter.

Decision 0007: Drawdown conventions
Status: Accepted — methodology 0.1.0

Problem
Drawdown statistics depend on whether the starting capital counts as a peak, on the sign convention, on how episodes are delimited, and on how durations are counted. Reconstructing drawdowns separately inside each ratio invites inconsistency.

Known definitions
`W_t = W_0 * prod_{i<=t}(1 + r_i)`, `D_t = W_t / max_{s<=t} W_s - 1`. Maximum drawdown `min D_t`; Ulcer Index `sqrt(mean(D_t^2))` (Martin & McCann, 1989); Pain Index `mean(|D_t|)` (Bacon); average drawdown either per episode or per period.

PerformanceAnalytics behavior
- `Drawdowns` computes `cumprod(1 + R) / cummax(c(1, cumprod(1 + R)))[-1] - 1`, i.e. the initial value 1 counts as a peak. `maxDrawdown` reports a positive magnitude; `AverageDrawdown` averages episode depths from `findDrawdowns`. Verified: `max_drawdown` and `average_drawdown` equal the negated PA values to 1e-9.
- `UlcerIndex` and `PainIndex` use `DrawdownPeak`, which divides returns by 100 (a percent-input assumption). On decimal inputs they do not match the textbook formula. Verified instead: OpenFactsheet `ulcer_index` and `pain_index` equal `sqrt(mean(Drawdowns(R)^2))` and `mean(|Drawdowns(R)|)`. `MartinRatio` inherits the issue; OpenFactsheet `martin_ratio` equals `Return.annualized / sqrt(mean(Drawdowns(R)^2))`.
- `findDrawdowns length` equals OpenFactsheet duration for recovered episodes and is one period longer for an episode still open at the end of the data (verified on all fixtures).

QuantStats behavior
`stats.max_drawdown` returns a signed value. Verified: agrees to 1e-9 on all fixtures.

Empyrical behavior
`max_drawdown` returns a signed value. Verified: agrees to 1e-9 on all fixtures.

Academic sources
- Magdon-Ismail, M., & Atiya, A. (2004). Maximum Drawdown. *Risk* 17(10).
- Martin & McCann (1989); Bacon (2021).

OpenFactsheet decision
- One engine (`metrics/path.py`): `W_0 = 1` precedes the first return and is the first peak, so a first-period loss is a drawdown.
- Drawdowns and depths are **signed** (negative = loss). Ulcer and Pain indices are non-negative magnitudes, as their originators define them.
- An episode starts at the last peak before wealth falls below it and ends at the first period in which wealth regains that peak (recovery). If it never recovers it is *ongoing* and runs to the last observation. Every decline below a prior peak is an episode, however small.
- Duration = periods from peak to recovery (`recovery - peak`); ongoing: `n - peak`, flagged as a lower bound.
- Average drawdown = mean of episode depths; 0 when no drawdown occurred.
- Time under water, Ulcer and Pain are computed over observations 1..n (the inception point is excluded).
- Returns below -100% make every wealth-path metric `not_applicable`.

Rationale
Counting the initial capital as a peak reflects the investor's experience from inception. One engine guarantees that every drawdown metric describes the same path.

Consequences
- PA's `UlcerIndex`, `PainIndex` and `MartinRatio` values differ from OpenFactsheet on decimal data; this is a PA input-convention issue, documented in reference implementations.
- Drawdown statistics grow with history length; path metrics use the `path_risk` minimum-data policy (0015).

Decision 0008: Sharpe ratio: arithmetic mean over sample volatility, square-root-of-time
Status: Accepted — methodology 0.1.0

Problem
The "annualized Sharpe ratio" has at least two common forms: arithmetic mean return over volatility scaled by `sqrt(P)`, and annualized geometric return over annualized volatility. They can differ substantially for volatile series.

Known definitions
- Ex-post Sharpe (Sharpe, 1994): `mean(r - rf) / std(r - rf)`, annualized by `sqrt(P)`.
- Geometric variant: `[(prod(1 + r - rf))^(P/n) - 1] / (std(r) * sqrt(P))`.

PerformanceAnalytics behavior
`SharpeRatio.annualized(R, Rf, scale, geometric = TRUE)` by default divides the **geometric** annualized excess return by `StdDev.annualized`. With `geometric = FALSE` it computes `mean * scale / (sd * sqrt(scale))`. Verified: OpenFactsheet equals the `geometric = FALSE` form to 1e-9 and differs from the PA default on all fixtures.

QuantStats behavior
`stats.sharpe(returns, rf = 0, periods = 252)` computes `mean / std(ddof = 1) * sqrt(periods)`. Verified with `periods = 12`: agrees to 1e-9.

Empyrical behavior
`sharpe_ratio(returns, period = "monthly")` computes `mean / std(ddof = 1) * sqrt(P)`. Verified: agrees to 1e-9.

Academic sources
- Sharpe, W. F. (1966). Mutual Fund Performance. *Journal of Business* 39(1).
- Sharpe, W. F. (1994). The Sharpe Ratio. *JPM* 21(1).
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *FAJ* 58(4).
- Getmansky, Lo & Makarov (2004). *JFE* 74(3).

OpenFactsheet decision
`sharpe_ratio = mean(r) / std(r, ddof = 1) * sqrt(P)` with `rf = 0` (0005), the canonical `P` (0002) and sample standard deviation (0003).

- Zero volatility ⟶ `not_applicable`.
- Irregular/unknown frequency ⟶ `not_applicable` (it is an annualized metric).
- The metric is `volatility_based`: high smoothing risk forces confidence to `low`; moderate caps it at `moderate`; strongly non-normal returns add a warning (0014).

Rationale
This is Sharpe's own ex-post definition and the form shared by QuantStats and empyrical. Using the arithmetic mean keeps the Sharpe, Information ratio, active return and alpha conventions consistent with each other (0012).

Consequences
- Values differ from PerformanceAnalytics' default `SharpeRatio.annualized`. The cross-library test asserts the difference so that it cannot disappear silently.
- Sampling uncertainty is reported separately (standard error with a 95% interval, Probabilistic Sharpe Ratio, minimum track record length); see 0019. The Deflated Sharpe Ratio is not computed because it needs the number of strategies tried.

Decision 0009: Calmar ratio over the full history

Status: Accepted — methodology 0.1.0

Problem

Young (1991) defined the Calmar ratio on a trailing 36-month window. Others use the full history, and some use the arithmetic rather than geometric return. A window choice is exactly the kind of period selection that can flatter results.

Known definitions

- Young (1991): annualized return over the last 36 months divided by the maximum drawdown over the same window.
- MAR ratio: the same quantity over the full history.

PerformanceAnalytics behavior

`CalmarRatio(R, scale) = Return.annualized(R, scale) / abs(maxDrawdown(R))` over whatever series is supplied (full history, geometric return). Verified: agrees to 1e-9 on all fixtures.

QuantStats behavior

`stats.calmar` divides its default CAGR (which assumes daily observations, see 0004) by the absolute maximum drawdown. Verified: `calmar == cagr_default / |max drawdown|`, so on monthly data it differs from OpenFactsheet unless the CAGR frequency is corrected.

Empyrical behavior

`calmar_ratio(returns, period) = annual_return / abs(max_drawdown)` over the supplied series. Verified: agrees to 1e-9.

Academic sources

- Young, T. W. (1991). Calmar Ratio: A Smoother Tool. *Futures* 20(1).
- Bacon (2021).

OpenFactsheet decision

`calmar_ratio = cagr / |max_drawdown|` over the **full supplied history**, using the canonical CAGR and signed maximum drawdown (0007).

- No drawdown (maximum drawdown 0) ⟶ `not_applicable` (zero divisor).
- Annualized metric: not applicable without a regular frequency.
- `martin_ratio = cagr / ulcer_index` follows the same full-history rule.

Rationale

The full history is the only window that requires no choice. Trailing windows may be added later as separately identified, objectively defined statistics, never as a replacement.

Consequences

- OpenFactsheet's "Calmar" is, strictly, Young's MAR ratio; the metric documentation says so.
- Values fall as track records lengthen because maximum drawdown grows with history; comparisons across different history lengths are flagged by `compare()`.

Decision 0010: Historical VaR, Expected Shortfall and tail ratio

Status: Accepted — methodology 0.1.0

Problem

Tail measures depend on the confidence level, on the estimator (historical, parametric, Cornish-Fisher), on the empirical quantile definition (there are nine common ones), on how ties at the quantile are handled, and on the sign convention. A marketer could pick any of these to shrink reported losses.

Known definitions

- Historical VaR at level `c`: the `1 - c` empirical quantile of returns.
- Historical ES: mean of returns at or beyond that quantile.
- Hyndman & Fan (1996) type 7: linear interpolation between order statistics (R and NumPy default).
- Tail ratio: `|Q_0.95| / |Q_0.05|`.

PerformanceAnalytics behavior

`VaR.historical` returns `-quantile(r, 0.05)` (type 7); the public `VaR(p = 0.95, method = "historical")` returns a negative number. `ES.historical` averages `r[r < q]` (strictly below the quantile) and falls back to VaR when that set is empty. Verified: `var_historical_95` and `expected_shortfall_historical_95` agree to 1e-9 on all fixtures (no fixture has a return exactly equal to the quantile).

QuantStats behavior

`stats.value_at_risk` is parametric: verified equal to the normal 5% quantile with the sample mean and sample standard deviation, and different from the historical VaR. `stats.cvar` conditions on that parametric threshold (function behavior observed, not asserted). `stats.tail_ratio` agrees with OpenFactsheet to 1e-9 on all fixtures (verified).

Empyrical behavior

`tail_ratio` = `|percentile 95| / |percentile 5|`; verified to 1e-9. `conditional_value_at_risk` averages returns at or below the 5th percentile (reported in library docs, not verified here).

Academic sources

- Jorion, P. (2006). *Value at Risk* (3rd ed.).
- Acerbi, C., & Tasche, D. (2002). On the Coherence of Expected Shortfall. *JBF* 26(7).
- Hyndman, R. J., & Fan, Y. (1996). Sample Quantiles in Statistical Packages.

OpenFactsheet decision

- Confidence level 95% for VaR and ES (`tail.var_confidence`, `tail.es_confidence`); not user-selectable in the canonical report.
- Historical (empirical) estimator only in 0.1.0; parametric and modified VaR are future supplementary metrics.
- Quantile: linear interpolation (`numpy.quantile(method="linear")`, R type 7).
- `var_historical_95 = Q_0.05(r)`; `expected_shortfall_historical_95 = mean(r[r <= Q_0.05])`. The `<=` rule guarantees a non-empty tail; it differs from PA only when an observation equals the quantile exactly.
- Both reported as **signed per-period returns** (negative = loss), consistent with drawdowns (0007).
- `tail_ratio = |Q_0.95| / |Q_0.05|`; `not_applicable` when `Q_0.05 = 0`.

Rationale

Historical estimates need no distributional assumption. Type 7 is the default of R and NumPy, so results reproduce in both. A fixed level prevents confidence shopping.

Consequences

- With monthly data a 95% tail estimate rests on about `n/20` points; the `tail` minimum-data policy marks estimates `low` confidence below 5 expected tail observations (n < 100) and `moderate` below 10 (n < 200) (0015).
- Values are per observation period, not annualized or horizon-scaled.
- The same estimators are applied to the drawdown path for Drawdown-at-Risk and CDaR (0021).

Decision 0011: Skewness and kurtosis estimators

Status: Accepted — methodology 0.1.0

Problem

Several sample skewness and kurtosis estimators are in use (biased moment estimators, bias-adjusted estimators, "excess" vs raw kurtosis). They differ noticeably in short samples, and library defaults disagree.

Known definitions

With `m_k = (1/n) sum (r_t - mean)^k` (Joanes & Gill, 1998):

- `g1 = m3 / m2^(3/2)` (biased moment skewness).
- `G1 = sqrt(n(n-1)) / (n-2) * g1` (adjusted Fisher-Pearson; Excel `SKEW`, pandas).
- `G2 = (n-1)/((n-2)(n-3)) * ((n+1) * m4/m2^2 - 3(n-1))` (bias-corrected excess kurtosis; Excel `KURT`, pandas).

PerformanceAnalytics behavior

- `skewness(method = "moment")` = `g1`. Verified: OpenFactsheet `skewness` equals `g1 * sqrt(n(n-1))/(n-2)` to 1e-9.
- `skewness(method = "sample")` uses the population standard deviation in a sample-style formula and equals `G1 * (n/(n-1))^1.5`, i.e. it is neither `g1` nor `G1`. Verified to 1e-9 on all fixtures.
- `kurtosis(method = "sample_excess")` = `G2`. Verified to 1e-9.

- `kurtosis` default `"excess"` is the biased moment estimator minus 3.

QuantStats behavior
`stats.skew` and `stats.kurtosis` call pandas `.skew()` / `.kurt()`, i.e. `G1` and `G2`. Verified to 1e-9 on all fixtures.

Empyrical behavior
No skewness/kurtosis functions in the inspected version.

Academic sources
- Joanes, D. N., & Gill, C. A. (1998). Comparing Measures of Sample Skewness and Kurtosis. *JRSS D* 47(1).

OpenFactsheet decision
- `skewness` = `G1` (`scipy.stats.skew(bias=False)`); requires n >= 3.
- `excess_kurtosis` = `G2` (`scipy.stats.kurtosis(fisher=True, bias=False)`); requires n >= 4; normal distribution = 0.
- Constant returns ⟶ `not_applicable`.
- Both are tagged `estimated`; direction is `context_dependent`.
- Minimum-data policies: skewness low confidence below 60 observations, moderate below 120; kurtosis low below 120, moderate below 240 (0015).

Rationale
`G1`/`G2` are the most widely reproduced estimators (Excel, pandas, SciPy with `bias=False`), so any reader can check the numbers.

Consequences
- OpenFactsheet skewness differs from PerformanceAnalytics' `"sample"` method by the factor `(n/(n-1))^1.5`; the cross-library test pins this relationship.
- The same moments drive the distribution-shape warning on volatility-based metrics (skewness <= -1 or excess kurtosis >= 3 with n >= 24).

Decision 0012: Benchmark alignment and benchmark-relative metrics
Status: Accepted — methodology 0.1.0

Problem
Benchmark-relative metrics require pairing fund and benchmark returns period by period. Date conventions differ (month-start vs month-end), histories overlap only partly, and libraries annualize alpha, active return, Treynor and capture ratios differently.

Known definitions
- CAPM/Jensen regression: `r_fund = a + b * r_bench + e` (excess returns).
- Tracking error: `std(r_fund - r_bench) * sqrt(P)`.
- Information ratio: active return / tracking error (arithmetic or geometric active return).
- Treynor: annualized (excess) return / beta.
- Up/down capture: fund vs benchmark performance in benchmark up/down periods (Morningstar compounds returns geometrically).

PerformanceAnalytics behavior
- `CAPM.beta`, `CAPM.alpha`: `lm` slope and per-period intercept on `na.omit(merge(...))`. Verified: beta and `alpha / 12` agree to 1e-9.
- `TrackingError`: `sd(Ra - Rb) * sqrt(scale)`. Verified.
- `InformationRatio` = `ActivePremium / TrackingError`, where `ActivePremium` is the difference of **geometric** annualized returns. Verified to differ from OpenFactsheet's arithmetic IR, which equals `mean(active) * 12 / (sd(active) * sqrt(12))` computed in R.
- `TreynorRatio` uses `Return.annualized` (geometric) / beta. Verified: PA's value equals the geometric annualized aligned fund return divided by beta, while OpenFactsheet's equals the arithmetic annualized mean divided by beta.
- `UpDownRatios(method = "Capture")` divides **sums** of returns; up periods are `Rb > 0`, down periods `Rb <= 0`. Verified: PA's up and down captures equal those sum ratios, and both differ from OpenFactsheet's geometric captures.

QuantStats behavior
`stats.information_ratio` reports the non-annualized per-period ratio. Verified: it equals OpenFactsheet's information ratio divided by sqrt(12).

Empyrical behavior
`beta` agrees to 1e-9 (verified). `alpha` annualizes the intercept geometrically, `(1 + a)^P - 1`; verified that empyrical equals `(1 + alpha_OFS / 12)^12 - 1`. `up_capture` / `down_capture` compound returns in up (`> 0`) / down (`< 0`) benchmark periods and annualize them; verified to 1e-9.

Academic sources
- Jensen (1968), *JoF* 23(2); Treynor (1965), *HBR* 43(1).
- Grinold & Kahn (2000); Goodwin (1998), *FAJ* 54(4).
- Morningstar Upside/Downside Capture Ratio methodology.

OpenFactsheet decision
Alignment (`compare/alignment.py`):

- Both dated: monthly/quarterly/yearly matched by **calendar period** (so month-start and month-end dates align), daily/weekly by exact date; only the intersection is used. Missing benchmark returns are **never filled**.
- The loss of observations is disclosed (`fund_only_periods`, `benchmark_only_periods`, overlap dates, coverage); coverage below 90% caps confidence at `moderate`.
- Different regular frequencies ⟶ `BenchmarkAlignmentError` (no resampling).
- Neither dated: paired by position only if lengths match; exactly one dated: lengths must match and the undated series is assumed to share the dates. Both cases add warnings; otherwise an error is raised.
- Benchmark metrics report the aligned observation count separately.

Metrics (rf = 0, 0005):

- `beta`, `alpha`: statsmodels OLS with intercept; `alpha = a * P` (arithmetic); details give per-period alpha, standard error, t and p (i.i.d. errors).
- `r_squared` (regression), `correlation` (Pearson).
- `tracking_error = std(active, ddof=1) * sqrt(P)`; `active_return = mean(active) * P`; `information_ratio = mean(active) / std(active) * sqrt(P)`.
- `treynor_ratio = mean(r_fund) * P / beta` on aligned periods; note added when beta < 0.3; beta = 0 ⟶ `not_applicable`.
- `upside_capture` / `downside_capture`: geometric annualized fund return over benchmark up (`> 0`) / down (`< 0`) periods divided by the same for the benchmark; zero-benchmark periods belong to neither.

Rationale
Calendar-period alignment handles real-world date conventions without inventing data. Arithmetic annualization keeps alpha, active return, IR and Sharpe (0008) mutually consistent. Geometric capture follows the Morningstar convention investors see in fund documents.

Consequences
- IR, active return, Treynor and capture values differ from PerformanceAnalytics; alpha differs from empyrical by the annualization method. The tests pin these relationships.
- A benchmark is never inferred; without one, all benchmark metrics are `not_applicable` with the reason stated.

Decision 0013: Percent-versus-decimal return-scale detection
Status: Accepted — methodology 0.1.0

Problem
`1.5, -0.8, 2.1` are probably percentages; `0.015, -0.008, 0.021` are probably decimals. Reading one as the other changes every metric by a factor of 100 (or more after compounding). A single threshold misclassifies low-volatility percent series and high-volatility decimal series.

Known definitions
No standard exists; libraries assume decimals.

PerformanceAnalytics behavior
Assumes decimals except `DrawdownPeak` (and therefore `UlcerIndex`, `PainIndex`, `MartinRatio`), which divides by 100, i.e. assumes percent.

QuantStats behavior
Assumes decimals; no detection.

Empyrical behavior
Assumes decimals; no detection.

Academic sources
None; this is an input-validation convention.

OpenFactsheet decision
A deterministic classifier (`normalization/scaling.py`) returns `detected_decimal`, `detected_percent` or raises `AmbiguousReturnScaleError`:

1. A scale declared by the caller (`return_scale="decimal" | "percent"`) always wins and is recorded as `declared_*`.
2. Explicit `%` labels in cells or column headers ⟶ percent.
3. All values zero ⟶ decimal (immaterial).
4. Let `m` be the median absolute non-zero value and `U` the plausible-decimal upper bound for the frequency: daily 0.05, weekly 0.10, monthly 0.20, quarterly 0.30, yearly 0.60, irregular/unknown 0.30 (`return_scale.decimal_median_abs_upper`).
   - `m > U`: percent, unless `m / 100 > U` as well, in which case an `InputError` is raised (implausible in either scale; if all values are positive the message suggests price/NAV levels).
   - `m <= U / 4` (`confident_divisor = 4`): decimal. Any values below -1 are kept and flagged as impossible, not rescaled.
   - `U / 4 < m <= U` (ambiguous zone): percent if any value is below -1 (impossible as a decimal return); otherwise `AmbiguousReturnScaleError` asking the caller to declare the scale.

Fund and benchmark are classified independently; a declared scale applies to both.

Rationale
The median absolute return is robust to a few extreme periods, and plausible magnitudes depend on frequency. Refusing to guess in the ambiguous zone follows the rule “do not silently guess in high-risk cases”.

Consequences
- Money-market-like percent series (e.g. 0.1% per month written as `0.1`) and very volatile decimal series (crypto) may require an explicit declaration.
- The evidence string is shown in the input diagnostics (`return_scale_evidence`), and a percent conversion is recorded as a transformation.
- Threshold changes are methodology changes.

Decision 0014: Return-smoothing risk: diagnose and warn, never unsmooth
Status: Accepted — methodology 0.1.0

Problem
Returns of illiquid or appraisal-priced portfolios (many hedge funds, private assets) are smoothed: reported marks lag true values. Smoothing induces positive serial correlation, understates volatility and inflates Sharpe-type ratios. Unsmoothing models exist but add model risk and would change reported returns.

Known definitions
- Sample autocorrelation `rho_k` (full-sample variance denominator).
- Getmansky, Lo & Makarov (2004): observed returns as a moving average of true returns; significant positive `rho_1` indicates smoothing/illiquidity.
- Ljung-Box Q test of joint zero autocorrelation.

PerformanceAnalytics behavior
Provides `table.Autocorrelation` and `Return.Geltner`/`Return.GLM`-style unsmoothing tools; no automatic warning. R's `acf` and `Box.test` were used as references: OpenFactsheet lag-1..3 autocorrelations agree to 1e-9 and the 3-lag Ljung-Box p-value to 1e-7 on all fixtures (verified).

QuantStats behavior
No smoothing diagnostics (reported in library docs, not verified here).

Empyrical behavior
No smoothing diagnostics.

Academic sources
- Getmansky, M., Lo, A. W., & Makarov, I. (2004). An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns. *JFE* 74(3).
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *FAJ* 58(4).
- Ljung, G. M., & Box, G. E. P. (1978). *Biometrika* 65(2).

OpenFactsheet decision
Smoothing risk (`diagnostics/smoothing.py`), with `z = rho_1 * sqrt(n)`:

Level	Rule
high	`rho_1 >= 0.30` and `z >= 1.96`
moderate	`rho_1 >= 0.15` and `z >= 1.645`
none	otherwise
not_assessable	fewer than 12 observations, or `rho_1` undefined

Effects on volatility-based metrics (volatility, annualized volatility, downside deviation, Sharpe, Sortino, tracking error, Information ratio): `high` forces confidence to `low` (status `low_confidence`) with a warning; `moderate` caps confidence at `moderate` with a warning. The interpretation adds that the figure is an optimistic bound. Path and tail metrics are not downgraded for smoothing.

Related stale-price rules: 10% or more exactly-zero returns caps volatility-based metrics at `moderate`; 3 or more identical consecutive returns is a finding. If smoothing is `none` but the Ljung-Box p-value is below 0.05, an informational finding is added. `autocorrelation_lag1..3` and `ljung_box_pvalue` (supplementary) are reported as metrics.

Returns are **never unsmoothed** in the canonical report.

Rationale
Warning preserves the reported data and makes the risk visible; unsmoothing would replace observed returns with model output. The thresholds combine an economically material effect size with statistical significance.

Consequences
- The thresholds are a transparent heuristic, versioned in `policy/defaults.py` (`smoothing`); changing them is a methodology change.
- An optional unsmoothing research module may be added outside the canonical report.

Decision 0015: Minimum-data policies and confidence
Status: Accepted — methodology 0.1.0

Problem
Every statistic can be computed from very little data, but many are unreliable until the sample is large. Hiding such metrics would be selective reporting; showing them without qualification overstates certainty.

Known definitions
Common practitioner rules of thumb: at least 36 monthly observations for risk-adjusted ratios; standard error of skewness about `sqrt(6/n)`, of kurtosis about `sqrt(24/n)`, of an autocorrelation about `1/sqrt(n)`. GIPS prohibits annualizing returns for periods shorter than one year.

PerformanceAnalytics behavior
No minimum-data policy; functions return a number or `NA`.

QuantStats behavior
No minimum-data policy.

Empyrical behavior
Returns `NaN` for too-short input; no confidence labels.

Academic sources
- Lo, A. W. (2002). *FAJ* 58(4).
- Joanes & Gill (1998); Ljung & Box (1978).
- CFA Institute (2020). GIPS.

OpenFactsheet decision
Each metric names a policy (`policy/thresholds.py`). Below the mathematical minimum the status is `insufficient_data` (no value). Otherwise the value is shown and confidence is set from the statistical thresholds:

Policy	Math min	Basis	Low below	Moderate below
observed_point	1	none	—	—
path_risk	1	years	3	5
location	1	years	3	5
dispersion	2	years	3	5
ratio_standard	2	years	3	5
skewness	3	observations	60	120
kurtosis	4	observations	120	240
tail	2	expected tail obs. (n x 5%)	5	10
autocorrelation	5	observations	36	60
ljung_box	10	observations	36	60
asymptotic_inference	4	observations	36	60
benchmark_regression	3	years of overlap	3	5
benchmark_simple	2	years of overlap	3	5

Additional rules (`policy/confidence.py`):
- Years basis with unknown years (undated, no declared frequency) ⟶ `low`.
- Annualized metric with less than one year of data ⟶ `low`, with a warning that the figure extrapolates a partial year (GIPS).
- Missing periods cap wealth-path and annualized metrics at `moderate`.
- Smoothing and stale-return rules: 0014.
- Benchmark coverage below 90% caps benchmark metrics at `moderate`.
- Status: `supplementary` for supplementary metrics; else `low_confidence` when confidence is `low`; else `applicable`.

Rationale
Separating “can it be computed” (applicability) from “how reliable is it” (confidence) lets the report compute everything while stating uncertainty. Thresholds are conventions, not laws of nature, so they are explicit and versioned.

Consequences
- Short track records show a full metric set, mostly labeled `low` confidence.
- Confidence describes estimation reliability, never investment quality.
- Changing any threshold requires a methodology version bump.

Decision 0016: Lower-partial-moment ratios and dispersion: target 0, full-sample denominators
Status: Accepted — methodology 0.1.0

Problem
Kappa 3, the upside potential ratio, downside potential (LPM1), semi-deviation and mean absolute deviation all need a target and a denominator, and each can be reported per period or annualized. Every one of these choices changes the numbers, so each has to be fixed canonically rather than left to the user.

Known definitions

Lower and upper partial moments of order k at target τ:

$$\mathrm{LPM}_k(\tau) = \frac{1}{n}\sum_{t=1}^{n}\max(\tau - r_t, 0)^k, \qquad \mathrm{UPM}_k(\tau) = \frac{1}{n}\sum_{t=1}^{n}\max(r_t - \tau, 0)^k$$

- Kappa of order k (Kaplan & Knowles, 2004): $K_k = (\bar{r} - \tau)/\mathrm{LPM}_k(\tau)^{1/k}$. The Sortino ratio is Kappa 2, and Omega − 1 is Kappa 1.
- Upside potential ratio (Sortino, van der Meer & Plantinga, 1999): $\mathrm{UPM}_1(\tau)/\sqrt{\mathrm{LPM}_2(\tau)}$.
- Semi-deviation: the downside deviation with the sample mean as target. Some sources divide by the number of below-mean periods instead of by n.
- Mean absolute deviation: $\frac{1}{n}\sum|\ r_t - \bar{r}|$.

PerformanceAnalytics behavior

- `Kappa(R, MAR, l) = (mean(R) - MAR) / ((1/n) Σ_{r<MAR} (MAR - r)^l)^(1/l)`, per period. **Verified:** `kappa_3_ratio = Kappa(MAR=0, l=3)`.
- `UpsidePotentialRatio(R, MAR = 0, method = c("subset", "full"))` defaults to `"subset"`, which divides by the number of gains and of losses. **Verified:** OpenFactsheet equals `method = "full"` and differs from the default (the difference is pinned by a test).
- `SemiDeviation(R) = DownsideDeviation(R, MAR = mean(R), method = "full")`, per period. **Verified:** `semi_deviation = SemiDeviation × sqrt(12)`.
- `DownsideDeviation(R, MAR = 0, potential = TRUE) = Σ_{r<0} (0 - r) / n`. **Verified** equal to `downside_potential`.
- `MeanAbsoluteDeviation(R) = Σ |R - mean(R)| / n`. **Verified.**

QuantStats behavior

No equivalents are used as references here.

Empyrical behavior

No equivalents are used as references here.

Academic sources

- Kaplan, P. D., & Knowles, J. A. (2004). Kappa: A Generalized Downside Risk-Adjusted Performance Measure. *Journal of Performance Measurement* 8(3).
- Sortino, F. A., van der Meer, R., & Plantinga, A. (1999). The Dutch Triangle. *Journal of Portfolio Management* 26(1).
- Sortino & Price (1994); Bacon (2021).

OpenFactsheet decision

- Target $\tau = 0$ per period (`lower_partial_moments.target`), the same as the Sortino target (0006); Kappa order 3 (`lower_partial_moments.kappa_order`).
- All partial moments divide by all n observations (`lower_partial_moments.denominator = "full_sample"`).
- `kappa_3_ratio`, `upside_potential_ratio`, `downside_potential` and `mean_absolute_deviation` are **per period** (not annualized).
- `semi_deviation` uses the sample mean as target and the n denominator, and is **annualized** by $\sqrt{P}$ like `downside_deviation`:

$$\mathrm{SemiDev}_{\mathrm{ann}} = \sqrt{P}\sqrt{\frac{1}{n}\sum_{t=1}^{n}\min(r_t - \bar{r}, 0)^2}$$

- Kappa 3 and the upside potential ratio are `not_applicable` when no return is below 0% (zero denominator).
- Kappa 3, the upside potential ratio, semi-deviation and MAD are volatility-based, so smoothing and stale-price diagnostics lower their confidence. Downside potential is not.

Rationale

The full-sample denominator is the original definition and matches Sortino and Omega, so these measures stay mutually consistent (Omega = 1 + mean / LPM1 at threshold 0). No convention exists for annualizing Kappa 3 or the upside potential ratio, so they are reported per period, like Omega. Semi-deviation is a volatility-type quantity and is annualized like the other deviations.

Consequences

- The upside potential ratio differs from PerformanceAnalytics' default output. The difference is documented and pinned.

- Per-period ratios are not comparable across data frequencies; the metric documentation says so.

Decision 0017: Drawdown-based ratios: Burke, Sterling, Pain; payoff ratio

Status: Accepted — methodology 0.1.0

Problem

Burke, Sterling and Pain ratios each divide an annualized return by a drawdown statistic. The literature does not agree on which drawdowns to use (episodes, runs of losing periods, averages of annual maxima), and the Sterling ratio adds an arbitrary 10%. The payoff ratio needs a rule for zero returns.

Known definitions

- **Burke** (Burke, 1994): $(r_p - r_f)/\sqrt{\sum_k D_k^2}$ over drawdowns D_k.
- **Sterling:** originally the average of annual maximum drawdowns minus 10%; the variant popularized by Kestner (1996) and Bacon is annualized return divided by (|max drawdown| + 10%).
- **Pain ratio** (Zephyr; see Bacon): $(r_p - r_f)/$ Pain Index.
- **Payoff ratio:** average winning period / |average losing period|.

PerformanceAnalytics behavior

- `BurkeRatio(R)` builds "drawdowns" from runs of consecutive negative returns, not from peak-to-recovery episodes, and multiplies returns by 0.01 (it assumes percent inputs). **Verified to differ.** OpenFactsheet agrees with `CAGR / sqrt(sum(depths^2))`, where the depths come from PA's own `findDrawdowns(R)$return < 0`.
- `SterlingRatio(R, excess = 0.1)` = `Return.annualized / |maxDrawdown + 0.1|`, where `maxDrawdown` is a positive magnitude. **Verified equal** to `sterling_ratio`.
- `PainRatio(R)` divides by `PainIndex`, which inherits the `DrawdownPeak` percent-input issue (0007). OpenFactsheet agrees with `CAGR / mean(abs(Drawdowns(R)))` (**verified**).

QuantStats behavior

`stats.payoff_ratio` = mean of positive returns / |mean of negative returns|. **Verified equal** to `payoff_ratio` on all fixtures.

Empyrical behavior

No equivalents are used as references here.

Academic sources

- Burke, G. (1994). A Sharper Sharpe Ratio. *Futures* 23(3).
- Kestner, L. N. (1996). Getting a Handle on True Performance. *Futures* 25(1).
- Bacon, C. R. (2021). *Practical Risk-Adjusted Performance Measurement.*

OpenFactsheet decision

All three drawdown ratios use the geometric CAGR, $r_f = 0$, the full history, and the canonical drawdown engine (0007), with episode depths $\delta_k = \min_{t \in E_k} D_t$:

$$\text{Burke} = \frac{\text{CAGR}}{\sqrt{\sum_{k=1}^{K} \delta_k^2}}, \qquad \text{Sterling} = \frac{\text{CAGR}}{|\text{MDD}| + 0.10}, \qquad \text{PR} = \frac{\text{CAGR}}{\text{PI}}$$

- The Sterling adjustment is `sterling.excess = 0.10` and is disclosed on every result as a convention, not an estimate.
- Burke and the Pain ratio are `not_applicable` when no drawdown occurred. Sterling is always defined, because its denominator is at least 0.10.
- Payoff ratio: arithmetic means of strictly positive and strictly negative periods. Zero returns are excluded from both. It is `not_applicable` without at least one gain and one loss.

Rationale

One episode engine keeps Burke consistent with maximum drawdown, the Ulcer Index and the Pain Index. Sterling follows the most widely implemented variant (PA/Bacon), which needs no choice of calendar window. The alternative, an average of annual maximum drawdowns, would require a partial-year rule and is left out.

Consequences

- Burke values differ from PerformanceAnalytics' `BurkeRatio`; the difference is pinned by a test.
- Sterling values are not comparable with Sterling ratios computed from averaged annual drawdowns.

Decision 0018: Parametric VaR: Gaussian and Cornish-Fisher with population moments

Status: Accepted — methodology 0.1.0

Problem

Parametric VaR needs a variance estimator (sample or population) and, in the Cornish-Fisher (modified) version, estimators for skewness and kurtosis. It is model-dependent, so it must not be presented as if it were an observed or historical estimate.

Known definitions

With $z = \Phi^{-1}(0.05)$, moment skewness $S = m_3/m_2^{3/2}$ and excess kurtosis $K = m_4/m_2^2 - 3$:

$$\mathrm{VaR}_{95}^{N} = \bar{r} + z\sqrt{m_2}, \qquad \mathrm{mVaR}_{95} = \bar{r} + h\sqrt{m_2}, \quad h = z + \frac{(z^2-1)S}{6} + \frac{(z^3-3z)K}{24} - \frac{(2z^3-5z)S^2}{36}$$

(Zangari, 1996; Favre & Galeano, 2002). Implementations differ in whether they use population moments (m_k, denominator n) or sample moments.

PerformanceAnalytics behavior

`VaR.Gaussian` uses `-mean(r) - qnorm(alpha) * sqrt(m2)` with the centered (population) second moment. `VaR.CornishFisher` uses population m_2, m_3, m_4 with `.skewEst` and `.kurtEst` (moment skewness and excess kurtosis). The public `VaR(method = ...)` returns a signed (negative) number. **Verified:** `gaussian_var_95` and `modified_var_95` equal `VaR(p = 0.95, method = "gaussian")` and `VaR(p = 0.95, method = "modified")` to 1e-9 on all fixtures.

QuantStats behavior

`stats.value_at_risk` is a normal-distribution VaR that uses the sample (n − 1) standard deviation. **Verified** to equal `norm.ppf(0.05, mean, std(ddof=1))`; it therefore differs slightly from `gaussian_var_95`.

Empyrical behavior

No parametric VaR is used as a reference here.

Academic sources

- Zangari, P. (1996). A VaR Methodology for Portfolios That Include Options. *RiskMetrics Monitor*.
- Favre, L., & Galeano, J.-A. (2002). Mean-Modified Value-at-Risk Optimization with Hedge Funds. *Journal of Alternative Investments* 5(2).
- Boudt, K., Peterson, B. G., & Croux, C. (2008). Estimation and Decomposition of Downside Risk for Portfolios with Non-Normal Returns. *Journal of Risk* 11(2).
- Jorion, P. (2006). *Value at Risk*.

OpenFactsheet decision

- Both VaRs use population moments (`parametric_var.moments = "population"`) at 95% (`parametric_var.confidence`). They are per period and signed (negative = loss), like the historical VaR (0010).
- Both carry status `supplementary` and nature `model_dependent`, and they are volatility-based (smoothing lowers their confidence). The historical VaR remains the headline tail estimate.
- Gaussian VaR uses the `dispersion` minimum-data policy. Modified VaR uses the `kurtosis` policy (at least 4 observations; low confidence below 120).
- Modified VaR is `not_applicable` when all returns are identical ($m_2 = 0$).

Rationale

Population moments follow Boudt, Peterson & Croux (2008), the published reference implementation. They also make the Cornish-Fisher quantile reduce *exactly* to the Gaussian one when $S = K = 0$, so the two parametric VaRs are mutually consistent. The small difference from the sample standard deviation used for volatility (0003) is disclosed here.

Consequences

- `gaussian_var_95` does not equal `mean + z × volatility` exactly (population vs sample standard deviation); the relative difference is about $1/(2n)$.
- For extreme skewness or kurtosis, the Cornish-Fisher expansion can become non-monotonic and unreliable. The metric documentation warns about this, and the value is shown for context, not as the canonical tail estimate.

Decision 0019: Sharpe-ratio inference: PSR, standard error, minimum track record length

Status: Accepted — methodology 0.1.0

Problem

A Sharpe ratio from a finite, non-normal sample is uncertain. Measuring that uncertainty requires choosing a variance formula for the estimator, a reference Sharpe ratio, confidence levels, moment estimators, and whether the result is per period or annualized.

Known definitions

For the per-period ratio $\widehat{SR} = \bar{r}/s$, the asymptotic i.i.d. variance allowing for skewness g_3 and (non-excess) kurtosis g_4 is (Mertens, 2002; Opdyke, 2007):

$$\operatorname{Var}(\widehat{SR}) \approx \frac{1 - g_3\widehat{SR} + \frac{g_4-1}{4}\widehat{SR}^2}{n-1}$$

Bailey & López de Prado (2012) build on this variance:

$$\mathrm{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - g_3\widehat{SR} + \frac{g_4-1}{4}\widehat{SR}^2}}\right), \qquad \mathrm{MinTRL} = 1 + \left(1 - g_3\widehat{SR} + \tfrac{g_4-1}{4}\widehat{SR}^2\right)\left(\frac{z_{1-\alpha}}{\widehat{SR} - SR^*}\right)^2$$

The Deflated Sharpe Ratio additionally corrects for the number of strategies tried.

PerformanceAnalytics behavior

`ProbSharpeRatio` and `MinTrackRecord` use the formulas above with moment skewness and moment kurtosis, but the default `ignore_kurtosis = TRUE` fixes $g_4 = 3$. **Verified:** with `refSR = 0` and `ignore_kurtosis = FALSE`, OpenFactsheet's `probabilistic_sharpe_ratio` and `minimum_track_record_length` agree to 1e-9 on the fixtures with a positive Sharpe ratio. When `refSR > SR`, PerformanceAnalytics stops with an error; on the negative fixture OpenFactsheet reports PSR < 0.5 and MinTRL as `not_applicable` (both asserted). PA's `sr_confidence_interval` uses `qnorm(p)` with p = 0.95, which gives a 90% two-sided interval on the per-period Sharpe ratio.

QuantStats behavior

Not used as a reference for these statistics.

Empyrical behavior

Not used as a reference for these statistics.

Academic sources

- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *Financial Analysts Journal* 58(4).
- Mertens, E. (2002). Comments on Variance of the IID Estimator in Lo (2002).
- Opdyke, J. D. (2007). Comparing Sharpe Ratios: So Where Are the p-Values? *Journal of Asset Management* 8(5).
- Bailey, D. H., & López de Prado, M. (2012). The Sharpe Ratio Efficient Frontier. *Journal of Risk* 15(2).

OpenFactsheet decision

- $\widehat{SR}$ = per-period mean / sample standard deviation, with $r_f = 0$. $g_3 = m_3/m_2^{3/2}$ and $g_4 = m_4/m_2^2$ are population moments, and kurtosis is **not** ignored (`sharpe_inference.moments`).
- `probabilistic_sharpe_ratio`: $SR^* = 0$ (`sharpe_inference.reference_sharpe`). It is defined for any sign of $\widehat{SR}$.
- `sharpe_ratio_standard_error`: $\sqrt{P}\sqrt{\text{factor}/(n-1)}$. Its details give the two-sided **95%** interval $\widehat{SR}\sqrt{P} \pm z_{0.975}\,\mathrm{SE}$ (`sharpe_inference.ci_level`).
- `minimum_track_record_length`: 95% one-sided (`z_{0.95}`, `sharpe_inference.min_track_record_confidence`), in periods, with years and sufficiency in the details. It is `not_applicable` when $\widehat{SR} \le 0$.
- All three are `not_applicable` if $s = 0$ or the variance factor is not positive. They use the new minimum-data policy `asymptotic_inference` (mathematical minimum 4; low confidence below 36 observations, moderate below 60).
- They are volatility-based: serial correlation is not modelled, so smoothing risk lowers their confidence (0014).
- The Deflated Sharpe Ratio is **not computed**: it requires the number of trials, which a single return series does not reveal.

Rationale

Keeping kurtosis follows the paper. Setting $g_4 = 3$ would overstate confidence for fat-tailed strategies, which contradicts "humble about inference". $SR^* = 0$ is the only reference value that needs no judgment. A true two-sided 95% interval is the conventional reading of "95% CI".

Consequences

- PSR and MinTRL differ from PerformanceAnalytics defaults on fat-tailed series. They agree once `ignore_kurtosis = FALSE` is set.
- The interval is wider than PerformanceAnalytics' 90% interval.
- The i.i.d. variance understates uncertainty for smoothed or serially correlated returns (Lo, 2002). The metric documentation and confidence labels disclose this.

Decision 0020: Mean-return inference and the Jarque-Bera normality test

Status: Accepted — methodology 0.1.0

Problem

A confidence interval for the mean return needs a choice between a normal and a Student-t quantile, and between one- and two-sided intervals. The Jarque-Bera test can be computed from bias-corrected or population moments, and the choice changes the statistic.

Known definitions

- Mean standard error $s/\sqrt{n}$. Student-t interval $\bar{r} \pm t_{0.975,\,n-1}\, s/\sqrt{n}$ and t-statistic $t = \bar{r}/(s/\sqrt{n})$ (Student, 1908), exact for normal i.i.d. data.
- Jarque & Bera (1987):

$$JB = \frac{n}{6}\left(g_1^2 + \frac{g_2^2}{4}\right), \quad g_1 = \frac{m_3}{m_2^{3/2}}, \quad g_2 = \frac{m_4}{m_2^2} - 3, \quad p = 1 - F_{\chi^2_2}(JB)$$

The original test uses population (biased) moments. Some variants substitute bias-corrected G_1 and G_2.

PerformanceAnalytics behavior

PerformanceAnalytics has no function for these statistics. The R harness instead uses R's own `t.test(r)` as an independent reference, and computes Jarque-Bera manually from `skewness(method = "moment")` and `kurtosis(method = "excess")`. **Verified:** the CI bounds, t-statistic and p-value of `mean_return_standard_error` equal `t.test` to 1e-9 (p-value to 1e-7), and the Jarque-Bera statistic and p-value equal the manual R computation.

QuantStats behavior

Not used as a reference for these statistics.

Empyrical behavior

Not used as a reference. SciPy is used instead: `scipy.stats.jarque_bera` agrees with `jarque_bera_pvalue` (statistic to 1e-9, p-value to 1e-7), and `scipy.stats.ttest_1samp(r, 0)` agrees with the reported p-value (**verified**).

Academic sources

- Student [Gosset, W. S.] (1908). The Probable Error of a Mean. *Biometrika* 6(1).
- Jarque, C. M., & Bera, A. K. (1987). A Test for Normality of Observations and Regression Residuals. *International Statistical Review* 55(2).

OpenFactsheet decision

- `mean_return_standard_error` (per period) is $s/\sqrt{n}$. Its details give the two-sided **95%** Student-t interval with $n - 1$ degrees of freedom (`mean_inference.ci_level`), the t-statistic and the two-sided p-value against a zero mean. It is `not_applicable` when $s = 0$, uses the `dispersion` minimum-data policy, and is volatility-based.
- `jarque_bera_pvalue` uses population moments and the asymptotic χ^2_2 p-value, and reports the statistic in its details. It is `supplementary`, `model_dependent`, uses the `skewness` minimum-data policy (low confidence below 60 observations), and is `not_applicable` when $m_2 = 0$.

Rationale
The Student-t interval is exact under normality and reduces to the normal interval for large n, so short histories get honestly wide intervals. Jarque-Bera follows the original definition, which matches SciPy and R practice, so users can reproduce the value.

Consequences
- Neither statistic models serial correlation. Smoothed series get intervals that are too narrow, which the smoothing diagnostics flag.
- The Jarque-Bera test has poor size and low power in small samples, and flags trivial departures in very large ones. It is a diagnostic, not a performance measure, and the report labels it supplementary.

Decision 0021: Drawdown tail (DaR, CDaR, EDaR) and entropic VaR
Status: Accepted — methodology 0.1.0

Problem
Tail measures of the drawdown path (Drawdown-at-Risk, Conditional Drawdown-at-Risk, Entropic Drawdown-at-Risk) and the entropic VaR of returns need four choices: which drawdown path (compounded or uncompounded, with or without the inception peak), which periods form the sample, which quantile estimator, and how to compute the entropic infimum.

Known definitions
With losses X and tail probability $a = 0.05$:

$$\mathrm{DaR}_{95} = Q_{0.95}(-D), \qquad \mathrm{CDaR}_{95} = \mathbb{E}[-D \mid -D \geq \mathrm{DaR}_{95}], \qquad \mathrm{EVaR}_a(X) = \inf_{z>0} z \ln \frac{\mathbb{E}[e^{X/z}]}{a}$$

and $\mathrm{EDaR}_a = \mathrm{EVaR}_a(-D)$ (Chekhlov, Uryasev & Zabarankin 2005; Ahmadi-Javid 2012; Cajas 2021). Libraries differ in the drawdown path (uncompounded cumulative sums vs compounded wealth) and in the discrete quantile / CVaR estimator.

PerformanceAnalytics behavior
`CDD` ("Conditional Drawdown") is the 5% quantile over all `findDrawdowns` runs (drawdown episodes and the zero-return runs between them), inverted to a positive number; it is not a conditional expectation of the drawdown path (verified in the R harness and pinned by a test). No DaR / EDaR / EVaR.

QuantStats behavior
No drawdown tail or entropic measures.

Empyrical and skfolio behavior
empyrical has none. skfolio 1.0.6 (`skfolio.measures`) was compared in `tests/cross_library/test_python_references.py::test_skfolio_drawdown_and_entropic_measures`:

- `get_drawdowns(r, compounded=True)` starts its running peak at W_1, so a first-period loss is not a drawdown; the default is `compounded=False` (uncompounded). **Verified** (path pinned).
- `drawdown_at_risk` equals `numpy.quantile(-D, 0.95, method="inverted_cdf")` (lower order statistic). **Verified.**
- `cdar` uses a fractional Rockafellar–Uryasev weighting; it equals the OpenFactsheet value exactly when $(1 - 0.95)\,n$ is an integer. **Verified** on the fixtures where it is ($n = 60$ and $n = 120$).
- `edar` and `evar` agree with OpenFactsheet to relative 1e-7 (solver tolerance) on all fixtures. **Verified.**

Academic sources
- Chekhlov, A., Uryasev, S., & Zabarankin, M. (2005). Drawdown Measure in Portfolio Optimization. *IJTAF* 8(1).
- Ahmadi-Javid, A. (2012). Entropic Value-at-Risk. *JOTA* 155(3).
- Cajas, D. (2021). Entropic Portfolio Optimization. SSRN 3792520.

OpenFactsheet decision
- Path: the canonical compounded drawdowns $D_t = W_t/M_t - 1$, $t = 1..n$, with $W_0 = 1$ counting as a peak (0007); the inception point itself is not a sample observation.
- `drawdown_at_risk_95` $= Q_{0.05}(D)$ with linear interpolation, and `cdar_95` $= $ mean of $D_t \leq Q_{0.05}(D)$ — the historical VaR / ES estimators of 0010. Signed (negative = loss).

- `edar_95` $= -\mathrm{EVaR}_{0.05}(-D)$ and `evar_95` $= -\mathrm{EVaR}_{0.05}(-r)$, computed by `utils.math.entropic_value_at_risk`: losses scaled to unit maximum absolute value (EVaR is positively homogeneous), bounded Brent search over $\log z \in [-25, 12]$ (`xatol` 1e-12), result capped at the maximum loss; zero losses give 0 and constant losses give that constant.
- Confidence 95% (`drawdown_risk.confidence`, `entropic.confidence`). All four use the `tail` minimum-data policy; the three drawdown measures are wealth-path metrics.

Rationale
One estimator for every historical tail measure keeps DaR and CDaR consistent with VaR and ES and makes CDaR literally "Expected Shortfall of the drawdown path". Counting W_0 as a peak matches the rest of the drawdown module.

Consequences
- The coherent ordering $\mathrm{MDD} \leq \mathrm{EDaR} \leq \mathrm{CDaR} \leq \mathrm{DaR} \leq 0$ is expected (EVaR dominates the fractional CVaR, which in turn is at least as severe as the `<=`-quantile mean); it is asserted in a unit test.
- With $n \leq 1/a = 20$ observations the empirical EVaR equals the worst loss, so `evar_95` equals the worst period and `edar_95` the maximum drawdown (asserted in `tests/unit/metrics/test_phase2b.py`; stated on the metric pages).
- Drawdown observations are strongly serially dependent: the 5% tail often comes from one episode.

Decision 0022: Modified (Cornish-Fisher) Expected Shortfall
Status: Accepted — methodology 0.1.0

Problem
A parametric Expected Shortfall that accounts for skewness and kurtosis complements the Cornish-Fisher VaR (0018). It needs a moment convention, an expansion order and a rule for cases where the expansion breaks down.

Known definitions
Boudt, Peterson & Croux (2008) with $a = 0.05$, $z = \Phi^{-1}(a)$, moment skewness $S = m_3/m_2^{3/2}$, excess kurtosis $K = m_4/m_2^2 - 3$ and the Cornish-Fisher quantile h of `modified_var_95`:

$$\mathrm{mES}_{95} = \bar{r} - \frac{\sqrt{m_2}}{a}\,\varphi(h)\left[1 + \frac{h^3 S}{6} + \frac{(h^6 - 9h^4 + 9h^2 + 3)S^2}{72} + \frac{(h^4 - 2h^2 - 1)K}{24}\right]$$

PerformanceAnalytics behavior
`ES.CornishFisher` implements this expression with population moments (`centeredmoment`, `.skewEst`, `.kurtEst`) and the full expansion by default (`c = 2`); `ES(method = "modified")` returns a signed (negative) number. **Verified:** `modified_es_95` equals `ES(p = 0.95, method = "modified")` to 1e-9 on all fixtures (`tests/cross_library/test_performanceanalytics.py::test_phase2b_agreement`).

QuantStats behavior
`stats.cvar` conditions on a parametric normal VaR threshold; it is not a Cornish-Fisher ES and is not compared.

Empyrical behavior
None.

Academic sources
- Boudt, K., Peterson, B. G., & Croux, C. (2008). Estimation and Decomposition of Downside Risk for Portfolios with Non-Normal Returns. *Journal of Risk* 11(2).
- Zangari, P. (1996); Favre, L., & Galeano, J.-A. (2002) — Cornish-Fisher VaR.

OpenFactsheet decision
- Population moments and the full second-order expansion (`modified_es.moments = "population"`, `modified_es.confidence = 0.95`), the same moment convention as the parametric VaRs of 0018.
- Signed, per period; status `supplementary`, nature `model_dependent`, volatility-based; `kurtosis` minimum-data policy (at least 4 observations, low confidence below 120).
- When the computed mES is *less severe* than the modified VaR (mES > mVaR in signed terms), a note is attached: the expansion is unreliable for these moments and the historical Expected Shortfall should be relied on.
- `not_applicable` when returns are numerically constant (variance zero or below floating-point resolution).

Rationale
The Boudt–Peterson–Croux form is the published reference implementation and is consistent with the Cornish-Fisher VaR already reported. Flagging the non-monotone case is more honest than silently clipping mES to mVaR.

Consequences
- The historical ES (0010) remains the headline tail estimate; mES is shown for context.
- For extreme skewness or kurtosis the expansion misbehaves; the note makes this visible in the report.

Decision 0023: Calendar-year statistics: complete years only
Status: Accepted — methodology 0.1.0

Problem
Best / worst calendar year and the share of positive years depend on how partial years are treated (CLAUDE.md §41, "monthly table partial years"). A partial first or last year — or a year with a missing month — is not a calendar-year return, and including it would let the inception date shape the statistics.

Known definitions
A calendar-year return compounds the period returns of the year:

$$R_y = \prod_{t \in y}(1 + r_t) - 1$$

GIPS requires annual returns for complete calendar years and prohibits presenting partial periods as annual returns.

PerformanceAnalytics behavior
`apply.yearly(R, Return.cumulative)` / `table.CalendarReturns` compound every year present, including partial years. **Verified** on the monthly fixtures: restricted to years with 12 monthly observations, the R values equal OpenFactsheet's `calendar_year_returns` to 1e-9, and best / worst match (`tests/cross_library/test_performanceanalytics.py::test_phase2b_agreement`).

QuantStats behavior
Yearly aggregation of all years present (reported in library docs, not verified here).

Empyrical behavior
`aggregate_returns(..., "yearly")` compounds all years present (reported in library docs, not verified here).

Academic sources
- CFA Institute (2020). *GIPS Standards* — annual returns for complete calendar years.

OpenFactsheet decision
A calendar year counts only when the observations cover it (`MetricContext.calendar_years`; key `calendar_years` in `policy/defaults.py`):

Frequency	Complete year
monthly	all 12 calendar months observed (a missing month makes it partial)
quarterly	all 4 calendar quarters observed
yearly	the year's observation
daily, weekly	first observation on or before 1 January + 7 days and last on or after 31 December − 7 days (`daily_weekly_edge_tolerance_days = 7`)

- Each observation belongs to the calendar year of its (period-end) date.
- Partial years are listed in `details.excluded_partial_years` and in a warning ("Partial calendar years excluded: …") and never enter the statistics.
- `best_calendar_year`, `worst_calendar_year` and `positive_calendar_years_pct` (a zero year counts as not positive) use complete years only; details carry every complete year's return.
- Undated series, or an irregular / unknown frequency ⟶ `not_applicable`; no complete year ⟶ `not_applicable`, naming the partial years.
- Wealth-path metrics; `path_risk` minimum-data policy.

Rationale
Complete years are objective and match GIPS. The 7-day tolerance for daily and weekly data absorbs holidays and week-end dating without accepting a year that is visibly missing weeks at either end.

Consequences
- A history starting in July has its first calendar year excluded; it still contributes to every full-history metric.
- For weekly data a week straddling 31 December is assigned to the year of its end date, so calendar-year returns are approximate at the boundary.
- Quarterly and daily rules are covered by unit tests (`tests/unit/metrics/test_phase2b.py`); only monthly years are compared with R.

Decision 0024: Rolling one-year windows
Status: Accepted — methodology 0.1.0

Problem
Rolling statistics depend on the window length and step (CLAUDE.md §12.8: rolling periods must come from canonical policy, never from the user). A chosen window or start date could be tuned to flatter a fund.

Known definitions
With window w observations, the rolling compounded return ending at t is

$$R_t^{(w)} = \prod_{i=t-w+1}^{t} (1 + r_i) - 1, \qquad t = w, \ldots, n$$

PerformanceAnalytics behavior
`apply.rolling` / `zoo::rollapply` compute rolling functions with a caller-chosen width. **Verified:** with width 12 on the monthly fixtures, `rollapply(R, 12, function(x) prod(1 + x) - 1, align = "right")` (padding removed) gives the same minimum, maximum, share of positive windows and number of windows as OpenFactsheet, to 1e-9 (`tests/cross_library/test_performanceanalytics.py::test_phase2b_agreement`).

QuantStats behavior
Rolling metrics with caller-chosen windows (reported in library docs, not verified here).

Empyrical behavior
`roll_*` functions with a caller-chosen window (reported in library docs, not verified here).

Academic sources
- Bacon, C. R. (2021). *Practical Risk-Adjusted Performance Measurement.*

OpenFactsheet decision
- Window $w = \text{round}(P \times 1)$ observations (`rolling.window_years = 1`): 12 monthly, 4 quarterly, 52 weekly, 252 business-daily, 365 calendar-daily, 1 yearly. Step: one observation (`rolling.step_periods = 1`).
- Every start date is used; `best_rolling_year`, `worst_rolling_year` and `positive_rolling_year_pct` (a zero window counts as not positive) summarize all $n - w + 1$ windows. Details give the number of windows, w, and the end date of the best / worst window.
- $n < w \longrightarrow$ `not_applicable`. The metrics are flagged annualized (they need a regular frequency) and wealth-path; `path_risk` minimum-data policy.

Rationale
A single, frequency-derived window with every start date leaves nothing to choose: no favorable start date or window length can be selected.

Consequences
- Windows overlap, so rolling statistics are strongly autocorrelated; they are descriptive, not probability estimates.
- Windows count observations, not calendar time: with missing periods a window spans more than one year (the gap warning of the confidence policy applies).
- The worst rolling year is typically worse than the worst calendar year (0023).

Decision 0025: Episode durations, recovery factor and path summaries

Status: Accepted — methodology 0.1.0

Problem

Duration statistics must say how open (unrecovered) episodes are treated; recovery factor has competing numerators; the geometric mean and interquartile range need a stated estimator.

Known definitions

For episodes $k = 1..K$ with peak π_k, trough τ_k (first minimum), recovery ρ_k and depth δ_k (0007):

$$\ell_k = \begin{cases} \rho_k - \pi_k & \text{recovered} \\ n - \pi_k & \text{open} \end{cases}, \qquad \mathrm{RF} = \frac{W_n - 1}{|\ \mathrm{MDD}|}, \qquad \bar{r}_g = W_n^{1/n} - 1$$

PerformanceAnalytics behavior

`findDrawdowns` returns per-episode `return`, `length`, `peaktotrough` and `recovery`. It counts an episode still open at the end of the data one period longer (`length` and `recovery`). **Verified:** after that +1 adjustment, the median depth, mean length, maximum peak-to-trough and mean recovery (recovered episodes only) equal OpenFactsheet's values to 1e-9; when the only episode is open, `average_recovery_duration` is asserted `not_applicable`. `mean.geometric` equals `geometric_mean_return` and R `IQR` (type 7) equals `interquartile_range` (`test_phase2b_agreement`).

QuantStats behavior

`stats.recovery_factor` divides the **arithmetic sum** of returns by the absolute maximum drawdown. **Verified** to differ from OpenFactsheet (`test_quantstats_recovery_factor_uses_arithmetic_sum`).

Empyrical behavior

None of these statistics are compared.

Academic sources

- Bacon, C. R. (2021). *Practical Risk-Adjusted Performance Measurement.*
- Hyndman, R. J., & Fan, Y. (1996). Sample Quantiles in Statistical Packages.

OpenFactsheet decision

- `median_drawdown`: median of episode depths, signed; 0 when $K = 0$.
- `average_drawdown_duration`: mean ℓ_k over **all** episodes, open ones measured to the last observation; a note flags the result as a lower bound when an episode is open; 0 when $K = 0$.
- `max_peak_to_trough_duration`: $\max_k(\tau_k - \pi_k)$; 0 when $K = 0$.
- `average_recovery_duration`: mean $\rho_k - \tau_k$ over **recovered** episodes only; open episodes are excluded and counted in details; `not_applicable` when no episode has recovered.
- `recovery_factor`: compounded cumulative return $/\ |\ \mathrm{MDD}|$, not annualized; `not_applicable` when MDD = 0.
- `geometric_mean_return`: $W_n^{1/n} - 1$ per period (-1 after a total loss).
- `interquartile_range`: $Q_{0.75} - Q_{0.25}$ with linear interpolation, per period (volatility-based).
- Durations are in observation periods; details add years when P is known.

Rationale

Open episodes have a known lower bound on duration but no recovery time, so the two averages treat them differently and say so. The compounded numerator of the recovery factor is the investor's actual net gain.

Consequences

- Average duration can rise as an open drawdown lengthens; the note makes this visible.
- The recovery factor grows with track-record length; compare only equal histories.

Decision 0026: Canonical charts and the drawdown-episode table

Status: Accepted — methodology 0.1.0

Problem

Charts are where selective presentation is easiest: a start date that skips a crash, a truncated value axis that turns a 2% difference into a cliff, a log scale that flattens a drawdown, a heatmap whose red is paler than its blue, or a "top drawdowns" table that quietly shows the recent ones. The canonical report needs charts, but every presentation choice that changes what a reader sees must be fixed by the methodology, not by the report author.

Known definitions

There is no academic standard for factsheet charts. Practitioner tear sheets converge on the same set:

Chart	pyfolio-reloaded	QuantStats	PerformanceAnalytics
Cumulative wealth	`plot_rolling_returns` (log option)	`snapshot`, `returns` (log option)	`chart.CumReturns` (`wealth.index`)
Underwater / drawdown	`plot_drawdown_underwater`	`drawdown`	`chart.Drawdown`
Top drawdown episodes	`show_worst_drawdown_periods` (top 5)	`drawdown_details` (all)	`table.Drawdowns` (top 5)
Calendar-year returns	`plot_annual_returns`	`yearly_returns`	`table.CalendarReturns`
Monthly heatmap	`plot_monthly_returns_heatmap`	`monthly_heatmap`	`table.CalendarReturns`
Return distribution	`plot_monthly_returns_dist`	`histogram`, `distribution`	`chart.Histogram`
Rolling statistics	rolling volatility, Sharpe, beta	rolling volatility, Sharpe, Sortino, beta	`charts.RollingPerformance`

PerformanceAnalytics behavior

`chart.CumReturns` plots `cumprod(1 + r)` from the first return (optionally with `begin = "axis"` to align series); `chart.Drawdown` plots the same compounded drawdown path OpenFactsheet uses (0007); `table.Drawdowns(top = 5)` sorts episodes by depth. Reported in library docs, not verified numerically here (charts are presentation of values validated elsewhere).

QuantStats behavior

The monthly heatmap uses a diverging colormap centred on zero but scaled to the data range, so the color for a loss and an equal gain can differ when the range is asymmetric. Cumulative-return plots offer a log scale and "compounded" or "summed" modes. Reported in library docs, not verified here.

Empyrical behavior

Empyrical has no charts; pyfolio-reloaded (built on it) shows the five worst drawdown periods and a log-scale option for cumulative returns.

Academic sources

- Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.) — the lie factor; data-ink.
- Cleveland, W. S., & McGill, R. (1984). Graphical Perception. *JASA* 79(387) — position on a common scale is read most accurately; length requires a zero baseline.
- Freedman, D., & Diaconis, P. (1981). On the histogram as a density estimator. *Z. Wahrscheinlichkeitstheorie* 57 — bin width $2\,\mathrm{IQR}\,n^{-1/3}$.
- CFA Institute, *Global Investment Performance Standards* (2020) — presentation of the full track record and of every required period.

OpenFactsheet decision

Eight charts, always in this order, each either drawn or replaced by a sentence stating why it cannot be drawn:

1. **Growth of 1** — the canonical wealth index $W_0 = 1, W_t = \prod_{s \le t}(1 + r_s)$ over the full history, linear scale.
2. **Calendar-year returns** — complete years from the calendar-year engine (0023); partial years are drawn outlined, marked $*$ and captioned with their coverage; bars grow from 0%.
3. **Monthly returns** (dated monthly data only) — every month, empty cells for missing months, a diverging scale **symmetric** at $\pm \max_t |\ r_t|$ and a compounded Year column.
4. **Rolling one-year volatility** — annualized sample standard deviation over the one-year windows of 0024; the value axis includes 0.
5. **Underwater chart** — the canonical drawdown path D_t (including inception), with the maximum drawdown marked.
6. **Distribution of returns** — every return counted (no trimming), Freedman–Diaconis bin width clipped to 10–40 bins, bin edges aligned at 0% so no bin mixes gains and losses, a normal curve with the sample mean and volatility, and the historical 95% VaR marked.
7. **Rolling one-year return** — exactly the window returns behind `best_rolling_year`, `worst_rolling_year` and `positive_rolling_year_pct`.
8. **Growth of 1 over the benchmark overlap** — only the aligned periods (0012), both series rebased to 1 immediately before the first common period, one linear axis; not drawn without a benchmark.

Rules common to all charts:

- the full analyzed history is shown; no start or end date is chosen;
- linear value scales; bars and magnitudes (volatility, counts) always include 0;
- one value axis per chart; straight segments between observations (no smoothing);
- identity colors (fund, benchmark) and polarity colors (gain, loss) are fixed theme tokens, validated for color-vision deficiency, in light and dark themes;
- every chart has a caption stating period, observation count and the conventions above, plus a data table with every plotted value; hover tooltips only repeat values that are also shown;
- when serial correlation suggests smoothing (0014), the wealth and rolling-volatility captions say the path may look smoother than the underlying value changes;
- the HTML file is self-contained: no external scripts, fonts or images.

Drawdown-episode table. Every renderer (terminal, Markdown, HTML) lists the **10 deepest** episodes of the canonical episode engine, deepest first, ties broken by the earlier peak, with peak, trough and recovery dates, depth and the three durations in periods; open episodes are labelled. The caption states how many episodes exist in total; the JSON report lists all of them.

Rationale

Every chart is a view of a quantity the metric engine already computes, so a chart can never disagree with the table beside it and adds no new tunable. The conventions that most often mislead — truncated axes, asymmetric color scales, log scales hiding drawdowns, dropped outliers, omitted partial years, a benchmark drawn from a flattering start — are fixed rather than left to the report author. Sorting episodes by depth means a limited table can only omit shallower episodes, never a deeper one. A linear scale is chosen over log because the audience includes retail investors and because the underwater chart already shows proportional losses.

Consequences

- Long, strongly compounding histories look back-loaded on the linear wealth chart; the underwater chart and calendar-year bars show the proportional picture.
- The heatmap of a series with one extreme month looks pale elsewhere; that is the honest consequence of a symmetric, data-wide scale.
- Charts are not drawn in Markdown or terminal output; those renderers point to the HTML report and `report.charts()` (SVG strings keyed by chart id).
- Adding, removing or restyling a canonical chart in a way that changes what is shown is a methodology change.

Decision 0027: Benchmark-relative additions: capture ratio, batting average, residual risk, appraisal ratio, M², regime betas

Status: Accepted — methodology 0.1.0

Problem

CLAUDE.md §12.9 lists benchmark-relative measures beyond those of decision 0012: capture ratio, batting average, bull/bear market sensitivity "if objectively defined", residual volatility, the appraisal ratio (§12.3) and M² (§12.3). Libraries disagree on denominators (m vs m − 2), on how alpha is annualized, and on how up/down periods are defined. Several of these ratios have denominators that can be zero, negative or pure rounding noise.

Known definitions

With aligned fund returns r_t, benchmark returns b_t, $t = 1..m$, and the single-index OLS fit $r_t = \hat{\alpha} + \hat{\beta} b_t + \hat{\varepsilon}_t$:

- **Residual (specific) risk:** $\hat{\sigma}_\varepsilon$, the standard deviation of $\hat{\varepsilon}_t$ (with m or m − 2 in the denominator).
- **Appraisal ratio** (Treynor & Black 1973): $\alpha / \sigma_\varepsilon$.
- **M²** (Modigliani & Modigliani 1997): the fund's Sharpe ratio times the benchmark's volatility, plus the risk-free rate: the fund's return at the benchmark's risk.
- **Batting average** (Morningstar): share of periods in which the fund beat the benchmark.
- **Capture ratio:** upside capture / downside capture.
- **Bull / bear beta** (Fabozzi & Francis 1977): betas estimated separately on periods when the benchmark rose / fell; **timing ratio** = bull / bear beta.

PerformanceAnalytics behavior

Function bodies of PerformanceAnalytics 2.0.4 were read (conventions only, no code copied) and values computed on the `benchmark_case` fixture in the R harness:

- `SpecificRisk`: `sqrt(sum((e - mean(e))^2) / m) * sqrt(P)`, the **m-denominator** residual standard deviation. Verified: equals OpenFactsheet's residual volatility x sqrt((m − 2) / m).
- `AppraisalRatio(method = "appraisal")`: `CAPM.jensenAlpha / SpecificRisk`, where `CAPM.jensenAlpha = Rp − Rf − beta (Rb − Rf)` uses **geometric annualized** fund and benchmark returns. Verified numerically.
- `Modigliani`: per-period `SharpeRatio(Ra, FUN = "StdDev") * StdDev(Rb) + mean(Rf)`. Verified: equals OpenFactsheet's M² / 12.
- `CAPM.beta.bull` / `CAPM.beta.bear`: `lm` with intercept on the subsets `Rb > 0` / `Rb < 0` (excess of Rf = 0). `TimingRatio` = bull / bear. All verified to 1e-9.
- `UpDownRatios(method = "Percent")`: share of up (`Rb > 0`) or down (`Rb < 0`) periods in which `Ra > Rb` (strict). Verified against the details of `batting_average`.

QuantStats behavior

No appraisal ratio, M², batting average, regime betas or capture ratio in QuantStats 0.0.81 (not verified further).

Empyrical behavior

`up_down_capture = up_capture / down_capture` with geometric annualized captures. Verified to 1e-9 against `capture_ratio`. empyrical has no appraisal ratio, M², batting average or regime betas.

Academic sources

- Treynor, J. L., & Black, F. (1973). *Journal of Business* 46(1).
- Modigliani, F., & Modigliani, L. (1997). *Journal of Portfolio Management* 23(2).
- Fabozzi, F. J., & Francis, J. C. (1977). *Journal of Finance* 32(4).
- Grinold & Kahn (2000); Bacon (2021); Morningstar capture and batting-average definitions.

OpenFactsheet decision

All metrics use the aligned overlap of decision 0012, rf = 0 (0005) and arithmetic annualization.

- `residual_volatility` $= \hat{\sigma}_\varepsilon \sqrt{P}$ with $\hat{\sigma}_\varepsilon^2 = \sum \hat{\varepsilon}_t^2 / (m-2)$ (the regression standard error, consistent with the OLS standard errors already reported for alpha and beta). Details give the systematic volatility $|\hat{\beta}|\, s_b \sqrt{P}$. Model-dependent.
- `appraisal_ratio` $= P\hat{\alpha} / (\hat{\sigma}_\varepsilon \sqrt{P})$. Undefined when $\hat{\sigma}_\varepsilon \le \sqrt{\epsilon_{\text{mach}}}\, s_r$, i.e. $1 - R^2$ below machine epsilon: the benchmark explains the fund exactly and the ratio would be rounding noise.
- `m_squared` $= P\,(\bar{r}/s_r)\, s_b$ on aligned periods (sample standard deviations). Details give the benchmark's annualized mean and the difference (M² excess). Undefined when fund or benchmark returns are constant (tested on the values, not on a computed standard deviation, which can be rounding noise).
- `batting_average` = share of aligned periods with $r_t > b_t$ strictly; ties do not count as beating. Details give up- and down-market batting averages (benchmark > 0 / < 0), ties and period counts.
- `capture_ratio` = `upside_capture / downside_capture` (geometric, decision 0012). Undefined when downside capture is zero or negative: the fund did not lose on average when the benchmark fell, and a ratio with a non-positive denominator would have a meaningless sign. Both captures remain reported.
- `bull_beta` / `bear_beta`: OLS with intercept on periods with $b_t > 0$ / $b_t < 0$ (zero-benchmark periods belong to neither), at least 3 periods and a non-constant benchmark in the subset; a note is added below 12 periods.
- `timing_ratio` = bull / bear beta; undefined unless bear beta > 0, for the same reason as the capture ratio.

Exact solution for a constant fund. When the fund's aligned returns are identical, the single-index fit is solved exactly (beta 0, alpha equal to the return, zero residuals) instead of numerically. A numerical solve leaves rounding noise such as beta = 1e−17, which previously produced a Treynor ratio of about −8.6e15; ratios dividing by beta or residual risk are now `not_applicable`. R² is undefined for a constant fund (no variance to explain) instead of an `error` status.

Rationale

The m − 2 residual denominator and the arithmetic alpha keep the appraisal ratio consistent with the alpha t-statistic and with alpha, Treynor and the Information ratio of decision 0012. Refusing ratios whose denominators are non-positive or below numerical resolution follows "approximately right rather than exactly wrong": the components are always shown, only the meaningless quotient is withheld, with the reason.

Consequences

- Residual volatility and the appraisal ratio differ from PerformanceAnalytics' `SpecificRisk` and `AppraisalRatio`; the tests pin both relationships.
- Regime betas condition on the sign of the benchmark, which truncates the regressor; they are descriptive, not a test of timing skill. Formal timing models (Treynor-Mazuy, Henriksson-Merton) are not part of methodology 0.1.0.
- The metric catalog grows from 82 to 90 metrics; no existing value changes.

Decision 0028: Factor models with caller-supplied factor returns

Status: Accepted — methodology 0.1.0

Problem

Multi-factor regressions (CAPM with factor data, Fama-French three- and five-factor, Carhart four-factor, custom factors) separate a fund's return into compensated exposures and an unexplained intercept (alpha). Several choices change the numbers: whether the dependent variable is an excess return, how standard errors treat serial correlation (common in hedge-fund and illiquid returns), how the Newey-West lag is chosen, which reference distribution is used, how factor files in percent are scaled, and what happens when factor data have gaps. Each must be fixed in advance, not chosen per fund.

Known definitions

For m aligned periods and k factors (notation: notation):

$$y_t = r_t - r_t^{\mathrm{RF}} = \alpha + \sum_{j=1}^{k} \beta_j f_{j,t} + \varepsilon_t$$

- Fama & French (1993) and Carhart (1997) regress **excess** fund returns on excess-return (market) and zero-investment (size, value, momentum) factors.
- Newey & West (1987): heteroskedasticity- and autocorrelation-consistent (HAC) covariance with Bartlett weights $1 - \ell/(q+1)$.
- Newey & West (1994) and statsmodels' default use the lag rule $q = \lfloor 4(m/100)^{2/9} \rfloor$; R `sandwich::NeweyWest` defaults to a data-dependent (automatic) bandwidth with prewhitening.

PerformanceAnalytics behavior

PerformanceAnalytics offers single-index CAPM functions (decision 0012) but no multi-factor regression with HAC inference. The independent oracle is R `lm + sandwich::NeweyWest(lag = q, prewhite = FALSE, adjust = TRUE) + lmtest::coeftest` (Student t with $m - k - 1$ df). **Verified** on the committed factor fixture (Fama-French-style factors in percent with an RF column and one missing HML month): coefficients, HAC standard errors, t-statistics, p-values, 95% intervals, R^2, adjusted R^2 and residual standard error agree to 10^{-8} relative for the three-factor and CAPM specifications (`tests/cross_library/test_factor_models.py`, reference generated by `validation/r/factor_reference.R`).

QuantStats behavior

QuantStats has no factor regression (it reports single-index alpha/beta only).

Empyrical behavior

empyrical has no multi-factor regression; pyfolio-reloaded's factor attribution (`perf_attrib`) takes pre-computed exposures, a different problem. Not compared.

Academic sources

- Fama, E. F., & French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds.
- Carhart, M. M. (1997). On Persistence in Mutual Fund Performance.
- Fama, E. F., & French, K. R. (2015). A Five-Factor Asset Pricing Model.
- Newey, W. K., & West, K. D. (1987; 1994). HAC covariance; automatic lag selection.
- Asness, C., Krail, R., & Liew, J. (2001). Do Hedge Funds Hedge? (stale prices bias betas down).

OpenFactsheet decision

- **Data are supplied, never downloaded.** `ofs.analyze(returns, factors=...)` accepts a DataFrame, CSV/Excel path or table text with one column per factor. Without factors, every factor-model metric is `not_applicable` with the reason.

- **Risk-free column.** A column named `RF` (any case) is the risk-free rate and is subtracted from fund returns; it is not a factor. Without it the dependent variable is the fund return (risk-free rate 0%, consistent with 0005), and the report says so.
- **One return scale for all factor columns.** The percent/decimal classifier of 0013 runs once on the pooled non-RF factor values; the decision (or a declared `return_scale`) applies to every column, including RF, so a small RF column is never read on a different scale.
- **Alignment.** The periods where the fund and every factor column have a value (calendar period for monthly/quarterly/yearly data, exact date otherwise); missing values are never filled; different regular frequencies are a `FactorAlignmentError` (no resampling). Undated data pair by position only with equal lengths, with a warning.
- **Estimator.** OLS with an intercept (statsmodels). Newey-West covariance, Bartlett kernel, lag $q = \lfloor 4(m/100)^{2/9} \rfloor$, no prewhitening, small-sample factor $m/(m-k-1)$; Student t with $m-k-1$ degrees of freedom; two-sided 95% intervals.
- **Metrics.** `factor_alpha` (per-period intercept × P, arithmetic, as in 0012), `factor_alpha_t_statistic`, `factor_r_squared` (adjusted in details) and `factor_residual_volatility` (regression standard error × √P). Every loading, with HAC standard error, t, p and interval, is in report section 11 and in `factor_analysis` in the JSON.
- **Identification.** At least $k+2$ aligned observations (`insufficient_data` below). A constant factor or perfectly collinear factors make the loadings unidentified (`not_applicable`). A constant fund, or factors that reproduce the fund exactly (residual sum of squares $\leq m\,(16\,\varepsilon \max|\,y\,|)^2$), is reported as an exact fit: loadings as solved, standard errors 0, t-statistics undefined.
- **Confidence.** Policy `factor_regression`: low below 3 years of overlap, moderate below 5; overlap below 90% of the fund history caps at moderate; smoothing risk high → low, moderate → capped at moderate (smoothed returns bias loadings toward zero and alpha upward). A maximum variance inflation factor of 10 or more, or serially correlated residuals (Ljung-Box p < 0.05, from 10 observations), adds a warning.
- **Model label.** Recognized column sets are labelled (e.g. "Fama-French three-factor model"); any other set is a "Custom factor model". The label never changes the estimation.

Rationale

Excess returns are the specification in which the factor literature defines alpha, and using the RF column the caller supplies is data, not a tunable. HAC inference is the practitioner norm for fund regressions because serial correlation is common; a fixed lag rule avoids the data-dependent bandwidth (and prewhitening) that would make results depend on an optimizer's choices. Student t with residual degrees of freedom is conservative in short samples and matches the R oracle. Joint scale detection prevents the one failure that would silently corrupt every coefficient: reading RF (tiny in percent) as decimals while the factors are percent.

Consequences

- Alpha and loadings describe average exposures over the aligned periods; they do not establish causality and change when a different factor set is supplied. Reports therefore always show the model, the factor names and the dependent variable next to the numbers.
- The benchmark section's single-index alpha (i.i.d. standard errors, raw returns, decision 0012) and the factor-model alpha can differ even for a market-only factor file: different dependent variable (excess of RF) and different inference.
- No factor data are bundled or fetched; users must obtain them (for example from the Kenneth R. French Data Library) and pass them explicitly.

Decision 0029: Serial correlation and annualization: variance ratio, Lo-adjusted Sharpe, profit factor

Status: Accepted — methodology 0.1.0

Problem

Square-root-of-time annualization assumes serially uncorrelated returns. Smoothed or illiquid returns (hedge funds, private assets) are positively autocorrelated, so annualized volatility is understated and the annualized Sharpe ratio overstated. CLAUDE.md lists "annualization under serial correlation" as a decision to make explicitly. The conventional ratios stay canonical (decision 0008); the question is which established correction to report beside them, at which horizon, and from which autocorrelation estimator. Separately, the profit factor is widely quoted and was missing from the return summary.

Known definitions

With sample autocorrelations $\hat{\rho}_k$ and horizon q:

$$\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\hat{\rho}_k, \qquad \eta(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\hat{\rho}_k}} = \sqrt{\frac{q}{\mathrm{VR}(q)}}$$

- Lo & MacKinlay (1988): the variance ratio of q-period to one-period returns; asymptotic i.i.d. variance of $\sqrt{n}(\mathrm{VR} - 1)$ is $2(2q-1)(q-1)/(3q)$. Their overlapping estimator is asymptotically equivalent to the autocorrelation form above.
- Lo (2002), "The Statistics of Sharpe Ratios": the q-period Sharpe ratio is $\eta(q)\,\mathrm{SR}$, with $\eta(q) = \sqrt{q}$ only without serial correlation; eq. (33) gives the AR(1) closed form.
- Profit factor: gross gains / gross losses (Pardo 2008); equal to one plus Schwager's gain-to-pain ratio.

PerformanceAnalytics behavior

No function implements Lo's $\eta(q)$ annualization or the variance ratio (`SmoothingIndex` and `Return.GLM` address smoothing through a different model; unsmoothing is outside the canonical report, decision 0014). No profit factor.

QuantStats behavior

`stats.profit_factor` = sum of non-negative returns / |sum of negative returns|. **Verified** to agree on all single-series fixtures (zeros add nothing to either sum). No variance ratio or Lo adjustment.

Empyrical behavior

None of these statistics.

Academic sources

- Lo, A. W., & MacKinlay, A. C. (1988). Stock Market Prices Do Not Follow Random Walks.
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *Financial Analysts Journal* 58(4).
- Getmansky, M., Lo, A. W., & Makarov, I. (2004). Serial correlation and illiquidity in hedge fund returns.
- Pardo, R. (2008). *The Evaluation and Optimization of Trading Strategies*.

OpenFactsheet decision

- **Horizon:** one year, $q = \mathrm{round}(P)$ (12 monthly, 4 quarterly, 52 weekly, 252 or 365 daily) — the horizon the annualized ratios refer to. Undefined for yearly data ($q = 1$), irregular or undated data.
- **Estimator:** the sample autocorrelations of the dependence metrics (full-sample denominator, statsmodels `acf(adjusted=False)`), on simple returns, lags $1..q-1$.
- **Minimum data:** two years ($n \geq 2q$); below that both statistics are `not_applicable` with the reason. Confidence policy `long_horizon_dependence` (years basis): low below 5 years and moderate below 10, because the standard error of $\mathrm{VR}(q)$ is about $\sqrt{4/(3y)}$ for y years.
- `variance_ratio_1y` (dependence, **supplementary**): $\mathrm{VR}(q)$ with the i.i.d. standard error, z-statistic and two-sided p-value in details.
- `sharpe_ratio_autocorrelation_adjusted` (risk-adjusted): per-period Sharpe ratio of decision 0008 times $\eta(q)$; equal to the conventional Sharpe ratio when all $\hat{\rho}_k = 0$; `not_applicable` when $q\,\mathrm{VR}(q) \leq 0$. Not flagged volatility-based: it is the correction for the bias that flag warns about.
- `profit_factor` (returns): $\sum_{r_t>0} r_t / \left|\sum_{r_t<0} r_t\right|$; `not_applicable` without a losing period.
- An interpretation note states both Sharpe ratios and the variance ratio.

Rationale

Lo's adjustment is the established, reference-backed correction for exactly the bias the confidence engine already warns about; reporting it next to the unchanged conventional ratio shows the size of the bias instead of only naming it. Using the same autocorrelation estimator as the lag-1–3 metrics keeps the report internally consistent, and the autocorrelation form makes $\eta(q) = \sqrt{q/\mathrm{VR}(q)}$ an identity, so the two new numbers cannot disagree. A one-year horizon is the only choice that matches the annualized figures; shorter horizons would not correct them. The minimum of two years and the low/moderate thresholds reflect how noisy $q - 1$ estimated autocorrelations are; the adjustment is shown with reduced confidence rather than hidden.

Consequences

- The adjusted Sharpe ratio can be higher than the conventional one when returns mean-revert ($\mathrm{VR} < 1$); the interpretation note says which.

- With daily data the adjustment uses 251 (or 364) autocorrelations; it is labelled low or moderate confidence unless the history is long.
- The profit factor duplicates the information in the gain-to-pain ratio (PF = 1 + GPR); both are reported because both are widely quoted.

Decision 0030: Monthly-return tables from PDF factsheets (embedded text)

Status: Accepted — methodology 0.1.0

Problem

Many track records exist only as PDF factsheets with a monthly-return table (years in rows, January to December in columns, often a YTD column). Reading them must not become a silent source of error. The main hazard is structural, not character recognition: plain text extraction drops empty cells, so the row of a fund launched in April reads like a row starting in January, which shifts every return by three months without any visible failure. Scale (percent or decimal), multiple tables per document, scanned pages and reconciliation with annual totals also need fixed rules.

Known definitions

There is no standard for factsheet tables. CLAUDE.md sections 26, 27 and 51 set the policy: prefer embedded text to OCR, keep extraction optional and pluggable, record confidence and provenance, show the reconstructed table, refuse canonical metrics when extraction confidence is materially inadequate, and never make correctness depend on an opaque model call.

PerformanceAnalytics behavior

Not applicable (no document ingestion).

QuantStats behavior

Not applicable.

Empyrical behavior

Not applicable.

Academic sources

None; this is a data-provenance decision. Candidate libraries were compared for the embedded-text step: pypdf (BSD-3-Clause, pure Python, layout-preserving text extraction), pdfplumber (MIT; depends on pdfminer.six), PyMuPDF (AGPL, excluded by the licensing rule of CLAUDE.md section 21), Camelot (MIT; needs Ghostscript or OpenCV for some modes). pypdf is the lightest dependency that keeps character positions.

OpenFactsheet decision

- **Optional extra** `openfactsheet[pdf]` (pypdf ≥ 4); the core library does not depend on it. `ofs.load` / `ofs.analyze` accept a `.pdf` path.
- **Embedded text only.** Pages are read with layout-preserving extraction. A document without any text (a scan) is refused with an explanation; OCR stays out of the core library.
- **Table detection.** A header line naming the twelve months in order, optionally followed by a total column (YTD, Year, Total, Annual, ...); the year rows beneath it.
- **Column assignment by position.** Each value belongs to the column whose header centre is nearest, within half the median column spacing — never to "the next free month".
- **Refusal (no metrics) when** a value falls under no column, two values fall under one column, a cell is not a number, a year repeats, or the compounded months disagree with the year's total beyond worst-case rounding. The error lists every problem and prints the reconstructed table so the user can correct it and supply it as text.
- **Confidence** `high` when every year with a total reconciles and nothing else is noted; `moderate` when the table has no total column (no cross-check), a total is an arithmetic sum, or months inside the history are blank (missing, never zero). Moderate extractions are used, with the reasons in the input warnings.
- **Scale.** `%` in any cell, or in the three lines above the header (e.g. a title "Monthly returns (%)"), is percent evidence for the classifier of decision 0013; otherwise magnitudes decide.
- **Provenance.** The input transformations name the page, extraction engine and version, value count and confidence; `provenance.extraction` records the engine, the SHA-256 of the file, the page, the checks, the warnings and the reconstructed table.

- **Several tables** in one document become several series (for example fund and benchmark); `analyze` then asks the caller to select one or to use `compare`.

Rationale
Position-based assignment removes the one error that text extraction makes invisible, and reconciling each year with its printed total is a strong, independent check: a shifted or misread month almost never compounds to the printed YTD. Refusing rather than guessing follows principle 1.7 (never silently fail) and section 26 (never calculate silently from low-confidence extraction). Supplied correctly, the same returns give identical metrics whether they came from a PDF or from a CSV (tested).

Consequences
- Tables whose columns are not aligned under their headers (unusual layouts, rotated tables, tables split across pages) are refused rather than guessed.
- A factsheet whose YTD column uses a different definition (e.g. net of a performance fee not reflected in the months) is refused; the message shows the discrepancy.
- Screenshots and scanned PDFs still need transcription until an optional OCR extractor is added; it would feed the same validation and refusal rules.

Decision 0031: Monthly-return tables by OCR (screenshots and scanned PDFs)
Status: Accepted — methodology 0.1.0

Problem
Many track records exist only as screenshots or scanned factsheets. Character recognition (OCR) turns pixels into text with errors that embedded PDF text does not have: misread characters, lost minus signs, low-confidence cells, headers read as nonsense. Metrics must never be calculated silently from such errors (CLAUDE.md sections 26, 27 and 51), analysis must stay local (section 37), and the result must be auditable.

Known definitions
No standard exists. Candidate open-source engines, evaluated September 2026:

Engine	Licence	Install	Notes
RapidOCR 3.x	Apache-2.0 (PaddleOCR models, Apache-2.0)	`pip` (ONNX Runtime)	Word boxes with scores; the wheel ships its default det/rec/cls models
Tesseract	Apache-2.0	system program + `pytesseract`	Offline; weaker on dense numeric tables
PaddleOCR	Apache-2.0	`pip` (PaddlePaddle, large)	Same models as RapidOCR with a heavier runtime
EasyOCR / docTR	Apache-2.0	`pip` (PyTorch, large)	Heavier; no advantage for printed tables
Surya	GPL / restrictive model licence	—	Excluded (section 21)

PerformanceAnalytics behavior
Not applicable (no document ingestion).

QuantStats behavior
Not applicable.

Empyrical behavior
Not applicable.

Academic sources
None; this is a data-provenance decision.

OpenFactsheet decision
- **Engines.** Pluggable `OCREngine` interface. The default is RapidOCR (optional extra `openfactsheet[ocr]`); Tesseract is an alternative (`ofs.load(path, ocr_engine="tesseract")`, extra `openfactsheet[tesseract]`).

- **Local only.** RapidOCR is given the model files bundled in its installed wheel as explicit paths, so its automatic model download can never run. Tests block the network while recognizing. The engine name, version and the SHA-256 of every model file are recorded in `provenance.extraction.ocr`.
- **Scanned PDFs.** Pages without embedded text are rendered with pypdfium2 at 300 DPI (`ocr.dpi`) and recognized. Pages with text keep using the text.
- **One reconstruction.** Recognized words are grouped into lines (vertical overlap of at least half the smaller height) and handed to the same table reconstruction as PDF text (decision 0030): columns by position, gaps missing and never zero, YTD reconciliation.
- **Stricter acceptance for OCR:**
  - any cell recognized with confidence below 0.90 (`ocr.min_cell_confidence`) refuses the table;
  - every year row must have a total cell, and the total must reconcile. A lost minus sign or digit is otherwise undetectable, so a recognized table without a total column is refused;
  - numeric cells that fail to parse get only the documented repairs (`ocr.numeric_repairs`: O/o ⟶ 0, l/I/| ⟶ 1, dash variants ⟶ minus). Each repair is recorded and lowers the extraction confidence to moderate;
  - at most three month headers may be unreadable. They are placed by the even spacing of the others (steps within 15%), and this is recorded and lowers confidence to moderate.
- **Refusal** raises `OCRExtractionError` with every problem and the reconstructed table.
- **Disclosure.** Accepted tables show the source, engine, confidence, checks and the table exactly as read in section 1 of every renderer. The reconciliation message states its tolerance: misreads smaller than the worst-case rounding error cannot be detected.

Rationale

The printed YTD figure is an independent check that a misread cell almost never passes. Requiring it for recognized text is what makes OCR acceptable for canonical metrics. An accepted table therefore carries the printed numbers, whichever engine read it, so the engine choice is recorded but cannot change a metric. Bundled models with recorded hashes keep recognition reproducible and private.

Consequences

- Screenshots without a YTD / total column are refused; they must be transcribed.
- Engine upgrades (new models) change the recorded hashes and may accept or refuse different images, but never produce different accepted values.
- OCR adds heavy optional dependencies (ONNX Runtime, OpenCV); the core install is unaffected.

Decision 0032: Security identifiers and the fund profile of a document

Status: Accepted — methodology 0.1.0

Problem

A factsheet says who the fund is: its name, objective, identifiers (ISIN, WKN, SEDOL, CUSIP, Valor, LEI, FIGI, tickers), currency, launch date and the benchmark it compares itself with. Reading these makes a report traceable to a specific fund. But a misread or guessed identifier would attribute metrics to the wrong security, and text such as “Index” appears in many sentences. The rules must be deterministic, evidence-based and unable to change any metric.

Known definitions

Identifier	Standard	Check
ISIN	ISO 6166	Letters ⟶ 10..35, Luhn over the digits
CUSIP	ANSI X9.6	Positions 2, 4, 6, 8 doubled, digit sums, mod 10
SEDOL	LSE	Weights 1, 3, 1, 7, 3, 9, no vowels, mod 10
LEI	ISO 17442	Letters ⟶ 10..35, number ≡ 1 (mod 97)
FIGI	OMG	Odd positions doubled, digit sums, mod 10; reserved prefixes excluded
WKN, Valor, Bloomberg ticker, RIC	—	No check digit

Embedded national codes:

- A DE ISIN is `DE000` + WKN + check digit.
- A US or CA ISIN contains the CUSIP.

- A GB or IE ISIN contains `00` + SEDOL.
- A CH or LI ISIN contains the zero-padded Valor.

The check digits catch every single-digit substitution. Because a letter becomes two digits before checking, some letter substitutions are **not** caught (for example `US0P78331005` passes, as the property tests show).

PerformanceAnalytics behavior
Not applicable.

QuantStats behavior
Not applicable.

Empyrical behavior
Not applicable.

Academic sources
ISO 6166 (ISIN), ISO 17442 (LEI), the ANSI CUSIP and LSE SEDOL specifications, and the Object Management Group FIGI standard (check-digit definitions).

OpenFactsheet decision
- **Fixed fields.** `DocumentProfile` holds only: fund name, objective, benchmark name(s), currency, launch date and identifiers. Each carries evidence (page, line, snippet, rule) and a confidence. No personal data (manager names, contacts) is extracted.
- **Identifiers:**
  - ISINs are recognized anywhere in the text when their check digit passes.
  - CUSIP, SEDOL, LEI and FIGI shapes are recognized only next to their label.
  - Codes without a check digit (WKN, Valor, Bloomberg ticker, RIC) are taken only immediately after their label.
  - A labelled code whose check digit fails is reported as invalid and never corrected.
  - An identifier matching the national code embedded in an ISIN is linked to that ISIN; a mismatch against the only ISIN is noted.
- **Labels.** Label vocabularies in English, German, French, Spanish and Italian are data (`documents/labels.py`), matched case-insensitively and without accents, longest label first, and only at the start of a line.
- **Fund name.** Taken from a name label (`high`). Otherwise from a title line on page 1 that contains a fund word and is not a sentence, label or table row (`moderate`, rule "title line"); for OCR the tallest such line wins.
- **Objective.** The text after an objective or policy heading, up to the next heading, blank line or table (at most eight lines).
- **Launch date:**
  - ISO dates are taken as printed.
  - Dotted dates are day-first.
  - A slash date is converted only when its day and month cannot be swapped; otherwise it is kept as printed with `low` confidence and a note.
- **Currency.** The first ISO 4217 code after a currency label.
- **Non-interference.** The profile never changes a metric. A benchmark named in the document is displayed, and the benchmark section says its returns were not supplied; it is never used as benchmark returns (CLAUDE.md §23).
- **Where it appears.** Section 1 of every renderer ("Source document") and the JSON key `document`.

Rationale
Check digits and labels are objective evidence. Accepting an unlabelled identifier only when its check digit passes keeps random codes out, and reporting failed check digits rather than fixing them prevents a misread code from being silently turned into another real security. Keeping the evidence with every field lets a reader verify each value against the source document.

Consequences
- Identifiers printed without labels in tables with several share classes are handled by share-class detection (decision 0033), not here.
- Some OCR misreads of letters in ISINs can pass the check digit. The OCR confidence floor and the cross-identifier links reduce, but cannot eliminate, this risk; the evidence snippet is always shown.

- Labels in other languages are not recognized until added to the vocabulary.

Decision 0033: Share classes, table roles and benchmark returns in documents

Status: Accepted — methodology 0.1.0

Problem

Factsheets often list several share classes (currencies, fee levels, distribution policies) but show one monthly-return table, and they frequently print the benchmark's returns next to the fund's: as a second table, or as a second row per year ("2023 Fund …", "Benchmark …"), sometimes with a third "Relative" row. Reading the wrong row as the fund, attributing the table to the wrong class, or silently using the printed benchmark would all misstate the analysis. CLAUDE.md §23 forbids fabricating or inferring a benchmark.

Known definitions

None are standard; layouts vary by issuer. The evidence a reader uses:

- the table's caption or footnote ("Performance of the A EUR Acc class", an ISIN in the footnote);
- the performance section heading;
- the number of classes;
- the currency the returns are stated in.

PerformanceAnalytics behavior

Not applicable.

QuantStats behavior

Not applicable.

Empyrical behavior

Not applicable.

Academic sources

None; this is a data-provenance decision.

OpenFactsheet decision

- **Series-labelled rows.** A table row may carry a label left of the January column. A labelled line without a year continues the previous year. Each distinct label becomes its own series, validated separately (positions, YTD reconciliation).
- **Roles.** Each series is `fund`, `benchmark`, `relative` or `unknown`:
  - A row label decides first: relative/difference words → relative; benchmark or index-provider words → benchmark (unless it also says fund); fund, class or portfolio words → fund; anything else → unknown.
  - The unlabelled series of a labelled table is the fund.
  - A single-series table is a benchmark only when its caption says so without naming the fund.
  - The role and its evidence are recorded in `provenance.extraction`.
- **Relative series** are not return series and are ignored, with a note.
- **Benchmark series are never used automatically:**
  - `ofs.load()` returns them as "`<label>` (benchmark, from document)" in a `ReturnTable`.
  - `ofs.analyze(document)` uses the single fund series and states that the benchmark returns were not used; benchmark-relative metrics need `benchmark=...` passed explicitly.
  - A benchmark series that cannot be read reliably is left out with a note, never refusing the fund.
  - A document whose only table is a benchmark is not analyzed as the fund.
  - Several fund tables (e.g. one per class) are left for the caller to choose.
- **Share classes** are detected from a class table (a header with a class word and "ISIN", then rows up to the first row without an ISIN) or from a line naming a class next to an ISIN (same or next line). Each class records its label, ISINs, currency and evidence.
- **Attribution rules**, in fixed order:
  1. the caption or footnote mentions exactly one class (label or ISIN);
  2. the performance heading above the table mentions exactly one class;
  3. the document has exactly one class;

4. the caption states exactly one currency and exactly one class has it.

The first deciding rule wins. Rules deciding differently, or no deciding rule with several classes, give `ambiguous` with the candidates. There is no fallback such as "the first class listed". Attribution describes the report and never changes a metric.

Rationale
Every rule uses evidence printed in the document, and the result carries that evidence, so a reader can check it. Refusing to guess keeps the report honest when a factsheet leaves the class unclear. Treating printed benchmark returns as data the user may choose to supply, rather than an automatic input, preserves the explicit-benchmark rule while still making them available.

Consequences
- Row labels outside the vocabularies are `unknown`; such series are returned by `ofs.load()` but `analyze()` then asks the caller to choose.
- The planned `share_class=` declaration was not added: ambiguity never affects the metrics, and the candidates are shown. It can be added later if needed.
- Class tables whose rows lack ISINs (identifiers only in other columns or pages) are not detected as classes.

Decision 0034: The reference manual
Status: Accepted — methodology 0.1.0 (documentation only; no reported number changes)

Problem
A reader of an OpenFactsheet result needs to know what every number means, how it was computed, what it assumes, when it misleads and where the definition comes from. The Markdown pages answer this, but they are many files, change with the `main` branch, and cannot be cited or archived as the documentation of the exact version that produced a result. R packages solve this with the reference manual CRAN builds for every release: one PDF, one topic per function, a fixed layout. Few Python packages offer an equivalent.

Known definitions
- **CRAN reference manual**: generated from the package's Rd files; a title page with the DESCRIPTION fields, a table of contents, one topic per documented object (Description, Usage, Arguments, Details, Value, References, See Also, Examples) and an index.
- **Sphinx PDF output**: generated from reStructuredText through LaTeX; needs a TeX installation and is not reproducible byte for byte.

PerformanceAnalytics behavior
Ships the standard CRAN reference manual (one topic per function, with the formulas and the papers they come from).

QuantStats behavior
No reference manual; the documentation is the README and docstrings.

Empyrical behavior
No reference manual; docstrings only.

Academic sources
None; this is a documentation decision.

OpenFactsheet decision
- **One PDF per version**, generated from the code, never edited by hand. It is laid out like a CRAN manual:
  - a title page with the package description;
  - contents;
  - **Overview and usage**;
  - **Functions and classes**: one topic per name in `openfactsheet.__all__` (plus `Comparison`), taken from signatures and Google-style docstrings;
  - **Metrics**: one topic per registered metric, grouped by report section. Each topic has the formula, usage, direction, whether it is annualized, inputs, interpretation, assumptions, when it misleads, minimum data,

the exact calculation and confidence criteria, reference implementations, references and the decisions that discuss it;
  - **Methodology**: the methodology, assumptions, criteria, notation, policies and JSON schema documents;
  - **Methodology decisions**: every decision note;
  - **Validation**;
  - a bibliography with the metrics citing each source;
  - an index.
- **Single source.** Each metric's documentation is one record (`metrics/catalog_docs.metric_record`) from which both its Markdown page and its manual topic are rendered. The hand-written documents are included as they are; links between them become links between manual topics. The data the manual is built from is committed in `docs/manual/data/`, and `scripts/generate_docs.py --check` fails when it is stale.
- **Toolchain: Typst** from PyPI (`pip install openfactsheet[docs]`). It needs no TeX installation, no network and no system fonts:
  - The two Typst packages used are vendored in `docs/manual/packages`: MiTeX (Apache-2.0) turns the registry's LaTeX formulas into Typst math, and cmarker (MIT) renders the Markdown.
  - Fonts are the ones embedded in Typst.
  - The timestamp is the last commit's (or `SOURCE_DATE_EPOCH`), so the same sources give the same bytes.
  - The PDF conforms to PDF/A-2b for archiving.
- **Nothing is skipped.** A formula MiTeX cannot translate, an internal link to a missing topic, or a missing named destination fails the build and CI; the build never falls back to plain text.
- **Stable link targets.** Every topic is a PDF named destination:
  - `metric-<id>` for metrics;
  - `api-<name>` for functions and classes;
  - `decision-<nnnn>` for decisions;
  - `doc-<name>` for documents;
  - `ref-<key>` for sources;
  - `category-<id>` for metric groups.

  `…/openfactsheet-manual.pdf#nameddest=metric-sharpe_ratio` opens a metric's topic.
- **The PDF is committed** as `docs/manual/openfactsheet-manual.pdf`, so the manual of every version is in the repository and readable on GitHub.
  - It records a SHA-256 digest of everything it was built from in its keywords: the template, the data, the documents, the vendored packages, the date and the Typst version.
  - The test suite and CI compare that digest with the current sources, so a change to a metric, docstring or document that is not followed by a rebuild fails.
  - The digest check needs no compiler and gives the same answer on every platform, unlike a byte comparison of compressed PDFs.
  - The date is the version's release date from `CHANGELOG.md` ("unreleased" before release), never the clock or the git history, so committing the PDF does not make it stale.
  - The Typst compiler is pinned exactly.
  - The accessibility structure tree Typst writes is packed into object streams (pikepdf), which keeps the file about 60% smaller.

Rationale

A manual built from the same records as the code cannot drift from it, and one versioned PDF can be cited, archived and linked to from a result. Typst gives a reproducible, dependency-light build that runs anywhere Python runs. Failing on anything that does not compile keeps the manual complete instead of silently degraded.

Consequences

- **Docstrings are now user documentation.** A test requires every argument of every public function and method to be documented.
- **Formulas must be written in the LaTeX subset MiTeX supports.** One formula (`factor_alpha_t_statistic`) changed `\big\lfloor` to `\left\lfloor`; the rendered formula is the same.
- The manual documents the version it was built from. Linking results to the manual of their own version, bundling it with the package and publishing it per version are follow-up work.

Decision 0035: Linking a result to its documentation

Status: Accepted — methodology 0.1.0 (presentation and metadata only; no reported number changes)

Problem

Decision 0034 built the reference manual and gave every topic a stable named destination, precisely so that "a result can link to exactly the page that documents it". Nothing used them: a report said what a number was called and what it was worth, and left the reader to find the definition. A factsheet that travels — pasted into an email, committed as JSON, opened months later — then arrives without its methodology, which is the opposite of what OpenFactsheet exists for (CLAUDE.md sections 1.4 and 43).

Two things have to be decided: **what** a result links to, and **which version** of it.

Known definitions

- **CRAN / R help**: `?SharpeRatio` opens the topic of the installed version. The documentation is bound to the version in use, not to a web page.
- **Sphinx / `__doc__` links**: HTML docs point at "latest" or "stable"; a result printed by an older version links to documentation that no longer describes it.
- **DOIs and permalinks**: a reference is pinned to an immutable version.

PerformanceAnalytics behavior

Prints numbers; the manual is consulted separately. Charts and tables carry no links.

QuantStats behavior

Tear sheets carry no links to documentation.

Empyrical behavior

Returns bare floats.

Academic sources

None; this is a documentation and provenance decision.

OpenFactsheet decision

- **Every result carries a link to the topic that defines it.** `MetricResult.documentation` is the URL of its manual topic (`…/openfactsheet-manual.pdf#nameddest=metric-sharpe_ratio`), and it is part of the JSON of every metric. A metric that could *not* be computed links to its definition too: the reason is only understandable next to it.
- **The report carries the links of its own version.** `FactsheetReport.documentation` is a `DocumentationLinks` object (the manual, the metric catalog, the documentation tree, and a topic by metric, category, API name, decision, document or bibliography key), and the JSON has a `documentation` block with the manual, the tag, the source ref and whether the version is released.
- **Links are pinned to the version that produced the result**, never to "latest". A released version links to its own release assets and to its sources at `v<version>`; before release there is no such tag, so results link to the rolling `manual-latest` pre-release built from `main` and say so in words. The two constants that express this (`MANUAL_TAG`, `SOURCE_REF`) are checked against `CHANGELOG.md` by the test suite, so a release cannot ship links to a manual that was never published for it.
- **Anchors have one source.** `openfactsheet.manual` builds every named destination (`metric_anchor`, `category_anchor`, `api_anchor`, `decision_anchor`, `document_anchor`, `reference_anchor`); no renderer, export or test spells one out.
- **Every renderer links what it can.** The HTML and Markdown reports link each metric name, including the names listed in sections whose metrics are not applicable (no benchmark supplied, no factors supplied) — all 97 in both. Plain text cannot carry a link per row, so the terminal report gives the manual of its version and the destination pattern, on one unwrapped line so that it stays copyable.
- **A link may not point at a topic that does not exist.** The test suite resolves every metric's and category's destination against the named destinations the committed manual data declares, and decision 0034's build already fails on a missing destination.

Rationale

The manual is the deliverable that explains the numbers; a result that does not reach it leaves the explanation optional. Pinning the link to the version that computed the number is what makes it an audit trail rather than a convenience: conventions change between methodology versions, and a reader checking an old report must see the

definition that was actually used. Keeping the link derivable from a metric's id (one anchor function, no table of URLs) means the links cannot drift from the registry as metrics are added.

Consequences

- The JSON report gains a `documentation` block and one `documentation` key per metric; the schema document and the golden reports record it. Schema version 0.1 is unchanged: the keys are additions.
- Reports are no longer self-contained with respect to the manual: the link is only resolvable online. The number, its status, its warnings and the assumptions it used remain in the report itself, as before.
- Bundling the manual with the package, so that a result can be read next to its documentation offline, remains follow-up work.

Validation

Reference implementations

OpenFactsheet implements every metric independently from its mathematical definition and then checks the numbers against established implementations. Agreement across languages and libraries is evidence of correctness; disagreements are either fixed or recorded here and in the methodology decisions, with a test that pins the relationship.

Libraries and licenses

Library	Version checked	License	Role
PerformanceAnalytics (R)	2.0.4 on R 4.3.1	GPL-2 | GPL-3	External numerical oracle only. Function bodies were read to understand conventions; **no code was copied** into OpenFactsheet.
empyrical-reloaded	0.5.12	Apache-2.0	Python cross-check (optional `validation` extra).
QuantStats	0.0.81	Apache-2.0	Python cross-check (optional `validation` extra).
skfolio	1.0.6 and 1.4.1+	BSD-3-Clause	Python cross-check for DaR / CDaR / EDaR / EVaR (optional `validation` extra). The drawdown-path convention changed in 1.4.1 (difference 15), so both are asserted.
R `stats` (base R)	R 4.3.1	GPL-2 | GPL-3	External oracle only: `acf`, `Box.test`, `cor`, `lm`, `t.test`, `pchisq` in the R harness.
statsmodels	installed version	BSD-3-Clause	Runtime dependency: OLS, ACF, Ljung-Box.
SciPy	installed version	BSD-3-Clause	Runtime dependency: skewness, kurtosis, normal / Student-t / chi-squared distributions; `jarque_bera` and `ttest_1samp` used as test references.
NumPy	installed version	BSD-3-Clause	Runtime dependency: numerical kernels, quantiles.
pandas	installed version	BSD-3-Clause	Runtime dependency: labeled time series and input handling.

Reference libraries are never runtime dependencies of the core package.

How the checks run

R harness. `validation/r/performanceanalytics_reference.R` reads the committed synthetic fixtures in `tests/fixtures/` and writes `tests/fixtures/reference/performanceanalytics.json`. The JSON is committed, so the Python test suite does not need R. Regenerate from the repository root with:

```
Rscript validation/r/performanceanalytics_reference.R
```

The benchmark case aligns fund (month-end dates) and benchmark (month-start dates) by calendar month in R, the same convention OpenFactsheet uses.

Python checks. `tests/cross_library/test_performanceanalytics.py` compares against the committed R values. `tests/cross_library/test_python_references.py` calls empyrical, QuantStats and skfolio directly and is skipped automatically when they are not installed (`pip install -e ".[validation]"`); its SciPy checks (Jarque-Bera, one-sample t-test) always run.

Where PerformanceAnalytics stops with an error (PSR and MinTRL for a negative Sharpe ratio), the harness records `null` and the tests assert OpenFactsheet's documented behavior instead.

Fixtures. `monthly_fund_10y` (120 months, fat-tailed), `negative_fund` (60 months), `smoothed_hedge_fund` (96 months, smoothed, large losses), `benchmark_case_fund` / `benchmark_case_benchmark` (102 months each, 96 overlapping). Generated once by `scripts/make_fixtures.py` and committed.

Tolerance for agreement: relative 1e-9 / absolute 1e-12 (Ljung-Box, Jarque-Bera and t-test p-values: relative 1e-7).

Per-metric agreement

"Agrees" means asserted equal within tolerance on every applicable fixture. "Differs (pinned)" means the test asserts the documented relationship or inequality. "—" means not compared.

Metric id	PerformanceAnalytics	empyrical	QuantStats	Notes
cumulative_return	Agrees (`Return.cumulative`)	Agrees (`cum_returns_final`)	—	
cagr	Agrees (`Return.annualized`, geometric)	Agrees (`annual_return`)	Agrees with `stats.cagr(periods=12)`; default differs (pinned)	QuantStats' default `periods` assumes daily observations.
annualized_arithmetic_mean	Agrees (`Return.annualized(geometric=FALSE)`)	—	—	
volatility	Agrees (`StdDev`)	—	—	
annualized_volatility	Agrees (`StdDev.annualized`)	Agrees (`annual_volatility`)	—	
downside_deviation	Agrees (`DownsideDeviation(MAR=0)` x sqrt(12))	Agrees (`downside_risk`)	—	PA reports per period.
sharpe_ratio	Agrees (`SharpeRatio.annualized(geometric=FALSE)`)	Agrees (`sharpe_ratio`)	Agrees (`stats.sharpe`)	PA default (`geometric=TRUE`) differs (pinned).
sortino_ratio	Agrees (`SortinoRatio` x sqrt(12))	Agrees (`sortino_ratio`)	—	PA reports per period.
calmar_ratio	Agrees (`CalmarRatio`)	Agrees (`calmar_ratio`)	Differs (pinned): `stats.calmar` = default CAGR / |MDD|	Inherits QuantStats' daily CAGR default.
martin_ratio	Agrees with `Return.annualized / sqrt(mean(Drawdowns^2))`	—	—	PA `MartinRatio` inherits the `DrawdownPeak` issue.
omega_ratio	Agrees (`Omega(L=0, method="simple")`)	Agrees (`omega_ratio`)	—	
max_drawdown	Agrees with `-maxDrawdown`	Agrees (`max_drawdown`)	Agrees (`stats.max_drawdown`)	PA reports a positive magnitude.
average_drawdown	Agrees with `-AverageDrawdown`	—	—	PA reports a positive magnitude.
longest_drawdown_duration	Differs (pinned): equal for recovered episodes, PA one period longer for an ongoing episode	—	—	`findDrawdowns` `length`.
ulcer_index	Agrees with `sqrt(mean(Drawdowns^2))`	—	—	PA `UlcerIndex` differs (percent-input assumption).
pain_index	Agrees with `mean(abs(Drawdowns))`	—	—	PA `PainIndex` same issue.
var_historical_95	Agrees (`VaR(p=0.95, method="historical")`)	—	Differs (pinned): `stats.value_at_risk` is parametric normal	Verified equal to the normal 5% quantile with sample mean/std.
expected_shortfall_historical_95	Agrees (`ES(p=0.95, method="historical")`)	—	—	PA uses `r < q`; OpenFactsheet `r <= q` (no ties in fixtures).
tail_ratio	—	Agrees (`tail_ratio`)	Agrees (`stats.tail_ratio`)	
skewness	Agrees with `skewness(method="moment")` x sqrt(n(n−1))/(n−2)	—	Agrees (`stats.skew`)	PA `"sample"` = G1 x (n/(n−1))^1.5 (pinned).
excess_kurtosis	Agrees (`kurtosis(method="sample_excess")`)	—	Agrees (`stats.kurtosis`)	
best_period / worst_period	—	—	Agrees with max / min of the series	

Metric id	PerformanceAnalytics	empyrical	QuantStats	Notes
positive_period_pct	—	—	Agrees (`stats.win_rate`)	Fixtures contain no zero returns; QuantStats excludes zeros from the denominator.
autocorrelation_lag1..3	Agrees with R `acf`	—	—	
ljung_box_pvalue	Agrees with R `Box.test(lag=3, type="Ljung-Box")`	—	—	
beta	Agrees (`CAPM.beta`)	Agrees (`beta`)	—	
alpha	Agrees with `CAPM.alpha` x 12	Differs (pinned): empyrical = (1 + alpha/12)^12 − 1	—	Arithmetic vs geometric annualization.
correlation	Agrees with R `cor`	—	—	
r_squared	Agrees with R `lm` R^2	—	—	
tracking_error	Agrees (`TrackingError`)	—	—	
information_ratio	Differs (pinned) from `InformationRatio`; agrees with arithmetic IR computed in R	—	Agrees after scaling: `stats.information_ratio` x sqrt(12)	PA uses geometric `ActivePremium`; QuantStats is per period.
active_return	Differs (pinned) from `ActivePremium`	—	—	Arithmetic vs difference of geometric returns.
upside_capture	Differs (pinned) from `UpDownRatios(Capture, Up)`	Agrees (`up_capture`)	—	PA divides sums of returns.
downside_capture	Differs (pinned) from `UpDownRatios(Capture, Down)`	Agrees (`down_capture`)	—	PA = sum ratio over `Rb <= 0` (verified).
treynor_ratio	Differs (pinned): PA = geometric annualized return / beta (verified)	—	—	OpenFactsheet = arithmetic annualized mean / beta (verified against PA's beta).
gain_to_pain_ratio	—	—	Agrees (`stats.gain_to_pain_ratio`)	Monthly fixtures.
payoff_ratio	—	—	Agrees (`stats.payoff_ratio`)	
kappa_3_ratio	Agrees (`Kappa(MAR=0, l=3)`)	—	—	Per period.
upside_potential_ratio	Agrees (`UpsidePotentialRatio(MAR=0, method="full")`); differs (pinned) from the default `method="subset"`	—	—	Per period.
semi_deviation	Agrees (`SemiDeviation` x sqrt(12))	—	—	PA reports per period.
downside_potential	Agrees (`DownsideDeviation(MAR=0, potential=TRUE)`)	—	—	Per period.
mean_absolute_deviation	Agrees (`MeanAbsoluteDeviation`)	—	—	
sterling_ratio	Agrees (`SterlingRatio(excess=0.1)`)	—	—	
burke_ratio	Differs (pinned) from `BurkeRatio`; agrees with `Return.annualized /`	—	—	PA uses runs of negative returns and a percent-input scaling.

Metric id	PerformanceAnalytics	empyrical	QuantStats	Notes
	`sqrt(sum(findDrawdowns depths^2))`			
pain_ratio	Agrees with `Return.annualized / mean(abs(Drawdowns))`	—	—	PA `PainRatio` inherits the `DrawdownPeak` issue.
gaussian_var_95	Agrees (`VaR(p=0.95, method="gaussian")`)	—	—	Population standard deviation; QuantStats `stats.value_at_risk` uses the sample std (see var_historical_95 row).
modified_var_95	Agrees (`VaR(p=0.95, method="modified")`)	—	—	Cornish-Fisher with population moments.
jarque_bera_pvalue	Agrees with a manual R computation from `skewness(method="moment")` and `kurtosis(method="excess")`	—	—	Also agrees with SciPy `stats.jarque_bera` (statistic 1e-9, p-value 1e-7).
probabilistic_sharpe_ratio	Agrees (`ProbSharpeRatio(refSR=0, ignore_kurtosis=FALSE)`) on fixtures with a positive Sharpe ratio	—	—	PA stops with an error when SR < 0; OpenFactsheet reports PSR < 0.5 (asserted).
minimum_track_record_length	Agrees (`MinTrackRecord(refSR=0, p=0.95, ignore_kurtosis=FALSE)`) on fixtures with a positive Sharpe ratio	—	—	`not_applicable` when SR <= 0 (asserted); PA stops with an error.
sharpe_ratio_standard_error	— (same variance formula; PA's interval is 90% two-sided)	—	—	Checked against a hand computation in unit tests.
mean_return_standard_error	Agrees with R `t.test` (CI bounds, t-statistic; p-value to 1e-7)	—	—	Also agrees with SciPy `stats.ttest_1samp` p-value.

Phase 2b metrics. Columns as above; skfolio results are in the Notes column.

Metric id	PerformanceAnalytics / R	empyrical	QuantStats	Notes
drawdown_at_risk_95	—	—	—	skfolio `drawdown_at_risk` differs (pinned): lower order statistic `quantile(-D, 0.95, "inverted_cdf")`; OpenFactsheet = linear `Q_0.05(D)` (asserted).
cdar_95	Differs (pinned): `CDD` = 5% quantile over all `findDrawdowns` runs (episodes and zero runs)	—	—	Agrees with skfolio `cdar` on the canonical path when 0.05·n is an integer (n = 60 and 120).
edar_95	—	—	—	Agrees with skfolio `edar` on the canonical path (rel 1e-7, solver tolerance).
evar_95	—	—	—	Agrees with skfolio `evar` (rel 1e-7).
modified_es_95	Agrees (`ES(p=0.95, method="modified")`)	—	—	Population moments, full expansion.
median_drawdown	Agrees with the median of `findDrawdowns` negative-episode depths	—	—	

Metric id	PerformanceAnalytics / R	empyrical	QuantStats	Notes
average_drawdown_duration	Agrees with the mean `findDrawdowns` length after subtracting 1 from an open last episode	—	—	Open-episode +1 convention (difference 3).
max_peak_to_trough_duration	Agrees with max `findDrawdowns peaktotrough`	—	—	
average_recovery_duration	Agrees with the mean `findDrawdowns recovery` over recovered episodes (open last episode dropped)	—	—	`not_applicable` asserted when the only episode is open (negative_fund).
recovery_factor	—	—	Differs (pinned): `stats.recovery_factor` = arithmetic sum of returns / |MDD|	OpenFactsheet = compounded cumulative return / |MDD| (asserted).
geometric_mean_return	Agrees (`mean.geometric`)	—	—	
interquartile_range	Agrees with R `IQR` (type 7)	—	—	
best_calendar_year	Agrees with R compounded returns of years with 12 monthly observations	—	—	Monthly fixtures; partial years excluded by OpenFactsheet.
worst_calendar_year	Agrees (same computation)	—	—	
positive_calendar_years_pct	Year set and every complete-year return agree (to 1e-9)	—	—	Share itself derived from the same years.
best_rolling_year	Agrees with the max of `zoo::rollapply(R, 12, prod(1 + x) - 1)`	—	—	Monthly fixtures.
worst_rolling_year	Agrees with the min of the same rolling series	—	—	
positive_rolling_year_pct	Agrees with the share of positive windows; window count agrees	—	—	

Benchmark-relative additions (decision 0027), on the `benchmark_case` fixture (96 aligned months).

Metric id	PerformanceAnalytics / R	empyrical	QuantStats	Notes
capture_ratio	— (ratio of PA's summed captures would differ as in item 8)	Agrees (`up_down_capture`)	—	
batting_average	Up/down components agree (`UpDownRatios(method="Percent", side="Up"/"Down")`)	—	—	The overall share has no PA counterpart; checked by hand in unit tests.
residual_volatility	Agrees with R `summary(lm(...))$sigma` x sqrt(12); differs (pinned) from `SpecificRisk`	—	—	PA divides the residual sum of squares by m, not m − 2 (verified).
appraisal_ratio	Differs (pinned) from `AppraisalRatio(method="appraisal")`	—	—	PA = `CAPM.jensenAlpha` (geometric) / `SpecificRisk` (verified); OpenFactsheet = alpha / residual_volatility (asserted).
m_squared	Agrees with `Modigliani` x 12	—	—	PA reports per period.
bull_beta	Agrees (`CAPM.beta.bull`)	—	—	
bear_beta	Agrees (`CAPM.beta.bear`)	—	—	
timing_ratio	Agrees (`TimingRatio`)	—	—	

Factor models (decision 0028), on the `factor_case` fixture (Fama-French-style factors in percent with an RF column and one missing HML month; 107 aligned months for the three-factor model, 108 for CAPM). Reference: R `lm` + `sandwich::NeweyWest(lag = floor(4 (m/100)^(2/9)), prewhite = FALSE, adjust = TRUE)`

- `lmtest::coeftest` / `coefci`, generated by `validation/r/factor_reference.R` into `tests/fixtures/reference/factor_models.json`.

Quantity	R reference	Notes
alpha and every loading	Agrees to 1e-9 relative (`coef(lm)`)	
Newey-West standard errors, t, p, 95% CI	Agree to 1e-8 relative	Bartlett kernel, fixed lag, no prewhitening, small-sample factor m/(m−k−1), Student t with m−k−1 df. `NeweyWest()` defaults (automatic bandwidth, prewhitening) differ by design.
factor_alpha	Agrees with `coef(lm)[1] × 12`	
factor_alpha_t_statistic	Agrees with `coeftest(...)[1, 3]`	
factor_r_squared	Agrees with `summary(lm)$r.squared` (adjusted: `$adj.r.squared`)	
factor_residual_volatility	Agrees with `summary(lm)$sigma × sqrt(12)`	

Serial-correlation additions (decision 0029):

Metric id	Reference	Notes
profit_factor	Agrees with QuantStats `stats.profit_factor` on all single-series fixtures	
variance_ratio_1y	Autocorrelations agree with statsmodels `acf(adjusted=False)` to 1e-12; the standard error equals Lo & MacKinlay's (1988) asymptotic variance	No maintained Python/R implementation of the autocorrelation form was available (R `vrtest` implements the overlapping estimator, asymptotically equivalent).
sharpe_ratio_autocorrelation_adjusted	Lo (2002) eq. (33) AR(1) closed form reproduced to 1e-12 from the autocorrelation sum	No library implementation identified.

The R harness records per-episode `findDrawdowns` columns, whether the last episode is open, complete-year returns, rolling 12-month returns, modified ES, `mean.geometric` and `IQR`; `test_phase2b_agreement` rebuilds each statistic from them. The skfolio risk measures are compared on OpenFactsheet's canonical drawdown path, which its own compounded path matches from 1.4.1 (difference 15).

R's base `t.test` is used as an additional independent reference for the mean-return interval; it is not part of PerformanceAnalytics.

Metrics without a reference comparison (median_return, arithmetic_mean, current_drawdown, drawdown_count, time_underwater_pct, negative_period_pct, streaks, zero_return_pct) are checked against hand computations in `tests/unit/metrics/test_formulas.py` and property tests.

Known convention differences

1. **Sharpe ratio (PA default).** `SharpeRatio.annualized` uses the annualized geometric return in the numerator unless `geometric = FALSE`. See 0008.
2. **Ulcer / Pain / Martin (PA).** `DrawdownPeak` divides returns by 100, which assumes percent inputs; on decimal inputs PA's values differ from the textbook formulas applied to PA's own `Drawdowns()` path. See 0007.
3. **Drawdown duration (PA).** `findDrawdowns` counts an episode still open at the end of the data one period longer than OpenFactsheet.
4. **Skewness (PA `"sample"`).** Uses the population standard deviation inside a sample-style formula: `G1 x (n/(n−1))^1.5`. See 0011.
5. **Expected Shortfall (PA).** Averages returns strictly below the quantile; OpenFactsheet includes a return equal to the quantile. Identical unless ties occur. See 0010.
6. **Alpha (empyrical).** Annualized geometrically, `(1 + a)^P − 1`; OpenFactsheet uses `a x P`.
7. **Information ratio, active return, Treynor (PA).** PA uses geometric annualized returns; OpenFactsheet uses arithmetic means.

8. **Capture ratios (PA).** PA divides sums of returns and treats zero-benchmark periods as down periods; OpenFactsheet compounds geometrically (Morningstar-style) and excludes zero-benchmark periods from both sides. See 0012.
9. **QuantStats defaults.** `stats.cagr` (and therefore `stats.calmar`) assumes daily observations unless `periods` is passed; `stats.value_at_risk` is parametric (normal); `stats.information_ratio` is not annualized. All three are verified in `tests/cross_library/test_python_references.py`. `stats.cvar` conditions on the parametric threshold (observed, not asserted).
10. **Upside potential ratio (PA default).** `UpsidePotentialRatio` defaults to `method = "subset"` (divides by the number of gains / losses); OpenFactsheet uses the full-sample denominator, equal to `method = "full"`. See 0016.
11. **Burke ratio (PA).** `BurkeRatio` treats runs of consecutive negative returns as drawdowns and multiplies returns by 0.01; OpenFactsheet uses peak-to-recovery episode depths. See 0017.
12. **Pain ratio (PA).** `PainRatio` inherits the `PainIndex` / `DrawdownPeak` percent-input issue (item 2).
13. **PSR / MinTRL (PA defaults).** `ProbSharpeRatio` and `MinTrackRecord` default to `ignore_kurtosis = TRUE` (kurtosis fixed at 3); OpenFactsheet keeps the sample kurtosis and agrees with `ignore_kurtosis = FALSE`. PA stops with an error when the reference Sharpe ratio exceeds the observed one; OpenFactsheet reports PSR below 0.5 and MinTRL as `not_applicable`. See 0019.
14. **Sharpe confidence interval (PA).** `ProbSharpeRatio(...)$sr_confidence_interval` uses `qnorm(p)` with p = 0.95, a 90% two-sided interval on the per-period Sharpe ratio; OpenFactsheet reports a two-sided 95% interval on the annualized Sharpe ratio (not numerically compared).
15. **Drawdown path (skfolio) — resolved upstream in 1.4.1.** Up to skfolio 1.4.0, `get_drawdowns(compounded=True)` started its running peak at W_1, so a first-period loss was not a drawdown. Since 1.4.1 the starting wealth is the first peak, which is OpenFactsheet's convention. The default remains `compounded=False` (uncompounded cumulative sums). OpenFactsheet counts W_0 = 1 as a peak and compounds; the test asserts the path of whichever version is installed, because the `validation` extra is not pinned. See 0021.
16. **Drawdown-at-Risk (skfolio).** `drawdown_at_risk` uses the lower order statistic (inverted-CDF quantile); OpenFactsheet uses the linear quantile of decision 0010 (asserted).
17. **CDaR (skfolio).** `cdar` uses a fractional Rockafellar-Uryasev weighting of the tail; it equals OpenFactsheet's mean of `D_t <= Q_0.05(D)` only when 0.05·n is an integer.
18. **Conditional Drawdown (PA).** `CDD` returns the 5% quantile over all `findDrawdowns` runs (the drawdown episodes and the zero-return runs between them), inverted to a positive number: neither a conditional expectation nor computed on the drawdown path, so it is not a CDaR reference (verified in the R harness and pinned by a test).
19. **Recovery factor (QuantStats).** `stats.recovery_factor` divides the arithmetic sum of returns by |MDD|; OpenFactsheet uses the compounded cumulative return (asserted). See 0025.
20. **Open drawdown episodes (PA).** As in item 3, `findDrawdowns` `length` and `recovery` count an open last episode one period longer; the tests subtract 1 (length) or drop the episode (recovery) before comparing.
21. **Specific risk and appraisal ratio (PA).** `SpecificRisk` is the population (m-denominator) standard deviation of the regression residuals, annualized; OpenFactsheet uses the regression standard error (m – 2), consistent with the OLS standard errors. `AppraisalRatio(method = "appraisal")` divides Jensen's alpha computed from geometric annualized returns by `SpecificRisk`; OpenFactsheet divides the arithmetic annualized OLS alpha by its residual volatility (both relationships asserted). See 0027.

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Status	Meaning
established	The originator and seminal work are well documented in the literature.
attributed	Commonly attributed to the named originator(s), but the primary source, its date or the exact original definition is not fully verified.
convention	A standard statistic or practitioner convention with no single originator.
uncertain	The originator is not reliably established; the entry names where the measure is documented instead.
pending	No curated attribution yet; the registry's references are listed.

Return summary

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
`cumulative_return`	Cumulative return	Standard statistic (no single originator)	—	yes	convention	—
`cagr`	Annualized return (CAGR)	Standard statistic (no single originator)	CFA Institute (2020)	yes	convention	Annualization conventions follow GIPS.
`arithmetic_mean`	Mean return per period	Standard statistic (no single originator)	—	yes	convention	—
`geometric_mean_return`	Geometric mean return per period	Standard statistic (no single originator)	—	yes	convention	—
`annualized_arithmetic_mean`	Annualized mean return (arithmetic)	Standard statistic (no single originator)	—	yes	convention	—
`median_return`	Median return per period	Standard statistic (no single originator)	—	yes	convention	—
`best_period`	Best period	Standard statistic (no single originator)	—	yes	convention	—
`worst_period`	Worst period	Standard statistic (no single originator)	—	yes	convention	—
`gain_to_pain_ratio`	Gain-to-pain ratio	Jack D. Schwager	Schwager (2012)	—	attributed	Popularized by Schwager; a similarly named trading statistic predates it.
`payoff_ratio`	Payoff ratio (average win / average loss)	Practitioner convention (no single originator)	—	yes	convention	—

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
profit_factor	Profit factor (gross gains / gross losses)	Practitioner convention (no single originator)	Pardo (2008)	—	convention	—

Risk summary

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
volatility	Volatility per period	Standard statistic (no single originator)	Markowitz (1952)	yes	convention	Variance as the measure of portfolio risk: Markowitz (1952).
annualized_volatility	Annualized volatility	Standard statistic (no single originator)	Markowitz (1952)	yes	convention	Square-root-of-time scaling; see Lo (2002) on its limits.
mean_absolute_deviation	Mean absolute deviation	Standard statistic (no single originator)	—	yes	convention	—
interquartile_range	Interquartile range	Standard statistic (no single originator)	—	yes	convention	—
downside_deviation	Downside deviation (annualized)	Harry Markowitz (semivariance); Frank A. Sortino and co-authors (deviation below a target)	Markowitz (1959); Sortino & van der Meer (1991); Sortino & Price (1994)	yes	established	—
semi_deviation	Semi-deviation (annualized)	Harry Markowitz	Markowitz (1959)	yes	established	—
downside_potential	Downside potential (LPM 1)	Vijay S. Bawa; Peter C. Fishburn (lower partial moments)	Bawa (1975); Fishburn (1977)	yes	attributed	The name ‘downside potential’ for the first lower partial moment follows Bacon.

Drawdowns

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
max_drawdown	Maximum drawdown	Practitioner drawdown statistic (no single originator)	—	yes	convention	Statistical properties: Magdon-Ismail & Atiya.
average_drawdown	Average drawdown (per episode)	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
current_drawdown	Current drawdown	Practitioner drawdown	—	yes	convention	—

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
		statistic (no single originator)				
longest_drawdown_duration	Longest drawdown duration	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
drawdown_count	Number of drawdown episodes	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
time_underwater_pct	Time under water	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
ulcer_index	Ulcer Index	Peter G. Martin	Martin & McCann (1989)	yes	established	Introduced by Martin in 1987; published in Martin & McCann (1989).
pain_index	Pain Index	Thomas Becker (Zephyr Associates)	—	yes	attributed	Attribution as given in Bacon; primary source not verified.
median_drawdown	Median drawdown (per episode)	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
average_drawdown_duration	Average drawdown duration	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
max_peak_to_trough_duration	Longest decline (peak to trough)	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
average_recovery_duration	Average recovery time (trough to recovery)	Practitioner drawdown statistic (no single originator)	—	yes	convention	—
drawdown_at_risk_95	Drawdown-at-Risk (95%)	Alexei Chekhlov, Stanislav Uryasev, Michael Zabarankin	Chekhlov et al. (2005)	—	established	—
cdar_95	Conditional Drawdown-at-Risk (95%)	Alexei Chekhlov, Stanislav Uryasev, Michael Zabarankin	Chekhlov et al. (2005)	—	established	—

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
edar_95	Entropic Drawdown-at-Risk (95%)	Dany Cajas (EDaR), building on Amir Ahmadi-Javid (EVaR)	Cajas (2021); Ahmadi-Javid (2012)	—	established	—

Risk-adjusted performance

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
sharpe_ratio	Sharpe ratio	William F. Sharpe	Sharpe (1966)	—	established	Introduced as the 'reward-to-variability ratio'; restated by Sharpe (1994).
sortino_ratio	Sortino ratio	Frank A. Sortino and co-authors	Sortino & van der Meer (1991); Sortino & Price (1994)	yes	established	—
calmar_ratio	Calmar ratio	Terry W. Young	Young (1991)	yes	established	—
martin_ratio	Martin ratio (Ulcer Performance Index)	Peter G. Martin	Martin & McCann (1989)	yes	established	Also known as the Ulcer Performance Index.
omega_ratio	Omega ratio	Con Keating, William F. Shadwick	Keating & Shadwick (2002)	yes	established	—
kappa_3_ratio	Kappa 3 ratio	Paul D. Kaplan, James A. Knowles	Kaplan & Knowles (2004)	yes	established	—
upside_potential_ratio	Upside potential ratio	Frank A. Sortino, Robert van der Meer, Auke Plantinga	Sortino et al. (1999)	yes	established	—
burke_ratio	Burke ratio	Gibbons Burke	Burke (1994)	yes	established	—
sterling_ratio	Sterling ratio	Attributed to Deane Sterling Jones	Kestner (1996)	yes	attributed	Several definitions are in use; the canonical variant is documented on the metric page.
pain_ratio	Pain ratio	Thomas Becker (Zephyr Associates)	—	yes	attributed	Attribution as given in Bacon; primary source not verified.
recovery_factor	Recovery factor	Practitioner convention (no single originator)	—	yes	convention	—
sharpe_ratio_autocorrelation_adjusted	Sharpe ratio (serial-correlation adjusted, Lo 2002)	Andrew W. Lo	Lo (2002)	—	established	—

Tail behavior

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
var_historical_95	Value at Risk (historical, 95%)	Industry risk-management practice (no single originator)	J.P. Morgan/ Reuters (1996)	—	convention	Popularized by J.P. Morgan's RiskMetrics.
expected_shortfall_historical_95	Expected Shortfall / CVaR (historical, 95%)	Carlo Acerbi, Dirk Tasche; R. Tyrrell Rockafellar, Stanislav Uryasev (CVaR)	Acerbi & Tasche (2002); Rockafellar & Uryasev (2000)	yes	established	—
tail_ratio	Tail ratio (95th / 5th percentile)	Practitioner convention (no single originator)	—	yes	convention	—
gaussian_var_95	Value at Risk (Gaussian, 95%)	Industry risk-management practice (no single originator)	J.P. Morgan/ Reuters (1996)	—	convention	Variance-covariance VaR as popularized by RiskMetrics.
modified_var_95	Modified VaR (Cornish-Fisher, 95%)	Peter Zangari; Laurent Favre, José-Antonio Galeano	Zangari (1996); Favre & Galeano (2002)	—	established	—
modified_es_95	Modified Expected Shortfall (Cornish-Fisher, 95%)	Kris Boudt, Brian G. Peterson, Christophe Croux	Boudt et al. (2008)	—	established	—
evar_95	Entropic Value at Risk (95%)	Amir Ahmadi-Javid	Ahmadi-Javid (2012)	—	established	—

Distribution

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
skewness	Skewness	Standard statistic (no single originator)	Joanes & Gill (1998)	—	convention	Moment statistic; estimator per Joanes & Gill.
excess_kurtosis	Excess kurtosis	Standard statistic (no single originator)	Joanes & Gill (1998)	—	convention	Moment statistic; estimator per Joanes & Gill.
jarque_bera_pvalue	Jarque-Bera normality p-value	Carlos M. Jarque, Anil K. Bera	Jarque & Bera (1987)	—	established	—

Consistency

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
positive_period_pct	Positive periods	Practitioner convention (no single originator)	—	yes	convention	—
negative_period_pct	Negative periods	Practitioner convention (no single originator)	—	yes	convention	—

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
longest_winning_streak	Longest run of positive periods	Practitioner convention (no single originator)	—	yes	convention	—
longest_losing_streak	Longest run of negative periods	Practitioner convention (no single originator)	—	yes	convention	—
best_calendar_year	Best calendar year	Practitioner convention (no single originator)	CFA Institute (2020)	yes	convention	—
worst_calendar_year	Worst calendar year	Practitioner convention (no single originator)	CFA Institute (2020)	yes	convention	—
positive_calendar_years_pct	Positive calendar years	Practitioner convention (no single originator)	CFA Institute (2020)	yes	convention	—
best_rolling_year	Best rolling one-year return	Practitioner convention (no single originator)	—	yes	convention	—
worst_rolling_year	Worst rolling one-year return	Practitioner convention (no single originator)	—	yes	convention	—
positive_rolling_year_pct	Positive rolling one-year windows	Practitioner convention (no single originator)	—	yes	convention	—

Serial dependence & smoothing

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
autocorrelation_lag1	Autocorrelation (lag 1)	Standard statistic (no single originator)	—	—	convention	Use as a smoothing diagnostic: Getmansky, Lo & Makarov.
autocorrelation_lag2	Autocorrelation (lag 2)	Standard statistic (no single originator)	—	—	convention	Use as a smoothing diagnostic: Getmansky, Lo & Makarov.
autocorrelation_lag3	Autocorrelation (lag 3)	Standard statistic (no single originator)	—	—	convention	Use as a smoothing diagnostic: Getmansky, Lo & Makarov.
ljung_box_pvalue	Ljung-Box p-value (3 lags)	Greta M. Ljung, George E. P. Box	Ljung & Box (1978)	—	established	—
zero_return_pct	Zero-return periods	Data-quality diagnostic (no single originator)	—	—	convention	Motivated by the illiquidity analysis of Getmansky, Lo & Makarov.
variance_ratio_1y	Variance ratio (one-year horizon)	Andrew W. Lo, A. Craig MacKinlay	Lo & MacKinlay (1988)	—	established	—

Benchmark-relative analysis

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
beta	Beta	William F. Sharpe; John Lintner (CAPM)	Sharpe (1964); Lintner (1965)	yes	established	Use in fund performance evaluation: Treynor (1965), Jensen (1968).
alpha	Alpha (annualized, CAPM regression)	Michael C. Jensen	Jensen (1968)	yes	established	—
r_squared	R-squared	Standard statistic (no single originator)	—	yes	convention	—
correlation	Correlation with benchmark	Standard statistic (no single originator)	—	yes	convention	—
tracking_error	Tracking error (annualized)	Practitioner convention (no single originator)	—	yes	convention	—
active_return	Active return (annualized, arithmetic)	Practitioner convention (no single originator)	—	—	convention	—
information_ratio	Information ratio	Jack L. Treynor, Fischer Black (appraisal ratio); Richard C. Grinold (information ratio)	Treynor & Black (1973); Grinold (1989)	—	attributed	The origin of the name and of the benchmark-relative form is not settled; see Goodwin (1998).
treynor_ratio	Treynor ratio	Jack L. Treynor	Treynor (1965)	yes	established	—
upside_capture	Upside capture ratio	Industry practice (Morningstar methodology)	Morningstar, Inc	yes	convention	—
downside_capture	Downside capture ratio	Industry practice (Morningstar methodology)	Morningstar, Inc	yes	convention	—
capture_ratio	Capture ratio (upside / downside capture)	Industry practice (Morningstar methodology)	Morningstar, Inc	yes	convention	—
batting_average	Batting average (share of periods beating the benchmark)	Practitioner convention (no single originator)	Morningstar, Inc	yes	convention	—
residual_volatility	Residual (specific) volatility (annualized)	Standard statistic (no single originator)	Treynor & Black (1973)	yes	convention	Denominator of the Treynor-Black appraisal ratio.
appraisal_ratio	Appraisal ratio	Jack L. Treynor, Fischer Black	Treynor & Black (1973)	yes	established	—

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
m_squared	M-squared (Modigliani risk-adjusted return)	Franco Modigliani, Leah Modigliani	Modigliani & Modigliani (1997)	yes	established	—
bull_beta	Bull-market beta	Frank J. Fabozzi, Jack C. Francis	Fabozzi & Francis (1977)	yes	established	—
bear_beta	Bear-market beta	Frank J. Fabozzi, Jack C. Francis	Fabozzi & Francis (1977)	yes	established	—
timing_ratio	Timing ratio (bull beta / bear beta)	Not reliably established	Fabozzi & Francis (1977)	yes	uncertain	Ratio of the Fabozzi-Francis bull and bear betas, as described in Bacon.

Factor analysis

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
factor_alpha	Factor-model alpha (annualized)	Michael C. Jensen (alpha); Eugene F. Fama, Kenneth R. French; Mark M. Carhart (factor models)	Jensen (1968); Fama & French (1993); Carhart (1997)	—	established	—
factor_alpha_t_statistic	Factor-model alpha t-statistic (Newey-West)	Whitney K. Newey, Kenneth D. West (HAC standard errors)	Newey & West (1987)	—	established	—
factor_r_squared	Factor-model R-squared	Standard statistic (no single originator)	Fama & French (1993)	—	convention	—
factor_residual_volatility	Factor-model residual volatility (annualized)	Standard statistic (no single originator)	Fama & French (1993)	—	convention	—

Statistical confidence

ID	Metric	Originator(s)	Seminal work	Bacon (secondary)	Status	Notes
probabilistic_sharpe_ratio	Probabilistic Sharpe Ratio (vs 0)	David H. Bailey, Marcos López de Prado	Bailey & Lopez de Prado (2012)	—	established	—
sharpe_ratio_standard_error	Sharpe ratio standard error (annualized)	Andrew W. Lo; Elmar Mertens (non-normal correction)	Lo (2002); Mertens (2002)	—	established	—
minimum_track_record_length	Minimum track record length	David H. Bailey, Marcos López de Prado	Bailey & Lopez de Prado (2012)	—	established	—
mean_return_standard_error	Mean return standard error (per period)	Standard statistic (no single originator)	Student (1908)	—	convention	—

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